

# PISA 2025 Assessment and Analytical Framework



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# Foreword

The OECD Programme for International Student Assessment (PISA) assesses the extent to which 15-year-old students near the end of their compulsory education have acquired the knowledge and skills that are essential for full participation in modern societies. The assessment does not just ascertain whether students can reproduce knowledge; it also examines how well students can extrapolate from what they have learned and can apply that knowledge in unfamiliar settings, both in and outside of school. This approach reflects the fact that modern economies reward individuals not for what they know, but for what they can do with what they know.

The assessment focuses on the core domains of reading, mathematics and science. Students' proficiency in an innovative domain is also assessed; in PISA 2025, this domain was Learning in the Digital World, which focused on students' capacity to engage in a self-regulated process of knowledge building and problem-solving using computational tools.

PISA was implemented for the first time in 2000 and has been implemented every three years since then, with the exception of the 2021 cycle that was postponed to 2022 because of the COVID-19 pandemic.

In PISA 2025, science was the major domain of assessment, as it was in 2006 and 2015. The revised framework and assessment place more emphasis on the evaluation of designs for scientific enquiry, as well as on the critical interpretation of scientific data and evidence. It includes a new competency area focused on students' ability to research, evaluate and use scientific information for decision making and action. It also broadens the contextual measures from attitudes to the concept of "science identity", which aims at describing students' engagement in science in a comprehensive way. Finally, it has a strengthened focus on sustainability and environmental education.

# Acknowledgements

PISA is the product of a collaborative effort between OECD and the governments of both OECD Member countries and its Partner countries/economies. The assessments are developed co-operatively, agreed by participating countries/economies, and implemented by national organisations. The co-operation of students, teachers and principals in participating schools has been crucial to the success of PISA during all stages of development and implementation.

The science framework was developed by the 2025 Science Expert Group (SEG), chaired by Jonathan Osborne (Stanford University, United States), with the following members: Doris Jorde (University of Oslo, Norway), Joe Krajcik (Michigan State University, United States), Yew Jin Lee (National Institute of Education, Singapore), Knut Neuman (Leibniz-Institute for Maths & Science Education, Germany), Nascira Ramia (Universidad San Francisco de Quito, Ecuador), Russell Tytler (Deakin University, Australia) and Young Shin Park (Chosun University, South Korea). The Environmental Sciences Expert Group was chaired by Peta White (Deakin University, Australia), with the following members: Nicole Ardoin (Stanford University, United States), Chris Eames (University of Waikato, New Zealand) and Martha Monroe (University of Florida, United States). Dave Leach (Oxford University Press) was Project Director on the development of the framework. The work of the PISA 2025 SEG builds on previous versions of the PISA Science Framework.

The Learning in the Digital World (LDW) Framework was developed by Natalie Foster, Emma Linsenmayer and Mario Piacentini of the OECD Secretariat, with the Learning in the Digital World Expert Group (LDWEG). The LDWEG included Gautam Biswas (Vanderbilt University, United States), Engin Bumbacher (University of Teacher Education, Vaud, Switzerland), Frank Goldhammer (Leibniz Institute for Research and Information in Education – DIPF, Germany), Sanna Järvelä (University of Oulu, Finland), Samuel Greiff (formerly University of Luxembourg, Luxembourg, now Technical University of Munich, Germany), Ido Roll (Technion – Israel Institute of Technology, Israel) and Chris Piech (Stanford University, United States). The framework also benefitted from the contributions of several researchers involved in the wider development of the LDW instrumentation, including Katarzyna Bobrowicz (University of Luxembourg, Luxembourg), Richard Davis (formerly Swiss Federal Institute of Technology Lausanne, now KTH Royal Institute of Technology, Sweden), Nicole (Hutchins) Meijer (Accingo, United States), Leonard Tetzlaff and Lothar Persic-Beck (Leibniz Institute for Research and Information in Education – DIPF, Germany) Anke Maria Weber (University of Luxembourg, Luxembourg) and Megan Welsh (University of California Davis, United States). Elisabeth McClure and Bo Stjerne Thomson (LEGO Foundation) also reviewed and provided comments on an early version of the framework.

The framework for the PISA 2025 questionnaires was developed by Daniel Salinas and Miyako Ikeda at the PISA Secretariat, with the guidance of and input from the questionnaire expert group. The questionnaire expert group was chaired by Therese Hopfenbeck (Melbourne Graduate School of Education, Australia). This group included Jonas Bertling (Education Testing Service), David Kaplan (University of Wisconsin-Madison, United States), Nordin Abd Razak, (Universiti Sains, Malaysia) and Nina Jude (Ruprecht-Karls-Universität Heidelberg, Germany). For certain content included in the

questionnaires, the Secretariat consulted external experts, including Elise Huillery, Alexandre Touw, Larry Suter, Mark Bray, Nutsa Kobakhidze, Carol Dweck, David S. Yeager, and Mary C. Murphy.

The publication was prepared by the OECD Secretariat, guided by Andreas Schleicher and Yuri Belfali. Juliana González Rodríguez and Yeonsoo Kim co-ordinated the production of the framework with Tue Halgreen, with contributions from Catalina Covacevich, Miyako Ikeda and Daniel Salinas. Sophie Limoges provided communications support, and Cassandra Morley edited the report.

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# 1 What is PISA?

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This chapter introduces the PISA 2025 Assessment and Analytical Framework. It describes what makes PISA unique, the key features of the PISA 2025 test, an overview of what is assessed in each domain, as well as the evolution of reporting student performance and the context questionnaires. It also presents how PISA is the result of a collaborative effort.

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The OECD Programme for International Student Assessment (PISA), now in its ninth cycle, seeks to determine what is important for citizens to know and be able to do. PISA assesses the extent to which 15-year-old students near the end of their compulsory education have acquired the knowledge and skills that are essential for full participation in modern societies.

The assessment focuses on the core school subjects of reading, mathematics, and science. Students' proficiency in an innovative domain is also assessed; in 2025, this domain was Learning in the Digital World. In 2025, PISA also introduced an optional assessment of English as a foreign language focused on reading, speaking and listening skills. The PISA assessment does not just ascertain whether students can reproduce knowledge; it also examines how well students can extrapolate from what they have learned and can apply that knowledge in unfamiliar settings, both in and outside of school. This approach reflects the fact that modern economies reward individuals not for what they know, but for what they can do with what they know.

PISA is an ongoing programme that monitors trends in the knowledge and skills that students around the world, and in demographic subgroups within each country, have acquired. In each round of PISA, one of the core domains is tested in more detail, taking up roughly one-half of the total testing time. The major domain in 2025 was science, as it was in 2006 and 2015. Reading was the major domain in 2000, 2009 and 2018, and mathematics was the major domain in 2003, 2012 and 2022.

With this alternating schedule of major domains, a thorough analysis of achievement in each of the three core areas is presented at regular intervals, while an analysis of trends is offered in every cycle for all three core domains.

Through questionnaires distributed to students and school principals, and optional questionnaires distributed to parents and teachers, PISA also gathers information about students' home background, their approaches to learning and their learning environments. Combined with the information gathered through the various questionnaires, the PISA assessment provides three main types of outcomes:

- Basic indicators that provide a profile of the knowledge and skills of students
- Indicators derived from the questionnaires that show how such skills relate to various demographic, social, economic and educational variables
- Indicators on trends that show changes in outcomes and their distributions, and in relationships between student-, school- and system-level background variables and outcomes

Policymakers around the world use PISA findings to gauge the knowledge and skills of the students in their own country/economy compared with those in other participating countries/economies, establish benchmarks for improvements in the education provided and/or in learning outcomes, and understand the relative strengths and weaknesses of their own education systems.

This publication presents the theory underlying the PISA 2025 assessment. It includes the frameworks for assessing the major domain, science (Chapter 2), and the innovative domain, Learning in the Digital World (Chapter 3). These chapters outline the knowledge content that students need to acquire in each domain, the processes that students need to be able to perform, and the contexts in which this knowledge and these skills are applied. They also discuss how each domain is assessed. The publication concludes with the frameworks for the questionnaires distributed to students, school principals, parents and teachers (Chapter 4). The frameworks for the reading and mathematics assessments are not included in this publication, as they received their last major updates when they were the major domain of assessment (2018 for reading, 2022 for mathematics). Links to access the full versions of these frameworks are included as Annexes (Annex A and Annex B, respectively). The framework for the Foreign Language Assessment will be made available by May 2027 and a link to the most recent framework for this domain can be found in Annex C.

### Box 1.1. Key features of PISA 2025

#### The content

PISA not only assesses whether students can reproduce knowledge, but also whether they can extrapolate from what they have learned and apply their knowledge to new situations. It emphasises the mastery of processes, the understanding of concepts, and the ability to function in various types of situations.

The PISA 2025 survey focused on science, with reading and mathematics as minor domains of assessment. For the first time, Learning in the Digital World was assessed as an innovative domain. PISA 2025 also included an assessment of English as a foreign language, which was optional for countries and economies.

#### The students

Over 760 000 students completed the assessment in 2025 across participating countries and economies, representing about 33 million 15-year-olds in the schools of the 91 countries and economies.

#### The assessment

PISA 2025 used computer-based tests, with the exception of countries that were unable to deliver the test digitally.

Each student took a two-hour test. Test items were a mixture of multiple-choice questions and questions requiring students to construct their own responses. The items were organised in groups based on a passage setting out a real-life situation. Over 15 hours of test items were used, with different students taking different combinations of test items.

Students also answered a background questionnaire that took 35 minutes to complete. The questionnaire sought information about the students themselves, their home lives, and their school and learning experiences. School principals completed a questionnaire that covered the school system and the learning environment.

To obtain additional information, some countries/economies decided to distribute a questionnaire to teachers to learn about their training and professional development, their teaching practices and their job satisfaction. In some countries/economies, optional questionnaires were distributed to parents, who were asked to provide information on their perceptions of and involvement in their child's school, their support for learning in the home, and their own engagement with science.

Countries/economies that took part in the Foreign Language Assessment also administered questionnaire modules focused on language learning in and outside of school.

### What makes PISA unique

PISA is the most comprehensive and rigorous international programme to assess student performance and to collect data on the student, family and institutional factors that can help explain differences in performance. Decisions about the scope and nature of the assessments and the background information to be collected are made by leading experts in participating countries, and are steered jointly by governments on the basis of shared, policy-driven interests. Substantial efforts and resources are devoted to achieving cultural and linguistic breadth and balance in the assessment materials. Stringent quality-

assurance mechanisms are applied in translation, sampling and data collection. As a consequence, results from PISA have a high degree of validity and reliability.

PISA's unique features include:

- **Policy orientation**, which links data on student learning outcomes with data on students' backgrounds and attitudes towards learning, and on key factors that shape their learning in and outside of school; this exposes differences in performance and identifies the characteristics of students, schools and education systems that perform well
- **Innovative concept of “literacy”**, which refers to students' capacity to apply knowledge and skills, and to analyse, reason and communicate effectively as they identify, interpret and solve problems in a variety of situations
- **Relevance to lifelong learning**, as PISA asks students to report on their motivation to learn, their beliefs about themselves and their learning strategies
- **Regularity**, which enables countries to monitor their progress in meeting key learning objectives
- **Breadth of coverage**, which, in PISA 2025, encompassed 38 OECD Member countries and 53 Partner countries and economies

## The PISA 2025 test

The PISA 2025 assessment was conducted principally via computer, as has been the case since 2015. Paper-based assessment instruments were provided for countries that could not test their students by computer; but the paper-based assessment was focused on reading, mathematics and science trend items (i.e. those items that had already been used in prior paper-based assessments) with limited new development. Regardless of the delivery mode, the assessment consisted of a cognitive testing session of 120 minutes, followed by a student questionnaire session of approximately 35 minutes.

The PISA 2025 design continued to leverage the innovations made possible by the change to computer delivery as the main mode of assessment in PISA 2015, such as the multi-stage adaptive testing (MSAT) design that was first introduced in PISA 2018. For the first time, all three PISA core domains were assessed with an MSAT design in PISA 2025. More detail on the MSAT design employed in PISA 2025 can be found in the *PISA 2025 Technical Report* (OECD, forthcoming).

## An overview of what is assessed in each domain

Box 1.2 presents definitions of the three domains assessed in PISA 2025. The definitions all emphasise the functional knowledge and skills that allow one to participate fully in society. Such participation requires more than just the ability to carry out tasks imposed externally by, for example, an employer; it also involves the capacity to participate in decision making. The more complex tasks in PISA require students to reflect on and evaluate material, not just answer questions that have one correct answer.

### Box 1.2. Definitions of the domains

**Science:** The science assessment in PISA covers the competencies to explain phenomena scientifically; to construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically; and to research, evaluate and use scientific information for decision making and action.

**Mathematics:** The mathematics assessment in PISA measures an individual's capacity to reason mathematically and to formulate, employ and interpret mathematics to solve problems in a variety of real-world contexts. The framework includes concepts, procedures, facts and tools to describe, explain and predict phenomena. It assists individuals in acknowledging the role that mathematics plays in the world and to make the well-founded judgements and decisions.

**Reading:** The reading assessment in PISA measures an individual's capacity to understand, use, evaluate, reflect on and engage with texts in order to achieve one's goals, develop one's knowledge and potential, and participate in society.

**Science** was the major domain assessed in PISA 2025. The assessment was renewed for this cycle, and measures how well countries are preparing their students with an understanding of science and how it produces reliable knowledge. This is crucial for citizens who need to make informed personal decisions about science-related phenomena – such as those related to health and the environment – to engage in action within their families, local communities and wider societies.

The assessment measures to what extent young people have the competencies to:

- **Explain phenomena scientifically:** Recognise, construct, apply and evaluate explanations for a range of natural and technological phenomena
- **Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically:** Appraise and evaluate ways of investigating questions scientifically, and interpret and evaluate scientific data critically
- **Research, evaluate and use scientific information for decision making and action:** Obtain scientific information on a specific global, local or personal science-related issue and evaluate its credibility, potential flaws and the implications for personal and communal decisions

**Mathematical literacy** is defined as students' ability to reason mathematically and to formulate, employ and interpret mathematics to solve problems in a variety of real-world contexts.

PISA assesses students' performance in mathematics through questions related to:

- **Mathematical reasoning and problem-solving processes:** Includes the mathematical processes that describe what individuals do to connect the context of the problem with mathematics and thus solve the problem.

Mathematical reasoning (both deductive and inductive) involves evaluating situations, choosing strategies, drawing logical conclusions, developing and describing solutions, and recognising how those solutions can be applied. It is enabled by some key understandings that underpin school mathematics, and is the core of mathematical literacy. Included among these key understandings are: understanding quantity, number systems and their algebraic properties; appreciating the power of abstraction and symbolic representation; seeing mathematical structures and their regularities; recognising functional relationships between quantities; using mathematical modelling as a lens onto the real world (e.g. those arising in the physical, biological, social, economic and behavioural sciences); and understanding variation as the heart of statistics. Regarding problem solving, PISA defines three categories of processes: formulating situations mathematically;

employing mathematical concepts, facts, procedures and reasoning; and interpreting, applying and evaluating mathematical outcomes.

- **Content:** There are four categories (change and relationships; quantity; space and shape; and uncertainty and data) that are closely aligned with the content that is typically found in national school mathematics curricula content strands, such as numbers, algebra, functions, geometry and data handling.
- **Context:** The aspect of an individual's world in which the problems are placed. The framework identifies four contexts: personal, occupational, societal and scientific.

**Reading literacy** is defined as students' ability to understand, use, evaluate, reflect on and engage with text to achieve their purposes.

PISA assesses students' performance in reading through questions that involve a variety of:

- **Processes (aspects):** Students are not assessed on the most basic reading skills, as it is assumed that most 15-year-old students will have acquired these. Rather, students are expected to demonstrate their proficiency in locating information, including both accessing and retrieving information within a piece of text, and searching for and selecting relevant text; understanding text, including both acquiring a representation of the literal meaning of text and constructing an integrated representation of text; and evaluating and reflecting on text, including both assessing its quality and credibility, and reflecting on content and form.
- **Text formats:** PISA uses both single-source and multiple-source texts; static and dynamic texts; continuous texts (organised in sentences and paragraphs); non-continuous texts (e.g. lists, forms, graphs or diagrams); and mixed texts.
- **Situations:** These are defined by the use for which the text was constructed. For example, a novel, personal letter or biography is written for people's personal use; official documents or announcements are for public use; a manual or report is for occupational use; and a textbook or worksheet is for educational use. Since some students may perform better in one type of reading situation than another, a range of reading situations is included in the test.

**Learning in the Digital World** was the innovative domain introduced in PISA 2025. This assessment is based on the understanding that education technologies can transform how students learn because they offer new opportunities to explore complex phenomena and to create digital representations of their ideas, which they can tinker with and share with others. Yet, not enough attention has been devoted to teaching the foundational skills and attitudes that students need to develop in order to be active and autonomous users of technology to confront open-ended, real-world problems.

The PISA 2025 Learning in the Digital World assessment focuses on two competencies that are essential to learning with technologies:

- **Self-regulated learning:** This refers to the monitoring and control of one's metacognitive, cognitive, behavioural, motivational and affective processes while learning.
- **Computational and scientific enquiry practices:** These refer to the capacity to use digital tools to explore systems, represent ideas and solve problems with computational logic.

**Foreign language skills (English)** were tested for the first time in PISA 2025. The framework defines foreign language proficiency as the ability to use a foreign language to communicate effectively. It requires a combination of communicative language competencies and general competencies that allow the foreign language learner to perform communicative language activities (reception, production, interaction and mediation). The PISA 2025 Foreign Language Assessment assessed 15-year-old students' foreign language reading comprehension, listening comprehension and spoken production in English.

## The evolution of reporting student performance in PISA

Results from PISA are reported using scales. Initially, the average score across OECD countries for all three of the core domains was 500 with a standard deviation of 100, which meant that two-thirds of students across OECD countries scored between 400 and 600 points. These scores represent degrees of proficiency in a particular domain. Scores in subsequent cycles of PISA are calibrated so as to be directly comparable to those in previous cycles; hence the average score across OECD countries in subsequent cycles has fluctuated slightly around the original 500.

Reading literacy was the major domain in 2000, and the reading scale was divided into five proficiency levels of knowledge and skills. The main advantage of this approach is that it is useful for describing what substantial numbers of students can do with tasks at different levels of difficulty. Since then, the assessment frameworks and item pools have been expanded to cover higher and lower difficulty levels, in some cases increasing the number of proficiency levels, as well as to cover subscales that can be reported separately.

## The context questionnaires

To gather contextual information, PISA asks students and the principals of their schools to respond to questionnaires. These take about 35 and 45 minutes, respectively, to complete. The responses to the questionnaires are analysed with the assessment results to provide at once a broader and more nuanced picture of student, school and system performance. Chapter 4 presents the questionnaire framework in detail. The questionnaires from all assessments since PISA's inception are available on the PISA website: [www.oecd.org/pisa/](http://www.oecd.org/pisa/).

The questionnaires seek information about:

- Students and their family background, including their economic, social and cultural capital
- Aspects of students' lives, such as their attitudes towards learning, their habits and life in and outside of school, and their family environment
- Aspects of schools, such as the quality of the schools' human and material resources, public and private management and funding, decision-making processes, staffing practices, and the school's curricular emphasis and extracurricular activities offered
- Context of instruction, including institutional structures and types, class size, classroom and school climate, and reading activities in class
- Aspects of learning, including students' interest, motivation and engagement

In PISA 2025, three additional questionnaires were offered as options:

- **ICT familiarity questionnaire**, focusing on the availability and use of ICT and on students' ability to carry out computer tasks and their attitudes towards computer use
- **Parent questionnaire**, focusing on parents' perceptions of and involvement in their child's school, their support for learning at home, school choice, their child's career expectations, and their background (immigrant/non-immigrant)
- **Teacher questionnaire**, which asks about teachers' initial training and professional development, their beliefs and attitudes, and their teaching practices

The contextual information collected through the student, school and optional questionnaires comprises only a part of the information available to PISA. Indicators describing the general structure of education systems (their demographic and economic contexts, such as their costs, enrolments, school and teacher

characteristics, and some classroom processes) and their effect on labour market outcomes are routinely developed and applied by the OECD (e.g. in the regular OECD publication, *Education at a Glance*).

## A collaborative project

PISA is the result of a collaboration between the OECD Secretariat, international contractors, experts and teams in participating countries and economies. The assessments are developed co-operatively, agreed by participating countries/economies, and implemented by national centres in each participating country/economy. The co-operation of students, teachers and principals in participating schools has been crucial to the success of PISA during all stages of development and implementation.

The PISA Governing Board (PGB), composed of representatives at the senior policy level from all participating countries/economies, determines the policy priorities for PISA. It also oversees adherence to these priorities during the implementation of the programme. The PGB sets priorities for developing indicators, establishing assessment instruments and reporting results. Experts from participating countries/economies also serve on working groups tasked with linking PISA policy objectives with the best available technical expertise in the different assessment domains. By participating in these expert groups, countries/economies ensure that the instruments are internationally valid and take into account differences in cultures and education systems.

Participating countries/economies implement PISA at the national level through National Centres managed by National Project Managers, subject to the agreed administration procedures. National Project Managers play a vital role in ensuring that the implementation is of high quality. They also verify and evaluate survey results, analyses, reports and publications.

The frameworks were developed under the guidance of expert groups (see Annex D). The OECD Secretariat has overall managerial responsibility for the programme, monitors its implementation on a day-to-day basis, acts as the secretariat for the PGB, builds consensus among countries, and serves as the interlocutor between the PGB and the international contractors charged with implementation. The OECD Secretariat is also responsible for producing the indicators, and for the analysis and preparation of the international reports and publications, in co-operation with the contractors and in close consultation with participating countries/economies at both the policy (PGB) and implementation (National Project Managers) levels.

# 2 PISA 2025 Science Framework

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Science was the major domain of assessment in PISA 2025. The science framework presented in this chapter reflects recent developments in science education and the growing importance of science for understanding complex global challenges. This chapter outlines the organisation of the domain in four interrelated aspects: 1) scientific competencies, including explaining phenomena, evaluating and designing scientific enquiry, and interpreting scientific information and using it for decision making and action; 2) the organisation of scientific knowledge in the framework, encompassing content, procedural and epistemic knowledge; 3) the role of environmental science competencies and “science identity”; and 4) the contexts in which students encounter science-related issues in personal, local, national and global settings. Items that illustrate the framework are presented at the end of the chapter.

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## Introduction

The fundamental goal of PISA is to measure students' ability to use their knowledge and understanding, which is demonstrated by their ability to manifest a particular set of competencies. The competencies developed by scientific education are perceived to be a key educational outcome (Rychen and Salganik, 2001<sup>[1]</sup>) and are defined in terms of the ability to use scientific knowledge and information interactively, i.e. "the ability to engage with science-related issues, and with the ideas of science, as a reflective citizen", and to use them for informed decision making. In this chapter, that outcome is framed in terms of three specific competencies that are major goals for science education for all students. These competencies form the basis for the 2025 international assessment of 15-year-olds and are a response to the question: What is important for young people to know, value and be able to do in situations requiring the use of scientific and technological knowledge?

This chapter provides a description and rationale for the framework that forms the basis of the instrument to assess the outcomes of science education, which is the major domain for PISA 2025. Previous PISA frameworks for the science assessment (OECD, 1999<sup>[2]</sup>; OECD, 2003<sup>[3]</sup>; OECD, 2006<sup>[4]</sup>; OECD, 2017<sup>[5]</sup>) have elaborated a conception of scientific literacy as the central construct for science assessment. In this chapter, the outcomes of scientific education are defined in terms of a set of three competencies that an individual would be expected to display. These competencies form the basis of the construct to be tested (William, 2010<sup>[6]</sup>). The framework for PISA 2025 refines and extends the previous competencies by building on the PISA 2015 framework, and the thinking of the expert group convened to develop this framework.

## Science & why it matters

Science education matters. At one level, there is the economic imperative faced by all OECD countries of ensuring that they educate the next generation of scientists. However, such a goal alone does not justify compulsory education in science for all students from kindergarten to age 15 or higher. Science earns its place at the curriculum table along with mathematics and literacy for two fundamental reasons. First, scientific knowledge represents a great cultural and intellectual achievement. In the space of a little over 400 years, our understanding of the living and non-living world has been totally transformed, improving our well-being and health in ways that were simply unimaginable then. Science has fundamentally changed our understanding of the universe we inhabit, the nature of the human body, the matter that surrounds us, how we came to be, the transmission of disease and its prevention and much, much more. Education is how this cultural achievement is shared with the next generation.

Somebody who has experienced an education in science should have some sense of the intellectual achievement that the sciences represent. They should be able to identify the major scientific ideas that have had a transformative effect on our culture and explain why. In addition, they should be able to point to the transcendent nature of this knowledge and the phenomena that it can explain; to wonder at its possible beauty, and to recognise the intellectual creativity that imagined the world not as it appears to be, but as one which often defies common sense (Wolpert, 1992<sup>[7]</sup>). For example, the idea that day and night is caused by a spinning Earth rather than a moving Sun or the idea that you look like your parents because every cell in your body carries a chemically coded message to enable our replication.

Second, a knowledge and understanding of science, and the ways in which it generates reliable knowledge, is crucial for citizens who need to make informed personal decisions about science-related phenomena such as health and the environment, to engage in action within their families, local communities and wider societies. This is particularly important in the 21st Century when humanity faces an uncertain future as it enters the Anthropocene, an era in which human impact is significantly changing Earth's systems, and a knowledge of science matters at the individual, regional and global levels as we seek to address these impacts.

It is science that informs us that life as we know it is in crisis, with climate change and biodiversity loss – arguably the greatest issues of our time – impacting all species, many irrevocably, and precipitating the sixth mass extinction (Dirzo et al., 2014<sup>[8]</sup>). The challenges we face as a population of 7 billion people with finite natural resources include the need for clean water, sufficient food supply, managing and preventing disease, producing energy sustainably, along with many others resulting predominantly from human-generated climate change (IPCC, 2021<sup>[9]</sup>). Dealing with all of these challenges will require a major contribution from science and technology, as well as other knowledge systems.

Education and schools are essential for preparing youth to deal with these challenges by developing their agency in the Anthropocene era. Scientifically educated citizens and societies understand how to evaluate and judge the credibility of scientific information and expertise to inform their actions; actions that are needed to bring about change both at the local level, where individuals and institutions may be faced with decisions about practices that affect their own health and food supplies, and at the national level. Science education can also inform and contribute to developing an ethic of care and justice that recognises the interdependence of living things, and by enabling systems thinking and creativity to design effective strategies to bring about the change required.

At the personal level, too, individuals are faced with choices about how to act; for example, whether to adopt a vegetarian diet, whether to farm sustainably, whether to get vaccinated, and many more. To make such choices requires a citizenry capable of evaluating the information, experts and other voices involved in these issues. And, as argued by the European Commission, the solutions to political and ethical dilemmas involving science and technology “cannot be the subject of informed debate unless young people possess certain scientific awareness” (European Commission, 1995, p. 28<sup>[10]</sup>). Moreover, while “this does

not mean turning everyone into a scientific expert”, it does mean providing them with a science education that would “enable them to fulfil an enlightened role in making choices which affect their environment and to understand in broad terms the social implications of debates between experts” (European Commission, 1995, p. 28<sub>[10]</sub>). This means understanding some of the social norms and practices science has established to ensure the knowledge it offers is trustworthy, such as the importance of scientific consensus, the markers of expertise, the limits to certainty and the mechanisms that the sciences have developed for avoiding error.

The explosion of information online has made the competency to critically evaluate scientific reports in the media and social networks ever more important, particularly when science is central to so many of these issues. And particularly when 85% of adolescents indicate that they use Google, two-thirds of them use YouTube, and about one-half use Wikipedia. Today at least one-quarter of young people receive news and up-to-date information via social media or online news sources. New technologies accelerate and amplify communication, both of information and disinformation. As a result, false news can travel faster, farther, and more broadly on X than established reliable sources (Vosoughi, Roy and Aral, 2018<sub>[11]</sub>). One unfortunate consequence is that scientific information has sometimes become politicised and challenged by those who either claim scientific credentials or by those who seek to deny its validity (Oreskes and Conway, 2010<sub>[12]</sub>; Michaels, 2020<sub>[13]</sub>).

Thus, a specific goal of science education should be to develop the competency to “research, evaluate and use scientific information for decision making and action”. This competency can only be developed by explicit teaching of some of the epistemic features and social practices that sustain the practice of science. For instance, this competency requires an understanding of why and when to trust science (Oreskes, 2019<sub>[14]</sub>). Today we live in a society where we are all epistemically dependent on expertise (Hardwig, 1985<sub>[15]</sub>; Norris, 1997<sub>[16]</sub>; Nichols, 2017<sub>[17]</sub>; Lynch, 2016<sub>[18]</sub>). Commonly we are faced with decisions about whether to purchase an electric car, avoid eating meat, or install solar panels. Non-experts simply lack the depth of knowledge needed to make an independent judgement (Hardwig, 1985<sub>[15]</sub>). Thus, the challenge that we are confronted with is evaluating expertise in a sea of misinformation or worse, disinformation. Bergstrom and West put it more bluntly, arguing that it requires the competency to identify and challenge “bullshit” (Bergstrom and West, 2020<sub>[19]</sub>). In the case of scientific information, fundamental to this competency is the knowledge that confidence in science is built by the emergence of a consensus among scientists. And, while the process of science is not infallible, and the history of the sciences can be seen as a history of error (Allchin, 2012<sub>[20]</sub>), science, as an institution and practice, has well-established mechanisms for vetting scientific claims and identifying mistakes, all of which help to establish trust and confidence in the knowledge it offers.

In addition, an education in science requires not just a knowledge of the concepts and theories of science, but also a knowledge of the common procedures and practices associated with scientific enquiry, and how these enable science to advance. Therefore, individuals who are scientifically educated have a knowledge of the major concepts and ideas that form the foundation of scientific and technological thought; how such knowledge has been derived; the common practices undertaken by scientists; the degree to which such knowledge is justified by evidence or theoretical explanations; and importantly, the social structures that enable the production of trustworthy knowledge.

Put simply, an *a priori* question of any scientific report is to ask whether there is any scientific consensus on this matter, and whether it has been peer reviewed. And, if the science is contested, what is the expertise and evidence of those who would contest the findings? Do they, for instance, have a conflict of interest? In short, to ask why critics should be trusted over the existing consensus. Thus, in the arguments around vaccination, we would expect scientifically educated students to ask: “What is the scientific consensus about vaccination?”, “What is the nature of the evidence on which that consensus is based?”, “What is the scientific expertise of those who question the consensus?”, and “What is the nature of the evidence that is used for the critique?”. Judgements of expertise require a knowledge of the mechanisms that society uses for validating expertise: i.e. “What qualifications does this individual have?”, “Which

institutions do they work for?”, and “What is their reputation in the field?”. While personal decisions may involve issues of values and other beliefs, it is still important to know and understand the relevant science.

In addition, a science education should be both broad and applied. Thus, the PISA science framework refers to a knowledge of science, science-based technology, and scientific and engineering practices. It should be noted, however, that science and technology differ in their purposes, processes and products. Technology asks: “What can we do with this knowledge?”, seeking the optimal solution to a human problem where there may be more than one optimal solution. In contrast, science seeks to answer three questions about the natural material world: 1) What exists?, 2) Why does any given phenomenon happen?, and 3) How do we know?

Nevertheless, science and technology are closely related. For instance, new scientific knowledge enables new technologies such as the advances in material science that led to the development of better batteries, solar panels, lighter airplanes and more. Likewise, new technologies can lead to new scientific knowledge through the development, for instance, of better telescopes, microscopes and digital instruments. Young people should also be able to recognise that, while science and technology are often a source of solutions, paradoxically, they can also be seen as a source of risk, generating new problems which, in turn, may require science and technology to resolve, e.g. the disposal of the batteries from electric cars or the extraction of cobalt required for their batteries and its environmental consequences. Therefore, individuals need to be able to consider the implications of the application of scientific knowledge and the issues it might pose for themselves or wider society.

Undoubtedly, many of the challenges of the 21st Century will require innovative solutions that have a basis in scientific thinking and scientific discovery. The recent development of novel mRNA vaccines and their potential for treating other diseases such as cancer is one such example. Another is the challenge of developing more efficient renewable energy supplies, including new technologies such as hydrogen. Societies will therefore require a cadre of well-educated individuals to undertake the research and technological innovation that will be essential to meet the economic, social and environmental challenges that the world currently faces. Yet, to engage with the wider society, such scientists will also need to be both knowledgeable about science and technology, and hold a deep understanding of the workings of the scientific community, its limitations and the consequences of its application.

## The outcomes of science education for 15-year-olds

Current thinking about the desired outcomes of science education is rooted strongly in the belief that an understanding of science is so important that it should be a feature of every young person's education (American Association for the Advancement of Science, 1990<sup>[21]</sup>; Confederacion de Sociedades Cientificas de España, 2011<sup>[22]</sup>; Millar and Osborne, 1998<sup>[23]</sup>; National Research Council, 2012<sup>[24]</sup>; Sekretariat der Ständigen Konferenz der Kultusminister der Länder in der Bundesrepublik Deutschland, Luchterhand, 2005<sup>[25]</sup>; 2005<sup>[26]</sup>; 2005<sup>[27]</sup>; Taiwan Ministry of Education, 1999<sup>[28]</sup>) (Ministry of Education, P. R. China, 2017<sup>[29]</sup>; 2017<sup>[30]</sup>; 2017<sup>[31]</sup>). Indeed, in many countries, science is an obligatory element of the school curriculum from kindergarten until the completion of compulsory education.

Many of the documents and policy statements cited above give pre-eminence to an education for citizenship. However, internationally, many of the curricula for school science are based on a view that the primary goal of science education should be the preparation of the next generation of scientists (Millar and Osborne, 1998<sup>[23]</sup>; Osborne and Dillon, 2008<sup>[32]</sup>). These two goals are not always compatible. Attempts to resolve the tension between the needs of the majority of students, who will not become scientists, and the needs of the minority who will become scientists, have led to an emphasis on teaching science through enquiry (National Research Council, 1996<sup>[33]</sup>; National Research Council, 2000<sup>[34]</sup>), new curriculum models (Millar, 2006<sup>[35]</sup>) and, more recently, an emphasis on developing competency with a set of eight scientific practices (National Research Council, 2012<sup>[24]</sup>) that address the needs of both groups. The emphasis in these frameworks and their associated curricula lies not on educating individuals to be producers of scientific knowledge. Rather, it is on educating young people to become informed critical users of scientific knowledge, a competency that all individuals are expected to need during their lifetimes (European Commission, 1995<sup>[10]</sup>).

To understand and engage in critical discussion, decision making and action about issues that involve science and technology requires three domain-specific competencies. The first is the competency to provide explanatory accounts of natural phenomena, technical artefacts and technologies and their implications for society. Such a competency requires a knowledge of the major explanatory ideas of science, the use of models, the questions that frame the practices and goals of science, and the social and ecological contexts in which science operates. The second is the competency to construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically; i.e. the ability to consider whether a research design is fit for purpose, how it might be improved and whether appropriate procedures have been used, coupled with the competency to interpret and evaluate data and evidence scientifically and evaluate whether the conclusions are warranted. The third is the competency to find and critically evaluate scientific information – and then use such knowledge for decision making that informs action. Thus, the outcomes of science education for 15-year-olds in PISA 2025 are defined by the three competencies in Box 2.1.

### Box 2.1. Science competencies: PISA 2025 definition

A scientifically educated person can engage in reasoned discourse about science, sustainability and technology to inform action. This requires the competencies to:

1. **Explain phenomena scientifically:** Recognise, construct, apply and evaluate explanations for a range of natural and technological phenomena.
2. **Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically:** Appraise and evaluate ways of investigating questions scientifically, and interpret and evaluate scientific data critically.
3. **Research, evaluate and use scientific information for decision making and action:** Obtain scientific information on a specific global, local or personal science-related issue and evaluate its credibility, potential flaws and the implications for personal and communal decisions.

All of these competencies require knowledge (see Box 2.2). Explaining phenomena scientifically, for instance, demands a knowledge of the content of science – referred to hereinafter as “content knowledge” – and, in addition, the nature of scientific explanations and the evidence supporting them. The second and third competencies require more than a knowledge of what we know. Rather, they depend on an understanding of how scientific knowledge is established and the degree of confidence with which it is held. Specific calls, therefore, have been made for teaching “the nature of science” (Lederman, 2007<sup>[36]</sup>), “ideas about science” (Millar and Osborne, 1998<sup>[23]</sup>) or more recently “scientific practices” (National Research Council, 2012<sup>[24]</sup>). Recognising and identifying the features that characterise scientific enquiry requires a knowledge of the standard procedures that are the foundation of the diverse methods and practices used to establish scientific knowledge – referred to here as “procedural knowledge”. Finally, the competencies require epistemic knowledge, i.e. an understanding of the rationale for the common practices of scientific enquiry, the status of the knowledge claims that are generated, the meaning of foundational terms such as theory, hypothesis and data, the importance of consensus, a knowledge of how to judge the proficiency of experts, the ways in which scientific information may be inappropriately used and presented, and how relevant scientific knowledge is in contributing to developing an answer or solution.

In addition to content knowledge, both procedural and epistemic knowledge are necessary to: identify questions that are amenable to scientific enquiry; judge whether appropriate procedures have been used to ensure that the claims are justified; and to distinguish the significance of scientific claims and evidence, matters of values, or economic considerations. Of significance in developing this definition of the outcomes of scientific education is that, in their lifetimes, individuals will need to acquire knowledge, not through scientific investigations, but by using resources such as libraries and the Internet. Procedural, epistemic and content knowledge are essential to deciding whether the many claims to knowledge that pervade scientific reports and contemporary media have been derived using appropriate procedures and are warranted.

### Box 2.2. Scientific knowledge: PISA 2025 definition

The PISA 2025 definition of scientific knowledge is based upon a view of scientific knowledge as consisting of three distinguishable yet related elements. The first of these and the most familiar is a knowledge of the facts, concepts, ideas and theories about the natural world that science has established. For instance, how plants synthesise complex molecules using light, water and carbon dioxide or the particulate nature of matter. This kind of knowledge is referred to as “**content knowledge**” or “knowledge of the content of science”.

Knowledge of the procedures that scientists use to establish scientific knowledge is referred to as “**procedural knowledge**”. This is a knowledge of the practices and concepts on which empirical enquiry is based, such as repeating measurements to minimise error and reduce uncertainty, the control of variables, and standard procedures for representing and communicating data (Millar et al., 1994<sup>[37]</sup>). These have been elaborated as a set of “concepts of evidence” (Gott, Duggan and Roberts, 2005<sup>[38]</sup>).

Furthermore, understanding science as a practice also requires “**epistemic knowledge**”, which refers to an understanding of the role of specific constructs and defining features essential to the process of knowledge building in science (Duschl, 2008<sup>[39]</sup>). Epistemic knowledge includes an understanding of the function that questions, observations, theories, hypotheses, models and arguments play in science, the values and issues that frame a question and drive scientific enquiry, a recognition of the variety of forms of scientific enquiry, and the role peer review and scientific consensus play in establishing knowledge that can be trusted.

A more detailed discussion of these three forms of knowledge is provided in the later section on scientific knowledge and in Box 2.6, Box 2.7 and Box 2.8.

People need all three forms of scientific knowledge to perform all three competencies that are the focus of this framework. Therefore, PISA 2025 will focus on assessing the extent to which 15-year-olds can display these competencies appropriately within a range of personal, local, national and global contexts. This perspective differs from that of many school science programmes, which are often dominated by content knowledge. Instead, the framework is based on a broader view of the kind of knowledge of science required by participating members of contemporary society.

In addition, the competency-based perspective also recognises that there is an affective element to a student’s display of these competencies, meaning that their sense of scientific identity, attitudes or disposition towards science will determine their level of interest, sustain their engagement, and will motivate them to take action (Osborne, Simon and Collins, 2003<sup>[40]</sup>; Moote et al., 2020<sup>[41]</sup>).

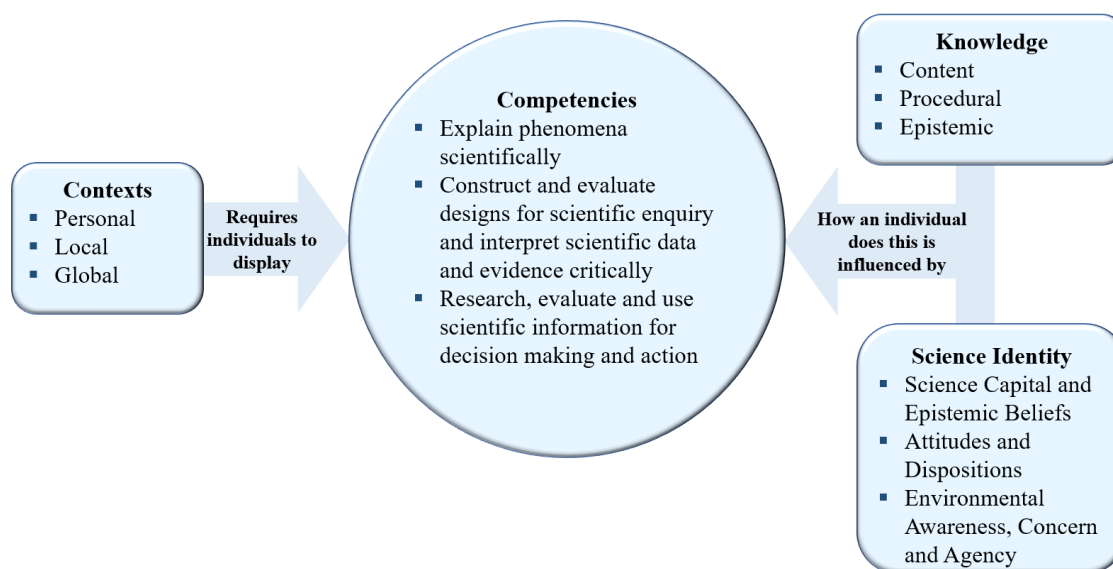
Furthermore, a science education should develop the competencies that enable a student to act should they so choose. Actions could be at the individual or household level, such as engaging in conservation behaviours or choosing to purchase or avoid products according to value-based criteria. Actions could also be taken with others to raise awareness within the community or make suggestions about solutions to environmental problems. Being willing to take these actions requires knowledge but also hope and a vision that solutions are possible, as well as efficacy – that is the belief that both individually and collectively that they can engage in the development of these solutions.

Thus, the scientifically educated person would commonly have an interest in scientific and environmental topics; engage with the issues posed by science, health, technology and sustainability; and feel that science has a meaningful connection and relevance to their life. Most of these individuals will not be employed as practising scientists. Rather, such individuals recognise that science, technology and research in this domain are essential elements of contemporary culture that frame much of our thinking

and multiple areas of life. Such attitudes and dispositions are assessed through a separate “non-cognitive” questionnaire.

Figure 2.1 summarises the outcomes of science education and their inter-relationships.

**Figure 2.1. Framework for the PISA 2025 science assessment**



### ***Explanatory notes***

The following remarks are offered to clarify the meaning of what it means to be a scientifically and environmentally educated person at the age of 15 for the purposes of the PISA 2025 assessment.

- The PISA science assessment places emphasis on the application of scientific knowledge in the context of everyday life situations.
- For the purposes of the PISA assessment, it should be noted that these competencies are only tested using the knowledge that 15-year-old students can reasonably be expected to have of the concepts and ideas of science (**content knowledge**), the procedures and strategies used in all forms of scientific enquiry (**procedural knowledge**), and the manner in which ideas are justified and warranted in science (**epistemic knowledge**).
- Finally, throughout this document, the term “natural world” is used to refer to phenomena associated with any object (living or non-living) or phenomenon occurring in the material world.
- Within this document, science is used as a generic term to refer to any or all of the natural sciences: physics, chemistry, biology, Earth, space and environmental sciences. The term “technology” is used in a very broad sense to mean any engagement in a systematic practice of design to achieve solutions to particular human problems. As such, it includes all types of human-made systems and processes and not just the limited sense that often equates technology with modern computational and communications devices. Technologies result when engineers apply their understanding of the natural world and of human behaviour to design ways to satisfy human needs and wants. In some countries, notably the United States, the term “engineering” is used as a synonym.
- Knowledge of and competency in addressing scientific problems is not limited to the realm of science. In addition, economics, political science, sociology, psychology and anthropology all play a role in crafting meaningful, appropriate, ethically just and viable solutions. This

assessment, however, focuses specifically on the scientific competencies that underpin those decisions.

- Three terms are used in the environmental section – “environmental”, “ecological” and “socio-ecological”. While there is some overlap between the domains indicated by these terms, environmental science draws on all the sciences including ecology. Ecology has a more refined focus and seeks to understand the interconnections within and between ecosystems. The term “socio-ecological” indicates a wider systems perspective on the close coupling of human and natural systems in relation to environmental challenges/crises, such as the sustainable management of aquifers that provide water to rivers and lakes, agriculture and urban communities (Young et al., 2006<sup>[42]</sup>; Eisenmenger et al., 2020<sup>[43]</sup>).

### **Science competencies: Rationale**

In this section, an elaboration is provided of each of the three competencies and why they are considered essential elements of the outcomes of any science education.

#### *Competency 1: Explain phenomena scientifically*

The cultural achievement of science is a set of explanatory theories that have transformed our understanding of the natural world, such as the idea that day and night is caused by a spinning Earth, or the idea that diseases can be caused by invisible micro-organisms. Moreover, such knowledge has enabled us to develop technologies that support human life, enabling, for example, the prevention of disease and rapid human communication across the globe. The competency to explain phenomena that occur in the material world is thus dependent on a knowledge of these major explanatory ideas of science (Harlen, 2010<sup>[44]</sup>).

Constructing explanations of scientific, technological and environmental phenomena, however, requires more than the ability to recall and use theories, explanatory ideas, information and facts (content knowledge). Offering a scientific explanation also requires an understanding of how such knowledge has been derived and the level of confidence we might hold about any scientific claims. For this competency, the individual requires a knowledge of the standard procedures and practices used in scientific enquiry to obtain such knowledge (procedural knowledge), and an understanding of their role and function in justifying the knowledge produced by science (epistemic knowledge).

The competency to explain phenomena requires the ability to draw on a range of constructs and unobservable entities, e.g. atoms, cells or chemical reactions, and to move between what is observable and their representations or models drawing on evidence (Braaten and Windschitl, 2011<sup>[45]</sup>). For instance, explanations may appeal to the theory of evolution to explain how new strains of a virus emerge, or explain differences in properties of elements using their position on the periodic table. Explanations are often causal in nature, connecting a phenomenon with a theory or model that has wide-ranging explanatory power, such as the kinetic molecular model of gases, or Newton’s laws. Explanations can be simple or can involve causal chains of argument. Explaining a rainbow, for instance, can be done simply by showing that droplets break light into its constituent colours; or can be complex, using principles of dispersion, total internal reflection and refraction, involving diagrams, to account for a rainbow’s shape and position. In biology, principles of structure and function and adaptation are used to explain structures and behaviours, such as differences in species found in different local habitats.

When explanations are tentative or speculative, however, they must be argued for (Osborne and Patterson, 2011<sup>[46]</sup>) using reasoning and must be consistent with evidence (Berland and Reiser, 2008<sup>[47]</sup>). Predictions and suggesting solutions require arguments to be made using theories, concepts or evidence. For instance, predicting what objects will float in water requires an understanding of relative density. Such knowledge can then enable the design of solutions. Explaining the societal or personal implications of

scientific knowledge is also an important aspect of scientific competency, for instance, explaining the basis of strategies for controlling evaporation in water conservation practice.

Constructing explanations often involves model-based reasoning (Lehrer and Schauble, 2005<sup>[48]</sup>), using models that can never be complete representations of reality. The Bohr model of the atom, for instance, is not an accurate model of the atom, there is much more to inheritance and the expression of phenotypes than the insights offered by Mendelian genetics; the ideal gas laws are what they say they are, – ideal. Such models work because they are “true enough” (Elgin, 2017<sup>[49]</sup>). Moreover, because models are representations, they must be constructed using diagrams, figures or mathematics (Tytler, Prain and Hubber, 2018<sup>[50]</sup>). How and what they represent must be explained, as well as their limitations. Current electricity is often represented as being like the flow of water in a pipe. And, while this is useful for explaining the function of an ammeter which is the equivalent of a flowmeter, it is not so useful for explaining the way a light turns on instantly when a switch is pressed. Understanding the central nature of models in explanation and how they can be used to communicate explanations appropriately are therefore important outcomes of any scientific education. In addition, many explanations require an understanding of systems, the elements they consist of and their interactions, e.g. climate change, water pollution, plastics and their impact on ecosystems.

To do this, scientists have developed an ontological zoo of unfamiliar entities and concepts, e.g. of electrons, genes, molecules, point charges, elements, compounds and concepts such as heat, current, acceleration amongst many, many more (Ogborn et al., 1996<sup>[51]</sup>), many of which are abstract. These are used with a range of styles of reasoning e.g. mathematical deduction, experimental exploration, hypothetical modelling, categorisation and classification, probabilistic and statistical thinking, and historically based evolutionary reasoning (Crombie, 1994<sup>[52]</sup>) to construct explanatory models of the material world (Lehrer and Schauble, 2005<sup>[48]</sup>) using three forms of argument – induction, deduction and abduction (inference to the best possible explanation).

*Competency 2: Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically*

A knowledge of science implies that students should understand the endeavour of scientific enquiry, which is to generate trustworthy knowledge about the natural and human-made world (Ziman, 1979<sup>[53]</sup>; Longino, 1990<sup>[54]</sup>; Oreskes, 2019<sup>[14]</sup>). Investigating the natural and material world can take many forms. There is the identification of the range and diversity of species, rocks or substance, a process of categorisation and classification. Controlled experiment studies where variables are identified and controlled enable the testing of well-defined hypotheses while simulations and modelling enable the testing of theoretical ideas against sets of observational data.

Data obtained by observation and experiment serve two functions. On the one hand, they can generate questions about the nature of the patterns and observations leading to the development of models and explanatory hypotheses that enable predictions. On the other hand, experiments, and the data they provide, are essential to testing the validity of predictions made by new hypotheses and models.

Scientists rarely work in isolation, and are members of research groups or teams that engage in extensive collaboration with colleagues both nationally and internationally. New knowledge claims are always perceived to be provisional and may be found to lack justification when subjected to critical review by peers. Peer review is a formal mechanism that the scientific community has established to ensure the objectivity of scientific knowledge (Longino, 1990<sup>[54]</sup>). Hence, scientists have a commitment to publish or report their findings and the methods used in obtaining the evidence. Doing so enables empirical studies, at least in principle, to be replicated and results confirmed or challenged.

Measurements, however, can never be absolutely precise. Rather, they are all subject to a degree of error. Much of the work of the experimental scientist is, therefore, devoted to the reduction of uncertainty by

repeating measurements, collecting larger samples, building instruments that are more accurate, and using statistical techniques that assess the degree of confidence in any result. Data then has to be analysed, selected and refined to find the underlying patterns. This requires a process of interpretation and evaluation, comparing findings with predictions and assessing the probability of the outcome being a chance event or evaluating alternative explanations. Students, therefore, need to develop an understanding that no measurement can be absolutely accurate and that there are standard ways of minimising error, e.g. repeating measurements, taking averages, eliminating outliers.

Uncertainty in the measurement means that our confidence in a finding is expressed in terms of the probability. Scientists treat every claim with some degree of tentativeness, whether 51% confidence (only slightly more likely to be true than false) or 99.99% confidence (highly likely to be true). Using this form of reasoning makes it possible for scientists, on the one hand, to be comfortable with uncertainty, and on the other hand, to change their position when new evidence emerges. Decision making invariably involves working with a degree of uncertainty. However, just because science cannot give us absolute certainty does not mean we do not trust the results. The degree of scientific consensus in the findings informs our potential actions, e.g. the scientific consensus on climate change (Oreskes, 2004<sup>[55]</sup>). All individuals should recognise this representation of the confidence in findings as a source of science's strength, recognising that there is always a finite possibility of the finding having occurred by chance. Large programmes of research with consistent findings provide us with confidence that the findings are very unlikely to have occurred by chance. All of this draws on a **procedural knowledge**.

Assessing scientific evidence may require an understanding of the concept of a normal distribution and the difference between a mean and a median. Furthermore, that confidence is commonly expressed with the use of an error bar, and that within any population there will be variability and outliers. In addition, they should understand that the larger the sample, the less likely there is to be error. Using this knowledge, the scientifically educated individual should be able to make evaluative judgements of statistical data asking, for instance, what meaning can be attached to the outliers. Moreover, they should understand that the notion of significance in science is based on a conception that a given finding is not a random effect. 15-year-olds should, for instance, be able to explain why using outliers for an argument is flawed, e.g. my granny lived to 98 and smoked 28 cigarettes a day.

Science has well-established procedures such as the use of controls that are the foundations of a logical argument to establish a causal relation. The use of controls enables the scientist to claim that any change in a perceived outcome can be attributed to a change in one specific feature. Failure to use such techniques leads to results where effects are confounded and cannot be trusted. Good explanatory models are also the foundation of any convincing causal relationship. For instance, the correlation between latitude and skin cancer is explained by the effect of UV on the cell. Double-blind trials are another means of establishing causality, enabling scientists to claim that the results have neither been influenced by the subjects of the experiment, nor by the experimenter themselves. Such ideas are fundamental to science, and all 15-year-olds should have met these ideas and be able to explain how they are essential to justifying scientific findings.

However, not all science is conducted using such methods. Some scientists spend their whole lives working at a desk developing explanatory models, often mathematical. Other scientists such as taxonomists, ecologists and epidemiologists are engaged in the process of identifying underlying patterns and interactions in the natural world that warrant a search for an explanation. Astrophysicists, climate scientists and geologists cannot conduct experiments on the natural world. Instead, by gathering data through detailed observations, their science relies on arguments that are an inference to the best explanation (abduction), examining a range of hypotheses and eliminating those which do not fit with the evidence. Knowing something about the diversity and range of methods in science is, therefore, essential for the informed citizen. The scientifically educated individual should also be able to distinguish between cause and correlation, e.g. the sale of ice cream and the incidence of shark deaths (both increase in the

summer but are causally unrelated) versus the incidence of lung cancer and smoking (which are causally related).

Facility in this competency requires a knowledge of the common procedures and practices used in science (**procedural knowledge**), and the function of these procedures in justifying any claims advanced by science (**epistemic knowledge**). This competency also draws on **content knowledge** to identify appropriate questions, assess and interpret findings. Procedural and epistemic knowledge serve two functions. First, such knowledge is required by individuals to appraise scientific investigations and decide whether they have followed appropriate procedures and whether the conclusions are warranted. Second, individuals who have this knowledge should be able to evaluate, at least in broad terms, whether a scientific question has been investigated appropriately.

Interpreting scientific data is such a core activity of science that all students should have some rudimentary understanding of the process. Initially, data interpretation begins with looking for patterns, constructing simple tables and graphical visualisations such as pie charts, bar graphs, scatterplots or Venn diagrams of the distribution of heights, eye colour and hair colour in a class, or the number of different species found in the school playground. At the higher level, it requires the use of more complex data sets and the use of the analytical tools offered by spreadsheets, more sophisticated data analysis tools designed for schools, statistical packages, basic computational models and algorithms.

Moreover, individuals are increasingly confronted with representations, often visual or graphical, of complex data sets that need to be read, interpreted and evaluated. Through science education, an individual should, therefore, develop a familiarity with common forms of representation of data, e.g. linear versus non-linear scales, and common flaws in data representations, e.g. inappropriate scales expressed in simple language. They should be able to translate between simple written textual accounts, diagrams and graphical representation. This knowledge and understanding are required to recognise what constitutes reliable and valid evidence and how to present data appropriately. Scientists make choices about how to represent the data in graphs, charts or, increasingly, in complex simulations or 3D visualisations to increase clarity as well as to persuade. Any relationships or patterns must then be recognised using a knowledge of standard patterns.

All individuals need to understand more than the procedures that have been applied to obtain any data set. That is, they need to be able to judge whether these procedures are appropriate, and whether the ensuing claims are justified (**epistemic knowledge**). For instance, many sets of data can be interpreted in multiple ways. Argumentation and critique, therefore, are essential to determining which is the most appropriate conclusion. Whether it is new theories, novel ways of collecting data, or fresh interpretations of old data, argumentation is the means that scientists and technologists use to make their case for new ideas. Disagreement amongst scientists is therefore normal rather than extraordinary. Indeed, it is a critical and sceptical disposition towards all empirical evidence that many would see as the hallmark of the professional scientist.

Fundamental to this competency is the understanding that it requires a set of scientific practices from asking an appropriate scientific question, using appropriate scientific ideas to design an appropriate experiment or field observation, analysing and interpreting data to engaging in argument and critique to evaluate the best interpretation of the data (Ford, 2008<sup>[56]</sup>). The idea of science as a set of practices engaged in by a community has emerged from the work of historians, philosophers, psychologists and sociologists over the past 40 years. Seeing science as a set of shared practices shows that theory development, reasoning and testing are components of a larger ensemble of activities that includes networks of participants and institutions (Longino, 1990<sup>[54]</sup>); specialised ways of talking and writing (Halliday and Martin, 1993<sup>[57]</sup>; Lemke, 1990<sup>[58]</sup>); the development of models to represent systems or phenomena (Nersessian, 2008<sup>[59]</sup>); the making of predictive inferences (Crombie, 1994<sup>[52]</sup>); and the construction of appropriate instrumentation enabling the testing of hypotheses by experiment or observation (Giere, Bickle and Mauldin, 2006<sup>[60]</sup>).

*Competency 3: Research, evaluate and use scientific information for decision making and action*

True knowledge is a collective good. Today, the Internet provides access to a sea of knowledge that was simply unimaginable even 20 years ago. The common assumption is that this is a good thing, and, in one sense, that is true. The Internet offers us answers to questions about everything from how to fix our broken bicycle to our concerns about health issues. In addition, it has brought together isolated individuals to pursue their common interests.

Across the globe, however, there is increasing concern about the ease with which people accept beliefs claimed to be “scientific” for which there is no substantive material evidence and for which there is good evidence to the contrary. Whether it is the idea that the Earth is flat, that vaccines cause autism, or that climate change is a hoax, the willingness of individuals to accept irrational beliefs – irrational in the sense that there is either no evidence or dubious evidence to support them – is of grave concern. For while true knowledge is a collective good, flawed or fake knowledge is both an individual and collective danger. For instance, the idea that vaccines are harmful endangers not only the lives of those who hold this idea, but the whole community who depends on a high level of vaccination to ensure its health. Obtaining and evaluating scientific information requires not just an understanding of the concepts but knowledge about science.

Essential to this competency is a knowledge and understanding of how trustworthy scientific knowledge is produced and “how monied interests try to ‘bend’ science, present pseudoscience as science, portray reliable science as ‘junk science,’ or foster an image of uncertainty even where scientific experts have reached a solid consensus” (Höttecke and Allchin, 2020, p. 4<sup>[61]</sup>). Such understanding requires and draws on epistemic knowledge of the social practices of science. First, the scientifically educated individual needs a basic understanding of the transformations that occur in the trajectory from laboratory to publication, or from “test tube to YouTube”. Second, an individual should be aware that there is a tendency of all individuals to seek confirmation for their biases which go unchallenged. Hence, they may exist in a “bubble” supported by news that reinforces the views and perceptions they already have. All scientific information should be approached with a policy of circumspection, which seeks to ask first if there is a conflict of interest, whether there is an established scientific consensus and whether the source has relevant expertise. In short, the process of evaluation starts not by interrogating the findings, as most competent outsiders do not have sufficient knowledge to do this, but rather examines the credibility of the source, and any potential prejudices of the channel of communication used in reporting the findings. Has, for instance, the finding been reported by somebody who has professional expertise in the relevant domain of science? In the current context, media literacy, especially digital media literacy, has become essential for all students (Höttecke and Allchin, 2020<sup>[61]</sup>; Kozyreva, Lewandowsky and Hertwig, 2020<sup>[62]</sup>; Bergstrom and West, 2020<sup>[19]</sup>), particularly when research shows students’ ability to evaluate sources is weak (Breakstone et al., 2021<sup>[63]</sup>).

At the core of this competency is an understanding that science is a communal enterprise and that science is not infallible. In many ways, error is the norm in science (Allchin, 2012<sup>[20]</sup>). However, while individual scientists or teams might be mistaken, the community is more trustworthy, and its goal is closure which is attained when there is overwhelming consensus. Moreover, that consensus is a consensus of experts (Collins and Evans, 2007<sup>[64]</sup>; Oreskes, 2004<sup>[55]</sup>; Oreskes, 2019<sup>[14]</sup>; Selinger and Crease, 2006<sup>[65]</sup>), as illustrated by the Intergovernmental Panel on Climate Change (IPCC) report on climate change, which synthesised the findings of scientific teams across the globe. Nor is science a democracy where one scientist’s views are as valuable as another. The views that matter are from those who have relevant background knowledge, skills in interpreting particular results, and awareness of potential flaws in reasoning. Thus, a petition claiming that vaccines are dangerous is meaningless if the signatories are not scientific experts on vaccination.

The scientifically educated individual also needs to understand that the image of the lone scientist making “great” discoveries does not represent reality. Rather, science is a communal and collaborative exercise where scientists vet each other’s ideas to identify flaws in their work, e.g. peer review (epistemic knowledge). The production of knowledge is dependent on a dialectic between construction and critique and that disagreement is both normative and productive (Ford, 2008<sup>[56]</sup>). The scientifically educated individual would understand why critique is essential to the construction of knowledge in science and why science-in-the-making, particularly when there is not a well-established consensus, will change and evolve. A good example is contemporary climate models, which are constantly evolving and improving to the point where there is an overwhelming consensus about the phenomena, increasing accuracy in their predictions and the identification of the human impact. Consequently, science-based recommendations change as more and better data are available. When forming a view on any scientific claim to inform their possible action, an informed citizen should begin by asking what is the scientific consensus, do those presenting the claims have appropriate scientific expertise and what data and evidence are they using to support their claims? Based on such information, they should then be able to critique arguments that are flawed, e.g. that reducing speed limits from 50 to 30 km/hr will have little effect, or that the risks of vaccines outweigh their benefits. Thus, this competency draws extensively on epistemic knowledge.

In evaluating action, students need to know that all activities have risks associated with them. Decisions about action often have to be taken in the absence of complete scientific knowledge and that uncertainty is an inherent feature of science (procedural and epistemic knowledge). Science rarely offers certain knowledge. Hence most decisions require an evaluation of risk (Beck, 1992<sup>[66]</sup>; Adam, Beck and van Loon, 2000<sup>[67]</sup>). When faced with potentially serious irreversible threats, e.g. deforestation in the Amazon or nuclear disasters, the precautionary principle is a common criterion to apply. Risk statistics also only measure fatalities and not injuries, and the decision to take risks is different depending on whether it is an individual risk, e.g. riding a bike, a population risk, e.g. the threat posed by a virus or new disease, a systemic risk, e.g. biodiversity loss, or a lifetime versus immediate risk, e.g. smoking ten cigarettes today versus smoking ten cigarettes a day. In addition, most decisions of this nature have a normative and social dimension which must also be weighed.

Making judgements about personal and societal policy can often involve dealing with unresolved and contested knowledge where scientists are working in complex and sometimes shifting environments, and also involves balancing competing interests and values separate from the science. Assessing risk in these situations can be difficult, involving on balance judgements about the status of scientific knowledge in relation to other values and interests. For instance, currently it is suggested that electric cars will reduce our dependence on fossil fuels and reduce pollution. Yet, electric cars currently require the mining of rare earth materials and have environmental and social impacts, which raise issues of social justice such as use of inhuman/exploitative practices to take advantage of local labour and their environment.

## Organisation of the domain

For purposes of assessment, the PISA 2025 definition of the outcomes of any science education may be characterised as consisting of four interrelated aspects (see Table 2.1). Each of these aspects is discussed further below.

**Table 2.1. The components of science education**

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Contexts</b>         | Personal, local, national and global issues, both current and historical, which demand some understanding of science and technology.                                                                                                                                                                                                                                                                                              |
| <b>2. Knowledge</b>        | An understanding of the major facts, concepts and explanatory theories that form the basis of scientific knowledge. Such knowledge includes both knowledge of the natural world and technological artefacts (content knowledge), knowledge of how such ideas are produced (procedural knowledge) and an understanding of the underlying rationale for these procedures and the justification for their use (epistemic knowledge). |
| <b>3. Competencies</b>     | The ability to explain phenomena scientifically, construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically, and research, evaluate and use scientific information for decision making.                                                                                                                                                                                          |
| <b>4. Science Identity</b> | A set of dispositions, agency and attitudes towards science and personal capital indicated by an interest in science and technology; valuing of scientific approaches to enquiry, where appropriate, and a perception and awareness of environmental issues.                                                                                                                                                                      |

### *Contexts for assessment items*

PISA 2025 will assess important scientific knowledge using contexts that raise issues and choices that are relevant to the science and environmental education of participating countries. Such contexts will not, however, be restricted to the common aspects of participants' national curricula. Rather, the assessment will require evidence of the successful use of the three science competencies in important situations reflecting personal, local, national and global contexts.

Assessment items will not be limited to school science contexts. In the PISA 2025 science assessment, the focus of the items will be on situations relating to the self, family and peer groups (personal), to the community (local and national), and to life across the world (global). Technology and environmentally based topics may be used as a common context. Also, appropriate to some topics are historical contexts which may be used to assess students' understanding of the processes and practices that are involved in advancing scientific knowledge.

Table 2.2 lists the applications of science and technology within personal, local, national and global settings that are primarily used as the contexts for assessment items. The applications will be drawn from a wide variety of life situations and will be generally consistent with the areas of application for scientific literacy in the previous PISA frameworks. The contexts will also be chosen in light of their relevance to students' interests and lives. The areas of application are: health and disease, natural resources, environmental quality, hazards, and the frontiers of science and technology. They are the areas in which scientific competency has particular value for individuals and communities in enhancing and sustaining quality of life, and in the development of public policy.

**Table 2.2. Contexts for the PISA 2025 science assessment**

|                   | Personal                                                                                                      | Local/National                                                                                                                 | Global                                                                                                            |
|-------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Health & Disease  | Maintenance of health, accidents, nutrition, vaccination                                                      | Control of disease, social transmission, food choices, obesity, community health                                               | Pandemics, food security, healthy lifestyles                                                                      |
| Natural Resources | Personal consumption of materials, types of food, and energy. Consuming locally produced foods. Choosing non- | Maintenance of human populations, quality of life, security, production and distribution of food, energy supply. Environmental | Renewable and non-renewable sources of energy, natural systems, population growth, sustainable use of species and |

|                                        |                         |                                                                                        |                                                                                                                                                           |                                                                                                                                              |
|----------------------------------------|-------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|                                        |                         | dairy and vegetarian diets                                                             | impact of mining and resource extraction. Production of renewable energy                                                                                  | land. Biodiversity and its value                                                                                                             |
| Environmental Impacts & Climate Change |                         | Sustainable practices of recycling and reduction of resource use                       | Population distribution, waste management, environmental impact. Use of regenerative agriculture                                                          | Environmental sustainability, management of pollution and air quality, loss of soil/biomass. Mass extinction of species. Ocean acidification |
| Hazards                                |                         | Risk assessments of lifestyle choices                                                  | Rapid changes (e.g. earthquakes, severe weather), slow and progressive changes (e.g. coastal erosion, sedimentation), risk assessment. Facial recognition | Threats posed by climate change, impact of modern communication, energy and its production, e.g. fracking, nuclear, gas                      |
| Contemporary Technological Challenges  | Scientific and Advances | Scientific aspects of the use of new technologies (e.g. gene editing, virtual reality) | New materials, devices and processes, genetic modifications, health technology, transport, use of artificial intelligence                                 | Exploration of space, origin and structure of the universe                                                                                   |

The PISA science assessment, however, is not an assessment of contexts. Rather, it assesses competencies and knowledge in specific contexts. The selection of these contexts, however, will be chosen based on the knowledge and understanding that students are likely to have acquired by the age of 15.

Sensitivity to linguistic and cultural differences will be a priority in item development and selection, not only for the sake of the validity of the assessment, but also to respect these differences in participating countries. In developing any international test, it is not possible, however, to include the differences in traditional and local knowledge about natural phenomena that exist between participating countries. This is not to deny, however, the contribution such knowledge can make and has made to their respective cultures.

### ***Science competencies: Performance outcomes***

In this section, an elaborated description of what kinds of performance and capability might be expected for each competency is provided. The set of science competencies in Box 2.3, 2.4 & 2.5 reflects a view that science is best seen as an ensemble of social and epistemic practices which are common across all sciences (National Research Council, 2012<sup>[24]</sup>). Hence, all these competencies are framed as actions. They are written in this manner to convey what a scientifically educated person should both understand and be capable of doing. Fluency with these competencies is, in part, what distinguishes the expert scientist from the novice. While it would be unreasonable to expect a 15-year-old student to have the expertise of a scientist, a scientifically educated student can be expected to appreciate their role and significance and undertake an approximation of the practice described.

### Box 2.3. PISA 2025 Science Competency 1

#### Explain phenomena scientifically

Produce and evaluate explanations and solutions for a range of natural and technological phenomena and problems demonstrating the ability to:

- Recall and apply appropriate scientific knowledge
- Use different forms of representations and translate between these forms
- Make and justify appropriate scientific predictions and solutions
- Identify, construct and evaluate models
- Recognise and develop explanatory hypotheses of phenomena in the material world
- Explain the potential implications of scientific knowledge for society

In the PISA 2025 science assessment, 15-year-olds who can demonstrate the competency of explaining phenomena scientifically can recall the appropriate knowledge in a given situation and apply it to construct an explanation for the phenomenon of interest. Explanations in science require different forms of representations and students need to be able to use written text, diagrams, charts and graphs. Explaining in science also extends to predicting what will happen and proposing solutions to science-related problems. For example, planning ways of mitigating the effects of future sea-level rises. A scientifically educated person should be expected to draw on standard scientific models to construct and/or evaluate representations to explain everyday phenomena such as why water evaporates faster on a warm day, how introducing a new organism might disrupt a habitat, and why gases are compressible but liquids are not, and use these explanations to make predictions.

Scientific content knowledge can also be used to recognise or develop tentative explanatory hypotheses in contexts where there is a lack of knowledge or data. Finally, 15-year-olds should be able to explain the potential implications of scientific knowledge for society. For instance, knowledge of the behaviour of viruses and bacteria to inform good social policy to prevent transmission, or knowledge of chemical processes to inform more sustainable solutions, e.g. developing long-lasting batteries.

### Box 2.4. PISA 2025 Science Competency 2

#### Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically

Construct and evaluate scientific investigations, ways of addressing questions scientifically and interpret the data demonstrating the ability to:

- Identify the question in a given scientific study
- Propose an appropriate experimental design
- Evaluate whether an experimental design is best suited to answer the question
- Interpret data presented in different representations, draw appropriate conclusions from data and evaluate their relative merits

The competency to “construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically” is reliant on the ability to discriminate questions that can appropriately be answered by scientific investigation from other types of questions that draw on different ways of knowing in the world. Evaluating questions also requires a judgement of the value of the outcome and their importance. For instance, the search for a malaria vaccine has been an ongoing programme of scientific research for

several decades and, given the number of people who are killed by malarial infections, any findings that suggested a vaccine would be achievable would be of substantial significance.

This competency requires both procedural and content knowledge of the key features of an experimental investigation and other forms of scientific enquiry. For example, in the case of an experimental investigation, what quantities could be measured, what variables could be changed, and which should be controlled. Then, what action should be taken so that accurate and precise data can be collected. In the case of developing simulations to model a phenomenon, e.g. climate change, the spread of a virus, or the effect of running on body temperature, the individual needs to be able to identify the relevant variables and evaluate their fit with empirical data. This competency also requires the ability to identify if an investigation is driven by an underlying theoretical premise or, alternatively, whether it seeks to determine identifiable patterns such as the work of an epidemiologist.

For this competency, the scientifically educated individual should be able to interpret and make sense of basic forms of scientific data and evidence that are used to make claims and draw conclusions from data represented in standard simple forms, e.g. Coronavirus charts, air quality data, or population statistics. They should also be able to make judgements about the conclusions that might be drawn from such evidence and whether they are warranted, drawing on their knowledge about the inherent uncertainty in data and the ways it might be misrepresented, whether it fits with the scientific consensus, and the credibility of the source. Those who possess this competency can interpret the meaning of scientific evidence and its implications for a specified audience in their own words, using diagrams or other representations as appropriate. This competency may require the use of basic mathematical tools to analyse or summarise data, and the ability to use standard methods to transform data into different representations.

This competency includes the ability to recognise how data sets can be transformed into different types of visual display and how to select an appropriate data representation to respond to a given question. This involves the ability to recognise how different forms of data display emphasise or hide different aspects of data patterns, for instance using logarithmic scales to perceive numbers ranging over many orders of magnitude (Harford, 2021<sup>[68]</sup>; Bergstrom and West, 2020<sup>[19]</sup>). Such knowledge is fundamental to data literacy and the data representations commonly encountered both online and in the media.

### Box 2.5. PISA 2025 Science Competency 3

#### Research, evaluate and use scientific information for decision making and action

Research and evaluate scientific claims and arguments in a variety of representations and draw appropriate conclusions demonstrating the ability to:

- Search, evaluate and communicate the relative merits of different sources of information (scientific, social, economic and ethical) that may have significance or merit in arriving at decisions on science-related issues, and whether they support an argument or a solution
- Distinguish among claims based on strong scientific evidence, expert vs. non-expert, and opinion, and provide reasons for the distinction
- Construct an argument to support an appropriate scientific conclusion from a set of data
- Critique standard flaws in science-related arguments using epistemic and procedural knowledge, e.g. poor assumptions, cause vs. correlation, faulty explanations, generalisations from limited data
- Justify decisions using scientific arguments, either individual or communal, that contribute to solving contemporary issues or sustainable development

The past decade has seen an explosion in the amount and flow of information and the ability of individuals to access this information. Unfortunately, as well as a flow of valid and reliable information there has been an increasing flow of misinformation, and worse, disinformation. When it comes to scientific information, both valid and mis-informed, all citizens need the competency to judge the credibility and value of the information that commonly surrounds any science-related issue. The prominence of information and misinformation surrounding the pandemic of 2020 is an excellent example, e.g. whether to wear masks, the dangers associated with Covid-19, and the value of possible cures/vaccines.

Most individuals' interaction with science will be through secondary sources or secondary data. To evaluate such reports, individuals need to understand how to evaluate the status of sources and expertise, the status of the publication in which the information is published, the role of peer review, standard issues in questioning the quality of the data, e.g. accuracy, precision and sample size, and common flaws in arguments (generalising from limited data, distinguishing cause from correlation).

A scientifically educated person should understand the importance of developing a sceptical disposition, particularly to media reports of single findings in science, recognising that all research builds on previous work, that the findings of any one study are always subject to uncertainty, and that the study may be biased by the sources of funding. This competency requires students to possess both procedural and epistemic knowledge but may also draw, to varying degrees, on their content knowledge of science.

Therefore, a goal of science education should be to develop the competency required for researching, evaluating and using scientific information for decision making and action for personal, local and global science-related issues. This requires the basic competencies used by fact checkers of "click restraint" and "lateral reading" used to check the credibility of sources (Breakstone et al., 2021<sup>[63]</sup>). 15-year-olds need to know how to use Wikipedia to establish whether there is any scientific consensus. In addition, the scientifically educated 15-year-old should be able to identify some of the assumptions, claims, evidence and reasoning in a scientific argument, and be able to construct arguments from the scientific evidence and information they obtain, e.g. for vaccination, carbon off-setting, water conservation, or air quality. They should also be able to identify common flaws. These include false assumptions, e.g. human behaviour does not contribute to climate change; distinguishing correlation from causation, e.g. stork population and birth rate in any country, sales of ice cream and shark deaths; faulty explanations, e.g. vaccines causing autism, the failure to distinguish between weather and climate (for example, "scientists

can't predict the weather in two weeks' time, how can they predict the climate in 20 years?"); and generalisations from limited data, e.g. small samples. In addition, there needs to be a recognition that science is only one factor amongst others such as economic, behavioural and values that inform decisions, and how scientific knowledge and practice interacts with these.

## **Scientific knowledge**

The three competencies developed by an education in science require three forms of knowledge that are discussed below.

### *Content knowledge*

Only a sample of the content domain of science can be assessed in the PISA 2025 science assessment. Hence, it is important that clear criteria are used to guide the selection of knowledge that is assessed. The knowledge to be assessed will be selected from the major fields of physics, chemistry, biology, Earth and space sciences, such that the knowledge:

- Has relevance to real-life situations
- Represents an important scientific concept or major explanatory theory that is well established and has enduring utility
- Is appropriate to the developmental level of 15-year-olds

Therefore, it will be assumed that students have some knowledge and understanding of the major explanatory ideas and theories of science, e.g. those to be found in Harlen (2010<sub>[44]</sub>). These include ideas such as our understanding of the history and scale of the Universe, the particle model of matter, and the theory of evolution by natural selection. These examples of major explanatory ideas are provided for illustrative purposes and there has been no attempt to list comprehensively all the ideas and theories that might be seen to be fundamental for a scientifically educated individual. Moreover, it is recognised that within any of these domains, for both content and the practices of science, there exists a hierarchy of progression, which should be attended to when formulating items for assessment (Alonzo and Gotwals, 2012<sub>[69]</sub>).

Box 2.6 shows the content knowledge categories and examples selected by applying these criteria. No claim is made that this is a comprehensive list, but such knowledge is required for understanding the natural world and for making sense of experiences in personal, local, national and global contexts. The framework uses the term “systems” instead of “sciences” in the descriptors of the content knowledge. The intention is to convey the idea that citizens must understand concepts from the physical and life sciences, earth and space sciences, and their application in contexts where the elements of knowledge are interdependent or interdisciplinary. Phenomena viewed as subsystems at one scale may themselves be viewed as whole systems at a smaller scale. For example, the transport system can be seen as an entity in itself, or as a subsystem of the human body; a molecule can be studied as a stable configuration of atoms but also as a subsystem of a cell or a gas. Hence, applying scientific knowledge and deploying scientific competencies requires consideration of which system and which boundaries apply to any particular context.

In addition, understanding many of the contemporary challenges facing humanity and the influence of humans on the planet requires a knowledge of the interdependence of living organisms. For example, the scientific knowledge to explain the importance of top predators, the significance of bees, or the impact of deforestation.

### Box 2.6. Knowledge of the content of science in PISA 2025

#### Physical systems that require knowledge of:

- Structure and properties of matter (e.g. particle model, bonds, changes of state, thermal and electrical conductivity)
- Chemical changes of matter (e.g. chemical reactions, energy transfer, acids/bases)
- Motion and forces (e.g. velocity, friction) and action at a distance (e.g. magnetic, gravitational and electrostatic forces and interactions)
- Energy and its transfer (e.g. conservation, dissipation, chemical reactions)
- Interactions between energy and matter (e.g. light and radio waves, sound and seismic waves, absorption of radiation by carbon dioxide)

#### Living systems that require knowledge of:

- The concept of organism (including animals, plants and micro-organisms, (e.g. viruses, bacteria))
- Genes (e.g. expression, heredity/inheritance, biotechnology) and their interaction with the environment
- Cells (including structure and function, energy, respiration (carbon oxidising), photosynthesis (carbon fixing), growth, etc.)
- Plant and animal systems, their health and maintenance (e.g. circulatory/transport, reproduction, respiration, transport, excretion, digestion/nutrition) and associated inter-relationships
- Biological evolution (biodiversity, genetic variation, adaptation and natural selection)
- Ecosystems (e.g. matter and energy flow, food chains, habitat, disruption, e.g. pollution)
- The biosphere (e.g. sustainability in the global ecosystem)
- Interactions of humans and their impact and effect on the environment, other species and sustainability

#### Earth and space systems that require knowledge of:

- Structures of the Earth's systems (e.g. atmosphere, hydrosphere, geosphere, e.g. plate tectonics, seismology)
- The finite nature of mineral resources, their use and the effects on the environment in their exploitation
- Energy in the Earth systems (e.g. sources, global warming, plate tectonics, geological cycles, water cycle)
- Water, supply and conservation (e.g. fresh water, aquifers)
- Interactions and change among Earth systems (e.g. climate change, geochemical cycles, constructive and destructive tectonic forces, ocean acidification)
- Earth's history (e.g. fossils, origin and evolution, erosion and deposition)
- Earth in space (e.g. moon phases, solar systems, galaxies)
- The origin of the Universe and the solar system (e.g. stellar evolution, formation of the planets, the Big Bang theory)

As well as content knowledge, the scientifically educated individual would understand that the construction of knowledge is dependent on a set of interdependent key practices (National Research Council, 2012<sup>[24]</sup>) which require scientists to:

- Ask questions about the material world
- Develop and using models
- Plan and carry out investigations
- Analyse and interpret data
- Use mathematics and computational thinking
- Construct explanations
- Engage in argument from evidence
- Obtain, evaluate and communicate information

These practices require either or both procedural and epistemic knowledge, which are elaborated next.

### *Procedural knowledge*

A fundamental goal of science is to generate explanatory accounts of the material world. Tentative explanatory accounts are first developed and then tested through empirical enquiry. Empirical enquiry is reliant on certain well-established concepts such as the notion of dependent and independent variables, the control of variables, types of measurement, forms of error, methods for minimising error, common patterns observed in data, and methods of presenting data. It is this knowledge of the concepts and procedures that are essential for scientific enquiry that underpin the collection, analysis and interpretation of scientific data. Such ideas form a body of procedural knowledge which has also been called “concepts of evidence” (Gott, Duggan and Roberts, 2005<sup>[38]</sup>; Millar et al., 1994<sup>[37]</sup>).

One can think of procedural knowledge as being a knowledge of the standard procedures and practices scientists use to obtain reliable and valid data. Such knowledge is required both to undertake scientific enquiry and engage in critical review of the evidence that might be used to support claims made from the data. It is expected, for instance, that students will know that scientific knowledge has differing degrees of certainty associated with it and can explain why, for instance, that there is a difference between the confidence associated with measurements of the speed of light (which has been measured many times with ever more accurate instrumentation), and measurements of fish stocks in the North Atlantic or the mountain lion population in California. The examples listed in Box 2.7 convey the general features of procedural knowledge that may be tested.

### Box 2.7. PISA 2025 procedural knowledge

#### Procedural knowledge

- The concept of variables including dependent, independent and control variables
- Concepts of measurement, e.g. quantitative (measurements), qualitative (observations), the use of a scale, categorical and continuous variables
- Ways of assessing and minimising uncertainty, such as repeating and averaging measurements
- Mechanisms to ensure the precision (closeness of agreement between repeated measures of the same quantity), and accuracy of data (the closeness of agreement between a measured quantity and a true value of the measure)
- Common ways of abstracting and representing data using tables, graphs and charts, and their appropriate use
- The control of variables strategy and its role in experimental design or the use of randomised controlled trials to avoid confounded findings and identify possible causal mechanisms
- Given a scientific question, what might be an appropriate design for its investigation, e.g. experimental, field based or pattern seeking; the role of controls to establish causality
- What processes of peer vetting are used by the scientific community to ensure knowledge claims are trustworthy

#### *Epistemic knowledge*

Epistemic knowledge is a knowledge of the constructs and defining features essential to the process of knowledge construction in science, and their role in justifying the knowledge produced by science. As such, epistemic knowledge provides a rationale for the procedures and practices in which scientists engage, a knowledge of the structures and defining features that guide scientific enquiry, and the foundation for the basis of belief in the claims that science makes about the natural world. In short, an explanation and justification of how claims in science are justified. The distinction between procedural and epistemic knowledge is exemplified by being able to explain what the control of variables strategy is, which is procedural knowledge, and being able to explain why the use of the control of variables strategy or replication of measurements is central to establishing knowledge in science, which is epistemic knowledge.

15-year-olds should, for instance, know that any particle model of matter is a simplified representation of matter, and be able to explain how the Bohr model is a limited model of what we know about the atom and its constituent parts but was based on the best evidence at the time. They would know that the construction of models, be they directly representational, abstract or mathematical, is a key feature of science and that such models are akin to maps rather than accurate pictures of the material world. Essentially, the aim of a model is to produce a representation that affords an understanding of a phenomenon, rather than replicate the phenomenon itself (Elgin, 2017<sup>[49]</sup>). Good models also enable the production of hypotheses and predictions.

Scientifically educated 15-year-olds will also understand that scientists draw on data to advance claims to knowledge, and that argument is a commonplace feature of the sciences. They should also be aware that there are different types of arguments in the sciences. For instance, some arguments in science are hypothetico-deductive (e.g. Copernicus' argument for the heliocentric system), some are inductive (the conservation of energy), and some are an inference to the best explanation (Darwin's theory of evolution or Wegener's argument for moving continents) based on the best available evidence.

The scientifically educated 15-year-old would have a basic understanding of the collaborative and communal nature of scientific enquiry and how it develops trustworthy knowledge. They would be able to explain how the control of variables strategy enables the production of reliable knowledge. In short, that an observed effect can be attributed to the independent variable. They would know the role and significance of peer review as the mechanism that the scientific community has established for testing claims to new knowledge and for achieving consensus; they would know that error and mistakes are an inherent feature of science; that all new scientific findings are vetted by other scientists who are experts in that domain; that only those that have been peer reviewed are worthy of trust; and that there is a hierarchy of journals within the community, e.g. *Science*, *Nature*, *Cell*, *the Lancet*, *New England Medical Journal*, etc. They would, for instance, be able to explain the purpose of peer review and replication in minimising error and producing trustworthy knowledge. They could explain that the conception of a “theory” as used in science is not the same as the notion of a “theory” in everyday language where it is used as a synonym for a “guess” or a “hunch”.

Box 2.8 represents what are considered to be the major features of epistemic knowledge. At its core, epistemic knowledge has four elements: a knowledge of the role of models in science; the role of data and evidence in science; the nature of scientific reasoning; and the collaborative and communal nature of scientific enquiry.

## Box 2.8. PISA 2025 epistemic knowledge

### Epistemic knowledge

The constructs and defining features of science, i.e. an understanding of:

- The nature of scientific observations, facts, hypotheses, models and theories
- The purpose and goals of science (to produce reliable explanations of the natural world and to predict future events) as distinguished from technology (to produce an optimal solution to human need)
- The values of science, e.g. a commitment to peer-reviewed publication, objectivity and the elimination of bias

More specifically, this requires an understanding of:

#### **Models**

- How understanding of the material world is constructed using physical, conceptual, system and mathematical models in science, e.g. particle model of matter
- The distinction between a model and reality, e.g. that a model is a representation of something that may be too small to see or too large to imagine, e.g. Bohr model of the atom
- How models enable predictions and explanations, e.g. Sun-Earth model of daily movements (continues over page)
- How the limitations of models (e.g. number of variables, simple v complex models, quality of data provided) constrain their use

#### **Data and evidence in scientific claims**

- How scientific claims are supported by data, methods, reasoning and evaluation in science
- How scientific evidence is generated, e.g. the nature of the practices undertaken by scientists
- How measurement error affects the degree of confidence in scientific knowledge

#### **The nature of scientific reasoning**

- Some of the different forms of empirical enquiry, e.g. experiment, field work and its role, controlled experiments, pattern seeking
- The types of reasoning (deduction, abduction, induction, probabilistic thinking) used in establishing knowledge and their goal (to test explanatory hypotheses, or identify patterns and entities) and examples of each, e.g. Newton's Laws of Motion\* (deduction), Mendelian Genetics (induction), Theory of Evolution (abduction)
- The ethical dilemmas raised in scientific practice, e.g. animal experimentation, conflicts of interest
- The role of scientific knowledge, along with other forms of knowledge, in identifying and addressing societal and technological issues, as well as its limits

#### **The collaborative and communal nature of the sciences**

- How specific scientific research is funded and supported, e.g. government, private, and the mechanisms for deciding
- The importance of consensus in warranting belief

- How peer review helps to establish confidence in scientific claims and is dependent on a scientific community
- Key scientific practices undertaken by scientists to produce shared knowledge, their role and their collaborative nature
- The limits to certainty and confidence in scientific findings, how it is expressed, the evolution of certainty and the role of consensus
- How scientific findings are communicated within the community and to the public (e.g. pre-prints, peer reviewed journals, public communication)

Epistemic knowledge is most likely to be tested in a pragmatic fashion in a context where a student is required to interpret and answer a question that requires some epistemic knowledge rather than assessing directly whether they understand the features in Box 2.8. For instance, students may be asked to identify whether the conclusions are justified by the data or what piece of evidence best supports the hypothesis advanced in an item and explain why.

### ***“Science Identity”***

Young people’s views of themselves as individuals who are both interested in and value science and scientific ways of thinking is captured by the construct of science identity. The inclusion of the identity construct as a major dimension for the PISA 2025 construct of science competency and outcomes is based on the principle that while scientific knowledge and competencies are important and valuable for young people’s futures, identity outcomes are also crucial for supporting agency and active citizenship in a rapidly changing world. Such outcomes are the extent to which young people feel meaningfully connected to science, recognise themselves and feel recognised by others as science interested/competent (Carlone and Johnson, 2007<sup>[70]</sup>), and engage with the sciences as critical consumers and decision-makers in their daily lives (Bell et al., 2018).

The aim of PISA is to assess 15-year-olds’ knowledge and competencies to engage scientifically with the world, which would be incomplete without a measure of young peoples’ sense of agency, attitudes and values in relation to science. In short, if the knowledge and competencies of the sciences are not valued as a way of thinking and being in the world, science education has failed to achieve one of its major goals. Given the inclusion of the new competency “research, evaluate and use scientific information for decision making and action”, which places even more emphasis on the capability of 15-year-olds to use such knowledge, consideration of these traits of a scientific identity becomes ever more important. It is not enough to “know” science if there is no meaningful connection that then translates into its common use.

The identity construct goes beyond considerations of short-term attitudes and affective responses to science and science classrooms events, to frame engagement with science and longer-term aspirations in terms of self-processes that are bounded by socio-cultural structures and interactions with others that shape a sense of self (Ashbacher et al., 2014). The construct has proved powerful in interpreting the experience of science and science classrooms of minority and culturally and ethnically diverse students (including Indigenous students and immigrant communities) (Calabrese Barton et al., 2013<sup>[71]</sup>). As such, it provides a strong critical and social justice perspective on the interactions of students with science in schools and provides an important perspective for PISA in reporting on science competencies at national and global levels (Archer et al., 2017<sup>[72]</sup>; Archer, Dewitt and Osborne, 2015<sup>[73]</sup>). For all students, their identification with science has been shown to mediate learning.

The identity construct is complex, however. From a measurement perspective, it encompasses a range of constructs related to perceptions, dispositions and values, and attitudes/affect. These include student self-concept (Jansen, Scherer and Schroeders, 2015<sup>[74]</sup>), which is a measure of their perception of their competency in science; self-efficacy (Bandura, 1997<sup>[75]</sup>), which is a measure of their perception of their

ability to perform science-related tasks in everyday life; student agency; motivation; epistemological beliefs; and science capital (Archer et al., 2015<sup>[76]</sup>). Self-efficacy and collective efficacy are key skills that help determine whether youth believe they can work on current issues and how well they are able to do so.

Science capital is a sociological construct referring to science-related forms of cultural and social capital (Bourdieu, 1992<sup>[77]</sup>; 2010<sup>[78]</sup>). From a measurement perspective, it is seen as a mix of four main components: 1) science-related knowledge and understanding; 2) science-related attitudes and dispositions (e.g. feeling a connection with science, seeing school science as relevant to my daily life); 3) engaging with science for leisure/pleasure (e.g. consuming science-related social media); and 4) science-related social capital (e.g. knowing science-interested people; being supported by significant others to develop and pursue science interests) (Archer et al., 2015<sup>[76]</sup>). Together, these elements of science capital support a young person's identification with science, that is, how far they can see themselves in their social context as being "a science person" and the extent to which they can critically appraise and use science within their wider lives. Attitudes, which were a primary dimension for PISA 2015, are subsumed under this frame of identity.

A major distinction within the attitudes literature (Gardner, 1975<sup>[79]</sup>; Osborne, Simon and Collins, 2003<sup>[40]</sup>; Tytler, 2014<sup>[80]</sup>) is that between attitudes towards science, and scientific attitudes. While the former is measured by the level of interest displayed in scientific issues and activities, the latter is a measure of a disposition to value ways of thinking and working scientifically, including respect for evidence as the basis for claim making, explanation and decisions.

Such a binary distinction, however, is inadequate. For instance, recently there has been interest in aesthetic dimensions of scientists' work (Wickman, 2006<sup>[81]</sup>) and of student responses to aspects of school science (Anderhag et al., 2016<sup>[82]</sup>; Jakobson and Wickman, 2008<sup>[83]</sup>; Wickman, Prain and Tytler, 2021<sup>[84]</sup>), which associate feelings such as appreciation, interest, wonder or even awe with scientific phenomena, specific living entities such as insects, the workings of a TV, the explanation for the patterns in moon phases, ways of transforming data, or the evidence-based nature of a scientific argument. The construct of a scientific aesthetic emphasises that there exists a continuity between meaning and feeling (Dewey, 1934<sup>[85]</sup>; Lemke, 1990<sup>[58]</sup>; Lemke, 2015<sup>[86]</sup>) and all conceptual work involves some aesthetic commitment. Richard Dawkins (1997), for example, describes the "spine-shivering, breath-catching awe ... that modern science can provide." Charles Darwin famously underscored the grandeur in the view of life suggested by his theory of natural selection (Darwin, 1968).

In addition, individuals who are prone to experiencing awe on a regular basis are known to have an increased tolerance for ambiguity and uncertainty (Shiota, Keltner, & John, 2006). Gottlieb et al. (2018) have shown that the disposition to experience awe predicts a more accurate understanding of how science works, rejection of creationism, and rejection of unwarranted teleological explanations more broadly. Developing a scientific identity can, therefore, be viewed as the development of a "taste" for science (Anderhag, 2016<sup>[87]</sup>), drawing on Bourdieu (1987<sup>[88]</sup>). This is consistent with views from the psychological literature showing the reciprocal relationship between cognition and affect (Hidi, Renninger and Krapp, 2004<sup>[89]</sup>; Schiepe-Tiska et al., 2016<sup>[90]</sup>).

Environmental crises are a pressing global concern. Threats to the environment are increasingly apparent in students' everyday lives, including an increasing focus in the media. Students are challenged to understand and respond to complex environmental issues, which involve scientific knowledge and may require personal decision making and action. Knowledge of and concern for environment challenges allied with a disposition of hope are important ingredients of agency in the Anthropocene (Li and Monroe, 2018<sup>[91]</sup>; Li and Monroe, 2017<sup>[92]</sup>; Ojala, 2012<sup>[93]</sup>), and part of the PISA 2025 identity construct.

The science identity construct in PISA 2025 draws on a range of the elements contributing to a scientific identity, reflecting the need to assess competencies relevant to life in the 21st Century and emphasising

particularly student ability to use scientific information to inform decision making and agency – elements which will be measured both in the cognitive and non-cognitive instruments.

### *Defining Science Identity for PISA 2025*

The PISA 2025 assessment will evaluate the following elements of science identity:

- Science capital and epistemic beliefs
  - Science capital (science-related knowledge, engagement with science, and social capital)
  - Epistemic beliefs, i.e. general values of science and scientific enquiry
- Science capital: Attitudes and dispositions
  - Science self-concept (sense of self in relation to science including future participation)
  - Science self-efficacy
  - Enjoyment of science
  - Instrumental motivation
- Environmental awareness, concern and agency

These areas were selected for measurement because they are important attributes of a scientifically educated individual.

Science capital is synthetic measure of the degree of knowledge of science that an individual has, their engagement in activities of a scientific nature both formally and informally, their attitudes and dispositions, and their knowledge and understanding of scientific work gained from outside of school, e.g. if their parents work in science or have scientific pursuits.

Epistemic beliefs shown in Box 2.9 are closely related to students' valuing of scientific perspectives and approaches to enquiry involving an appreciation of the objects, products and processes that drive scientific exploration. Evidence of this dimension of identity could include a commitment to evidence as the basis of belief for explanations of the material world; being comfortable with uncertainty and the notion of risk; valuing evidence-based argument and debate as a means of establishing the validity of any idea; and a commitment to the scientific approach to enquiry when appropriate.

As part of epistemic beliefs, a critical disposition is required for the competency "research, evaluate and use scientific information for decision making and action". Elements of critical disposition in science that would be highly desirable for young people to develop would be: the capacity and confidence to be critical consumers of science; the disposition to use science as a part of their intellectual toolkit in making decisions that involve multiple forms of knowledge; a recognition of competing values and knowledge claims about science-related issues; a concern with issues of equity associated with science and technology development and its deployment; and presenting a considered reasoned stance on science-related issues that values scientific evidence.

Of relevance here is a critical orientation towards the role of science in identifying and dealing with environmental/sustainability issues, including designing scientific solutions that contribute to equity and social justice. This is reflected in a concern for the environment and actions that will sustain the planet, and a willingness to undertake appropriate scientifically informed actions. In particular, this disposition is demonstrated by the ability to recognise the complexity of many environmental issues and identify the competing scientific principles and social considerations in deciding on or advocating for appropriate practices.

Attitudes and disposition towards science shown in Box 2.10 play a significant role in students' interest, attention, and response to science and technology, and to issues that affect them in particular. One goal

of science education is to develop attitudes that lead students to engage with scientific and science-related issues. Such attitudes are needed to address the personal, local, national and global issues that young people will encounter in their lives. Interest in science also has an established relationship with achievement, and aspirations to further study, careers and lifelong learning in science. For instance, there is a considerable body of literature that shows that for most individuals who pursue scientific careers, interest in science was established by age 14 (Ormerod and Duckworth, 1975<sup>[94]</sup>; Tai et al., 2006<sup>[95]</sup>). Policy concerns in many OECD countries about the number of students, particularly females, choosing to pursue the study of physics, engineering and computer science make the measurement of attitudes towards science an important aspect of the PISA assessment.

For PISA 2025, the earlier constructs of environmental awareness and environmental optimism have been modified and extended to environmental awareness, environmental concern, and environmental agency, as shown in Box 2.11. Given the major challenges represented by threats of anthropogenic climate change, biodiversity loss and, more recently, the global pandemic, and the significant movements based on youth concern for intergenerational justice seen, for instance, in the youth strikes for climate, there is a pressing need to explore identity dimensions related to environmentally related competencies, concerns and agency. To what extent do young people acknowledge that science is core to providing solutions to these crises and to what extent do they see themselves as having any agency to engage with and act on these issues at the personal, local and global levels as appropriate?

## Box 2.9. Science capital and epistemic beliefs

### Science capital

This dimension of scientific identity is indicated by:

- An understanding of the nature of some scientific work
- The general level of knowledge of scientific ideas
- A feeling of connection to science and the personal relevance of science
- Engagement with science-related activities at home and in school
- Knowledge of and support from science-interested people

### Epistemic beliefs

This dimension of scientific identity is indicated by:

- A commitment to evidence as the basis of belief for explanations of the material world
- A commitment to scientific approaches to enquiry when appropriate
- A valuing of critique as a means of establishing the validity of any idea
- Developing an interest in scientific phenomena and associated models and explanations
- Trust in claims made by a consensus of scientists and domain-specific experts compared to other sources of information
- A recognition that uncertainty is an inherent feature of any scientific inquiry and its implications
- A recognition that scientific knowledge evolves and changes
- Understanding that science can make an important contribution to solving social and environmental problems

### Box 2.10. Attitudes and dispositions

This is a dimension of science identity indicated by:

- A willingness to engage with science-related issues and consider the issues critically using both science and other forms of knowledge or values
- How closely the individual identifies with science: recognition by self and others of competency to engage with science-related phenomena
- How capable the student perceives they are at the sciences
- The level of interest students have in pursuing scientific careers or the study of a science after school
- The range of extracurricular and out-of-school science activities that students engage in
- How much students like learning about the sciences both in and out of school

### Box 2.11. Environmental awareness, concern and agency

This construct is indicated by:

- Taking a critical, evidence-informed perspective on personal and socially relevant environmental issues (including environmental awareness, concern and agency)
- Awareness of environmental issues and recognition of the scientific and social complexity underlying environmentally sustainable actions
- A concern for the environment and sustainable living, and the issues of equity and social justice they raise
- Critical evaluation of the role of science and other factors in sustainability practices
- A disposition to take and promote environmentally sustainable practices
- A sense of personal agency that is informed by scientific and environmental understanding

Further detail on some of these constructs can be found in the PISA 2025 Questionnaire Framework and in the book *Science-Related Outcomes: Attitudes, Motivation, Value Beliefs, Strategies* (Schiepe-Tiska et al., 2016<sup>[90]</sup>), written by the team that developed the science framework for PISA 2015.

## Assessment features

### ***Cognitive demand***

A key feature of the PISA 2025 framework is the definition of levels of cognitive demand within the assessment of science and across all three competencies of the framework. In assessment frameworks, item difficulty, which is empirically derived, is often confused with cognitive demand. Empirical item difficulty is estimated from the proportion of the test taker population that is successful in solving the item correctly and thus assesses the overall facility of the test taker population with an item, whereas cognitive demand refers to the type of mental processing required (Davis and Buckendahl, 2011<sup>[96]</sup>). Care needs to be taken to ensure that the depth of knowledge required, i.e. the cognitive demand required of students, is understood explicitly by the item developers and users of the PISA framework. For instance, an item can have high difficulty because the knowledge it is testing is not well known but the cognitive demand is simply recall. Conversely, an item can be cognitively demanding because it requires the individual to relate and consider many items of knowledge, each of which are easily recalled. Thus, not only should the PISA test instrument discriminate in terms of performance between easier and harder test items, but the test also needs to provide information on how students across the ability range can deal with problems at different levels of cognitive demand (Brookhart and Nitko, 2011<sup>[97]</sup>).

The competencies are articulated using a range of terms defining cognitive demand through the use of verbs such as “recognise”, “interpret”, “analyse” and “evaluate”. However, in themselves, these verbs do not necessarily indicate a hierarchical order of difficulty that is dependent on the level of knowledge required to answer any item. Various classifications of cognitive demand schemes have been developed and evaluated since Bloom's Taxonomy was first published (Bloom, 1956<sup>[98]</sup>); and the frameworks described briefly below have served to develop the knowledge and competencies in the PISA 2025 Framework. These have been largely based on categorisations of knowledge types and associated cognitive processes that are used to describe educational objectives or assessment tasks.

Bloom's Revised Taxonomy (Anderson and Krathwohl, 2001<sup>[99]</sup>) identifies four categories of knowledge: factual, conceptual, procedural and metacognitive. This categorisation considers these forms of knowledge to be hierarchical and distinct from the six categories of performance used in Bloom's first taxonomy: remembering, understanding, applying, analysing, evaluating and creating. In Anderson and Krathwohl's framework, these two dimensions are now seen to be independent of each other, allowing for lower levels of knowledge to be crossed with higher-order skills and vice versa.

Another schema can be found in the framework based on “depth of knowledge” developed by Webb (1997<sup>[100]</sup>) specifically to address the disparity between assessments and the expectations of student learning. For Webb, levels of depth can be determined by taking into account the complexity of both the content and the performance required. His schema consists of four major categories: Level 1 (recall), Level 2 (using skills and/or conceptual knowledge), Level 3 (strategic thinking), and Level 4 (extended thinking). Each category is populated with a large number of verbs that can be used to describe cognitive performances. Some of these appear at more than one level. This framework offers a more holistic view of learning and assessment tasks and requires an analysis of both the content and cognitive process demanded by any task. Webb's depth of knowledge (DOK) approach is a simpler but more operational version of the SOLO Taxonomy (Biggs and Collis, 1982<sup>[101]</sup>) which describes a continuum of student understanding through five distinct stages of pre-structural, unistructural, multi-structural, relational and extended abstract understanding.

More recently, Tekkumru-Kisa, Stein and Doyle (2020<sup>[102]</sup>) have published a schema that discusses the levels of demands of tasks and distinguishes the level of thinking needed to engage in different types of tasks and scientific practices. This provides some insights into the nature of higher-order tasks and the demands they make, which are particularly salient for this framework and items that might assess the competencies at higher levels.

In drawing up such a framework, it is recognised that there are challenges in developing test items based on a cognitive hierarchy. The three main challenges are that:

- Too much effort is made to fit test items into particular cognitive frameworks, which can lead to poorly developed items.
- Misclassification between intended and actual demand with frameworks defining rigorous, cognitively demanding goals, and items that may operationalise the standard in a much less cognitively demanding way.
- Without a well-defined and understood cognitive framework, item writing and development often focuses on item difficulty and uses a limited range of cognitive processes and knowledge types, which are then only described and interpreted post hoc, rather than building from a theory of increasing competency.

The approach taken for the PISA 2025 Framework is to draw on these frameworks in developing our definition of cognitive demand. As the competencies are the central feature of the framework, the cognitive framework needs to assess and report on them across the student ability range. Webb's Depth of Knowledge Levels offer a taxonomy for cognitive demand that requires items to identify both the cognitive demand from the verbal cues that are used, e.g. analyse, arrange, compare, and the expectations of the depth of knowledge required.

Figure 2.2. PISA 2025 framework for cognitive demand

|           |                      | Competencies                     |                                                                                                             |                                                                                  | Depth of Knowledge |        |      |
|-----------|----------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------|--------|------|
|           |                      | Explain phenomena scientifically | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically | Research, evaluate and use scientific information for decision making and action | Low                | Medium | High |
| Knowledge | Content Knowledge    |                                  |                                                                                                             |                                                                                  |                    |        |      |
|           | Procedural Knowledge |                                  |                                                                                                             |                                                                                  |                    |        |      |
|           | Epistemic Knowledge  |                                  |                                                                                                             |                                                                                  |                    |        |      |

The grid in Figure 2.2 above provides a framework for mapping items against the two dimensions of knowledge and competencies. In addition, each item can also be mapped using a third dimension based on a depth of knowledge taxonomy. This provides a means of operationalising cognitive demand, as each item can be categorised as making demands that are:

- **Low (L):** Carrying out a one-step procedure, for example, one that requires the recall and use of a fact, term, principle or concept, or locating a single point of information from a graph or table. A one-step procedure might also involve sorting using a single criterion, classifying with easily observable, macroscopic features, identifying one element of evidence that does or does not support a claim, or using everyday or simple school science concepts for explanations in familiar contexts, e.g. why a metal spoon gets hot when placed in a hot drink.
- **Medium (M):** Use an application of any of the three forms of (content, procedural, epistemic) knowledge to describe or explain phenomena, select appropriate procedures involving two or more steps, organise/display data, interpret or use simple data sets or graphs to support or question a claim, construct an argument from limited evidence in familiar contexts, or use standard models to explain in familiar contexts. Cognitively, this would require either the use of two or more steps in the reasoning using one idea, or relating two ideas/pieces of information in one step generally in familiar contexts. Identify from two or more pieces of evidence their appropriate or inappropriate use.
- **High (H):** Analyse more complex information or data, synthesise or evaluate evidence, justify, reason given various sources, develop a plan or sequence of steps to investigate and respond to a problem, or critique a flawed argument using complex or abstract concepts. Cognitively, this would require either the use of two or more steps in the reasoning, the use of two or more ideas, the evaluation of divergent claims, the consideration of rebuttals or qualifiers often in unfamiliar contexts, and the ability to make connections among two or more representations to develop meaning.

Thus, items that merely require recall of one piece of information make low cognitive demands, even if the knowledge itself might be quite complex. In contrast, items that require recall of more than one piece of knowledge and require a comparison and evaluation made of the competing merits of their relevance would be seen as having high cognitive demand. The difficulty of any item, therefore, is a combination both of the degree of complexity and range or depth of knowledge it requires, and the cognitive operations that are required to process the item.

Therefore, the factors that determine the demand of items assessing science achievement include:

- The number and degree of complexity of elements of knowledge demanded by the item
- The level of familiarity and prior knowledge that students may have of the content, procedural and epistemic knowledge involved
- The cognitive operation required by the item, e.g. recall, analysis, evaluation
- The extent to which forming a response is dependent on models or abstract scientific ideas

This four-factor approach allows for a broader measure of scientific competence across a wider range of student ability. Categorising the cognitive processes required for the competencies that form the basis of scientific literacy together with a consideration of the depth of knowledge required offers a model for assessing the level of demand of individual items. In addition, its relative simplicity offers a framework for minimising the problems encountered in applying such frameworks. The use of this cognitive framework will also facilitate the development of an a priori definition of the descriptive parameters of the reporting proficiency scales (see Figure 2.3. A tool for constructing and analysing assessment units and items for the Cognitive Questionnaire

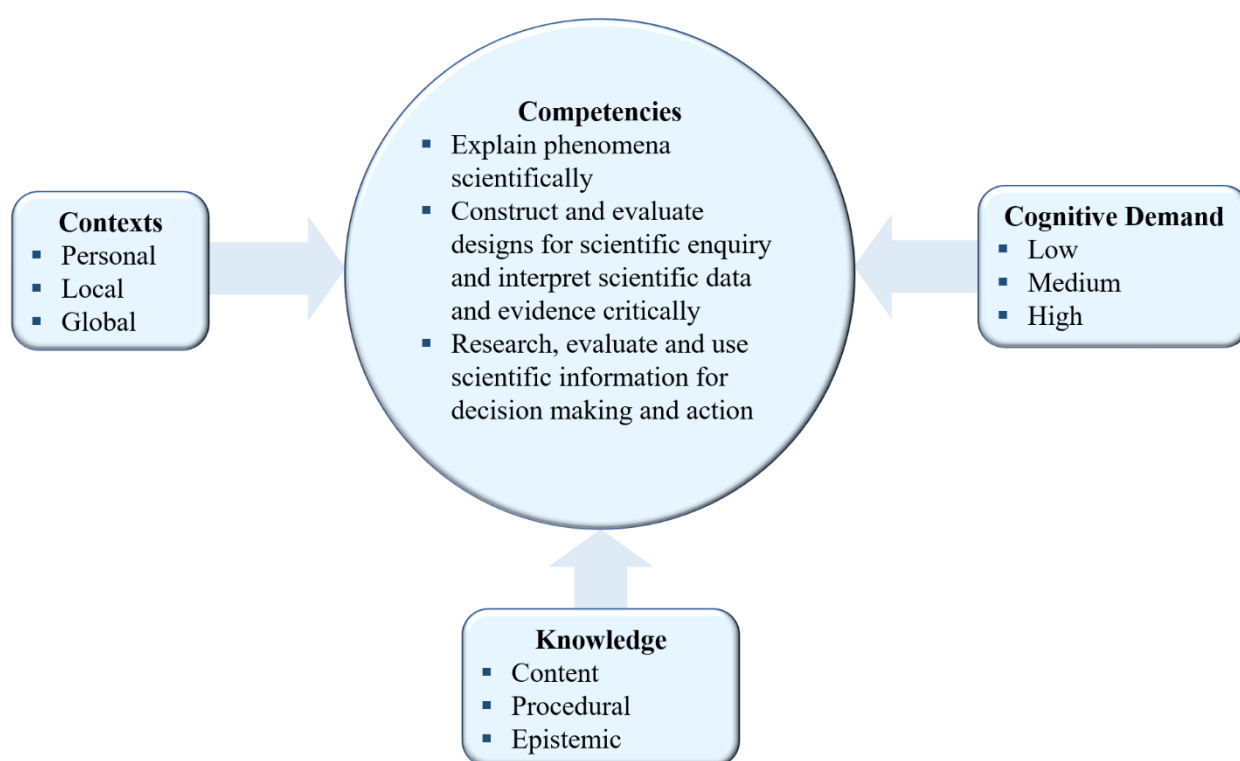
### ***Test characteristics***

In accordance with the PISA definition of science outcomes, test questions (items) will require the use and application of the scientific competencies and knowledge within a context.

Figure 2.3 is a variation of Figure 2.1 that presents the basic components of the PISA framework for the 2025 scientific competency assessment in a way that can be used to relate the framework with the structure and the content of assessment units. This may be used both synthetically as a tool to plan assessment exercises, and analytically as a tool to study the results of standard assessment exercises. As a starting point to construct assessment units, it shows the need to consider the contexts that will serve as stimulus material, the competencies required to respond to the questions or issues, the knowledge central to the exercise, and the cognitive demand.

A test unit is defined by specific stimulus material, which may be a brief written passage, or text accompanying a table, chart, graph or diagram. In units created for PISA 2015, the stimulus material included non-static stimulus material, such as animations and interactive simulations, for the first time. These will also be included in the test for PISA 2025 using any enhanced facilities available. The items are a set of independently scored questions of various types, as illustrated by the examples already discussed.

**Figure 2.3. A tool for constructing and analysing assessment units and items for the Cognitive Questionnaire**



The reason PISA employs this unit structure is to facilitate the employment of contexts that are as realistic as possible, reflecting the complexity of real situations, while making efficient use of testing time. Using situations about which several questions can be posed, rather than asking separate questions about a larger number of different situations, reduces the overall time required for a student to become familiar with the material in each question. However, the need to make each score point independent of others within a unit needs to be considered. It is also necessary to recognise that, because this approach reduces the number of different assessment contexts, it is important to ensure that there is an adequate range of contexts so that bias due to the choice of contexts is minimised.

PISA 2025 test units will require the use of all three scientific competencies and draw on all three forms of science knowledge. In most cases, each test unit will assess multiple competencies and knowledge categories. Individual items, however, may only assess only one form of knowledge and one competency.

The need for students to read texts to understand and answer written questions on science raises an issue of the level of reading comprehension that will be required. Stimulus material and questions will use language that is as clear, simple and brief, and as syntactically simplified as possible while still conveying the appropriate meaning. The number of concepts introduced per paragraph will be limited. Questions within the domain of science that assess reading or mathematical literacy will be avoided.

### *Item response formats*

Three classes of items will be used to assess the competencies and scientific knowledge identified in the framework. About one-third of the items will be in each of the three classes:

- **Simple multiple-choice** items calling for:
  - Selection of a single response from four options
  - Selection of a “hot spot,” an answer that is a selectable element within a graphic or text
- **Complex multiple-choice** items calling for:
  - Responses to a series of related “yes/no” questions that are treated for scoring as a single item (the typical format in PISA 2006)
  - Selection of more than one response from a list
  - Completion of a sentence by selecting drop-down choices to fill multiple blanks
  - Drag-and-drop responses, allowing students to move elements on screen to complete a task of matching, ordering or categorising.
- **Constructed response** items calling for written or drawn responses:
  - Constructed response items in science typically call for written responses ranging from a phrase to a short paragraph (e.g. two to four sentences of explanation). A small number of constructed response items call for drawing (e.g. of a graph or diagram). For computer delivery, any such items will be supported by simple drawing editors that are specific to the response required.

Also, in PISA 2025, as in PISA 2015, some responses will be captured by interactive tasks using simulations, for example, a student’s choices for manipulating variables in a simulated scientific enquiry, or require the construction of an explanation for the observed behaviour of the simulated system. Responses to these interactive tasks will likely be scored as complex multiple-choice items. Some kinds of responses to interactive tasks may be sufficiently open-ended that they will be treated as constructed response.

### **Assessment structure**

For PISA 2025, computer-based assessment will be the primary mode of delivery for all domains. All new science items will be developed for the computer-based assessment and, where possible, for paper-based assessment. A paper-based assessment instrument will be provided for countries choosing not to test their students by computer.

The desired balance between the three knowledge components: content, procedural and epistemic knowledge is shown in Table 2.3 in terms of percentages of score points. Table 2.3 also shows the target distribution of score points among the various knowledge categories. These weightings are broadly consistent with the previous framework and reflect a consensus view amongst the experts consulted in the writing of this framework.

The elements of the cognitive science framework that contribute to agency in the Anthropocene will be measured using the scale provided below in the Environmental section to provide a measure of the extent to which students have this competency.

**Table 2.3. Target distribution of score points for knowledge**

| Knowledge types            | Systems  |        |               |                    |
|----------------------------|----------|--------|---------------|--------------------|
|                            | Physical | Living | Earth & space | Total over systems |
| Content                    | 15-20%   | 15-20% | 10-15%        | 38-48%             |
| Procedural                 | 10-13%   | 10-13% | 7-10%         | 27-33%             |
| Epistemic                  | 8-11%    | 8-11%  | 7-10%         | 24-30%             |
| Total over knowledge types | 37%      | 37%    | 26%           | 100%               |

The target balance for scientific competencies is provided in Table 2.4. These weightings have been chosen so that the assessment is evenly split between items that draw predominantly on content knowledge and items that draw predominantly on procedural and/or epistemic knowledge.

Item contexts will be spread across personal, local/national and global settings roughly in the ratio 1:2:1, as was the case in PISA 2015. A wide selection of areas of application will be used for units, subject to satisfying as far as possible the various constraints imposed by the distribution of score points shown in Table 2.3 and Table 2.4.

**Table 2.4. Target distribution of score points for scientific competencies**

| Scientific competencies                                                                          | % of score points (approx.) |
|--------------------------------------------------------------------------------------------------|-----------------------------|
| Explaining phenomena scientifically                                                              | 36-44%                      |
| Construct and evaluate designs for scientific enquiry and interpret data and evidence critically | 24-36%                      |
| Research, evaluate and use scientific information for decision making and action                 | 24-36%                      |
| <b>Total</b>                                                                                     | <b>100%</b>                 |

## Reporting scale

To meet the aims of PISA, the development of scales of student achievement is essential. A descriptive scale of levels of competence needs to be based on a theory of how the competence develops, not just on a post-hoc interpretation of what items of increasing difficulty seem to be measuring. The PISA 2006 framework therefore explicitly defined the parameters of increasing competence and progression, allowing item developers to design items representing this growth in ability (Kane, 2006<sup>[103]</sup>; Mislevy and Haertel, 2006<sup>[104]</sup>). Initial draft descriptions of the scales are offered below, though it is recognised that these may need to be modified as data are accumulated after field testing of the items. Although comparability with the PISA 2015 scale descriptors (OECD, 2017<sup>[5]</sup>) has been maximised in order to enable trend analyses, the new elements of the PISA 2025 framework such as the new competency “research, evaluate and use scientific information for decision making” have also been incorporated.

## Trend reporting

To report trends over the years, a number of items are repeated in each cycle. These enable student overall performance on the test to be compared with previous years. Only the trend in overall performance can be reported and not the trends in individual competencies, especially as these change each twelve-year cycle.

Table 2.5. Reporting scale proposed for PISA 2025

| Level | Descriptor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6     | At level 6, working in unfamiliar contexts, students can draw on a range of scientific theories, principles, and concepts of high demand from different disciplines to build models, consider their limitations, and use those models to construct or evaluate scientific explanations of complex phenomena. They can apply those explanations to make predictions not only about phenomena but also about potential future developments or implications for society. Students are able to make judgements requiring multiple criteria to identify and explain the purposes of scientific enquiries of different types and justify how these purposes align with the scientific question being investigated. They can apply epistemic and procedural knowledge to evaluate and justify designs of multi-faceted enquiries. They can interpret and align data in a variety of representational forms from data sets. Students can evaluate interpretations of complex data sets drawing on procedural and epistemic knowledge to make reasoned judgements about their accuracy and precision. When required to evaluate sources of high cognitive demand, containing multi-modal information, students can identify those sources which are most credible based on one or more scientific criteria or one or more fact-checking procedures. From these they can draw appropriate conclusions. They can provide a justification for their choice drawing on content, procedural or epistemic knowledge of science, and/or social, ethical or economic considerations. They can identify the premises in an argument or one or more flaws in sources of scientific information – either in their credibility, their use of evidence, or in the arguments from the evidence. Based on their evaluation, they can provide justifications considering multiple issues for possible decisions and actions. |
| 5     | At level 5, students can draw on a range of scientific theories, principles, and concepts of medium to high cognitive demand to identify and construct scientific explanations of familiar phenomena in all contexts. They can use these explanations to make predictions. They can identify both the strength and the limitation of a model. Drawing on procedural and epistemic knowledge, students can identify and explain the purposes of enquiries of different types and distinguish the independent, dependent and controlled variables for a scientific investigation. They can apply epistemic and procedural knowledge to evaluate alternative experimental/investigative designs and justify their choices. They can interpret complex data representations, evaluate with reasons whether a given interpretation is flawed and provide a more appropriate interpretation would be. Drawing on multiple sources of information of medium to high cognitive demand containing multi-modal information, students can identify those sources which are most credible based on one or more scientific criteria or standard fact-checking procedures. They can provide a justification for their choice drawing on either content, procedural or epistemic scientific knowledge and either a social, ethical or economic consideration. They can identify the premise in an argument or a flaw in a source – either in its credibility, its use of evidence, or in the arguments it uses. Based on their evaluation, they can provide a reasoned justification for possible decisions and actions.                                                                                                                                                                                                                                                                                           |
| 4     | At level 4, students can construct and evaluate scientific explanations of phenomena by drawing on a range of scientific principles and various representations of medium to high cognitive demand. Given a model, they can identify either a strength or a limitation. Drawing on procedural and epistemic knowledge they can propose experimental or investigative designs involving two or more independent variables in a limited context. They can justify a design for an enquiry using procedural or epistemic knowledge. They can interpret straightforward data representations and evaluate the validity of scientific claims based on the evidence provided. Given a need for information to inform decision making or action, students can draw on multiple sources of medium cognitive demand, containing multi-modal information, to identify which is most credible using a basic fact-checking procedure or another science-based criterion. They can identify the evidence supporting a claim and whether it is a scientific, economic or social issue. They can provide a justification for their choice. Given several possible errors in a source or its interpretation, they can identify the error and describe the flaw.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3     | At level 3, students can construct or evaluate scientific explanations and models of phenomena with relevant cueing or support, by drawing on scientific principles and concepts, and representations of medium cognitive demand. Given a simple model, they can identify either a strength or a limitation of the model. They can provide a justification for a simple experimental design requiring control of variables or sampling of a population and distinguish between a scientific and non-scientific question using elements of procedural and epistemic knowledge. Given a research design, they can identify the research question or given an interpretation of a set of data, they are able to identify a flaw in the interpretation using procedural or epistemic knowledge. Alternatively, offered a set of simple data presented in a tabular or graphical representation, they can provide a valid interpretation. Given a need for information for decision making or action from sources of medium cognitive demand, students can identify which sources of evidence are relevant and justify their choice. They can use one or more criteria to judge whether a source is credible and provide an evidence-based justification for their choice. They can decide a scientific method that is appropriate to answering a given question.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2     | At level 2, students can identify an appropriate scientific explanation from a non-scientific explanation for everyday/common scientific phenomena in familiar personal, local or global contexts, by drawing on appropriate scientific concepts of low to medium cognitive demand. They can offer a simple explanation of an everyday or familiar scientific phenomenon that draws on basic school scientific principles and concepts. They can evaluate designs for simple enquiries drawing on elements of procedural knowledge and identify appropriate interpretations of data sets with simple relationships and identify outliers and possible reasons for their occurrence. Using their epistemic knowledge, they can identify appropriate explanations for variations in measurement. Given a need for information for decision making or action, students can identify a relevant source of information from several of low to medium cognitive demand that is needed to inform action on a given scientific problem and justify its use. Using a single criterion e.g. relevant expertise, scientific consensus, or basic science, they can identify whether the source is credible. They can identify the evidence from straightforward graphical or tabular data that supports a claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1a    | At level 1a, in familiar personal, local or global contexts, students can identify a scientific explanation of a simple phenomenon drawing on basic scientific knowledge, information or evidence of low cognitive demand. Students can choose the most appropriate experimental design involving control of one variable by drawing on low-level procedural knowledge. Students can identify one relevant piece of evidence that is needed to inform action on a given scientific problem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1b    | At Level 1b, in familiar personal, or local contexts, students can recognise a scientific explanation of an everyday macroscopic phenomenon. Drawing on low-level procedural knowledge, they can identify from two experimental designs which would be the better to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| answer a given question. They can select from different interpretations of a simple data which set/graphical display is better. Students can identify one relevant source of evidence needed to inform action on a given scientific issue from several. |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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The proposed level descriptors are based on the 2025 framework described in this chapter and offer a qualitative description of the differences between levels of performance. The factors used to determine the demand of items, assessing science achievement, that have been incorporated into this outline of the proficiency scales include:

- The number and degree of complexity of elements of knowledge demanded by the item
- The level of familiarity and prior knowledge that students may have of the content, procedural and epistemic knowledge involved
- The cognitive operation required by the item, e.g. recall, analysis, construction, evaluation
- The extent to which forming a response is dependent on models or abstract scientific ideas

The intention of the test is to distribute items equally across all levels.

## Environmental science

### *Introduction*

Human impact is significantly changing Earth's systems (IPCC, 2021<sup>[9]</sup>). These changes began with industrialisation in the 1800s and have increased exponentially since 1950 (Lewis and Maslin, 2015<sup>[105]</sup>). While humans may have the highest living standards and life expectancy ever (Pinker, 2018<sup>[106]</sup>; Rosling, Rosling and Rosling Rönnlund, 2018<sup>[107]</sup>), the overwhelming majority of other living organisms are in crisis – a crisis that is threatening humanity, as well. Climate change and biodiversity loss is impacting all species, many irrevocably, and precipitating the sixth mass extinction (Dirzo et al., 2014<sup>[8]</sup>). Human impact in the Anthropocene has led to significant disruptions to the systems within the biosphere, hydrosphere, geosphere and atmosphere (IPCC, 2021<sup>[9]</sup>). Thus, humanity faces an uncertain future. For many people, and young people in particular, climate change is seen as the greatest challenge of our time. To meet this challenge, scientific knowledge and reasoning is an essential element for decision making individually, communally and globally to mitigate impacts and adapt to more sustainable practices and systems (Steffen et al., 2011<sup>[108]</sup>).

With an increasing population of more than 7 billion people and finite natural resources, challenges include: ensuring clean air and water, ensuring food security, managing diseases, generating renewable energy, striving for health and well-being, and managing our own living choices responsibly to ensure ample resources for all species and future generations (IPCC, 2021<sup>[9]</sup>; Barnosky et al., 2012<sup>[109]</sup>; Rockström et al., 2009<sup>[110]</sup>). Dealing with these challenges, and the many others resulting from human-induced climate change and environmental impact, will require that young people are able to understand and act on contributions from science and technology, alongside other disciplines and knowledge systems (Schipper, Dubash and Mulugetta, 2021<sup>[111]</sup>).

Scientifically informed 15-year-olds will have to evaluate the sources of information about these issues, as well as use creative and systems thinking to explore and consider appropriate courses of action to regenerate and sustain Earth's systems (Young et al., 2006). Scientific knowledge is important in informing the decisions and actions that contribute to individuals and communities making informed, sustainable living choices and developing the critical thinking, media literacy and hopefulness required to address this challenge (Monroe et al., 2017<sup>[112]</sup>).

In addition, an appreciation of diverse knowledge systems and respect for cultural heritage also contributes to potential solutions (Reyes-García et al., 2018<sup>[113]</sup>); (Salomon et al., 2019). Young people need to be aware of how systems of governance and power might frame and impact issues that are social, environmental and ecological (Berkes and Folke, 1998<sup>[114]</sup>; Muller, Hemming and Rigney, 2019<sup>[115]</sup>; Young et al., 2006<sup>[42]</sup>). Young people will benefit from working across generations to address socio-ecological inequities and to create and sustain healthy communities (Thiery et al., 2021). This will require education to support young people to develop an ethic of care and justice (Merrett, 2004<sup>[116]</sup>; Skovdal and Evans, 2016<sup>[117]</sup>) based on a worldview that can be enhanced through a science education that presents an eco-centric worldview, which includes humans as part of the environment rather than separate from it. Such a systems thinking perspective is necessary to look beyond patterns and linear relationships to support the design and enactment of sustainable living choices. For instance, systems thinking is usefully applied when considering the impact of personal choices (such as whether to adopt a predominantly vegetable-based diet or use public transport); local choices (e.g. working toward reducing the availability of single-use plastic); community actions (e.g. collaborating with others to engage in civic actions to change the regional transportation system); and global choices (e.g. supporting international policy to reduce fossil-fuel dependence).

Agency will be required for 15-year-olds to enact the necessary changes to meet their goals (OECD, 2019<sup>[118]</sup>). Agency involves undertaking critical appraisal of complex systemic issues and evaluating whether evidence-based claims on these issues are made by legitimate experts. It involves using their

evaluation to make decisions about setting goals to bring about change and how to take responsible action, as well as making decisions by examining and reasoning with the evidence in a scientific way. The ability to make decisions to act responsibly for themselves, and with others, is a measure of agency in the Anthropocene. For example, demonstrating agency in the Anthropocene involves reflecting on personal lifestyle choices and implementing change, influencing others to reflect and change, and providing feedback to organisations and governments about changes required. These actions contribute to better management of resources (such as in circular economies where wastes are eliminated as materials are (re)cycled).

Science education is critical in providing young people with a basic understanding of Earth's systems and their interactions with human systems. Understanding the degree to which these socio-ecological issues are complex and their interactions through the use of appropriate tools (such as systems mapping) is essential to prepare young people to address contemporary challenges, such as mitigating and adapting to climate change. In these uncertain times, young people also need the following set of attitudes and dispositions to work individually, with others, and across generations for systemic change and sustainability:

- **Systems thinking**, which is the ability to recognise complex interactions among relevant variables and understand the consequences of changes to those variables
- **Self-efficacy**, which refers to the belief that one can act
- **Collective efficacy**, which is believing that one's group can meet their goals
- **Outcome expectancy**, which is the belief that one's actions will make an impact on the issue of interest
- **Agency**, which is the perception that one influences one's own actions and circumstances
- **Hope**, which is the sense that there is a way toward a possible future that is worth achieving

These components are intertwined, as the ability to recognise complex systems requires consideration of how any intervention might improve a situation, the belief that one has the agency and efficacy to take the desired actions, and that achieving any goal contributes to a more hopeful and desirable vision of the future (Ajzen, 1985<sup>[119]</sup>; Snyder, Rand and Sigmon, 2002<sup>[120]</sup>). And those who believe their group can work effectively tend to have a greater sense of their own self-efficacy (Jugert et al., 2016<sup>[121]</sup>). Similarly, outcome expectancy is a core element of both hope and efficacy.

Systems thinking capabilities are important across all areas of science education and environmental issues provide important examples of the need to consider an issue at the systemic level. Systems can be ecological or social, or a combination of both. While many educators teach young people to identify the function of components of a system (such as planets and stars or veins and lymph nodes), the interactions between these components often create new structures and functions. Seeking to understand a system and its complex relationships enables recognition of how and when changes in one variable in a system can affect others and its potential mitigation.

Hope in particular has been demonstrated to be an essential attitude for addressing complex socio-ecological issues. Without hope, the belief that the current predicament will not change or improve may result in anxiety, depression and helplessness (Peterson et al., 1993). The spatial and temporal scale of contemporary environmental issues such as climate change and biodiversity loss have led to a definition of hope that includes actions that can be taken with others (Li and Monroe, 2017<sup>[92]</sup>; 2018<sup>[91]</sup>) or collective efficacy. This arises when individuals work with communities to effect change (Ojala, 2012<sup>[93]</sup>; Li and Monroe, 2017<sup>[92]</sup>; 2018<sup>[91]</sup>; Ardoin, Bowers and Wheaton, 2022<sup>[122]</sup>). Coupling a sense of hope with knowledge about the complexity of interconnected Earth systems will enable environmental and social challenges to be addressed (Ojala, 2015<sup>[123]</sup>). Key to this outcome is the belief that possible solutions and pathways exist that can be taken by individuals, communities, organisations, businesses and

governments (Li & Monroe, 2019). Thus, measuring whether young people have a sense of hope about the future is important in assessing the degree to which they have agency in the Anthropocene.

### ***Agency in the Anthropocene***

The central construct of the environmental-related outcomes of students' science education to be measured is defined as Agency in the Anthropocene (Box 2.12). A full elaboration of this construct and its associated competencies can be found in the supporting document *Agency in the Anthropocene* (White et al., 2023<sup>[124]</sup>).

#### **Box 2.12. Agency in the Anthropocene: Definition**

Agency in the Anthropocene requires understanding that human impact has already significantly altered Earth's systems, and continues to do so. Young people with Agency in the Anthropocene believe that their actions will be appreciated, approved and effective as they work to mitigate climate change, biodiversity loss, water scarcity and other complex issues and crises. Agency in the Anthropocene refers to ways of being and acting within the world that position people as part of (rather than separate from) ecosystems, acknowledging and respecting all species and the interdependence of life. Those with Agency in the Anthropocene acknowledge the many ways societies may have created injustices and work to empower all people to contribute to community and ecosystem well-being. They demonstrate hope, resilience and efficacy in the face of crises that are both social and ecological. Moreover, they respect and evaluate multiple perspectives and diverse knowledge systems and demonstrate their ability to engage with other young people and adults, across generations, in civic processes that lead to improved community well-being and sustainable futures. Young people with Agency in the Anthropocene work individually and with others across a range of scales, from local to global, to understand and address complex challenges that face all beings in our communities.

### ***Competencies for Agency in the Anthropocene***

The Anthropocene represents a time of significant challenge to our social structures and Earth systems. Addressing these challenges will require that we consider several changes: change in the ways that we as humans interact with each other and our environment; change in our environment, overall; change in our technologies; and change in our value systems. A young person growing up into this anthropocentric world requires three essential competencies that underpin the concept of Anthropocene Agency in PISA 2025, elements of which will be measured by the PISA 2025 science assessment, defined in Box 2.13.

### Box 2.13. Competencies for Agency in the Anthropocene

A 15-year-old student who demonstrates Agency in the Anthropocene can:

1. Explain the impact of human interactions with Earth's systems
1. Make informed decisions to act based on evaluation of diverse sources of evidence and application of creative and systems thinking to regenerate and sustain the environment
2. Demonstrate hope and respect for diverse perspectives in seeking solutions to socio-ecological crises

### ***Agency in the Anthropocene competencies in action***

A range of abilities underpin each of these competencies required for Agency in the Anthropocene. Those abilities are described in more detail below. The competencies are a mix of both cognitive and non-cognitive elements, reflecting the nature of agency in the Anthropocene.

#### *Competency 1: Explain the impact of human interactions with Earth's systems*

Elements of this competency are measured by Science Competency 1 (Explain phenomena scientifically). However, this competency focusses on human interactions to explore a student's understanding of human impact on Earth's systems. This competency requires both content and procedural knowledge.

A 15-year-old student who can explain the impact of human interactions with Earth's systems can:

1. Explain physical, living and Earth's systems that are relevant to the environment and how they interact with each other
2. Research and apply knowledge of human interactions with these systems over time
3. Apply this knowledge to explain both negative and positive human impacts within these systems over time
4. Explain how social, cultural or economic factors contribute to these impacts

#### *Competency 2: Make informed decisions to act based on evaluation of diverse sources of evidence and application of creative and systems thinking to regenerate and sustain the environment*

This competency draws on elements that are measured by Science Competency 2 (Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically) and Science Competency 3 (Research, evaluate and use scientific information for decision making and action). This competency requires content, procedural and epistemic knowledge.

A 15-year-old student who can make informed decisions to act based on critical appraisal of diverse sources of evidence and the application of creative, systems and intergenerational thinking to regenerate and sustain the environment can:

1. Access and critically appraise evidence from diverse ways of knowing and sources
2. Evaluate and design potential solutions to socio-ecological issues using creative and systems thinking, taking into account implications for current and future generations
3. Engage, individually and collectively, in civic processes to make informed, consensual decisions

4. Set goals, collaborate with other young people and adults across generations, and act for regenerative and enduring socio-ecological change at a range of scales (local to global)

*Competency 3: Demonstrate hope and respect for diverse perspectives in seeking solutions to socio-ecological crises*

This competency contains elements that are measured by the concept of “Science Identity”, including epistemic beliefs; dispositions of care and concern towards other people, other species, and the planet; and feelings of efficacy and agency in addressing socio-ecological crises. This competency requires content, procedural and epistemic knowledge.

A 15-year-old student who uses an ethic of care and justice, and demonstrates resilience, hope, efficacy and a respect for diverse perspectives in seeking solutions to social and environmental challenge can:

1. Evaluate actions drawing on an ethic of care for each other and all species based on a worldview where humans are part of the environment rather than separate from it (being eco-centric)
2. Acknowledge the many ways societies have created injustices and work to empower all people to contribute to community and ecosystem well-being
3. Exhibit resilience, hope and efficacy, individually and collectively, in responding to socio-ecological crises
4. Respect diverse perspectives on issues and seek solutions to regenerate impacted communities and ecosystems (Reyes-García et al., 2018<sup>[113]</sup>)

### **Reporting scale**

This section elaborates the environmental literacy proficiency scale based only on the cognitive elements to be measured by the Science Assessment Framework.

It should be noted that not all of the competencies of Agency in the Anthropocene defined above can be measured by the cognitive test. Instead, items from the science cognitive test that have an environmental focus and match the description of the competencies above will be used to construct a scale that is a measure of elements of Agency in the Anthropocene. The non-cognitive attitude questionnaire will measure other elements independently.

**Table 2.6. Reporting scale for the environmental scale in PISA 2025**

| Level | Descriptor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6     | At Level 6, students can combine scientific knowledge and systems thinking of high cognitive demand to explain how Earth systems operate, interact, and change over time, including long-term and large-scale human impacts. They can evaluate the quality of evidence and justify claims relating to both negative and positive impacts on Earth systems using scientific arguments based on evidence. Students can evaluate solutions to complex challenges considering environmental, social and economic dimensions. They can assess when compromise is required, anticipate unintended consequences, and consider impacts on current and future generations. Students can provide justifications for decisions and actions related to socio-ecological challenges, using evidence from multiple perspectives. |
| 5     | At Level 5, students can draw on scientific understanding of medium to high cognitive demand to explain complex interactions within and between Earth systems and the cumulative impacts of human activities over time. They can appraise information from more than one source to assess reliability, bias, and strength of evidence. Students can evaluate and compare alternative solutions to socio-ecological challenges using scientific or environmental reasoning. They can predict likely environmental outcomes based on trends in data and explain interactions among Earth systems resulting from human intervention. Students can justify decisions regarding environmental and climate-related risks using environmental or social considerations supported by evidence.                             |
| 4     | At Level 4, students can draw effectively on scientific ideas of medium to high cognitive demand to explain how Earth systems function, interact, and respond to human influence. They can analyse graphical, experimental, or modelled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | evidence to evaluate socio-ecological challenges. Students can identify contributing factors to socio-ecological change and explain how individual and community actions can influence broader systems. They can propose actions to reduce environmental and climate-related risks and evaluate their benefits and limitations. Students can justify decisions using combinations of environmental, social, cultural, and/or economic reasons, demonstrating the ability to reason about Earth systems.                                                                                                                                                                                                                                     |
| 3  | At Level 3, students can draw on knowledge of moderate cognitive demand to explain how human activities result in changes across Earth systems. They can interpret and compare data to identify patterns, trends, or relationships related to environmental impacts. Students can evaluate both the positive and negative impacts of proposed solutions to environmental and climate-related risks. They can recognise potential bias and comment on the credibility of information sources. They can explain how environmental changes may affect communities and ecosystems over time. Students can justify decisions and actions using environmental reasoning combined with one additional social, cultural, or economic consideration. |
| 2  | At Level 2, students can use knowledge of low to medium cognitive demand to describe how human activities influence Earth systems, including intended positive actions or unintended negative consequences. They can interpret simple data, diagrams, or examples to explain these impacts. Students can identify the advantage or disadvantage of a proposed environmental solution and recognise basic features of reliable information. They can explain how a practical solution to a socio-ecological challenge may affect themselves or their family. Students can select the best justification for a decision that resolves environmental and climate-related risks using simple, clear reasoning.                                  |
| 1a | At Level 1a, students can use knowledge of low cognitive demand to recognise a straightforward example of how human activity has had either a negative or positive impact on an Earth system. They can refer to a single piece of information or evidence to support a claim.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1b | At Level 1b, students can use knowledge of low cognitive demand to recognise a straightforward example of how human activity has had either a negative or positive impact on an Earth system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### ***Assessment for the environmental scale***

This scale will be constructed using content, procedural and epistemic knowledge questions in the science framework that are clearly related to any science that can be considered to be of an environmental or ecological nature. A similar assessment of environmental competence was done in 2006 for the “Green at Fifteen Project” (OECD, 2009<sup>[125]</sup>). Because of the cognitive focus of the science test, it will only be possible to measure Competency 1 and Competency 2. To measure this construct fully, however, it will also be necessary to ask questions about the following elements which will be included in the non-cognitive questionnaire:

- The science needed to respond to claims about environmental/health made by people or interest groups on the basis of other values/knowledges (e.g. responding to a person who refuses vaccination on the basis that there’s a percentage of people who develop serious side effects, or who argues that a number of vaccinated people have died)
- Identifying the science knowledge, including possible investigations, relevant for responding to different positions on environmental/health actions, including deciding on personal actions (e.g. identifying the science research needed to respond to the different concerns of farmers concerned about re-introduction of top predators into a local national park, or in deciding about culling introduced species such as brumbies; the question might for instance involve a matching of science investigations with a list of concerns or claims)
- Identifying which of different positions in an environmental controversy are scientifically based, and which are based on other knowledges and values
- Weighing different alternatives given the science knowledge relevant to a personal or community health/environmental issue, e.g. should I use plastic cups in catering for a large party, or glass cups and wash them in a dishwasher – the science would relate to the energetics of dishwashing, water and detergent use, recycling figures on plastic, etc. – it might simply ask which ideas are relevant for the decision
- Identifying the socio-ecological considerations that legitimately frame scientific research in an area (e.g. in scientific developmental research into mobile phones technical design, which issues might be expected to frame decisions, and what sort of principles are involved (economic, cultural, ethical, environmental), from a list – the sourcing of rare metals from exploitative practices, the

costs of extraction, the possibility of recycling of materials, the advertising campaign associated with the phone, the opinion of uses on phone colour)

- Predicting the consequences across different parts of the socio-ecological system that would flow from particular decisions, e.g. given a mapping of a complex system relating to an aquifer used by a population in multiple ways but increasingly contaminated by pesticide use by agriculture on which the community depends for food and livelihood, predict what environmental consequences might flow from a particular decision related to water use, or pesticide use, that might depend on knowing the pathways through related economic/recreational systems leading back to environmental impact

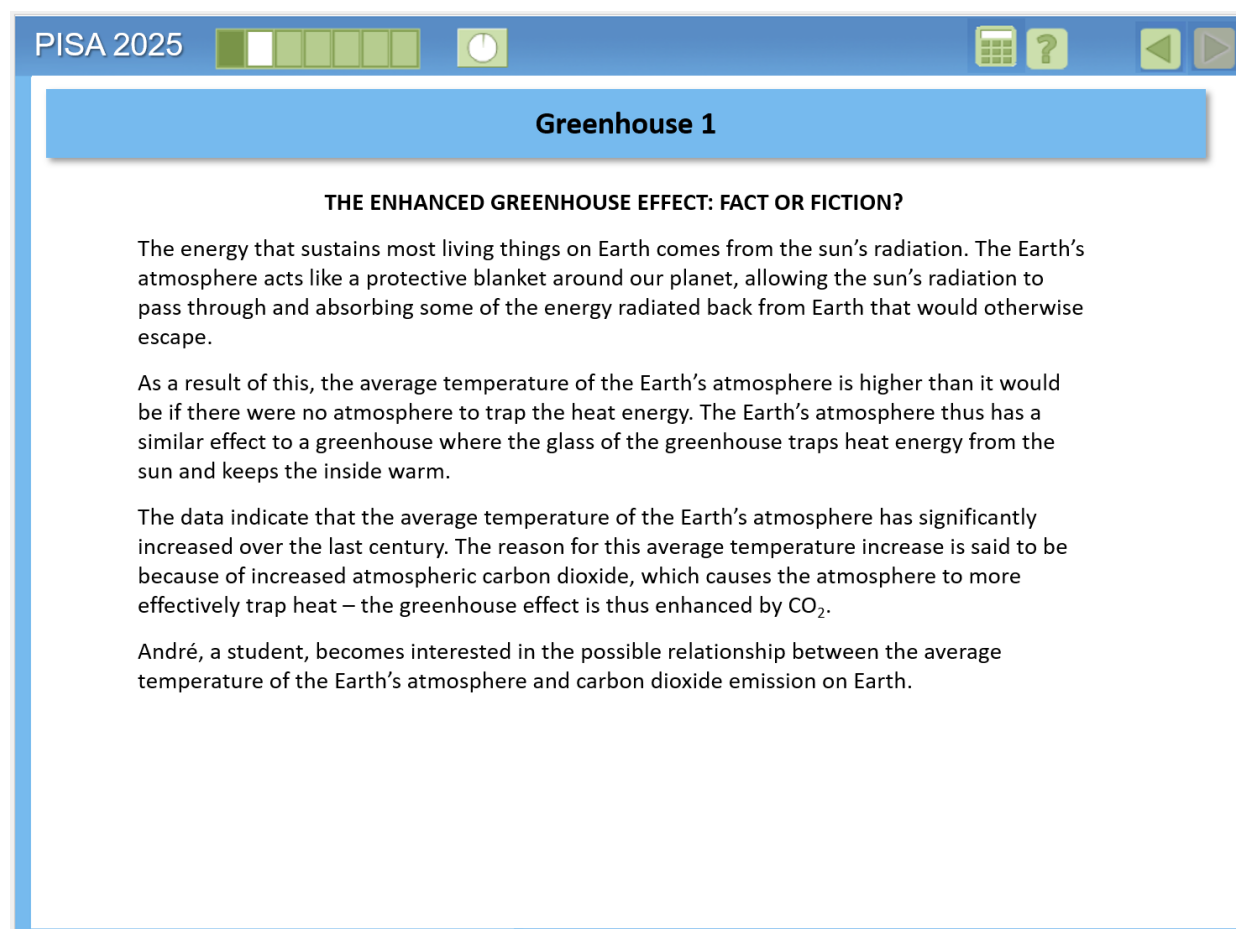
## Sample items

In this section, nine examples of science units are presented. Questions from the unit are shown in the manner they might be transposed and presented on screen.

### Science Example 1: Greenhouse

Science Example 1 is titled Greenhouse, and deals with the increase of the average temperature of the Earth's atmosphere. The stimulus material consists of a short text introducing the term "Greenhouse effect" and includes graphical information on the average temperature of the Earth's atmosphere and the carbon dioxide emission on the Earth over time.

The area of application is environment quality within a global setting.



PISA 2025

**Greenhouse 1**

**THE ENHANCED GREENHOUSE EFFECT: FACT OR FICTION?**

The energy that sustains most living things on Earth comes from the sun's radiation. The Earth's atmosphere acts like a protective blanket around our planet, allowing the sun's radiation to pass through and absorbing some of the energy radiated back from Earth that would otherwise escape.

As a result of this, the average temperature of the Earth's atmosphere is higher than it would be if there were no atmosphere to trap the heat energy. The Earth's atmosphere thus has a similar effect to a greenhouse where the glass of the greenhouse traps heat energy from the sun and keeps the inside warm.

The data indicate that the average temperature of the Earth's atmosphere has significantly increased over the last century. The reason for this average temperature increase is said to be because of increased atmospheric carbon dioxide, which causes the atmosphere to more effectively trap heat – the greenhouse effect is thus enhanced by CO<sub>2</sub>.

André, a student, becomes interested in the possible relationship between the average temperature of the Earth's atmosphere and carbon dioxide emission on Earth.

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### Greenhouse 2

André comes across two graphs on a Government website.

He concludes from these two graphs that it is certain that the increase in the average temperature of the Earth's atmosphere is due to the increase in the carbon dioxide emission.

**Q1.** What about the graphs supports André's conclusion?

Another student, Jeanne, disagrees with André's conclusion after comparing the two graphs.

**Q2.** Give an example of a part of the graphs that does not support André's conclusion. Explain your answer.

#### Government Website

The top graph, titled 'Carbon dioxide emission (thousand millions of tonnes per year)', shows a line starting near 0 in 1960 and rising steadily to over 20 by 1990. The bottom graph, titled 'Average temperature of the Earth's atmosphere (°C)', shows a line fluctuating between 14.6 and 15.4 degrees Celsius from 1960 to 1990, with a slight overall upward trend.

**Table 2.7. Framework categorisation for Greenhouse Question 1**

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                                              |
| Knowledge type       | Procedural                                                                                                  |
| Competency           | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically |
| Context              | Environmental, global                                                                                       |
| Cognitive demand     | Medium                                                                                                      |

Question 1 requires students to understand the data that is represented in the two graphs and construct an interpretation of their meaning. This question requires an interpretation of graphs involving a few linked steps. The question is categorised as medium cognitive.

**Table 2.8. Framework categorisation for Greenhouse Question 2**

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                                              |
| Knowledge type       | Procedural                                                                                                  |
| Competency           | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically |
| Context              | Environmental, global                                                                                       |
| Cognitive demand     | Medium                                                                                                      |

Question 2 also requires students to understand the data that is represented in the two graphs and construct an interpretation of their meaning. This question requires an interpretation of graphs involving a few linked steps. The question is categorised as medium cognitive, though is a bit more demanding as it requires students to identify a specific feature of the graph.

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**Greenhouse 3**

André persists in his conclusion that the average temperature rise of the Earth's atmosphere is caused by the increase in the carbon dioxide emission. But Jeanne thinks that his conclusion is premature. She says: "Before accepting this conclusion, you must be sure that other factors have not caused it."

**Q3.** Name one other factor that could have caused the mean temperature on the Earth to vary:

**Government Website**

Carbon dioxide emission (thousand millions of tonnes per year)

Average temperature of the Earth's atmosphere (°C)

**Table 2.9. Framework categorisation for Greenhouse Question 3**

|                      |                                  |
|----------------------|----------------------------------|
| Framework categories | 2025 Framework                   |
| Knowledge type       | Content                          |
| Competency           | Explain Phenomena Scientifically |
| Context              | Environmental, global            |
| Cognitive demand     | Low                              |

Question 3 requires students to use scientific knowledge to provide an alternative factor that might explain global warming. This question is considered to be of medium cognitive demand.

PISA 2025 

### Greenhouse 4

**Q4.** For scientists to decide whether the rise in average temperatures is caused by increased emissions of carbon dioxide, which (one or more) of the following would be considered important to establishing a strong case:

- ☐ Providing a scientific explanation for how carbon dioxide would affect the Earth's temperature.
- ☐ Researching community attitudes to climate change claims and renewable energy policies.
- ☐ Showing that the increase in carbon dioxide in the air is consistent with the increase in human activity.
- ☐ Developing models that explain how the increase in atmospheric carbon dioxide would cause a rise in global temperature.
- ☐ Showing how the temperature of the Earth has always fluctuated.

**Table 2.10. Framework categorisation for Greenhouse Question 4**

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                   |
| Knowledge type       | Epistemic                                                                        |
| Competency           | Research, evaluate and use scientific information for decision making and action |
| Context              | Environmental, global                                                            |
| Cognitive demand     | High                                                                             |

Question 4 requires students to draw on their epistemic knowledge to make a judgement about the nature of scientific evidence and which is most important in constructing an argument. The question is categorised as being of high cognitive demand.

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**Greenhouse 5**

**Q5.** In pursuing their different ideas and trying to find out who is correct, André and Jeanne research different sources of information, listed below. Tick those which you think are most trustworthy.

- ☐ An article in an energy company magazine discussing the science of climate.
- ☐ A report from a National Scientific Society summarising scientific work on climate change over a decade.
- ☐ An article in a peer reviewed scientific journal about modelling climate change.
- ☐ An article in the peer-reviewed journal *Science* summarising the consensus amongst scientists.
- ☐ An opinion piece in a magazine featuring a debate between two scientists.
- ☐ A debate on social media discussing how some scientists disagree about explanations for the temperature rise.

**Table 2.11. Framework categorisation for Greenhouse Question 5**

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                   |
| Knowledge type       | Epistemic                                                                        |
| Competency           | Research, evaluate and use scientific information for decision making and action |
| Context              | Environmental, global                                                            |
| Cognitive demand     | Medium                                                                           |

Question 5 requires students to draw on their epistemic knowledge to make a judgement about the nature of scientific evidence and which is most trustworthy in constructing an argument. The question is categorised as being of high cognitive demand.

The screenshot shows a digital interface for the PISA 2025 assessment. At the top, there is a blue header bar with the text 'PISA 2025' on the left and several navigation icons (a grid, a question mark, and arrows) on the right. Below the header, a blue box contains the title 'Greenhouse 6'. The main content area is a white box with a blue border. It contains the text of Question 6, which asks about the reliability of scientific findings. Below the question text is a list of six options, each preceded by a square checkbox.

**PISA 2025**

**Greenhouse 6**

**Q6.** In debating the issue, André and Jeanne discuss whether scientists' studies about climate change are influenced by political and social pressures and opinions. Which two of the following scientific practices are the most important for ensuring that scientific findings are reliable?

- ☐ Scientists argue about their ideas and the meaning of the data they collect. When sufficient findings agree the findings are considered to be trustworthy.
- ☐ Scientists invent models that are tested against the evidence they collect.
- ☐ Scientists work with private industry to make sure measurements are carefully made.
- ☐ Before being published, scientific papers are peer reviewed by other scientists who will reject the manuscript if they are not convinced by the findings.
- ☐ Scientists use everyday language in their reports to make sure the public is not confused.
- ☐ Scientists make sure their data are correct before they publish.

**Table 2.12. Framework categorisation for Greenhouse Question 6**

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                   |
| Knowledge type       | Epistemic                                                                        |
| Competency           | Research, evaluate and use scientific information for decision making and action |
| Context              | Environmental, global                                                            |
| Cognitive demand     | High                                                                             |

Question 6 requires students to draw on their epistemic knowledge to make a judgement about the nature of scientific evidence and which is most trustworthy in constructing an argument. The question is categorised as being of high cognitive demand.

The screenshots above illustrate how the Greenhouse question would be presented in an onscreen environment. The text and graphs are essentially unchanged, with students using page turners on the top right of the screen to view graphs and text as required. As the original questions were open responses, the onscreen version also necessitates an open response format in order to replicate the paper version as closely as possible, ensuring comparability between delivery modes and therefore protecting trend.

### **Science Example 2: Smoking**

This 2015 exemplar unit explores various forms of evidence linked to the harmful effects of smoking and the methods used to help people to stop smoking. New science items for 2025 will only be developed for computer-based delivery and therefore this exemplar is only shown in an onscreen format.

All onscreen standard question types in the PISA 2015 computer platform have a vertical split screen with the stimuli presented on the right-hand side and the questions and answer mechanisms on the left-hand side.

### Smoking: Question 1

This question requires students to interpret given evidence using their knowledge of scientific concepts. They need to read the information in the stimulus about early research into the potential harmful effects of smoking, and then select two options from the menu to answer the question.

Smoking 1



John and Rose are researching cigarette smoking for a school project.

Read John's research on the right. Then respond to the question below

Select **two** reasons from the list below that suggest why cigarette companies could claim there was **no** evidence that **tar** from cigarettes smoke caused cancer in humans.

☐ Humans are immune to tar

☐ Experiments were carried out with mice

☐ Chemicals from smoking decrease the effect of tar

☐ Humans may react differently from mice

☐ Filter-tip cigarettes remove all tar from smoke

#### John's Research

In the 1950s research studies found the tar from cigarette smoke caused cancer in mice. Tobacco companies claimed there was no evidence that smoking caused cancer in humans. They also began to produce filter-tip cigarettes.



In this question, students have to apply content knowledge using the competency of explaining phenomena scientifically. The context is categorised as health and disease in a local/national setting. The cognitive demand requires the use and application of conceptual knowledge and is therefore categorised as a medium level of demand.

**Table 2.13. Framework categorisation for Smoking question 1**

|                      |                                    |
|----------------------|------------------------------------|
| Framework categories | 2025 Framework                     |
| Knowledge type       | Content                            |
| Competency           | Explain phenomena scientifically   |
| Context              | Health and disease, local/national |
| Cognitive demand     | Medium                             |

### Smoking: Question 2

This question explores students' understanding of data.

The right-hand side of the screen shows authentic data of cigarette consumption and deaths from lung cancer in men over an extended period of time. Students are asked to select the best descriptor of the data by clicking on one of the radio buttons next to answer statements on the left-hand side of the screen.

PISA 2025

### Smoking 2

Rose found a graph while doing research on smoking.

Refer to Rose's research on the right. Then select the best response to the question below.

Which statement best describes the data shown in the graph?

- ☐ The graph shows that all men who smoked cigarettes developed lung cancer
- ☐ The graph shows that more men smoked cigarettes in the 1940s than in 2010
- ☐ There is no link between cigarettes smoked and deaths from lung cancer
- ☐ There is a positive link between cigarettes smoked and deaths from lung cancer

#### Rose's Research

| Year | Cigarette Consumption (men) (Per Person Per Year) | Lung Cancer Deaths (men) (Per 100,000 People) |
|------|---------------------------------------------------|-----------------------------------------------|
| 1900 | 1000                                              | 0                                             |
| 1920 | 2500                                              | 5                                             |
| 1940 | 3500                                              | 10                                            |
| 1950 | 4000                                              | 20                                            |
| 1960 | 3500                                              | 140                                           |
| 1980 | 3000                                              | 140                                           |

This unit tests content knowledge using the competency of “construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically”.

The context is health and disease applied to a local/national setting. As students need to interpret the relationship between two graphs, the cognitive demand is categorised as medium.

**Table 2.14. Framework categorisation for Smoking Question 2**

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                                              |
| Knowledge type       | Procedural                                                                                                  |
| Competency           | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically |
| Context              | Health and disease, local/national                                                                          |
| Cognitive demand     | Medium                                                                                                      |

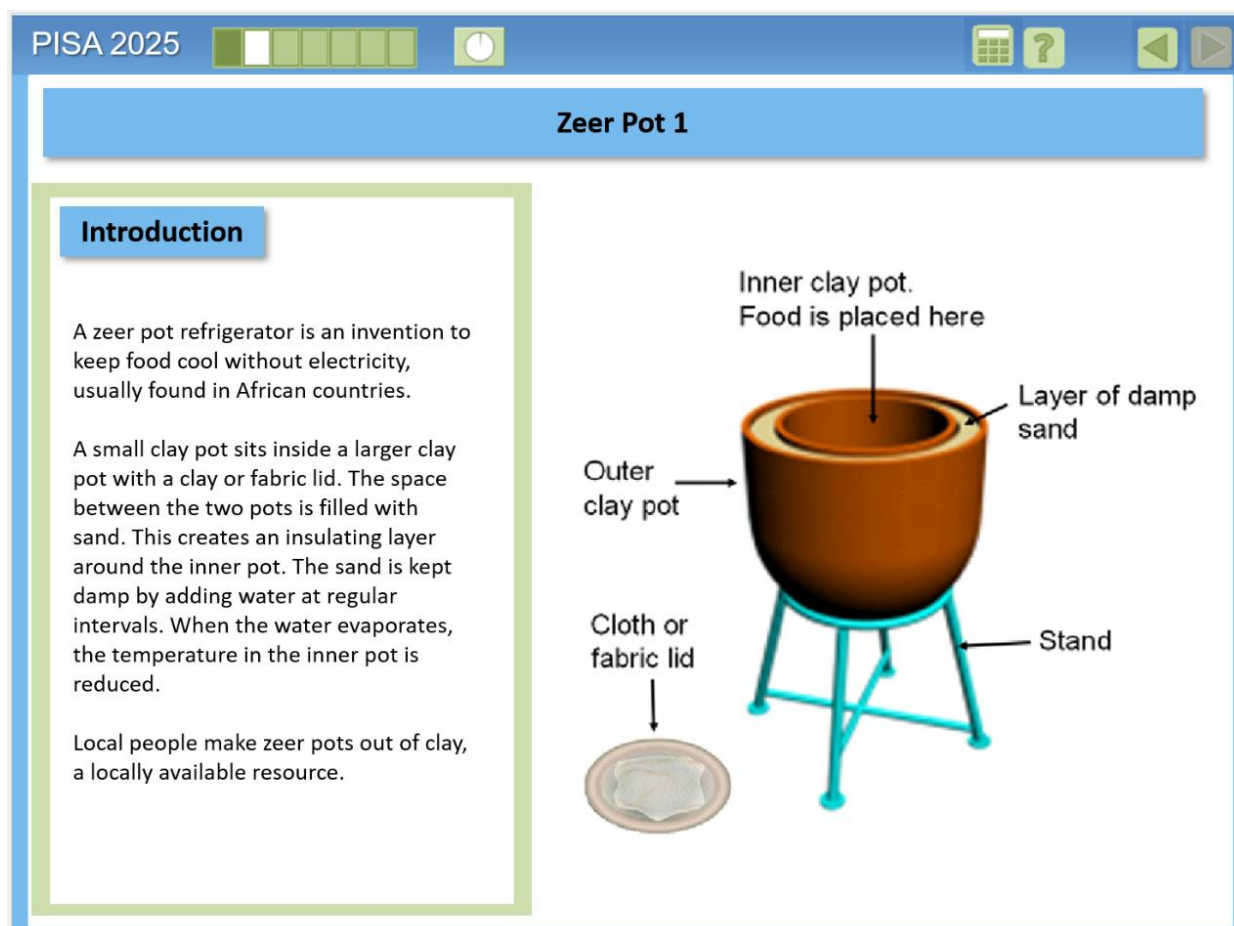
### Science Example 3: Zeer pot

This exemplar unit demonstrates a new feature of science assessment that was introduced in 2015; the use of interactive tasks using simulations of scientific enquiry to explore and assess scientific knowledge and competencies.

This unit is focused on an authentic low-cost cooling container called a Zeer pot, developed for localised needs in Africa, using readily available local resources. Cost and lack of electricity limits the use of refrigerators in these regions, while the hot climate necessitates food to be kept cool to prolong the length of time food can be kept before bacterial growth renders it a risk to health.

The first screenshot of this simulation introduces what a Zeer pot looks like and how it works. Students are not expected to understand how the process of evaporation causes cooling, just that it does.

### *Zeer pot: Stimulus*



Using this simulation, students are asked to investigate the conditions that will produce the most effective cooling effects (4°C) for keeping food fresh in the Zeer pot. The simulator keeps certain conditions constant (the air temperature and the humidity) but includes this information to enhance the authentic contextual setting. In the first question, students are asked to investigate the optimal conditions to keep the maximum amount of food fresh in the Zeer pot by altering the thickness of the sand layer and the moisture conditions.

## Zeer pot: Question 1

PISA 2025

### Zeer Pot 2

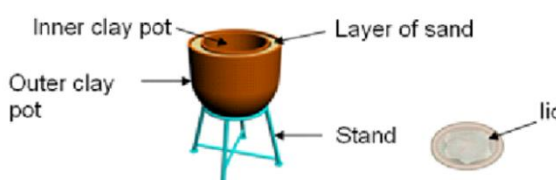
You have been asked to investigate the best design of a zeer pot for a family to keep their food fresh.

Food is best kept at a temperature of 4°C to maximise freshness and minimise bacterial growth.

Use the simulator opposite to work out the maximum amount of food that can be kept fresh (at 4°C) by varying the thickness and moisture condition of the sand layer.

You can run a number of simulations, and repeat or remove any data finding.

**Q1.** Maximum amount of food kept fresh at 4°C is  kg



| Thickness of Sand Layer (cm) | Amount of Food (kg) | Sand moisture (Damp/Dry) | Temperature (°C) |
|------------------------------|---------------------|--------------------------|------------------|
|                              |                     |                          |                  |

**Constant variables**

Air Temp: 35°C, Humidity: 20%

Thickness of sand layer (cm): 1 2 3 4 5

Amount of Food (kg): 0 4 8 12 16 20

Sand moisture: Damp Dry

Record Data Clear Data

When students have set their conditions (which also alter the visual display of the onscreen Zeer pot), they press the record data button which then runs the simulation and populates the data chart. They need to run a number of data simulations and can remove data or repeat any simulations as required. This screen then records their response to the maximum amount of food kept fresh at 4°C. Their approaches to the design and evaluation of this form of scientific enquiry can be assessed in subsequent questions.

The knowledge categorisation for this item is procedural and the competence is “evaluate designs for scientific enquiry and interpret scientific data and evidence critically”. The context categorisation is Natural Resources, although it also has links to Health and Disease. The cognitive demand of this question is categorised as high because students are given a complex situation, and they need to develop a systematic sequence of investigations to answer the question.

**Table 2.15. Framework categorisation for Zeer pot Question 1**

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                                              |
| Knowledge type       | Procedural                                                                                                  |
| Competency           | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically |
| Context              | Natural resources                                                                                           |
| Cognitive demand     | High                                                                                                        |

### Science Example 4: Running in hot weather

This unit presents a scientific enquiry about thermoregulation in the context of long-distance runners training in a location where weather conditions are sometimes hot and/or humid. The simulation allows students to manipulate the air temperature and air humidity levels, as well as whether or not the simulated runner drinks water. For each trial, data associated with the selected variables are displayed, including: air temperature, air humidity, drinking water (yes/no), sweat volume, water loss and body temperature. The runner's sweat volume, water loss and body temperature are also displayed on the top panel in the simulation panel. When the conditions trigger dehydration or heat stroke, those health dangers are highlighted with red flags.

#### *Running in hot weather: Stimulus*

PISA 2025

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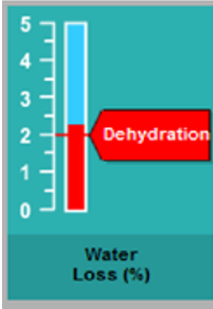
**Running in Hot Weather 1**

**INTRODUCTION**

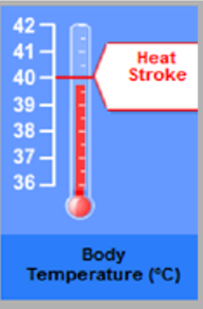
During long-distance running, body temperature rises and sweating occurs.

If runners do not drink enough to replace the water they lose through sweating, they can experience dehydration. Water loss of 2% of body mass and above is considered to be a state of dehydration. This percentage is labelled on the water loss meter shown below.

If the body temperature rises to 40°C and above, runners can experience a life-threatening condition called heat stroke. This temperature is labelled on the body temperature thermometer shown below.



**Water Loss (%)**



**Body Temperature (°C)**

Before beginning the unit, students are introduced to the simulation controls and asked to practice setting each control. Help messages are displayed if students do not perform the requested actions within one minute. If students time-out by not acting within two minutes, they are shown what the simulation would look like if the controls were set as specified in the provided instructions. As explained in the orientation that students take before beginning the Science section, reminders about how to use the controls, as well as how to select or delete a row of data, are available on each question screen by clicking on the “How to Run the Simulation” tab in the left panel.

### Running in hot weather: Stimulus

PISA 2025

**Running in Hot Weather 2**

**Introduction**

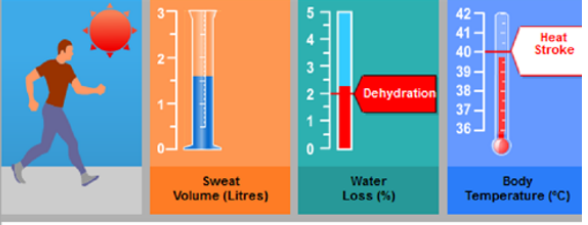
This simulation is based on a model that calculates the volume of sweat, water loss, and body temperature of a runner after a one-hour run.

To see how all the controls in this simulation work, follow these steps:

1. Move the slider for **Air Temperature**
2. Move the slider for **Air Humidity**
3. Click on either “Yes” or “No” for **Drinking Water**
4. Click on the “Run” button to see the results.

Notice that a water loss of 2% and above causes dehydration, and that a body temperature of 40°C and above causes heat stroke. The results will also display in the table.

**Note:** the results shown in the simulation are based on a simplified mathematical model of how the body functions for a particular individual after running for one hour in different conditions.



Air Temperature (°C) 20 25 30 35 40

Air Humidity (%) 20 40 60

Drinking Water ☐ Yes ☒ No

Run

| Air temperature (°C) | Air Humidity (%) | Drinking Water | Sweat Volume (Litres) | Water Loss (%) | Body Temperature (°C) |
|----------------------|------------------|----------------|-----------------------|----------------|-----------------------|
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |

In this question, students are provided with the specific values for each of the variables in the simulation. They must set the controls as specified and run the simulation once. A red flag is displayed indicating that, under these conditions, the runner would suffer from water loss leading to dehydration. This is the easiest question in the unit, requiring students to carry out a straightforward procedure, identify the flagged condition in the display as shown below, and interpret the display to correctly identify water loss as the cause of the runner’s dehydration.

Running in hot weather: Question 1

PISA 2025

Running in Hot Weather 3

**▶ Run the simulation**

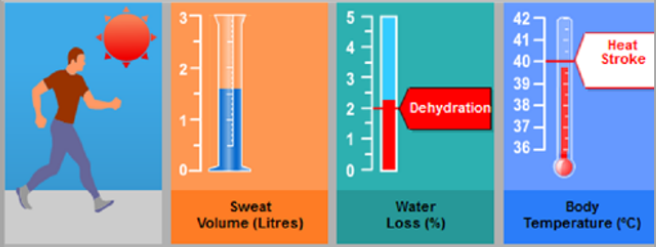
Run the simulation to collect data based on the information below. Click on a choice and then select data in the table to answer the question.

A runner runs for an hour on a hot, humid day (air temperature 35°C, air humidity of 60%) without drinking any water. This runner is at risk of both dehydration and heat stroke.

**Q1.** What would be the effect of drinking water during the run on the runner's risk of dehydration and heat stroke?

- ☐ Drinking water would reduce the risk of heat stroke but not dehydration.
- ☐ Drinking water would reduce the risk of dehydration but not heat stroke.
- ☐ Drinking water would reduce the risk of both heat stroke and dehydration.
- ☐ Drinking water would not reduce the risk of either heat stroke or dehydration.

Select two rows of data in the table to support your answer.



Air Temperature (°C) 20 25 30 35 40

Air Humidity (%) 20 40 60

Drinking Water ☐ Yes ☒ No

Run

| Air temperature (°C) | Air Humidity (%) | Drinking Water | Sweat Volume (Litres) | Water Loss (%) | Body Temperature (°C) |
|----------------------|------------------|----------------|-----------------------|----------------|-----------------------|
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |
|                      |                  |                |                       |                |                       |

There are a further four questions of this nature in this simulation. The full simulation can be found [here](#).

**Table 2.16. Framework categorisation for Running in hot weather Question 1**

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                                              |
| Knowledge type       | Procedural                                                                                                  |
| Competency           | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically |
| Context              | Health and disease                                                                                          |
| Cognitive demand     | Medium                                                                                                      |

**Science Example 5: Data dinosaurs**

In this question, students are presented with a set of data on dinosaurs. The tool allows them to plot one category against another to search for patterns in the data. The question tests their ability to determine which data sets should be identified and plotted to answer the questions posed.

## Data dinosaurs: Question 1

PISA 2025

### Data Dinosaurs 1

To use the data tool, drag the attributes that you want to look at to the axes on the graph. If you want to replace what an axis is showing just drag a new attribute to the graph.

Organise information in the graph to allow you to answer the following questions.

a. Of the dinosaurs that walk with lizard hips, how many can walk on either 2 or 4 legs?

b. What type of diet does the longest dinosaur have?

c. Write an observation about the weight of dinosaurs and the geological period in which they lived.

The second question presents them with a graph of the data and asks them to evaluate the conclusion critically.

Data dinosaurs: Question 2

## Data Dinosaurs 2

Palaeontologists have researched various attributes of dinosaurs and provided you with the data within a visualisation tool. This tool allows you to look at the relationship between various attributes of dinosaurs. One claim made by palaeontologists is that dinosaurs got lighter over time, and they produced the plot below to prove this.

Is there a strong relationship between the weight of dinosaurs and the age in which they lived?

Yes ☐
No ☐

Explain your answer.

**Table 2.17. Framework categorisation for Data dinosaurs Question 2**

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                                              |
| Knowledge type       | Procedural                                                                                                  |
| Competency           | Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically |
| Context              | Contemporary scientific and technological issues (global)                                                   |
| Cognitive demand     | Medium                                                                                                      |

### Science Example 6: Dangers of smoking

In this question, students are asked to use epistemic knowledge to assess which types of evidence best support a claim.

*Dangers of smoking: Question 1*

PISA 2025

**Smoking 3**

**Evidence of danger of smoking**

Smoking is said to be dangerous, but how do we know? Tick the boxes below that can be used as scientific evidence that smoking is dangerous

- ☐ A close friend or relative of yours has been smoking for a long time and has got ill
- ☐ Statistics show that smokers on average die at younger ages than non-smokers
- ☐ There are campaigns against smoking
- ☐ The newspapers have stories about people made ill by smoking
- ☐ Smokers are proved to be more often ill and off work than non-smokers
- ☐ Smoking will be banned in restaurants and cafés
- ☐ Statistics show that people with more education tend to smoke less

**Table 2.18. Framework categorisation for Dangers of smoking question**

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                                   |
| Knowledge type       | Epistemic                                                                        |
| Competency           | Research, evaluate and use scientific information for decision making and action |
| Context              | Health hazards (global)                                                          |
| Cognitive demand     | Medium                                                                           |

**Science Example 7: Who should we believe?**

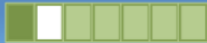
The following question is an example of the kind of question that might be used to test epistemic knowledge.

Who should we believe? Question 1

PISA 2025

## Top predators: Stimulus

PISA 2025



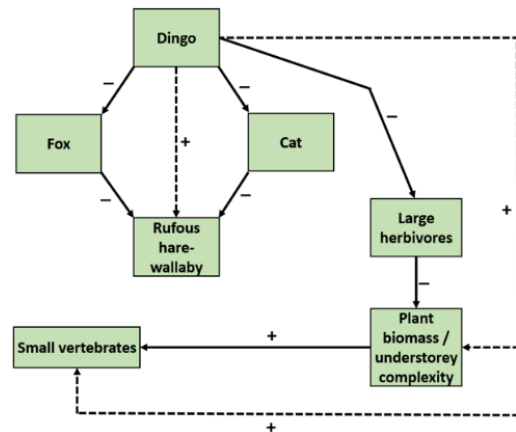
## Top Predators 1

In the Gariwerd National Park, Australia, the introduction of predators has changed the biodiversity.

It has been suggested that if an Australian native dog (dingo) was reintroduced it would alter the balance of the ecosystem.

The diagram shows how reintroducing the dingo might do this.

The solid lines indicate direct effects. The dotted lines are indirect effects. The '+' sign shows a positive effect. The '-' sign shows a negative effect.

**Biodiversity Diagram**

## Top predators: Questions 1 &amp; 2

PISA 2025

**Top Predators 2**

**Q1.** Select which of the following effects are shown by the diagram if dingos are reintroduced:

- ☐ Rufous hare-wallaby populations will decrease
- ☐ Small vertebrate numbers will increase
- ☐ Large herbivore numbers will decrease
- ☐ Plant biomass will increase

**Q2.** Which of the following is shown by the diagram?

- ☐ Fox and cat populations will decrease because they are preyed on by the dingo
- ☐ The rufous hare-wallaby will increase in numbers so numbers of foxes and cats will increase
- ☐ Dingos will have direct effects on numbers of foxes and cats and large herbivores
- ☐ Plant biomass will increase because dingos will reduce the numbers of herbivores.

**Biodiversity Diagram**

```

graph TD
    Dingo[Dingo] -- "-" --> Fox[Fox]
    Dingo -- "-" --> Cat[Cat]
    Dingo -- "-" --> LargeHerbivores[Large herbivores]
    Dingo -- "+" --> RufousHareWallaby[Rufous hare-wallaby]
    Fox -- "-" --> RufousHareWallaby
    Cat -- "-" --> RufousHareWallaby
    LargeHerbivores -- "-" --> PlantBiomass[Plant biomass / understorey complexity]
    PlantBiomass -- "+" --> SmallVertebrates[Small vertebrates]
    SmallVertebrates -- "+" --> Dingo
    
```

The diagram illustrates the following relationships:

- Dingo has negative effects (-) on Fox, Cat, and Large herbivores, and a positive effect (+) on Rufous hare-wallaby.
- Fox and Cat have negative effects (-) on Rufous hare-wallaby.
- Large herbivores have a negative effect (-) on Plant biomass / understorey complexity.
- Plant biomass / understorey complexity has a positive effect (+) on Small vertebrates.
- Small vertebrates have a positive effect (+) on Dingo.

Table 2.20. Framework categorisation for Top predators Question 1

|                      |                                                |
|----------------------|------------------------------------------------|
| Framework categories | 2025 Framework                                 |
| Knowledge type       | Content                                        |
| Competency           | Explain phenomena scientifically               |
| Context              | Environmental impacts & climate change (local) |
| Cognitive demand     | Medium                                         |

Table 2.21. Framework categorisation for Top predators Question 2

|                      |                                                |
|----------------------|------------------------------------------------|
| Framework categories | 2025 Framework                                 |
| Knowledge type       | Content                                        |
| Competency           | Explain phenomena scientifically               |
| Context              | Environmental impacts & climate change (local) |
| Cognitive demand     | Low                                            |

## Top predators: Question 3

PISA 2025

**Top Predators 3**

There is opposition to the suggestion from farmers and tourist agencies. They fear the dingos will attack sheep on their farms. Also, they will present a danger to tourists walking in the park.

**Q3.** The following objections have been written in media releases. Which of these could be investigated by scientists?

- ☐ Dingos are just wild dogs and pests
- ☐ Dingos carry disease that can be dangerous to cattle
- ☐ Dingo predation will endanger the koala population
- ☐ Dingos will frighten tourists away
- ☐ Dingos will breed and overrun the park

**Biodiversity Diagram**

```

graph TD
    Dingo -- "-" --> Fox
    Dingo -- "-" --> Cat
    Dingo -- "-" --> LargeHerbivores[Large herbivores]
    Dingo -- "+" --> RufousHareWallaby[Rufous hare-wallaby]
    Fox -- "-" --> RufousHareWallaby
    Cat -- "-" --> RufousHareWallaby
    LargeHerbivores -- "-" --> PlantBiomass[Plant biomass / understorey complexity]
    PlantBiomass -- "+" --> SmallVertebrates[Small vertebrates]
    SmallVertebrates -- "+" --> Dingo
    LargeHerbivores -- "+" --> Dingo
  
```

The diagram illustrates the following relationships:

- Dingo has negative effects (-) on Fox, Cat, and Large herbivores.
- Dingo has a positive effect (+) on Rufous hare-wallaby.
- Fox and Cat have negative effects (-) on Rufous hare-wallaby.
- Large herbivores have a negative effect (-) on Plant biomass / understorey complexity.
- Plant biomass / understorey complexity has a positive effect (+) on Small vertebrates.
- Small vertebrates have a positive effect (+) on Dingo.
- Large herbivores have a positive effect (+) on Dingo.

Table 2.22. Framework categorisation for Top predators Question 3

| Framework categories | 2025 Framework                                                      |
|----------------------|---------------------------------------------------------------------|
| Knowledge type       | Epistemic & content                                                 |
| Competency           | Research, evaluate & use information for decision making and action |
| Context              | Environmental impacts & climate change (local)                      |
| Cognitive demand     | Medium                                                              |

*Top predators: Question 4*

PISA 2025

### Top Predators 4

**Q4.** In debating this issue, which of the following arguments could be made drawing on scientific evidence (S) and which could be made that does not draw on scientific evidence (N)

- S

The arguments for introducing dingos based on the experience of wolf

N

introduction are possibly flawed because of the different environment conditions.
- S

The dingo should not be considered a native species because it was originally

N

brought into Australia by the indigenous population
- S

Dingos, if allowed to roam in packs, regulate their own numbers and should not

N

pose a danger to farm animals
- S

Dingos have important cultural value for the indigenous people who run the park

N

**Table 2.23. Framework categorisation for Top predators Question 4**

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                      |
| Knowledge type       | Epistemic                                                           |
| Competency           | Research, evaluate & use information for decision making and action |
| Context              | Environmental impacts & climate change (local)                      |
| Cognitive demand     | High                                                                |

## Top predators: Question 5

PISA 2025 





**Top Predators 5**

**Q5.** The following statements are made by different people involved in the controversy. Identify, for each statement, whether it represents: a scientific value and idea (S) **OR** represents other sets of values that are important to the issue (O)

S  
O

The arguments for introducing dingos based on the experience of wolf introduction are possibly flawed because of the different environment conditions.

S  
O

The dingo should not be considered a native species because it was originally brought into Australia by the indigenous population

S  
O

Dingos, if allowed to roam in packs, regulate their own numbers and should not pose a danger to farm animals

S  
O

Dingos have important cultural value for the indigenous people who run the park

Table 2.24. Framework categorisation for Top predators Question 5

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                      |
| Knowledge type       | Epistemic & content                                                 |
| Competency           | Research, evaluate & use information for decision making and action |
| Context              | Environmental impacts & climate change (local)                      |
| Cognitive demand     | High                                                                |

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Top predators: Question 6

PISA 2025

**Top Predators 6**

**Sources of Evidence**

Consider the statements below from various sources

**Q6.** Identify those sources that you think can be trusted.

- ☐ Scientific studies of biodiversity in the area reported in peer-reviewed papers
- ☐ An article discussing the science, in a magazine of the Farmers Federation
- ☐ An article in 'tourism weekly' about the park and the issue
- ☐ A government report that summarises the relevant scientific reports in reputable journals
- ☐ A blog post that describes stories of dingos and their behaviour in different parts of the country
- ☐ A local newspaper report of evidence presented at the local council meeting
- ☐ A twitter post about the issue that describes the failure of top predator introductions around the world

**Table 2.25. Framework categorisation for Top predators Question 6**

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                      |
| Knowledge type       | Epistemic                                                           |
| Competency           | Research, evaluate & use information for decision making and action |
| Context              | Environmental impacts & climate change (local)                      |
| Cognitive demand     | Medium                                                              |

Science **Example 9: The environmental impact of eating meat**

*Environmental impact of eating meat: Questions 1 & 2*

PISA 2025

The Environmental Impact of Eating Meat 1

Celia and Anton are discussing whether they should consider reducing the amount of meat in their diet for environmental reasons, and switch to a more vegetable-based diet. They consider the following information:

It takes 326 sq metres to produce a kg of beef, 12 sq metres to produce a kg of poultry meat, 2.8 sq metres for rice, and less than 1 sq metre for many vegetables.

Celia and Anton are aware that to maintain health the food they eat needs to contain an appropriate balance of food types - protein, carbohydrates and fats as well as a variety of trace nutrients.

**Q1.** What is the major food type provided by eating meat?

**Q2.** Name a plant-based food that could also provide the same type of food.

Table 2.26. Framework categorisation for Environmental impact of eating meat Question 1

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Framework categories | 2025 Framework                                  |
| Knowledge type       | Content                                         |
| Competency           | Explain phenomena scientifically                |
| Context              | Environmental impacts & climate change (global) |
| Cognitive demand     | Low                                             |

Table 2.27. Framework categorisation for Environmental impact of eating meat Question 2

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Framework categories | 2025 Framework                                  |
| Knowledge type       | Content                                         |
| Competency           | Explain phenomena scientifically                |
| Context              | Environmental impacts & climate change (global) |
| Cognitive demand     | Low                                             |

*Environmental impact of eating meat: Question 3*

PISA 2025

**The Environmental Impact of Eating Meat 2**

Anton argues that beef comes from cows and breeding cows for meat poses a threat to the environment.

**Q3.** Tick those reasons that might justify Anton's argument

- ☐ Cows take a long time to grow to full size
- ☐ Cows produce methane which is a greenhouse gas
- ☐ Cows take up too much land for the amount of food they produce, compared to vegetables
- ☐ Cows are sacred animals in some societies
- ☐ The skin of cows is useful for leather

**Table 2.28. Framework categorisation for Environmental impact of eating meat Question 3**

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| Framework categories | 2025 Framework                                                      |
| Knowledge type       | Content, epistemic & socio-ecological                               |
| Competency           | Research, evaluate & use information for decision making and action |
| Context              | Environmental impacts & climate change (global)                     |
| Cognitive demand     | Medium                                                              |

*Environmental impact of eating meat: Question 4*

PISA 2025

**The Environmental Impact of Eating Meat 3**

Celia says: ‘traditionally, humans have been omnivorous, consuming both meat and food such as grains, legumes, fruits. Meat, as part of a diet, delivers important food types and trace nutrients.

Anton replies: ‘an informed vegetarian diet can provide all these foods too! Since the world’s population is expanding, we need to reduce our forest clearance to provide pasture for cows and use our agricultural land more efficiently.’

**Q4.** Which of the following claims concerning ‘should we eat meat?’ can be justified using scientific evidence (S), and which is based on other types of knowledge or values (O)?

☐ S

☐ O

Our teeth are designed to eat meat

☐ S

☐ O

Some of our ceremonies involve eating meat and need to be maintained

☐ S

☐ O

A vegetarian human diet can provide all the food types and nutrients we need.

☐ S

☐ O

Meat tastes good we should not give it up

☐ S

☐ O

There is not enough available land to sustain current levels of meat production for a growing population.

☐ S

☐ O

Food production – especially meat production - is a major contributor to greenhouse gases, for instance through cows producing methane.

☐ S

☐ O

Meat is much more expensive than vegetables

☐ S

☐ O

Meat is a ready source of many of our nutritional needs

☐ S

☐ O

Meat production requires the extensive use of fertilizers. The overuse of fertilizers can pollute the land

☐ S

☐ O

To maximise efficiency of production hormones and drugs are sometimes used to make animals grow quickly and to keep them healthy in closely confined spaces. This drugs and hormones in this meat can negatively affect the health of people.

**Table 2.29. Framework categorisation for Environmental impact of eating meat Question 4**

| Framework categories | 2025 Framework                                                      |
|----------------------|---------------------------------------------------------------------|
| Knowledge type       | Content & epistemic                                                 |
| Competency           | Research, evaluate & use information for decision making and action |
| Context              | Environmental impacts & climate change (global)                     |
| Cognitive demand     | High                                                                |

## The Evolution of the Science Assessment Framework in PISA

In PISA 2000 and 2003, the science assessment framework's primary focus was on scientific literacy, which was defined as follows:

*“Scientific literacy is the capacity to use scientific knowledge, to identify questions and to draw evidence-based conclusions in order to understand and help make decisions about the natural world and the changes made to it through human activity” (OECD, 2000<sup>[126]</sup>; OECD, 2003<sup>[3]</sup>).*

In 2000 and 2003, the definition embedded knowledge of science and understandings about science within the one term, “scientific knowledge”. The 2006 definition separated and elaborated the term “scientific knowledge” by resolving it into two components: “knowledge of science” and “knowledge about science” (OECD, 2006<sup>[4]</sup>). Both definitions, however, referred to the application of scientific knowledge to understanding, and making informed decisions about, the natural world. In PISA 2006, the definition was enhanced by the addition of knowledge of the relationship between science and technology, an aspect that was assumed but not elaborated in the 2003 definition.

The PISA 2015 definition of scientific literacy was an evolution of these ideas. The major difference is that the notion of “knowledge about science” was specified more clearly and split into two components: procedural knowledge and epistemic knowledge.

In 2006, the PISA framework was also expanded to include attitudinal aspects of students' responses to scientific and technological issues within the construct of scientific literacy. In 2006, attitudes were measured in two ways: through the student questionnaire and through items embedded in the student test. Discrepancies were found between the results from the embedded questions and those from the background questionnaire with respect to “interest in science” for all students and the gender difference on these issues (OECD, 2009<sup>[125]</sup>; Drechsel, Carstensen and Prenzel, 2011<sup>[127]</sup>). More importantly, embedded items extended the length of the test. Hence for the 2015 framework, attitudinal aspects were only measured through the student questionnaire and there were no embedded attitudinal items. As to the constructs measured within this domain, the first (Interest in science) and third (Environmental awareness) remained the same as in 2006. The second “Support for scientific enquiry”, however, was changed to a measure of “Valuing scientific approaches to enquiry”, which was essentially a change in terminology to better reflect what was measured.

In PISA 2015, the contexts for assessment were changed from “Personal, Social and Global” in the 2006 Assessment to “Personal, Local/National and Global” to make the headings more coherent. This has been retained for the 2025 assessment.

In developing the framework for PISA 2025, four major changes have been made. First, it was decided to merge the two previous competencies “Evaluate and design scientific enquiry” and “Interpret data and evidence scientifically” into one competency now called “Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically”. The change of language was simply to place more emphasis on the evaluation of designs, as few adults are likely to be engaged in designing experiments. This change was made as both competencies were felt to be part of the process of engaging in enquiry. In addition, the change in the societal context which, as outlined above, is now dominated by information sources on the Internet, many of them scientific, placed a new emphasis on educating students to “research, evaluate and use scientific information for decision making and action”. Hence, the addition of this third new competency.

The second change was a shift from the specific focus on science literacy to a definition which, while encompassing this concept, is broader. Previously the framework has used the term “science literacy” as the outcome of science education. The 2025 framework has chosen to phase out this term to avoid any confusion. This change brings it into line with the frameworks for mathematics and reading.

The third is to change the affective factors influencing the competency from a focus on attitudes towards science to a focus on measuring a broader concept of “science identity”, which has been shown to be more comprehensive in describing students' engagement in science.

The fourth is the focus on education for sustainability and the development of a scale to measure elements of Agency in the Anthropocene.

In addition, the body of defined content knowledge has been revised to make it more coherent with more emphasis on the major ideas of science while the definitions of procedural and epistemic knowledge have been extended and clarified.

In summary, the PISA 2025 definition builds on and develops the PISA 2006 and PISA 2015 definitions, broadening the competencies and clarifying the ideas and knowledge required.

## Summary

Science was the major domain assessed in PISA 2025 and the 2025 definition builds on and develops the 2015 definition of scientific literacy. In particular, two of the competencies: “Evaluate and design scientific enquiry” and “Interpret data and evidence scientifically” in the 2015 framework have been merged into one competency: “Construct and evaluate designs for scientific enquiry and interpret scientific data and evidence critically”, with the addition of a third new competency: “Research, evaluate and use scientific information for decision making and action”. All of the competencies have been further elaborated, as has the concepts of procedural and epistemic knowledge, which were introduced in 2015. In addition, the 2025 framework has elaborated the conception of the cognitive demand required of items to provide greater guidance to item writers, specifically to produce more items at both ends of the spectrum of cognitive abilities. The 2025 framework therefore is an evolution of the conception of scientific literacy that is a response to the contemporary context where there is ever greater reliance on the evaluation and use of scientific information. In addition, it has built on and elaborated some of the earlier ideas found in previous frameworks. Given the multiple definitions of scientific literacy, this term has now been dropped and the framework is now referred to as the science framework.

The PISA 2025 definition of the outcomes of science education has its origin in the consideration of what 15-year-old students should know, value and be able to do as “preparedness for life” in modern society. Central to the definition and the assessment of the outcomes of science education are the competencies that are characteristic of science and scientific enquiry. The ability of students to make use of these competencies depends on their scientific knowledge, both their content knowledge of the natural world and their procedural and epistemic knowledge. In addition, it depends on a willingness to engage with science-related topics. Their attitudes towards science-related issues are measured separately in the background questionnaire. In addition, the 2025 framework has elaborated a scale for students’ understanding of socio-ecological issues and the ways in which this might be measured.

This framework describes and illustrates the scientific competencies and knowledge that will be assessed in PISA 2025 (see Box 2.14), and the contexts for test items.

### Box 2.14. Major components of the PISA 2025 Science Framework

| Competencies                                                                                                                                                                                                                                                                                 | Knowledge                                                                                                                                                                                                                                                                                                | Science Identity                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Explain phenomena scientifically</li> <li>• Construct and evaluate designs for scientific enquiry and interpret data and evidence critically</li> <li>• Research, evaluate and use scientific information for decision making and action</li> </ul> | <ul style="list-style-type: none"> <li>• Knowledge of the content of science:               <ul style="list-style-type: none"> <li>○ Physical systems</li> <li>○ Living systems</li> <li>○ Earth and space systems</li> </ul> </li> <li>• Procedural knowledge</li> <li>• Epistemic knowledge</li> </ul> | <ul style="list-style-type: none"> <li>• A critical science disposition</li> <li>• Valuing scientific perspectives and approaches to enquiry</li> <li>• Environmental concern, awareness and agency</li> <li>• Elements of science identity</li> </ul> |

Test items were grouped into units with each unit beginning with stimulus material that establishes the context for items. A combination of item types was used. Computer-based delivery of the test since 2015 has offered the opportunity for several novel item formats, including animations and interactive simulations. This improves the validity of the test and the ease of scoring.

The ratio of items assessing students’ content knowledge of science to items assessing procedural and epistemic knowledge of science is about 3:2. Approximately 40% of the items test the competency to

explain phenomena scientifically, 30% the competency to construct and evaluate designs for scientific enquiry and interpret data and evidence critically, and 30% their competency to research, evaluate and use scientific information for decision making. The cognitive demand of items consists of a range of low, medium and hard. The combination of these weightings and a range of items of varying cognitive demand enable proficiency levels that describe performance in the three competencies that define the outcomes of scientific education.

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# **3**

## **PISA 2025 Learning in the Digital World Assessment Framework**

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In PISA 2025, the innovative domain assessment was Learning in the Digital World (LDW). The LDW framework presented in this chapter is grounded in a social constructivist approach to learning and problem-solving, recognising the potential of digital environments as especially productive contexts for learning. The chapter outlines the rationale for assessing LDW in PISA 2025 and the interrelations between learning, problem solving and computational thinking in digital environments. The framework then defines LDW and the two constructs that are the focus of the LDW assessment: 1) computational problem-solving practices and 2) self-regulated learning processes. The chapter describes the unique task model and unit structure of the LDW assessment, including the task affordances that enable learning on the test, and concludes with the approach to developing evidence rules for both constructs. Illustrative examples of the LDW assessment items are presented throughout the chapter.

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## Introduction

### ***How is learning changing in an increasingly digital world?***

Every education system aims to develop students' lifelong learning skills. Learning is an active and participatory process, and students can develop their learning skills if given opportunities to actively explore and experiment with their ideas. In recent years, many disciplines – particularly those in the broader STEM (science, technology, engineering and mathematics) domain – have shifted from “learning about” towards “doing”, promoting more active and inquiry-based contexts of learning and practice (OECD, 2019<sup>[1]</sup>).

Digital learning technologies are fundamentally transforming how people learn. The digital world enables new forms of autonomous learning, providing access to vast resources, tools for analysing and organising information, and opportunities to collaborate with others. Computational tools, in particular, enable new and more powerful forms of learning and problem solving that combine human inquiry with computing power. Such tools allow learners to build tangible and executable representations of their ideas – for example, computer programs or models – that generate immediate feedback. These tools radically expand how learners can explore their emergent understanding, experiment with their ideas and iteratively develop solutions to problems, in turn providing opportunities for more active and authentic learning experiences. In the education community, computational tools and environments like *Scratch*, *NetLogo* and *Code.org* have established global learning communities, where millions of secondary (and primary) education students engage with, create and share personalised objects of their learning.

At the system level, many countries have significantly invested in integrating digital technology into teaching and learning, accelerated by the global COVID-19 pandemic. However, the global education community lacks evidence on whether such investments have been productive for learning and whether students have developed the competences they need to learn autonomously with technology. Instruction on how to harness digital and computational technologies for learning often focuses on narrow aspects of computer literacy, but students can encounter significant challenges related to how they manage their own learning processes in open, interactive and inquiry-based environments. Being able to self-regulate one's learning - i.e. monitoring and controlling one's metacognitive, cognitive, behavioural, motivational and affective processes while learning (Panadero, 2017<sup>[2]</sup>) - is therefore particularly important in digital environments that invite students to construct their own understanding using computational tools.

### ***What will the PISA 2025 Learning in the Digital World assessment measure?***

The PISA 2025 Learning in the Digital World assessment will measure how well students can make progress towards their learning goals – framed as understanding and solving – by using computational tools in an autonomous learning process. In each 30-minute unit, students will work on a series of increasingly complex tasks in an open-ended, scaffolded, and interactive digital environment that provides them with learning resources and feedback. The assessment is grounded in a social constructivist approach to learning that emphasises the active and iterative process of discovery through interactions with tools and resources.

The PISA 2025 Results will provide insights into the extent to which students around the world can solve problems using computational tools, as well as how they do so by engaging in a productive process of self-regulated learning. This dual assessment focus – on problem-solving skills and self-regulated learning processes – aligns with the goals of 21<sup>st</sup>-century education to develop lifelong, autonomous learners (OECD, 2019<sup>[3]</sup>). The assessment also aims to produce multidimensional measures of performance in PISA that better reflect what students can achieve where opportunities to learn are available.

## Perspectives on learning in the digital world

### ***Theories of learning grounded in social constructivist theory***

Constructivist learning theories define learning as an active process of knowledge (re)construction (Bodner, 1986<sup>[4]</sup>; Collins, Brown and Newman, 1989<sup>[5]</sup>; Piaget, 1971<sup>[6]</sup>; 1976<sup>[7]</sup>), where new information is incorporated into one's pre-existing knowledge and mental schemas rather than simply received, memorised and retrieved (Ackermann, 2001<sup>[8]</sup>). This fundamental idea is reflected in modern approaches to teaching and instructional design (e.g. student-centred pedagogy, learning-by-doing, inquiry-based learning, project-based learning, etc.) that stress the active role of learners in building their own understanding and meaning, and the importance of their prior knowledge and experiences in sense-making.

Social constructivism extends these ideas by viewing learning as a situated process, emphasising the importance of social context and interactions with other people or objects. An individual's "zone of proximal development" (Vygotsky, 1978<sup>[9]</sup>) represents the gap between what an individual knows and can do independently, and what they are able to learn with the help of others, especially those who are more knowledgeable.

"Constructionism" shares these views of learning but emphasises the importance of building knowledge constructions that are external, tangible, shared and personally meaningful (Papert, 1986<sup>[10]</sup>). During the process of artefact-building, a learner puts ideas, concepts and skills into action, and these will further develop through interaction and iteration.

Together, these approaches to learning all emphasise the active role of learners in constructing their own knowledge and understanding, the importance of authentic learning experiences, the social negotiation of knowledge through tools or people, and the role of educators as facilitators in this process (Kaffash et al., 2010<sup>[11]</sup>).

### ***The digital world as a productive context for social constructivist learning***

In the broadest sense, digital technologies encompass a diverse set of tools and resources that are used to create, store, manage, communicate, share or exchange information. For teaching and learning, some digital tools and environments are more transformative than others (OECD, 2019<sup>[12]</sup>; Lloyd, 2005<sup>[13]</sup>; Mioduser et al., 2003<sup>[14]</sup>; Salomon and Perkins, 2005<sup>[15]</sup>). For example, digital technologies (e.g. e-books, word processing software) may substitute or functionally enhance otherwise traditional teaching and learning experiences whereas others can enable new forms of active and participatory learning emphasised by social constructivist theories of learning.

Open-ended, digital learning environments that incorporate knowledge-building tools and resources are particularly productive for supporting student-centred inquiry processes, problem-solving and decision making. They provide learners with interactive tools they can use to discover, interpret and create new meaning, including by manipulating and building computational artefacts, and environments in which they can access abundant resources, receive immediate feedback, and engage in more authentic and experiential contexts of practice.

### ***Learning, problem solving and computational thinking***

Using computational tools for learning and problem solving requires learners to instruct a computer to do something. Learners must understand the reasoning behind the instructions they give to a computer to investigate, model or solve a particular issue, employing relevant concepts and strategies purposefully and meaningfully. In doing so, learners apply computational thinking, i.e. an approach to problem solving that draws upon fundamental practices in computer science and that promotes a deep understanding of the

entire problem solving process (Wing, 2006<sup>[16]</sup>). In translating their abstract ideas into a tangible computational form, learners develop a deeper understanding of their own conceptual models and thought processes. In turn, these tangible representations enable educators to interpret where students are in their learning process and what knowledge they mobilised to get there (Valente and Blikstein, 2019<sup>[17]</sup>).

In education systems across the world, science and mathematics curricula increasingly integrate computational thinking as a key competency. In general, these disciplines share a core set of evidence-based reasoning and problem-solving practices, often discussed under the umbrella of scientific inquiry (Pedaste et al., 2015<sup>[18]</sup>), and for which computational tools provide new and more powerful opportunities. For example, computational models can simulate complex phenomena and support the experimentation of ideas that would be neither practical nor feasible to replicate in the real world (Weintrop et al., 2016<sup>[19]</sup>). Using computational tools for inquiry-based learning readily lends itself to learning and applying concepts and practices drawn from the wider STEM disciplines (Valente and Blikstein, 2019<sup>[17]</sup>; Zhang and Biswas, 2019<sup>[20]</sup>; Brennan and Resnick, 2012<sup>[21]</sup>; Sengupta et al., 2013<sup>[22]</sup>; Basu and Biswas, 2016<sup>[23]</sup>; Basu, Biswas and Kinnebrew, 2017<sup>[24]</sup>) and supports contemporary approaches to STEM education that actively engage students with scientific ideas and practices in real-world contexts (Krajcik, 2015<sup>[25]</sup>). As such, several educational researchers have focused on developing digital learning environments that integrate computational tools aimed at fostering knowledge and skills in STEM contexts (Biswas, Segedy and Bunchongchit, 2015<sup>[26]</sup>; Wieman, Adams and Perkins, 2008<sup>[27]</sup>; de Jong, Sotiriou and Gillet, 2014<sup>[28]</sup>; Basu et al., 2012<sup>[29]</sup>; Maloney et al., 2010<sup>[30]</sup>; Sengupta and Wilensky, 2011<sup>[31]</sup>; Wilensky, 1999<sup>[32]</sup>).

This type of constructionist learning in digital environments also requires students to engage in a process of self-regulated learning. In open digital environments with unstructured information and opportunities for exploration and distraction, students need to purposefully direct their actions, monitor their knowledge gaps, iteratively build and debug their models or computational artefacts, identify and correct their errors by seeking and interpreting feedback from the environment, and adapt their strategies when stuck (Järvelä and Hadwin, 2015<sup>[33]</sup>). Metacognition is particularly relevant in student-centred learning activities, yet research has demonstrated that students frequently lack the metacognitive understanding necessary to engage in optimal inquiry learning (Dedić, 2014<sup>[34]</sup>; Keselman and Kuhn, 2002<sup>[35]</sup>; Kuhn et al., 2000<sup>[36]</sup>).

## The PISA definition of learning in the digital world

Learning in the digital world in PISA 2025 is defined as “the capacity to engage in an iterative and self-regulated process of knowledge building and problem solving using computational tools and practices”.

The definition recognises learning in the digital world as an iterative, self-regulated process that requires learners to be active participants in their learning. It recognises knowledge building and problem solving as particular forms of constructivist learning. In this definition, problem solving does not refer to the retrieval and reproduction of existing knowledge to solve a problem but rather to the process of using external tools and resources to incrementally advance one’s knowledge and reach a particular goal. During this iterative process, students are expected to self-regulate their learning by activating cognitive, metacognitive and affective processes.

In the assessment, learners engage in knowledge building and problem solving using computational tools. These tools allow learners to construct, modify and refine tangible and executable representations of their emergent knowledge and understanding, as well as build functional solutions, by applying computational problem-solving practices. Computational representations can take many forms but in the PISA 2025 assessment they include models (e.g. concept maps, flow charts) and algorithmic solutions (e.g. programs).

## The PISA 2025 assessment and measurement approach

The PISA 2025 learning in the digital world assessment has two main data collection instruments: a cognitive test and a questionnaire module.

In the test component, students complete two extended (30-minute) units that are structured as learning modules, during which they solve increasingly complex tasks and have access to learning resources such as tutorials, feedback and examples. Students' performance on the test is summarised by a computational problem-solving scale. The computational problem-solving scale describes the extent to which students can make progress towards their learning goals and build functional solutions using computational tools, such as modelling or programming tools, and using computational problem-solving practices, such as conducting experiments, analysing data, and building and debugging computational artefacts. Students' performance on the test is supported by self-regulated learning, which includes how they maintain their motivation and task engagement, monitor their progress and adapt their strategies, and evaluate their own performance.

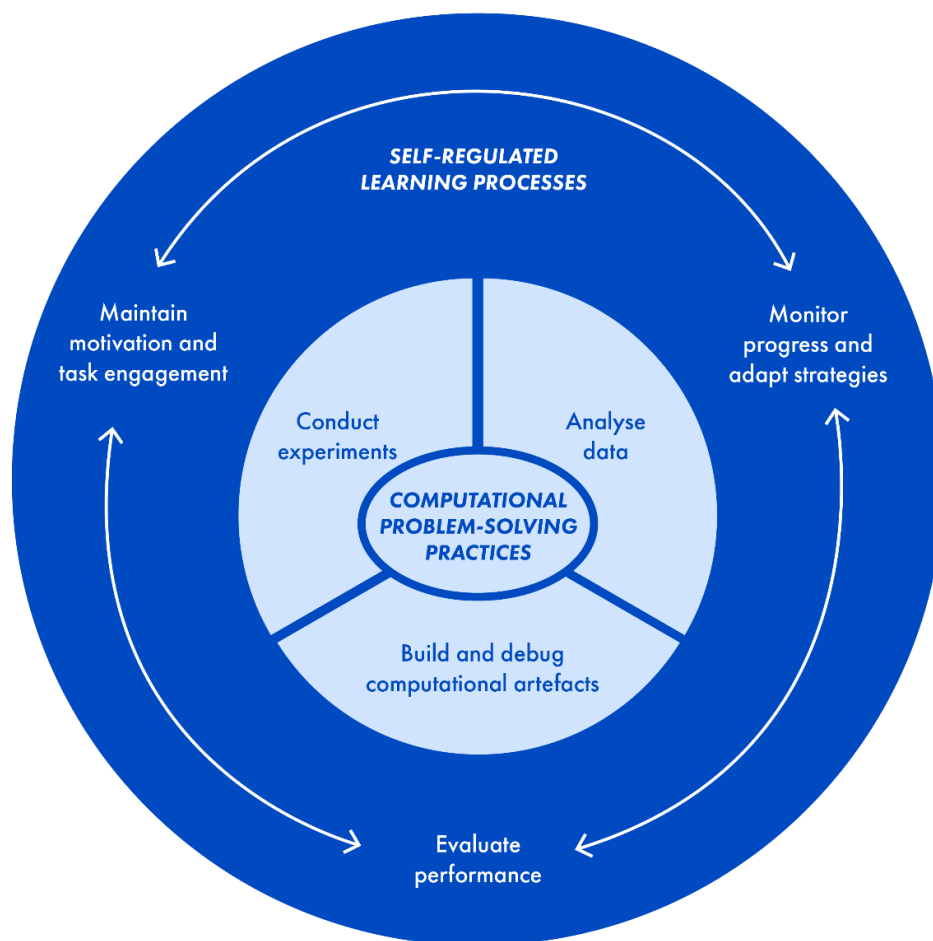
Evidence-centred design (ECD) (Mislevy, Steinberg and Almond, 2003<sup>[37]</sup>) provides a principled design process for developing innovative and coherent assessments that are built on evidence-based arguments. ECD makes explicit the connections between student actions and work products with latent multidimensional competences by developing three interconnected models: the competency model, which defines the constructs of interest; the task model, which defines the types of tasks that can provide evidence about the constructs of interest; and the evidence model, which describes how potential evidence can be interpreted and summarised across tasks. In the following sections, we describe each of these models in detail.

The PISA questionnaire module will complement data collected in the cognitive test and will focus on students' use of digital tools for learning, inside and outside of the classroom, as well as their attitudes towards information and communications technology (ICT) and their learning strategies.

### ***Competency model***

The competency model operationalises the learning in the digital world construct, as previously defined in this framework, for the purposes of measurement. The dual focus of the learning in the digital world assessment means that the competency model contains two distinct main components: 1) computational problem-solving practices and 2) self-regulated learning processes. Each of these components is further broken down into sub-components and their associated practices or processes, respectively (see Figure 3.1).

Figure 3.1. Competency model for the PISA 2025 learning in the digital world assessment



### *Computational problem-solving practices*

Several frameworks address the synergies between inquiry-based learning and computational thinking and problem solving, identifying a set of interrelated computational practices that support learning and problem solving using computational tools (Weintrop et al., 2016<sup>[19]</sup>; Zhang and Biswas, 2019<sup>[20]</sup>). We refer to these as “computational problem-solving practices”, which include elements related to the knowledge of computational concepts and strategies. These computational problem-solving practices are broken down into three distinct sub-components for the purpose of measurement in PISA.

#### **Conduct experiments**

Being able to determine the relationships between variables or agents within a system through systematic experimentation is an important computational problem-solving practice. It is underpinned by knowledge about certain concepts, such as dependent and independent variables, and strategies for collecting data for analysis, such as the control of variables.

Conduct experiments refers to students’ ability to:

- Use computational models, simulations and other tools to generate data
- Identify relevant hypotheses and conduct controlled experiments

## Analyse data

Being able to analyse data to understand how systems work is another important foundation for building computational models. It is underpinned by knowledge of mathematical and graphical representations of the relationships between variables, knowledge of different types of data, and the ability to select and use appropriate tools and methods to clean, manage, explore and analyse data.

Analyse data refers to students' ability to:

- Use data to make conclusions about how different elements of a system relate to each other
- Use data to make predictions about how a system will change over time

## Build and debug computational artefacts

Being able to translate ideas into instructions that can be executed by computers is fundamental to computational problem solving. This involves reframing problems so that they can be rendered suitable for computation through reflective and considered strategies, such as decomposition and abstraction, which simplify the problem by eliminating less relevant elements and enable the incremental development of modular solutions whose components can be tested independently (Weintrop et al., 2016<sup>[19]</sup>). Building and debugging computational artefacts is also underpinned by knowledge of fundamental computational operations such as repetition, iteration and conditional branching, and understanding of the outcomes of varying the sequence in which these operations are executed (i.e. control flow).

Build and debug computational artefacts refers to students' ability to:

- Identify sub-goals and address the constituent parts of a problem independently
- Identify and address repeated patterns through the same computational procedure
- Implement a generalisable sequence of solution steps by using control flow structures, such as repetitions and conditional statements
- Create an abstract representation of a system that can be executed by a computer
- Adapt and/or debug algorithms and computational models

## *Self-regulated learning processes*

Self-regulated learning involves a large umbrella of processes that support students to learn and problem solve in open, interactive digital environments. While several theories exist with overlapping phases, components and sub-processes (Panadero, 2017<sup>[2]</sup>), self-regulated learning processes are broken down into three distinct sub-components for measurement purposes in PISA 2025. However, as shown in Figure 3.1, these processes are iterative and closely interrelated. For example, students' capacity to regulate their motivation and engagement influences their capacity to enact cognitive and metacognitive strategies; and students' capacity to engage in a reflective process of evaluation following a learning task will influence their motivation as well as the cognitive and metacognitive strategies they choose to apply to a future task.

## Monitor progress and adapt strategies

Students can exercise control over their learning and problem-solving process by using cognitive and metacognitive learning strategies. Cognitive strategies refer to how students make progress towards their learning goal, such as choices about how to approach tasks and how to translate ideas into goal-oriented actions. As a result, cognitive strategies are usually domain- and sometimes even task-specific. Metacognition refers to students' awareness and understanding of the task demands and conditions, including how these change over time, and metacognitive strategies include those for self-monitoring progress towards learning goals. Metacognition supports students' cognitive strategies as it enables

individuals to identify their knowledge gaps, seek feedback when needed, and use that information to enact and adapt their strategies based on their emergent understanding of the task goals and their progress towards achieving them.

Monitor progress and adapt strategies refers to students' ability to:

- Evaluate their understanding of the task goals and identify knowledge gaps
- Enact goal-oriented strategies for making progress towards completing the task
- Seek feedback to monitor their progress towards task goals, for example through systematic testing, modification and debugging of their computational artefacts
- Act on feedback from the learning environment, adapting their cognitive strategies if necessary
- Engage in appropriate help-seeking behaviours, when needed

### **Maintain motivation and task engagement**

A learner's motivation will influence their desire to engage in nearly all types of learning activities (Pintrich and De Groot, 1990<sup>[38]</sup>; Bandura, 2001<sup>[39]</sup>), and research has found a positive relationship between motivation and deeper learning outcomes (Järvelä, Järvenoja and Muukkonen, 2021<sup>[40]</sup>; Saab, van Joolingen and van Hout-Wolters, 2009<sup>[41]</sup>). Learners' affective states (i.e. their emotions, feelings and attitudes) also closely interact with their motivation and engagement with learning tasks (Efklides, 2011<sup>[42]</sup>). The motivational and affective components of self-regulated learning describe the processes through which individuals manage their motivation, affective states and behaviours while learning, such as persisting in the face of perceived difficulty or exerting effort towards achieving one's learning goal (Järvenoja et al., 2018<sup>[43]</sup>; Fredricks, Blumenfeld and Paris, 2004<sup>[44]</sup>; Kim et al., 2015<sup>[45]</sup>).

Maintain motivation and task engagement refers to students' ability to:

- Be motivated to engage with learning tasks, spending a reasonable amount of time and executing sufficient actions to make progress towards their learning goals
- Invest effort in making progress towards their learning goals
- Demonstrate willingness to adapt or modify their computational artefact after receiving feedback
- Execute goal-oriented actions and behaviours despite experiencing negative affective states (e.g. frustration, boredom) or perceiving negative task conditions (e.g. difficulty)

### **Evaluate performance**

A self-regulated learning cycle also includes a reflective phase, during which learners evaluate their successes or failures. These reflective and evaluative metacognitive processes support learning by helping inform students about their future performance on similar tasks.

Evaluate performance refers to students' ability to:

- Evaluate their success in achieving or making some progress towards their learning goal, for example by appraising the quality of their computational artefact with respect to the task requirements
- Reflect on good and/or successful examples of how to achieve the learning goal, after completing the task, with a view to comparing their own strategies and computational artefacts to a "more knowledgeable" other
- Judge their capacity to successfully engage in similar and more complex learning and problem-solving tasks

## ***Other factors influencing student performance on the test***

### *Prior knowledge*

Students' prior knowledge of a given topic or type of task, as well as their knowledge of and basic familiarity with digital tools, may influence their performance on the assessment. A student's prior knowledge and understanding of a topic will inform their initial ideas about how to address a given task; for example, students who have existing knowledge of the different types of relationships between variables may conduct more targeted experiments, or students who are familiar with programming may have a greater understanding of the procedures that govern the behaviour of a computational artefact. Conversely, insufficient prior knowledge may lead to data misinterpretation, flawed experimentation behaviours, or confusion about task requirements (de Jong, 2006<sup>[46]</sup>; Glaser et al., 1992<sup>[47]</sup>; Schauble et al., 1991<sup>[48]</sup>; van Riesen et al., 2018<sup>[49]</sup>; Quintana et al., 2004<sup>[50]</sup>).

Prior knowledge therefore not only likely influences success in inquiry-based problem-solving tasks, but also the cognitive and metacognitive strategies that students enact. The more that a learner knows about a particular subject or problem-solving task, the more complex the strategies are that they are able to draw upon and the more effective those strategies are likely to be (Alexander, Graham and Harris, 1998<sup>[51]</sup>; de Boer, Donker-Bergstra and Kostens, 2012<sup>[52]</sup>).

In addition to prior content knowledge, being able to use computational tools and digital learning environments presupposes some functional digital literacy (e.g. how to navigate an interface, click on affordances, etc.). However, digital literacy does not necessarily translate into better learning outcomes using digital tools; for example, Wecker et al. (2007<sup>[53]</sup>) found that students who were more familiar with computers acquired significantly less knowledge in a digital science inquiry environment because they tended to exhibit more shallow information processing strategies (e.g. browsing).

Students' prior knowledge about the computational problem-solving practices targeted in each unit was measured through a short battery of discrete items during the first part of the unit. Students' performance in these items will not contribute to the overall computational problem-solving scale but will rather be used to contextualise students' performance on the scale relative to their prior knowledge at the beginning of the test. Students' functional digital literacy will not be explicitly measured within the test but can be inferred from their interactions with the different affordances of the digital learning environments; for example, whether they can execute the scaffolded instructions in the tutorial.

### *Opportunity to learn and self-efficacy with digital tools and resources*

#### **ICT use in school for learning**

Information about students' familiarity with and experience using digital learning tools was also collected via items included in the student questionnaire module. Findings from previous PISA cycles have revealed substantial disparities in the availability of digital resources across schools, schools' capacity to enhance teaching and learning using digital devices, and students' ability to use technology effectively. Disparities in the availability of devices and digital resources also persist, particularly in disadvantaged schools, in turn influencing students' opportunities to learn in digital contexts.

#### **ICT self-efficacy**

A student's exposure to and opportunity to learn with digital tools and resources will also likely influence their self-efficacy (i.e. their confidence in their ability to carry out specific tasks). Self-efficacy has found to be strongly associated with students' performance (Bandura, 1993<sup>[54]</sup>), in general, as well as in computer-based learning digital environments (Moos and Azevedo, 2009<sup>[55]</sup>). For example, students who feel confident using ICT may engage more readily with ICT-mediated learning experiences and persevere

through difficulties they might encounter while using technology. Research has shown that ICT self-efficacy is associated with better learning outcomes (Fraillon et al., 2014<sup>[56]</sup>; Thompson, Meriac and Cope, 2002<sup>[57]</sup>).

### *Attitudes and beliefs*

Student performance in computational problem solving and self-regulated learning may also be influenced by their attitudes and beliefs related to learning and problem-solving, as well as more general dispositions. These factors may particularly interact with students' motivation and engagement with the learning in the digital world tasks. Information about relevant attitudes and beliefs can be drawn from two sources: the PISA student questionnaire module (for more generalised student attitudes, beliefs and dispositions), which adapts existing scales within the literature; and within-unit self-report items (focused on task-specific attitudes and beliefs).

#### **Task interest and self-efficacy**

Task interest and self-efficacy, among other factors such as the perceived cost of effort or the extent to which a task is valued, likely influence an individual's motivation to engage with any given task (Flake et al., 2015<sup>[58]</sup>; Eccles et al., 1983<sup>[59]</sup>). Within each unit, students are asked to self-report their interest in the learning and problem-solving tasks, as well as their perceptions of task difficulty and their confidence in being able to tackle a similar but more complex task.

#### **Growth mindset**

Growth mindset describes the belief that one's abilities can be developed through effort and feedback. Students with a growth mindset may therefore be encouraged to view difficulty or struggle as a normal part of learning and to persist in their learning when challenged (Dweck, 2006<sup>[60]</sup>). In contrast, students with a fixed mindset believe that their abilities cannot be improved and interpret failure or struggle as evidence of limited ability, in turn increasing the likelihood that they disengage when tasks become demanding.

#### **Intrinsic motivation to learn**

Intrinsic motivation refers to engaging in learning for enjoyment or interest, such as enjoying learning new things (Deci and Ryan, 2000<sup>[61]</sup>), and it is often associated with deeper engagement. Other related constructs, such as curiosity, can also reflect students' inclination to seek out new information and explore new ideas or problems.

#### **Perseverance**

Perseverance, as a general disposition, refers to students' ability to persist with challenging tasks despite facing setbacks. Research has shown how low motivation and perseverance relate to avoidant behaviours associated with stress, anxiety and low self-efficacy (Dweck, 2006<sup>[60]</sup>; OECD, 2023<sup>[62]</sup>; OECD, 2024<sup>[63]</sup>). These behaviours might reflect forms of disengagement that can undermine students' capacity to engage in self-regulated learning or to solve complex problems.

#### **General learning strategies**

Students may routinely engage in more or less proactive and deliberate learning strategies, in general, such as setting learning goals, planning their work, and seeking help when needed. Such behaviours reflect students' agency in managing their own learning rather than being passive recipients of instruction. Students who act proactively in their own learning are often better able to benefit from cognitively demanding instruction tasks and to sustain their learning motivation over time (Zimmerman, 2002<sup>[64]</sup>).

## **Task model and unit structure**

The task model in ECD describes the design features of the test tasks, including their objectives, student response types (i.e. desired outputs and evidence-bearing opportunities), tools, resources and affordances. These design features can be modified or varied, within the constraints of the overall test architecture and infrastructure, to elicit the observables needed to make claims about students' ability in the test construct (Mislevy et al., 1999<sup>[65]</sup>).

### *Task objectives and response types*

One major constraint of the PISA assessment design is the limited amount of testing time available per student. Students only completed one hour's worth of test material for the learning in the digital world assessment, meaning the test must focus on the types of learning and problem solving using computational tools that students can reasonably achieve within the given testing time. For this reason, "learning goals" in each unit are framed as intermediate steps towards understanding a system or solving a problem.

Students make progress towards the task learning goal by developing a computational model of a system or building and debugging a computational artefact that solves a problem. Students' computational artefacts thus represent their response to the task, and the quality of their computational artefacts together with certain goal-directed actions (e.g. experimentation behaviours) within the wider digital learning environment provide evidence of their computational problem-solving practices. Other student actions and behaviours (e.g. use of the task affordances), can provide evidence of their self-regulated learning processes. Well-defined records of "events" (i.e. event data models) in the learning environment that are generated by students' interactions (e.g. number and characteristics of experiments, edits/additions to a computational artefact, feedback requests, etc.) are recorded in the test platform and will be parsed and interpreted according to detailed evidence models.

### *Computational tools*

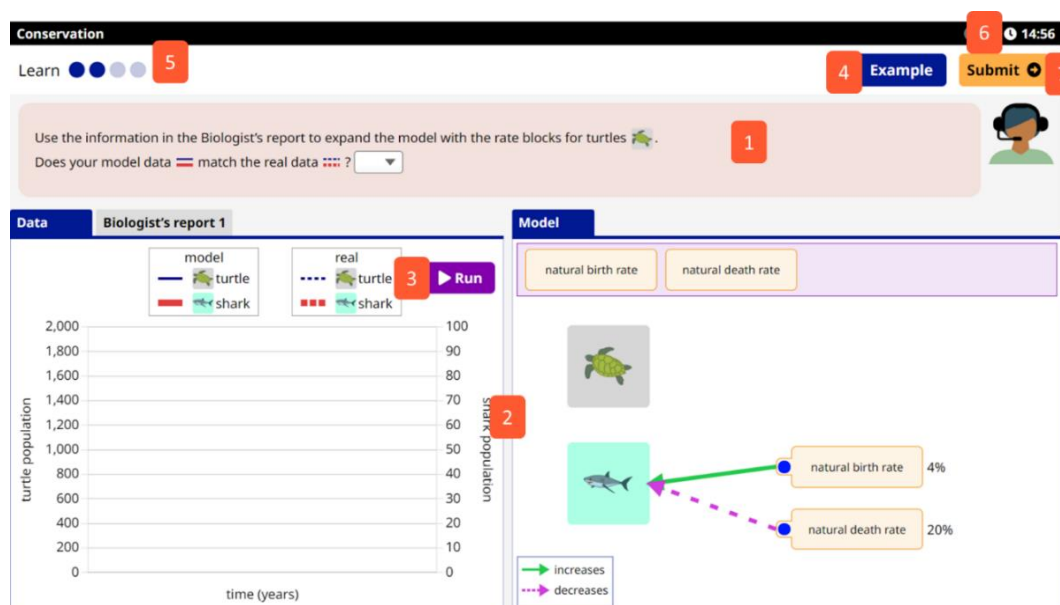
Each task on the test provides students with a computational tool (or tools), such as block-based programming tools, (executable) concept map builders, flowcharts and simulations. Each task also has a workspace in which to build an executable computational representation (e.g. a model, an algorithm) that can be used to represent and manipulate the relationships between variables in a system, generate data to validate hypotheses or make predictions, or control one or more agents within a simulation. In short, each tool and workspace enables students to build an executable computational artefact that serves to advance their understanding of a phenomenon or to solve a complex problem.

The computational tools integrated into the tasks have been designed to be accessible and intuitive. For example, block-based programming environments allow individuals without any prior knowledge of programming languages to build functional algorithms: command blocks within a pre-determined library can be dragged and dropped into the workspace, where they can be attached to other blocks to create an algorithm. Flowcharts represent another simple computational tool with intuitive, pre-defined commands and structures that can be used to model the behaviour of agents or control a simulation.

### *Task elements, affordances and learning resources*

Each task also includes a set of general elements, affordances and resources. In addition to supporting students' test experience by providing a similar structure across tasks, some of the affordances and resources provided to students serve two key design objectives for the assessment: 1) to support students to make progress on the test towards their learning goals, especially those students with limited prior knowledge; and 2) to provide evidence-bearing opportunities related to students' self-regulated learning processes as they work on the tasks.

Figure 3.2. Key task elements, affordances and resources, as illustrated by the released “Conservation” unit



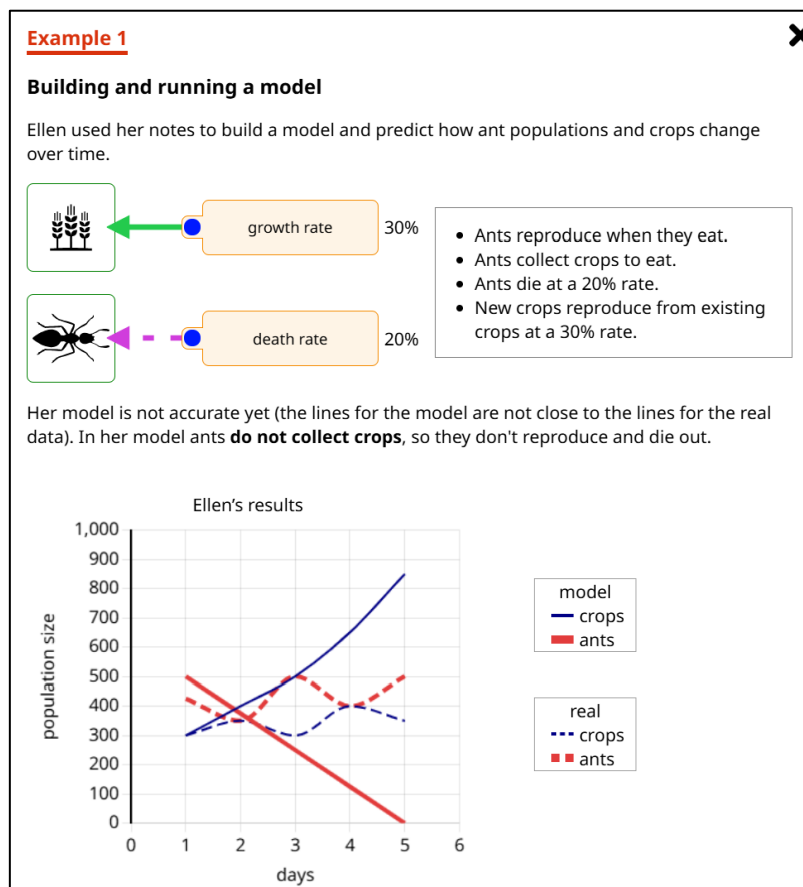
Note: The “Conservation” unit was developed as part of the PISA 2025 Learning in the Digital World test development but publicly released prior to the PISA 2025 Main Study administration. You can access an interactive version of the released unit at: <https://pisa2025ldw.pilaproject.org/>

The key task elements, affordances and learning resources provided within each interactive task are enumerated in Figure 3.2 and described below. They include:

1. **Task instructions and question space:** This space contains the task instructions. The task goal may be expressed verbally, or it may be represented visually (e.g. displaying the desired final state of the environment). In some tasks, students may need to respond to questions in this space after building and interacting with their computational artefacts.
2. **Interactive workspace:** Students build their computational artefact (e.g. algorithm, model, simulation) and view its output (e.g. graphs, data tables) in the task workspace. The layout of the interactive workspace depends on the computational tool(s) in each unit. In Figure 3.2, for example, the workspace contains a graphing tool (“Data” tab), a biologist’s report (“Biologist’s report 1” tab) and a concept mapping tool (“Model” tab).
3. **Test or feedback function:** Students can seek immediate feedback on the quality of their computational artefact by using the test or feedback functionality. The exact form of feedback varies depending on the type of computational tool and artefact (e.g. visual feedback in a block-based programming or dynamic simulation environment, or written feedback in a static modelling environment). In Figure 3.2, for example, the data from the real phenomenon and from the students’ model can be visualised and compared using a graphing tool under the “Data” tab.
4. **Examples:** In each task, students can view at least one functional and annotated solution to a similar task (see Figure 3.3).
5. **Unit phase progression bar:** The unit phase progression bar provides students with a visual cue about the number of items remaining (light blue dots) in each phase of the unit and on the items that have already been completed (darker blue dots).
6. **Timer:** A digital clock indicates the time remaining in each phase of the unit (i.e. across all tasks indicated in the unit phase progression bar).

7. "Submit" button: Students click here to submit their solution to the task and move on to the next item.

Figure 3.3. Worked example available to students in a task within the released "Conservation" unit



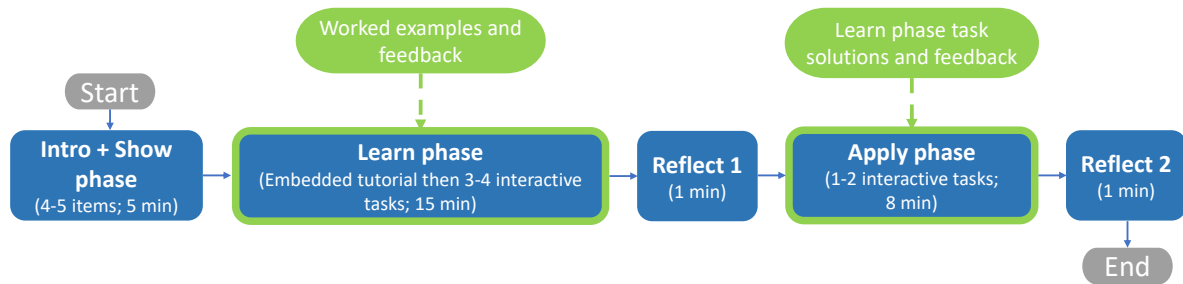
### Task sequence (unit structure)

While each unit is contextualised within a different scenario and integrates one or more computational tools, all units share the same internal organisation. The tasks are structured into a series of unit phases, according to their difficulty and level of scaffolding. This organisation of the unit allows students to familiarise themselves with the learning environment and tools and provides students with multiple opportunities to learn computational concepts and practices and to apply what they have learned in previous tasks to more complex tasks. Organising the test unit as a sequential learning progression aims to replicate authentic learning experience, where students progressively move to their zone of proximal development.

There are four main phases (i.e. groups of tasks) within each test unit, described in Figure 3.4. These include: 1) a short introduction to the unit context and overall learning goal ("Intro", i.e. introducing the problem that students will be asked to solve or learn about), followed by four to five short, discrete pre-test items that measure students' prior knowledge of relevant computational problem solving practices ("Show"); 2) a scaffolded learning phase in which students work through an embedded tutorial, demonstrating the affordances and resources in the learning environment and how to use the computational tools in the unit, followed by three to four interactive tasks focused on a particular computational problem-solving practice ("Learn"); 3) an application phase where students are presented with one or two more open, complex tasks and for which they must apply the computational problem-

solving practices they have learned in the previous tasks to a more difficult task (“Apply”); and 4) two short reflection phases, one following the “Learn” phase and one at the conclusion of the unit, with a series of self-report items focused on students’ affect, self-evaluation and perceptions of the tasks (“Reflect”).

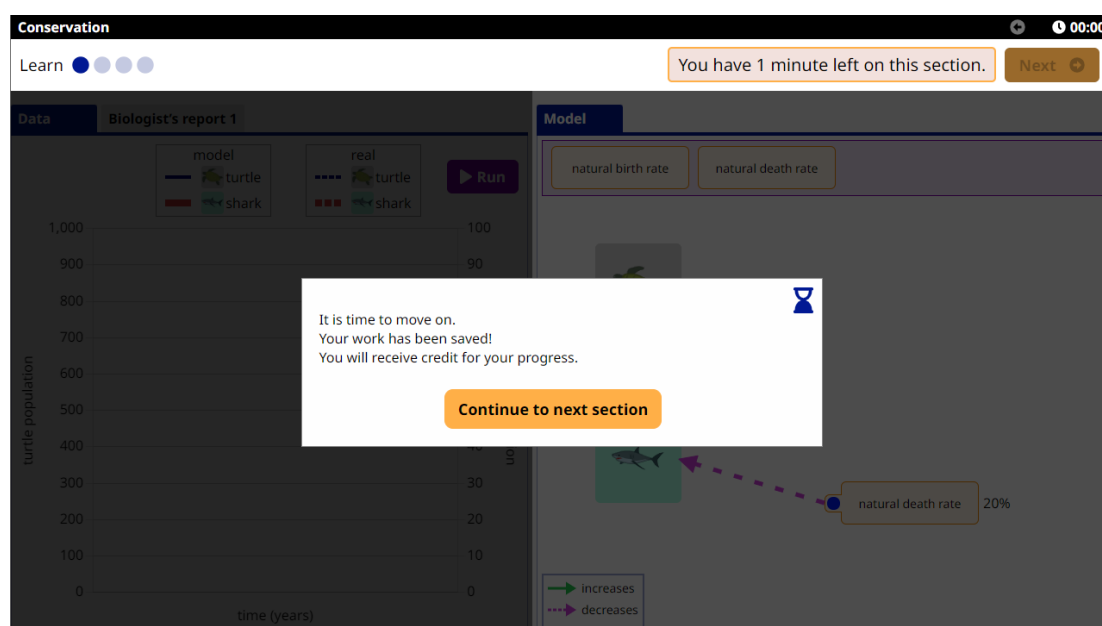
**Figure 3.4. Unit structure and flow of tasks within the test units**



The tasks in the “Learn” and “Apply” phases are interactive in the sense that students have access to computational tools and workspaces in which to build, modify and interact with computational artefacts, as well as access to feedback on their progress and examples to help them. The pre-test items in the “Show” phase do not include these affordances and resources.

Students have an allotted amount of time to complete or make their way through each phase of the unit. After students have completed each phase or the maximum time allowed for that phase has elapsed, they progress to the next phase of the unit. Breaking the extended learning unit down into independent phases, where students are automatically progressed to the next phase at regular intervals, ensures that all students move through the whole unit (i.e. students are not trapped in a task for the entire duration of the unit). Students are given an unobtrusive one-minute warning before the time runs out in each phase (see orange box in the top right-hand corner of Figure 3.5), before being shown a pop-up screen informing them that they will now be moved on to the next phase of the unit (also Figure 3.5). Any progress students make in the tasks in the previous phase will be saved and students will be awarded credit for their progress on the task.

Figure 3.5. Example of the forced progression pop-up screen after time has elapsed



The unit structure and flow of tasks within each phase of a unit, as well as the specific features, affordances and resources of tasks within each unit phase, is further described below and illustrated using tasks from the released “Conservation” unit.

### “Introduction” and “Show” phase

Each unit starts with an introduction page that presents the overall learning goal(s) of the unit and briefly describes the unit context and problem scenario. In the “Conservation” unit (see Figure 3.6), students are told that the unit focuses on a marine ecosystem and that they will learn how to build and use a model to understand and predict populations within the ecosystem. The page also introduces a gender-neutral computer agent that is present throughout the unit to guide students (e.g. introducing the unit context and goals, asking students questions, guiding students through the interface, etc.). The agent explains to the student that before they learn how to build a model, they must show the agent what they already know.

The introduction screen is then followed by a series of short, discrete pre-test items (“Show”) that serve to measure students’ initial knowledge and understanding of the target computational problem-solving practices in the unit. For example, in the “Conservation” unit, the pre-test items focus on students’ ability to read and understand graphs, draw inferences from datasets, and make predictions using data. Students do not have access to learning resources or feedback during this phase. For efficiency, the response types in this phase of the unit are limited to multiple-choice, true/false statements, or simple interactive responses (e.g. hot spots). Figure 3.7 shows an example pre-test item for the “Conservation” unit.

Figure 3.6. Introduction page of the “Conservation” unit

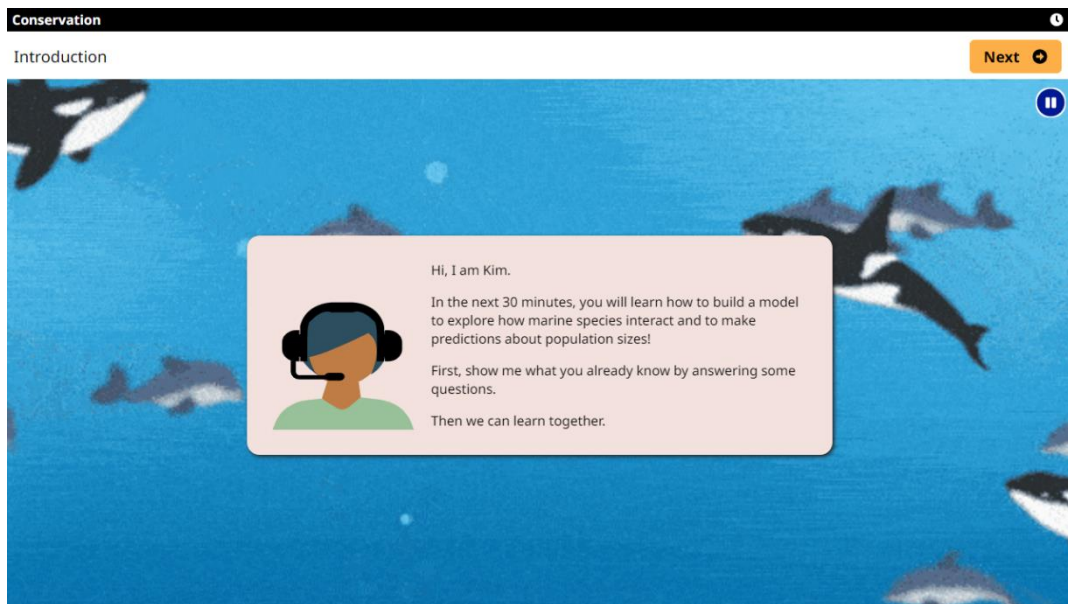
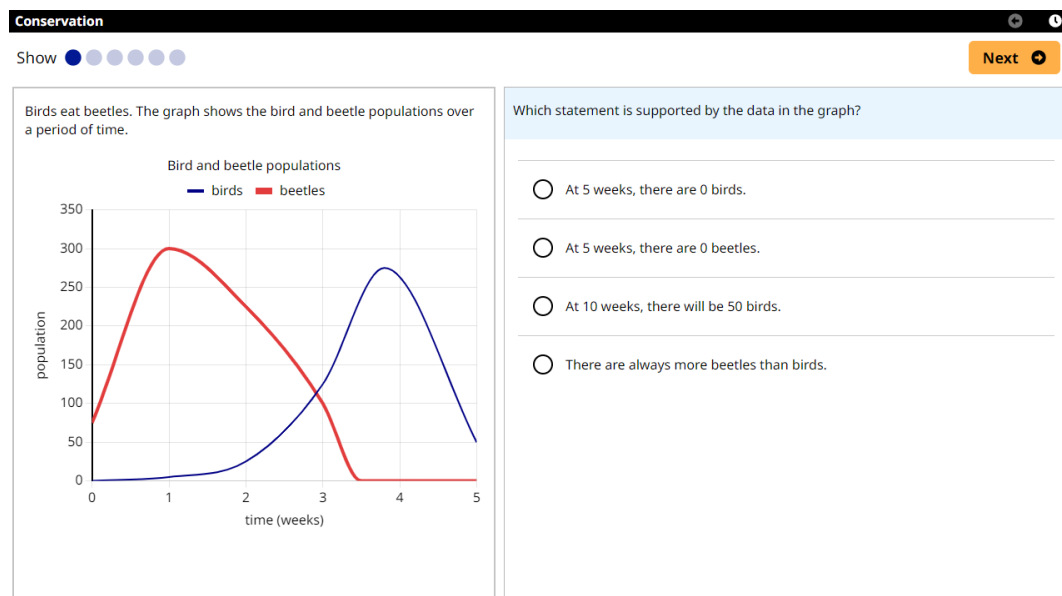


Figure 3.7. Example multiple-choice pre-test item on graph literacy in the “Conservation” unit

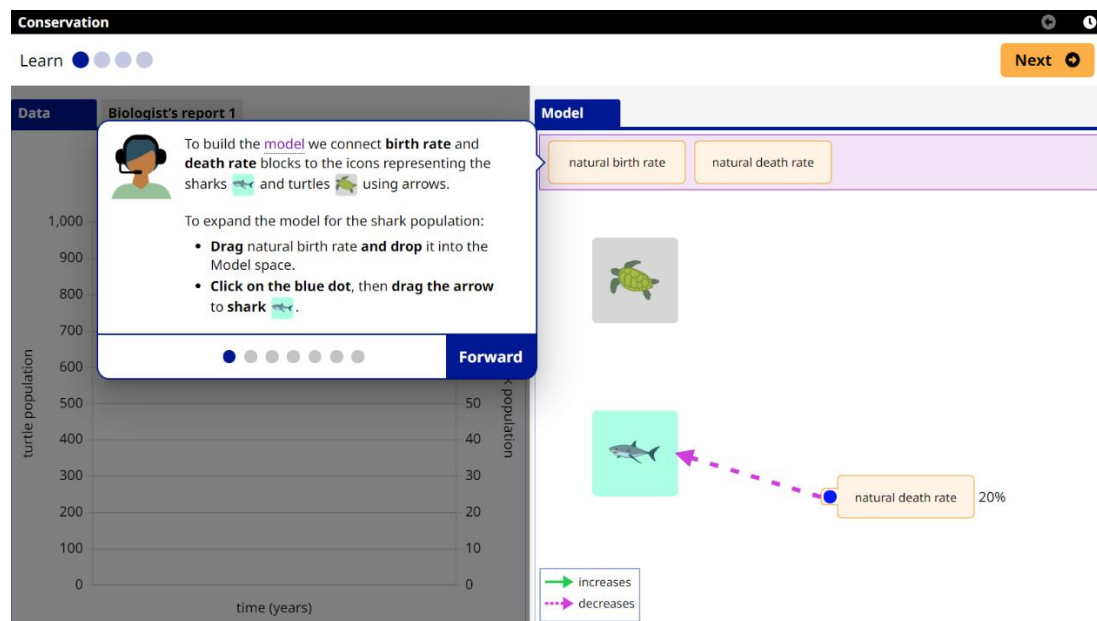


### “Learn” phase

The “Learn” phase of the unit begins with a heavily scaffolded tutorial, led by the tutor agent, that walks students through the interface, affordances and basic functionality of the computational tool(s) in the digital learning environment (see Figure 3.8). To reduce cognitive and reading load, part of the tutorial may include viewing short videos. During the tutorial, students are instructed to perform certain actions (e.g. drag-and-drop a block into the workspace, access the example, etc.) by the tutor agent to ensure that they understand how to navigate around and use the affordances of the workspace. If a student tries to move forward without carrying out an action, the system will automatically demonstrate the action to the student before moving them to the next step of the tutorial. This assistive mechanism ensures that all

students have at least some familiarity with and the opportunity to learn about the digital environment for each unit.

**Figure 3.8. Screenshot of scaffolded tutorial segment from the “Conservation” unit**

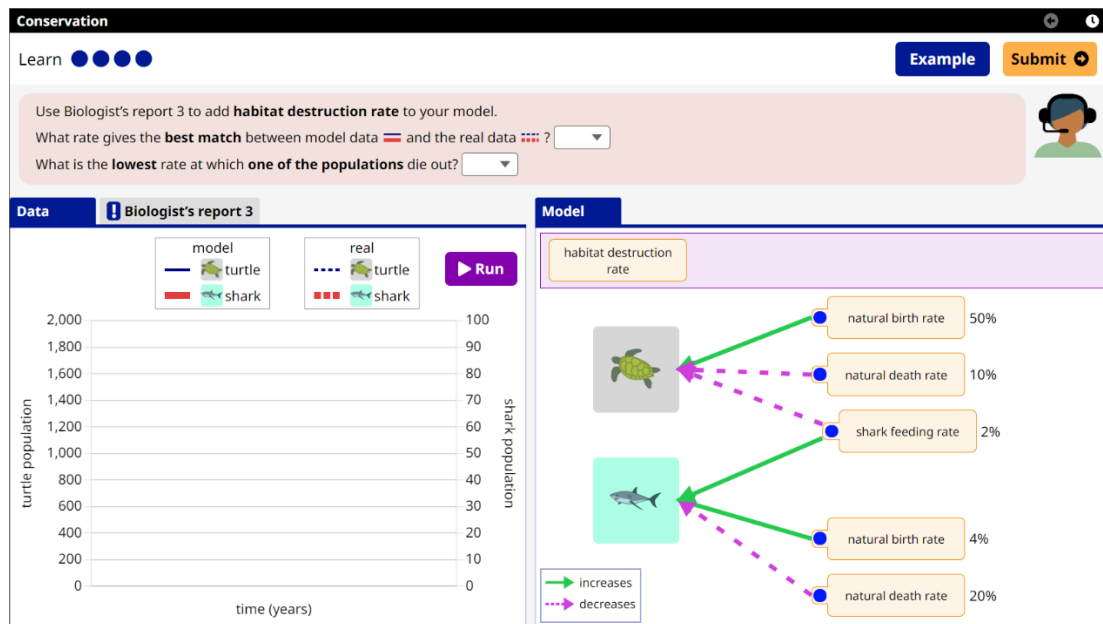


Following the scaffolded tutorial, the “Learn” phase then consists of a series of open, interactive tasks in which students learn and use computational problem-solving practices to solve progressively more challenging tasks. The practices targeted by each of the “Learn” phase tasks are those that the student will need to apply in the next phase of the unit. For example, in the “Conservation” unit, students will learn to integrate information into a computational model (i.e. how to connect and experiment with variables), to interpret lines in a time graph, and to conduct experiments to build a model that approximates real data.

The learning goals (i.e. the targeted practice and the task problem) in each task are specific and achievable in the space of a few minutes, and students’ progress towards achieving the goal is measurable. A simple “Learn” phase task in the “Conservation” unit, for example, asks students to complete an existing model of the ecosystem by referring to observations from the biologist’s report (see Figure 3.2). The students must connect the turtle population to the existing model, choosing the value of the two variables provided, and then respond to a simple question regarding the accuracy of their model. In the last task in the unit phase, students investigate the effect of an external shock to the ecosystem (Figure 3.9), adding a new variable to the model and conducting experiments to understand and predict its effects.

Throughout the “Learn” phase tasks, students have access to worked examples and feedback to support their learning and problem-solving process. An example of a worked example in the “Conservation” unit is shown in Figure 3.3. In this unit, feedback on the accuracy of students’ model is provided visually by executing the model and observing its output in the “Data” tab.

Figure 3.9. Last task in the “Learn” phase of the “Conservation” unit



After students submit their response to each task in the “Learn” phase, they respond to a short, multiple-choice item asking them to self-evaluate their performance in the task (see Figure 3.10). Once completed, they also view an annotated solution to the task and can compare it to their own solution to support their learning (see Figure 3.11).

Figure 3.10. Self-evaluation pop-up after student submits their solution to a task in the “Learn” phase

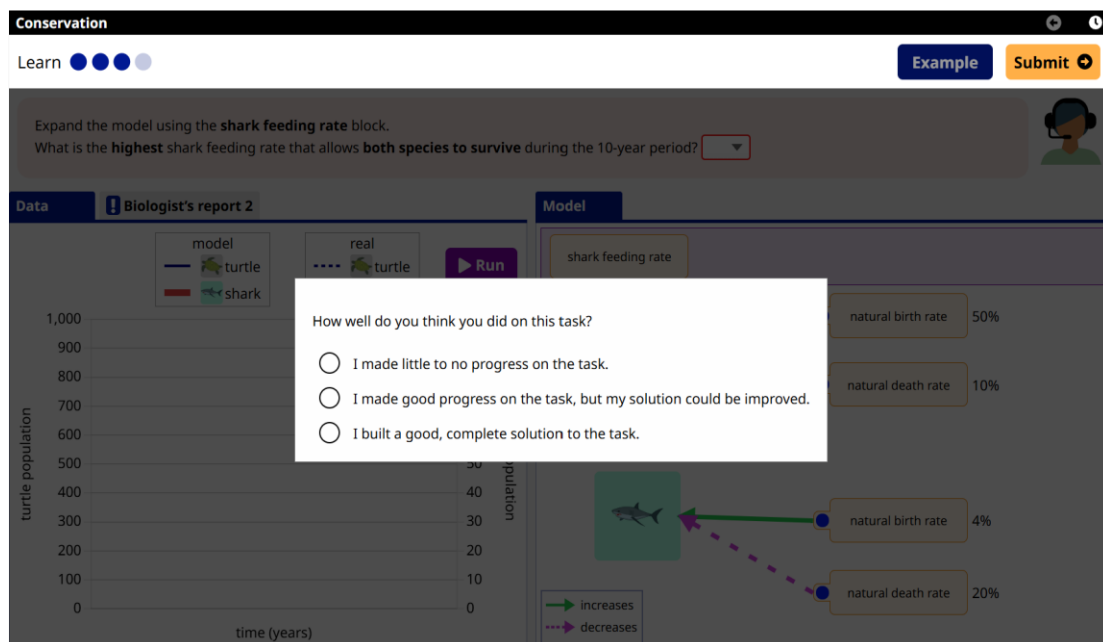
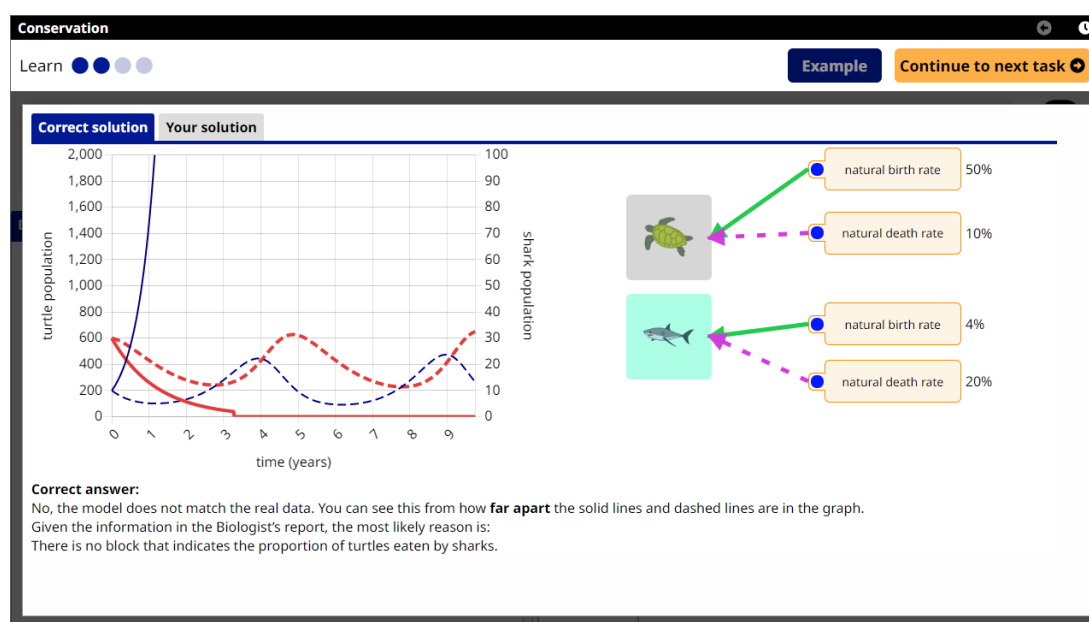


Figure 3.11. Annotated solution screen in the “Conservation” unit



### “Reflect 1” phase

After the “Learn” phase, students complete a short first “Reflect” phase consisting of two sets of self-report items. The first set of questions asks students to report their affective states and the strength of their feelings (see Figure 3.12). The second set of self-report items asks students about how difficult they found the tasks in the “Learn” phase of the unit, and how confident they feel about tackling a more complex problem (i.e. their task self-efficacy). The items in this phase of the unit intend to capture information to contextualise students’ self-regulated learning processes, particularly how they regulate their motivational and affective processes.

Figure 3.12. Self-report item on students’ affective states

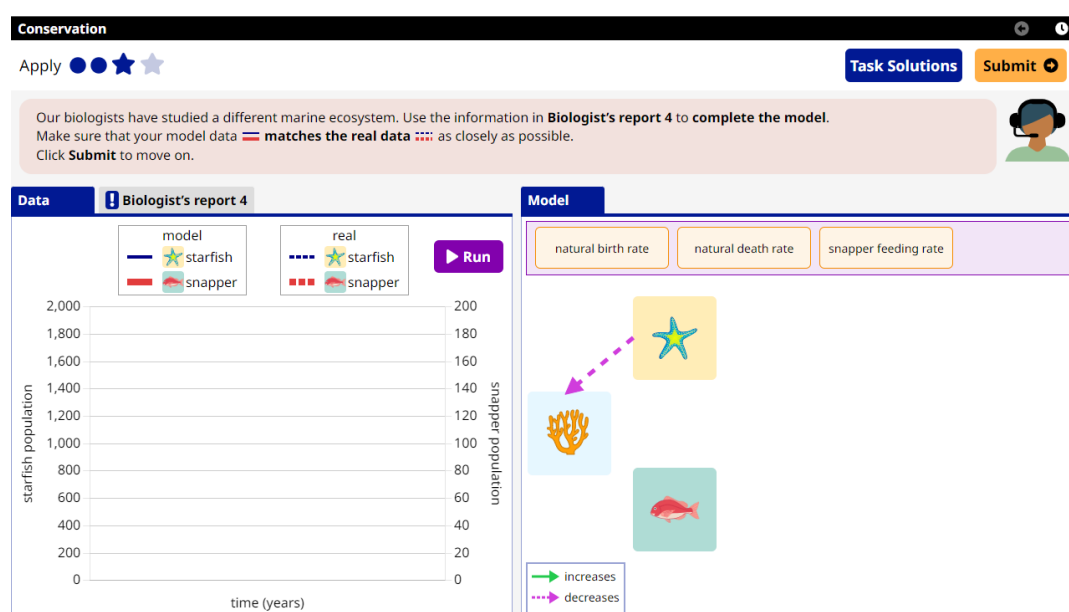
The screenshot shows the 'Reflect' phase of the 'Conservation' unit. At the top, there are tabs for 'Learn', 'Reflect', and 'Next'. The 'Reflect' tab is active, showing a question: 'Now that you have been learning for some time, how are you feeling? For each line, select the option that best describes how you currently feel.' Below the question, there are four rows of self-report items, each with four radio button options. A small icon of a person wearing a headset is visible on the right side of the form.

|   |                                                          |                                    |                                         |                                                       |
|---|----------------------------------------------------------|------------------------------------|-----------------------------------------|-------------------------------------------------------|
| 1 | <input type="radio"/> Very stressed                      | <input type="radio"/> Stressed     | <input type="radio"/> Relaxed           | <input type="radio"/> Very relaxed                    |
| 2 | <input type="radio"/> Very dissatisfied with my solution | <input type="radio"/> Dissatisfied | <input type="radio"/> Satisfied         | <input type="radio"/> Very satisfied with my solution |
| 3 | <input type="radio"/> Very confused                      | <input type="radio"/> Confused     | <input type="radio"/> Not very confused | <input type="radio"/> Not at all confused             |
| 4 | <input type="radio"/> Very bored                         | <input type="radio"/> Bored        | <input type="radio"/> Interested        | <input type="radio"/> Very interested                 |

### “Apply” phase

The final interactive task phase of the unit asks students to apply the computational problem-solving practices that they learned and used during the “Learn” phase to a more complex task. For certain units, this phase is structured in two sub-tasks to ensure students have sufficient opportunities to provide evidence of their ability to apply different computational problem-solving practices. For example, in the “Conservation” unit, the first sub-task of the “Apply” phase asks students to build an accurate model of a different marine ecosystem that contains other species (see Figure 3.13). Like in the separate “Learn” phase tasks, they must identify variables to add to the model and set values for these variables (“build and debug computational artefacts”), run the model to generate data (“conduct experiments”), and check that the data are accurate compared to the real data (“analyse data”). Students are exposed to all of these practices in a simpler context in the first two “Learn” tasks of the unit.

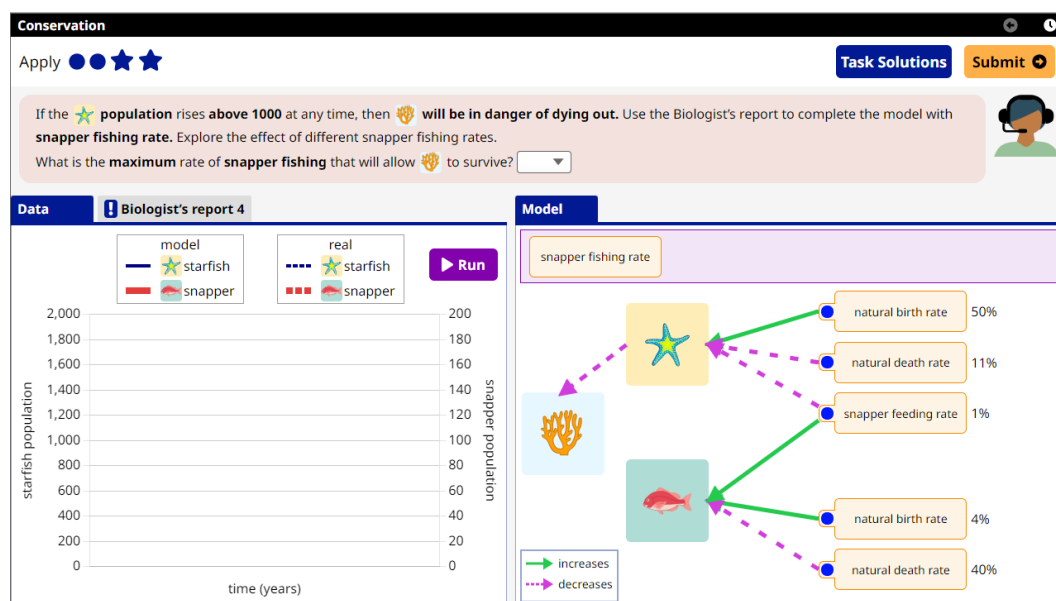
Figure 3.13. Part 1 of the “Apply” phase in the “Conservation” unit



In the second sub-task of this phase (see Figure 3.14), students must investigate how a new variable affects the system (similar to the third “Learn” phase task). All students enter this sub-task with an accurate model even if they did not successfully build it in the previous sub-task, in order to reduce dependencies between the sub-tasks. They must demonstrate all three sub-facets of the competency model for computational problem-solving practices by correctly adding the fishing rate variable to the model (“build and debug computational artefacts”), testing different rates (“conduct experiments”), and identifying the highest fishing rate at which the coral population will survive based on their results (“analyse data”).

In general, the “Apply” phase tasks have been designed according to a “low floor, high ceiling” principle, meaning that students should be able to make at least some progress towards the learning goal but most students will not be able to solve the task on their first attempt. In this phase of the unit, instead of worked examples, students can access the annotated task solutions from the “Learn” phase of the unit (see Figure 3.11 for an example annotated solution) to help students recognise how to correctly apply the targeted computational problem-solving practices needed to make progress in the task. Feedback is still available to students in the “Apply” phase tasks, as it is in the “Learn” phase.

Figure 3.14. Part 2 of the “Apply” phase in the “Conservation” unit



### “Reflect 2” phase

The unit ends with a second “Reflect” phase that includes another set of situated self-report items aiming to gather evidence about students’ self-regulated learning processes throughout the unit. This time, there are three sets of questions: one set asking students to self-evaluate their performance in the “Apply” task(s); a second set asking students about their affective states and the strength of their feelings, as in “Reflect 1” (see Figure 3.12), and a third set that asks about students’ perception of difficulty and the amount of effort they exerted throughout the unit.

### Assembling the tasks and test units

The learning in the digital world tasks are organised into clusters of 30 minutes, according to the above task sequencing. Each 30-minute cluster (i.e. one test unit) was placed in multiple computer-based test formats according to a rotated test design. This means that a proportion of the students that took PISA in 2025 sat up to one hour of test material for the Learning in the Digital World assessment. Although some units do not provide evidence for all components of the competency model, the test assembly provides adequate coverage of the competency model at the population level. Students who completed two units of the assessment sat test units that included different computational tools (for example, one unit in a simulation-based environment and another unit in a block-based programming environment).

### Evidence model

Every task has a detailed set of evidence or scoring rules describing how student responses, products (i.e. computational artefacts) and interactions with the digital learning environment will be parsed into observables that can provide evidence about the target constructs of interest. These scoring rules are derived from theory-based assumptions about what constitutes evidence of productive computational problem-solving practices and self-regulated learning processes. The scoring rules for each task have been iteratively refined and validated through multiple cognitive laboratories, pilot studies and the Field Trial.

### *Scoring approach for computational problem-solving practices*

Students' capacity to engage in computational problem solving is measured by the quality of their computational artefacts and the extent to which they make progress towards their learning goals, as defined in the tasks in the "Learn" and "Apply" phases of the unit. Evidence of students' computational problem-solving practices is derived through both response data (students' answers to the task questions or their computational artefacts) and response process data (their actions during the tasks). For the purpose of scoring the tasks, each broader learning goal is broken down into a series of constituent (implicit) task sub-goals that students must address in order to produce a high-quality, functional solution to that task. Each task sub-goal is mapped to a specific sub-component of the competency model to establish a valid chain of evidentiary reasoning.

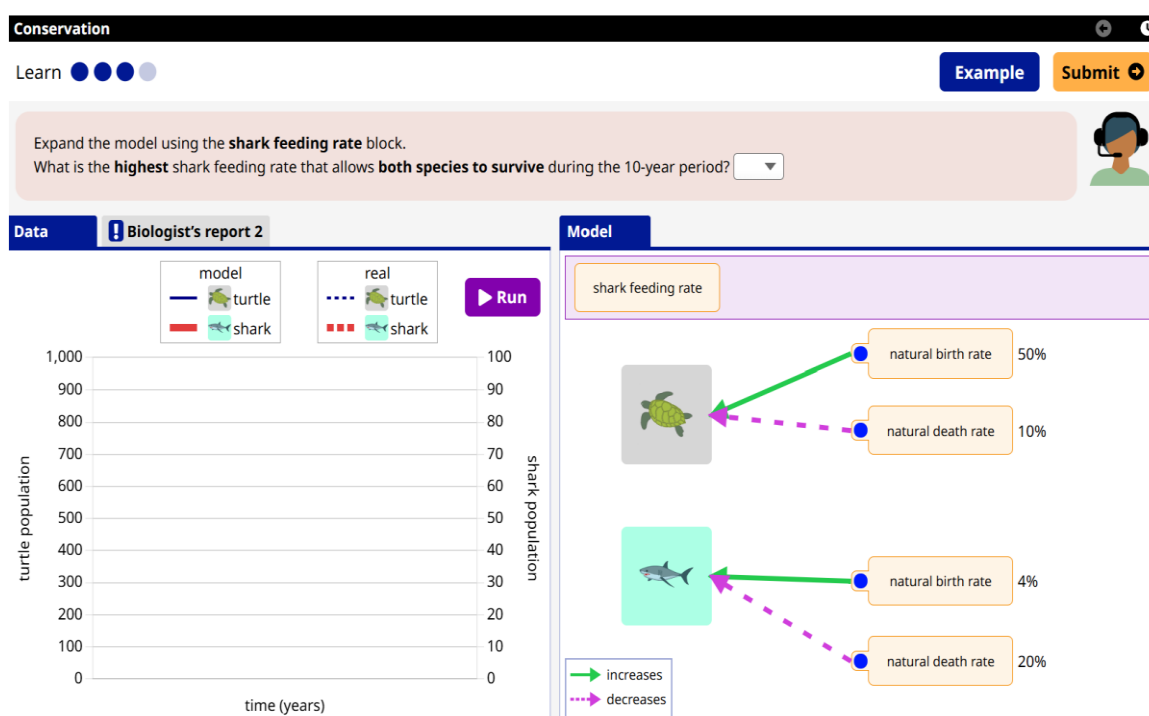
The evidence rules for each task use a polytomous scoring model, where students achieve full credit for successfully solving the task (i.e. achieving the learning or problem-solving goal) and achieve different levels of partial credit for successfully addressing various sub-goals within the task (i.e. making productive progress towards the learning or problem-solving goal despite not building a complete solution). The evidence rules specify how observable student actions, behaviours or outputs demonstrate achievement for each task sub-goal, how these sub-goals provide evidence about the target practices in the competency model, the level of credit that should be associated with (partially) achieving different sub-goals, and how progress towards the various sub-goals can be summarised in a single scoring rubric. Depending on the complexity of the task, there may be several partial credit categories corresponding to different solution approaches and the degree of progress that students make towards the overall learning goal. This scoring approach rewards students who demonstrate mastery of the constructs of interest while also rewarding, to a lesser degree, students who appear to be heading in a productive direction.

The scoring rules and partial credit approach are best illustrated through a concrete example task. Note, however, that the following example is described for illustrative purposes only and has not been subject to the same iterative refinement and rigorous validation procedures as the units retained in the final test item pool. For a more detailed description of how the evidence rules for the PISA 2025 Learning in the Digital World test were developed, refined, validated and automated for scoring, see Foster et al. (2026<sup>[66]</sup>).

#### **Example scoring rules for a task in the "Conservation" unit**

The third task in the "Learn" phase of the "Conservation" unit is shown in Figure 3.15. Students are first instructed to expand the model shown in the workspace by adding the shark feeding rate variable; to do this, they must infer the relationship between shark feeding rate and the two species' populations from the biologist's report. Once the student has correctly added the variable into the model, they must then conduct experiments to generate and analyse data from which they can answer the task question: "What is the highest shark feeding rate that allows both species to survive during the 10-year period?".

Figure 3.15. Example task in “Learn” phase of the “Conservation” unit



The first step in defining the scoring rules for this task is to identify the constituent sub-goals that students must achieve and that can provide evidence of a distinct sub-component of the competency model. As part of this process, it is possible to define observables (i.e. student actions or products) that provide evidence about the extent to which students fully or partially achieve each of these sub-goals. Some task sub-goals may inevitably be interdependent; for example, it is not possible to conduct meaningful experiments, which is a key step towards solving the task, without first adding the relevant variable to the model. While some interdependence between the task sub-goals is inevitable, to the extent possible, the associated observables for each sub-goal should be independent. The high-level sub-goals and associated observables, as well as their mapping to the framework, are described in Table 3.1 for the example task from “Conservation” unit.

Once the task sub-goals have been identified, as well as potential observables associated with different degrees of progress towards the various sub-goals, these can be organised within a hierarchical scoring rubric. The scoring rubric should reflect the various ways that students may make progress towards the learning and problem-solving goal, as well as the interdependence between any of the sub-goals. Table 3.2 describes such a scoring rubric for the example task in the “Conservation” unit. For the units included in the Learning in the Digital World assessment, the assumptions underlying the hierarchy of scores and evidence associated with each score band were empirically validated during the Field Trial.

Similar extended partial credit scoring rubrics have been developed for each of the “Learn” and “Apply” tasks included in the test. Students’ scores across these tasks are accumulated via a statistical model (e.g. item response theory) and contribute to students’ performance on the computational problem-solving scale.

**Table 3.1. High-level description of task sub-goals, framework mapping, and evidence**

|                                                                    | Sub-goal 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sub-goal 2                                                                                                                                                                                                                                                                                                                                                      | Sub-goal 3                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High-level description                                             | The student adds the shark feeding rate block into the model and links it correctly to the turtle and shark variables.                                                                                                                                                                                                                                                                                                                                                                                                                       | The student conducts multiple different experiments, including shark feeding rate, in the model.                                                                                                                                                                                                                                                                | The student analyses their dataset, generated from a correct model, and selects 3% as their response to the question.                                                                                                                                                                                                                                                                                          |
| Framework mapping                                                  | Build and debug computational artefacts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Conduct experiments                                                                                                                                                                                                                                                                                                                                             | Analyse data                                                                                                                                                                                                                                                                                                                                                                                                   |
| Levels of achievement of sub-goal (associated observable evidence) | <p>Not achieved – The student does not add the shark feeding rate block to the model.</p> <p>Partial (low) – The student links the shark feeding rate block to both the shark and the turtle, but not using the correct relationship arrows</p> <p>Partial (high) – The student links the shark feeding rate block to both the shark and the turtle, but only correctly to one.</p> <p>Achieved – The student adds the shark feeding rate block and links it correctly to the turtle (negative arrow) and to the shark (positive arrow).</p> | <p>Not achieved – The student does not conduct multiple experiments.</p> <p>Partial – The student conducts at least one experiment (i.e. generates data) with the shark feeding rate block set to any rate.</p> <p>Achieved – The student conducts at least two experiments (i.e. generates data) with the shark feeding rate block set at different rates.</p> | <p>Not achieved – The student does not select 3% or selects 3% but does not generate data with the correct model.</p> <p>Partial – The student generates data from a correct model, with the shark feeding rate set at any value, and selects 3%.</p> <p>Achieved – The student generates data from a correct model, including with the shark feeding rate set at 3% and 4%, respectively, and selects 3%.</p> |

Note: The task sub-goals and observable evidence described in this table are for illustrative purposes only and have not been empirically validated.

**Table 3.2. Scoring rubric for example task in the “Conservation” unit**

| Score | Description                                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Student correctly adds the shark feeding rate block to the model, conducts at least two different experiments that include the variable set to 3% and 4%, respectively, and selects 3%.               |
| 4     | Student correctly adds the shark feeding rate block to the model and EITHER conducts at least two different experiments OR conducts one experiment with the variable set to any value and selects 3%. |
| 3     | Student correctly adds the shark feeding rate block to the model and conducts at least one experiment with the variable set to any value.                                                             |
| 2     | Student adds the shark feeding rate block to the model and correctly links it to either the shark or turtle population, and conducts at least one experiment with the variable set to any value.      |
| 1     | Student incorrectly adds the shark feeding rate block to the model and conducts at least one experiment with the variable set to any value.                                                           |
| 0     | All other cases                                                                                                                                                                                       |

Note: The scoring rubric described here is for illustrative purposes only and has not been empirically validated.

### *Developing indicators of self-regulated learning processes*

Student actions and behaviours on the tasks, as well as their responses to the embedded self-report items in the unit, will be used to generate indicators of their self-regulated learning (SRL) processes. These indicators will not contribute to students' scores on the tasks but will be used to contextualise their

performance on the computational problem-solving scale (including relative to their performance on the pre-test items), as well as to provide insights into the types of self-regulated learning behaviours that students engaged in while learning and problem-solving.

These SRL indicators are mostly derived from student response process data and will provide information about each of the three SRL sub-components as defined in the competency model (maintain motivation and task engagement; monitor progress and adapt strategies; and evaluate performance). Example indicators for each sub-component are further described below.

Defining evidence rules for self-regulated learning processes is complex because, in many cases, these adaptive behaviours depend on the students' ability level and/or are influenced by other contextual factors (e.g. prior actions or solution states, time constraints, etc.). Consider, for example, help-seeking; students who know how to reach the solution do not need to seek help, meaning that there is no evidence-bearing opportunity for these students. As such, the simple presence or absence of behaviours cannot necessarily be interpreted as evidence of self-regulation (or a lack thereof). This means that any evidence rules for self-regulated learning processes must take into account these contextual complexities, acknowledging when and for whom there are evidence-bearing opportunities to observe a particular behaviour, and their logic must be extensively tested and validated against genuine student behaviours on the task.

All of the indicators were initially defined from theory-based assumptions about “good” self-regulated learning behaviours in the context of the test tasks. For example, theory assumes that good self-regulated learners would seek feedback as they work on complex learning tasks to regularly monitor their progress and, if needed, adapt their approach to the task. Once encoded, the indicators were tested and iteratively refined using a large-scale dataset from the PISA 2025 Field Trial. Following the PISA 2025 Main Study, data-driven approaches may also be explored to create additional self-regulated learning indicators. For example, sequence-mining techniques may be able to identify differences in learning strategies between different types of learners, identifying qualitatively similar patterns of actions and combining them into distinct “behaviour” categories.

### **Indicators related to students' engagement, motivation and affective regulation**

Indicators related to students' motivational and affective states will be developed using information collected via the situated self-report items in the “Reflect” phases of the units, sometimes together with student response process data. The situated self-report items asked students how they are feeling (Figure 3.12), their interest in the tasks, and the level of effort they exerted in the unit. Simple descriptive indicators will summarise the percentage of students who report feeling a positive affective state while learning, for example, or those who report investing effort in the unit (see Table 3.3). More complex indicators combine data points across items, for example students who reported overall positive affective states or whose overall affective profile remained consistent or improved throughout the task.

Other self-regulated learning indicators related to students' task engagement are derived uniquely from their response process data. For example, one relatively complex indicator of task engagement considers various student behaviours, including their total time spent on task, the total number of actions, the minimum number of relevant actions (i.e. the number of relevant actions a student needed to execute to achieve a partial credit score on the task), and their maximum pausing time (i.e. maximum time between any actions). Minimum thresholds for each of the four behaviours are defined on a task-by-task basis, and if a student exceeded the minimum threshold on at least two of the four behaviours, then they are identified as being at least somewhat engaged with the task.

Table 3.3. Selected indicators for students' motivational and affective regulation processes

| Indicator               | Description                                                                                                              | Code/value                                                                                                                                                                                                                                                                                                                                                                                                                  | Data source(s)                                                        | Framework alignment                                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Simple indicator        |                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                       |                                                                                       |
| Task interest           | Student reports general positive interest in the learning tasks after the "Learn" phase of the unit                      | 1 = Student reports feeling "interested" or "very interested" 0 = Student reports feeling "bored" or "very bored"                                                                                                                                                                                                                                                                                                           | Situated self-report item in "Reflect phase 1"                        | Maintain motivation and task engagement (Sub-component: Manage affect while learning) |
| Effort                  | Student reports that they put effort into completing the tasks                                                           | 1 = Student reports that they made "moderate effort" or "a lot of effort" 0 = Student reports that they made "no effort" or "limited effort"                                                                                                                                                                                                                                                                                | Situated self-report item in "Reflect phase 2"                        | Maintain motivation and task engagement (Sub-component: Manage effort)                |
| More complex indicators |                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                       |                                                                                       |
| Overall affect          | Student reports an overall positive affective disposition after the "Learn" phase of the unit                            | 4 = Student reports very positive affect (13+ points)<br>3 = Student reports mostly positive affect (10-13 points)<br>2 = Student reports mixed affect (10 points)<br>1 = Student reports mostly negative affect (6-9 points)<br>0 = Student reports very negative affect (4-6 points)<br><br>Note: Each affect response option corresponds to a "score" point between 1-4, with 1 representing the most negative response. | Situated self-report items in "Reflect phase 1"                       | Maintain motivation and task engagement (Sub-component: Manage affect while learning) |
| Affective transition    | Student reports the same (or more positive) overall affective disposition before and after the "Apply" phase of the unit | 2 = Student overall reports more positive affective states<br>1 = Student overall reports similar affective states<br>0 = Student overall reports more negative affective states<br><br>Note: Each affect response option corresponds to a "score" point between 1-4, with 1 representing the most negative response.                                                                                                       | Situated self-report items in "Reflect phase 1" and "Reflect phase 2" | Maintain motivation and task engagement (Sub-component: Manage affect while learning) |
| Task engagement         | Student is at least somewhat engaged in the task                                                                         | 1 = Student exceeds minimum threshold in at least two behaviours<br>0 = Student does not exceed minimum threshold in at least two behaviours<br><br>Behaviours:<br>1) Total time on task (above 20 <sup>th</sup> percentile)<br>2) Longest inactivity period (below 80 <sup>th</sup> percentile)<br>3) Total actions (above 50% of the median)<br>4) Relevant events (task-specific threshold)                              | Response process data (time data, event data)                         | Maintain motivation and task engagement (Sub-component: Task engagement)              |

### Indicators related to students' cognitive and metacognitive strategies

Several indicators have been constructed relating to students' monitoring and help-seeking behaviours, largely derived from their response process data while working on the tasks. For example, students' monitoring of their work can simply be captured by whether the student seeks feedback at any point during the task; a more complex version of the indicator describes the number of times students use a feedback affordance relative to their other actions on the task (i.e. as a percentage of their total actions) (see Table 3.4).

**Table 3.4. Selected indicators for students' metacognitive and cognitive strategies**

| Indicator                            | Description                                                                                                  | Code/value                                                                                                                                                                                                                                                                                                           | Data source(s)                                                  | Framework alignment                                                     |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| Simple indicator                     |                                                                                                              |                                                                                                                                                                                                                                                                                                                      |                                                                 |                                                                         |
| Use of feedback                      | Student seeks feedback on the quality of their computational artefact                                        | 1 = Student uses feedback affordance at least once<br>0 = Student does not use feedback affordance at least once                                                                                                                                                                                                     | Response process data (event data)                              | Monitor progress and adapt strategies (Sub-component: Monitor progress) |
| More complex indicators              |                                                                                                              |                                                                                                                                                                                                                                                                                                                      |                                                                 |                                                                         |
| Systematic monitoring                | Student regularly seeks feedback on the quality of their computational artefact                              | Numeric value (% feedback events out of total number of events)                                                                                                                                                                                                                                                      | Response process data (event data)                              | Monitor progress and adapt strategies (Sub-component: Monitor progress) |
| Help when needed                     | Student views the worked example during a task in which they struggled                                       | 1 = Student reports that they their task solution could be improved and viewed the worked example<br>0 = Student reports that they their task solution could be improved but did not view the worked example<br><br>Note: Students that report that they built a complete, functional solution are coded as missing. | Situated task evaluation and response process data (event data) | Monitor progress and adapt strategies (Sub-component: Help-seeking)     |
| Average events between feedback      | Student makes more or less modifications to their computational artefact between their monitoring behaviours | Numeric value (average number of relevant modification events between two different feedback events)                                                                                                                                                                                                                 | Response process data (event data)                              | Monitor progress and adapt strategies (Sub-component: Strategy)         |
| Use of control of variables strategy | Student tends to use the control of variables strategy when conducting experiments                           | Numeric value (% times that student varies only one variable between data generation attempts, out of the total number of attempts)                                                                                                                                                                                  | Response process data (event data)                              | Maintain motivation and task engagement (Sub-component: Strategy)       |

Some indicators also combine response process data with students' situated self-reports to facilitate the interpretation of their present or absent task behaviours. For example, one indicator describes whether a student used the worked example affordance (i.e. help resource) if they also reported that they struggled on the task (see Table 3.4). In these types of indicators, a significant proportion of students will record missing values as they did not encounter an evidence-bearing opportunity (i.e. they did not report struggling on the task).

Other indicators related to these self-regulated learning processes focus on extracting patterns of student actions or behaviours to infer their cognitive and metacognitive strategies. For example, some students may choose to make several changes to their computational artefact between monitoring instances, whereas others may take a more incremental approach to building and debugging their artefacts. Or in

tasks where students are asked to conduct experiments, it is possible to identify the kinds of strategic experimentation behaviours they use by identifying the number of variables they change between data generation attempts.

### Indicators related to students' reflective and evaluative processes

Finally, several indicators were created related to students' reflective and evaluative processes. After completing each learning and problem-solving task, students were asked to self-evaluate their progress on the task. Simple indicators describe differences in students' self-evaluations, whereas more complex indicators determine the extent to which students' self-evaluations align with their actual performance on the task (see Table 3.5). Similar indicators were also constructed related to whether students' task self-efficacy, related to tackling a more complex task in the "Apply" phase, is aligned with their earlier performance on the tasks in the "Learn" phase.

**Table 3.5. Selected indicators for students' reflective and evaluative processes**

| Indicator                       | Description                                                                                              | Code/value                                                                                                                                                                                                                                                                                                                                                  | Data source(s)                                   | Framework alignment                                             |
|---------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|
| Simple indicator                |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                             |                                                  |                                                                 |
| Self-evaluation                 | Student self-evaluates their performance on the task                                                     | 2 = Student self-reports that they built a good, complete solution to the task<br>1 = Student self-reports that they made good progress but their solution could have been improved<br>0 = Student self-reports that they made little to no progress on the task                                                                                            | Situated self-evaluation item                    | Evaluate performance (Sub-component: Evaluate task progress)    |
| View task solution              | Student spends time reflecting on the correct solution to the task after having submitted their solution | 2 = Student spends over time threshold viewing correct solution<br>1 = Student does not spend over time threshold viewing correct solution but scores highly on the task (full or high partial credit)<br>0 = All other cases                                                                                                                               | Response process data (time data) and task score | Evaluate performance (Sub-component: Evaluate solution quality) |
| More complex indicators         |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                             |                                                  |                                                                 |
| Self-evaluation and performance | Student self-evaluates their performance on the task correctly                                           | 2 = Student's self-evaluation is fully aligned with their performance<br>1 = Student's self-evaluation is partially aligned with their performance<br>0 = Student's self-evaluation is not at all aligned with their performance<br><br>Note: For each task, score ranges are identified for full, partial and no alignment for each self-evaluation option | Situated self-evaluation item and task score     | Evaluate performance (Sub-component: Evaluate task progress)    |

In addition, after each task, students were given the opportunity to review the correct solution to the task, comparing the correct solution to their own task solution. One indicator describes students who spend over a certain time threshold (e.g. 10 seconds) viewing the correct solution following the task, as it is assumed that reviewing the correct solution is a good reflective behaviour, especially for students who did not make significant progress in the task (see Table 3.5).

## Reporting

Evidence of student performance in computational problem-solving and self-regulated learning will be aggregated across test units for reporting. The reporting strategy aims to produce multidimensional information about students' capacity to engage in learning in the digital world. Students' competency in computational problem solving will be reported using a performance scale, organised by achievement levels, as is traditionally done in PISA. Initial results related to students' computational problem-solving performance across countries and economies will be reported in the PISA 2025 Results (Volume I) report, due to be released in 2026.

Beyond reporting on students' computational problem-solving performance, this assessment will also report on students' capacity to learn within digital learning environments. Information derived from indicators of their self-regulated learning processes, combined with evidence on students' learning during the test, will be reported in the PISA 2025 Results (Volume II), due to be released in early 2027. Several options will be explored to summarise this complex information into an accessible format for policymakers and the general public, including creating student profiles that describe typical clusters of behaviours.

Estimates of students' learning throughout the test can also be produced, for example by examining the performance of students in computational problem-solving relative to their pre-test scores. Other more complex modelling approaches, such as Dynamic Bayesian Network (Levy, 2019<sup>[67]</sup>) models, could also be explored to derive estimates of student learning.

## References

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# **4**

## **PISA 2025 Questionnaire Framework**

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This chapter presents the framework for the core questionnaires for students and schools used in PISA 2025. These questionnaires provide essential contextual information for interpreting student performance results and measure additional educational constructs that are relevant for policy and research in their own right. The chapter outlines the goals and rationale guiding the selection of questionnaire content for the PISA 2025 cycle and describes how the constructs included relate to one another and to student achievement. It also sets out guidelines for the re-administration of questions from previous PISA cycles and for the development of new constructs introduced in PISA 2025.

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## Introduction

Since its inception in the late 1990s, the Programme for International Student Assessment (PISA) has been making important contributions to education policy discussions within the Organisation for Economic Co-operation and Development (OECD) and partner countries and economies. PISA continues to yield indicators on the levels of performance and equity of educational systems, set benchmarks for international comparison and monitor trends over time. PISA also builds a sustainable database that allows researchers worldwide to study basic as well as policy-oriented questions on education, including those related to society and economy. The OECD and the PISA Governing Board (PGB) continue to look for ways to increase the scientific quality and policy relevance of the PISA questionnaires to meet these needs.

Since the first cycle of PISA in 2000, the student and school questionnaires have performed two interrelated purposes in service of the broader goal of evaluating educational systems:

- First, the questionnaires provide a context for interpreting the PISA performance results both within and between education systems.
- Second, the questionnaires aim to provide reliable and valid measurement of additional educational constructs, which can inform policy and research in their own right.

The OECD document *Definition and Selection of Competencies: Theoretical and Conceptual Foundations* (DeSeCo) provides a framework that can guide the longer-term extension of assessments into new competency domains (OECD, 2005<sup>[1]</sup>). Over the eight cycles of PISA to date, education policy discussions have shifted from a heavy focus on the first objective (i.e. interpreting performance results) to increased focus on the second aim as well (i.e. additional constructs). This development corresponds to a shift in policymakers' views of the core goals for education systems in the 21st Century. Whereas teaching primarily focused on clearly defined subject knowledge and skills, now it fosters broader skills as well, such as creativity, communication, collaboration and learning to learn. These help individuals face the demands of a technology-rich and truly global society (United Nations, 2015<sup>[2]</sup>). Students' well-being and social and emotional skills are increasingly seen as educational outcomes in their own right (OECD, 2019<sup>[3]</sup>). There is now a growing recognition that other factors and competencies aside from subject-specific knowledge play a vital role in students' success in school and beyond. In order to understand and guide policy decisions regarding student development, the PISA 2025 questionnaires strengthened the measurement of the contexts that promote learning in these areas, as well as an array of general constructs of policy relevance.

PISA 2025 continues the overall strategic priority implemented in PISA 2022 with regards to the allocation of questionnaire content between general and domain-specific constructs. For PISA 2022, the PGB recommended re-balancing questionnaire content toward a larger focus on general constructs and a slightly reduced focus on domain-specific constructs. Specifically, the PGB suggested that 40% of the content be devoted to domain-specific constructs and the remaining 60% focused on general constructs (split between 20% devoted to measuring the PISA index of Economic, Social and Cultural Status (ESCS) and 40% focused on other general constructs) (PISA Governing Board, 2017<sup>[4]</sup>). By contrast, the balance of questionnaire content across domain-specific constructs, ESCS, and general constructs in 2018 was 50%, 17% and 33%, respectively. In terms of time, **the 2025 Main Survey (MS) set the upper limit of testing time for the student questionnaire (STQ) at 35 minutes**. Approximately 7 minutes of the STQ are devoted to ESCS, 14 minutes to domain-specific constructs and 14 minutes to general constructs.

Furthermore, PISA 2025 questionnaires rely on survey design principles developed for the 2022 cycle; these principles concern question types, question wording, response options and scaled indices of previous surveys, as well as within-construct matrix sampling. Survey design principles are explained in detail in Chapter 5 of the *PISA 2022 Assessment and Analytical Framework* (OECD, 2023<sup>[5]</sup>).

Within the boundaries of these overall strategic priorities, two key areas of consideration guide the development of the PISA 2025 questionnaire framework: 1) re-administration of questions from previous PISA cycles and 2) new development.

#### **Box 4.1. PISA 2025 Questionnaire Framework: Constructs, item development and field trial**

One of the aims of the PISA 2025 Questionnaire Framework is to guide item development for the student and school questionnaires. To this end, it identifies a set of constructs, organised into modules that were tested in the PISA 2025 Field Trial and implemented in the Main Survey. For each construct, the framework provides a conceptual definition and a supporting review of the relevant literature. Some constructs originally recommended were not retained following the Field Trial; these are listed in Table 4.7.

The PISA Secretariat drafted the PISA 2025 Questionnaire Framework and provided measurement recommendations to the external contractor responsible for item development. For constructs that have been administered in previous PISA cycles and for which no updates were recommended, the Secretariat advised measuring them in a manner consistent with past cycles to allow for trend analyses. By contrast, for constructs not previously included in PISA (i.e. new developments), the Secretariat provided measurement recommendations based on consultations with external experts and a review of the extant research literature.

These measurement recommendations are intended to support item development. However, they are not prescriptive: the contractors in charge of item development remain responsible for finding the best ways to measure the constructs included in the Framework. The PISA 2025 questionnaires were subject to review by the PISA Secretariat, the Questionnaire Expert Group, and the PGB, and ultimately approved by the PGB.

### Box 4.2. PISA contextual questionnaires: Core and optional from 2000 to 2025

In PISA 2025, PISA collected data using five different questionnaires (see Table 4.1). All countries and economies participating in PISA implemented the student questionnaire (STQ) and the school questionnaire (SCQ). This PISA 2025 Questionnaire Framework covers the content of the STQ and SCQ. The remaining three questionnaires are optional (i.e. countries/economies can decide whether to implement them or not) and are not covered in this chapter.

PISA has collected data using the STQ and SCQ in every cycle since PISA 2000. Until PISA 2009, these core questionnaires were implemented using paper (paper-based, or PB). In PISA 2012, some countries and economies implemented the SCQ using computers (computer-based, or CB), and starting in PISA 2015, most countries and economies have implemented the STQ and the SCQ on computers with some countries and economies still using paper.

Questionnaires that are optional for PISA-participating countries and economies have been used since the beginning of PISA, yet they have grown in number and scope. In PISA 2000, only the Information and Communication Technology Familiarity Questionnaire (ICQ) was implemented on a paper-based mode (it has been offered in every PISA cycle and since PISA 2015 it is offered in computer-based mode only). A variety of questionnaires collecting additional information from students, parents and teachers have been added to PISA throughout the years in an effort to collect data on areas that are relevant for the improvement of educational practices and policies.

**Table 4.1. PISA core and optional questionnaires from 2000 to 2025**

|                                        | PISA<br>2000 | PISA<br>2003 | PISA<br>2006 | PISA<br>2009 | PISA<br>2012 | PISA<br>2015 | PISA<br>2018 | PISA<br>2022 | PISA<br>2025 |
|----------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>Core questionnaires</b>             |              |              |              |              |              |              |              |              |              |
| Student questionnaire (STQ)            | PB           | PB           | PB           | PB           | PB           | CB and<br>PB | CB and<br>PB | CB and<br>PB | CB and<br>PB |
| School questionnaire (SCQ)             | PB           | PB           | PB           | PB           | CB and<br>PB | CB and<br>PB | CB and<br>PB | CB and<br>PB | CB and<br>PB |
| <b>Optional questionnaires</b>         |              |              |              |              |              |              |              |              |              |
| ICT Familiarity Questionnaire (ICQ)    | PB           | PB           | PB           | PB           | PB           | CB           | CB           | CB           | CB           |
| Educational Career Questionnaire (ECQ) | -            | PB           | -            | PB           | PB           | CB           | CB           | -            | -            |
| Parent Questionnaire (PAQ)             | -            | -            | PB           | PB           | PB           | PB           | PB           | PB           | PB           |
| Teacher Questionnaire (TCQ)            | -            | -            | -            | -            | -            | CB           | CB           | CB           | CB           |
| Student Well-Being Questionnaire (WBQ) | -            | -            | -            | -            | -            | -            | CB           | CB           | -            |
| Financial Literacy Questionnaire (FLQ) | -            | -            | -            | -            | -            | -            | CB           | CB           | -            |

Notes: PB refers to paper-based mode of administration, CB refers to computer-based mode of administration.

In PISA 2000, there was an additional optional questionnaire: Cross-Curriculum Competency (CCC) questionnaire.

In PISA 2025, there was an additional add-in optional component related to the Foreign language assessment in the STQ, SCQ, PAQ and TCQ.

### **Guidelines for re-administration of questions from previous years**

A key research design feature of PISA has been the cyclical change of focus in the cognitive assessment. Science was the major domain of cognitive assessment in PISA 2006 and 2015, and again in 2025. Reading was the major assessment domain in PISA 2000, 2009 and 2018. Mathematics was the major domain in PISA 2003, 2012 and 2022. The major domain is the primary focus of domain-specific content in the associated PISA questionnaire (e.g. various science-related constructs marked the focus of the 2006 and 2015 questionnaires).

In order to describe educational constructs of interest over time at the country level, it is desirable to maintain a stable set of questionnaire measures that can be used as major reporting variables across PISA cycles. Given the cyclical nature of PISA, measurement stability can be considered at two levels:

- ***First, there is the issue of stability of measures across cycles of three years (i.e. administration of items for constructs that may appear in every cycle, e.g. ESCS).***
- ***Second, stability is desirable in measuring domain-specific constructs across cycles of nine/ten years (e.g. science-specific constructs assessed in the 2015 and/or 2006 cycles).***

A priority of PISA 2025 is to retain a reasonable number of questions that have been administered in previous PISA questionnaires. Table 4.2 summarises guidelines applied to decide whether to retain or delete previously administered PISA items in the PISA 2025 Field Trial (FT).

**Table 4.2. Guidelines for retention or deletion of questions from previous PISA cycles**

|    |                                                                                                                                                               |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <b><i>Retain questions that best explain variations in academic achievement within and across countries</i></b>                                               |
| 2. | <b><i>Retain questions that are of highest policy relevance and/or necessary to establish or extend trend lines, which can inform policy and research</i></b> |
| 3. | <b><i>Where possible and sensible, carry forward constructs intact, or with only minor changes that improve measurement precision</i></b>                     |
| 4. | <b><i>Delete or revise questions that are outdated (e.g. questions that make reference to resources or technologies that are no longer in use)</i></b>        |
| 5. | <b><i>Delete or revise questions that do not meet PISA 2025 psychometric criteria established by the Technical Advisory Group (TAG)</i></b>                   |
| 6. | <b><i>Delete or shorten questions that provide information that is redundant with other questions or items within a matrix question</i></b>                   |

Although this Framework is focused on the STQ and SCQ, and not on optional questionnaires, the goal of maintaining a stable set of questionnaire measures that can be used for trend analysis across PISA cycles applies also to the optional questionnaires.

### **Guidelines for new development**

The goal of continuity must be carefully balanced with innovations that take into account new developments and emerging priorities in educational systems as well as ongoing assessment designs in PISA. These include: introduction of new technologies (e.g. computer-based assessment (CBA)); introduction of new innovative domains of assessment (e.g. collaborative problem-solving in 2015, global competency in 2018, creative thinking in 2022, or learning in the digital world in 2025); continuous updates of the assessment and analytical frameworks for each content area across cycles (e.g. expansion of the PISA 2022 mathematics framework); and the emergence of new policy priorities (e.g. measuring student health and well-being as well as other social and emotional characteristics).

For PISA 2025, new developments come from the new science framework, which refines and extends the conception of scientific literacy described in previous PISA science frameworks; in particular, by drawing on the PISA 2015 framework which was itself developed from those used as the basis for assessment in 2006, 2009 and 2012 (OECD, 1999<sup>[6]</sup>; OECD, 2003<sup>[7]</sup>; OECD, 2006<sup>[8]</sup>; OECD, 2017<sup>[9]</sup>). The PISA 2025 Science Framework recognises that there is an affective element to a student's display of scientific competencies; that is, their sense of "science identity", which is defined as attitudes or disposition towards science that will determine their level of interest, sustain their engagement, and may motivate them to take action (Osborne, Simon and Collins, 2003<sup>[10]</sup>; Moote et al., 2020<sup>[11]</sup>). Science identity also encompasses personal capital indicated by an interest in science and technology; valuing of scientific approaches to enquiry, where appropriate, and a perception and awareness of environmental issues. Thus, a major new development included in this PISA 2025 Questionnaire Framework is a new module on science identity that includes new constructs such as science capital and environmental agency.

Other areas of the questionnaire framework that have been further refined and developed in the PISA 2025 Questionnaire Framework include teacher diversity (see Module 6), growth mindset (see Module 8), students' educational and occupational expectations for the future (see Module 9), and additional instruction outside of regular school hours (see Module 10).

Additionally, Learning in the Digital World (LDW) was the innovative domain of assessment in PISA 2025, and a defined section of the PISA 2025 questionnaires was devoted to constructs that contribute to the understanding of students' performance in this innovative domain. Please note that information regarding the specific LDW-related contextual constructs is provided in the separate LDW Framework for PISA 2025 (see Chapter 3 of this publication).

New development should make use of informed practices in survey methodology (e.g. principles regarding item types, response options, balancing of scales, length of matrix questions) and technological capabilities (e.g. routing, matrix sampling) to the extent that they will enhance measurement. PISA 2025 continued to use the survey design principles used in PISA 2022.

While this framework focuses on the conceptual underpinnings of the PISA questionnaires for students and schools, additional frameworks that are not part of this chapter provide in-depth theoretical foundation for additional questionnaires included in PISA 2025 as part of international options (i.e. frameworks for Foreign Language Assessment; Information and Communication Technology (ICT) Familiarity; and Teacher Well-being).

Table 4.3 summarises guidelines for considering the addition of new items for existing constructs as well as entirely new constructs in PISA 2025.

**Table 4.3. Guidelines for new development (addition of new items for existing constructs as well as entirely new constructs)**

|    |                                                                                                                                                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <i>Develop questions for new constructs that are central to the educational research literature and the PGB's priorities</i>                                                                                                                                               |
| 2. | <i>Develop new questions that are relevant to changes to the PISA science framework</i>                                                                                                                                                                                    |
| 3. | <i>Develop new questions for constructs that are related to the innovative domain assessed in PISA 2025 (i.e. learning in the digital world)</i>                                                                                                                           |
| 4. | <i>Develop new questions to replace previously used questions that do not comply with PISA 2025 psychometric criteria; substantially violate PISA 2025 survey design principles; and/or require updates to accurately describe students' living and learning realities</i> |
| 5. | <i>Develop new questions to replace previously used questions that do not offer sufficient flexibility to meet country- or region-specific needs of all participating education systems</i>                                                                                |

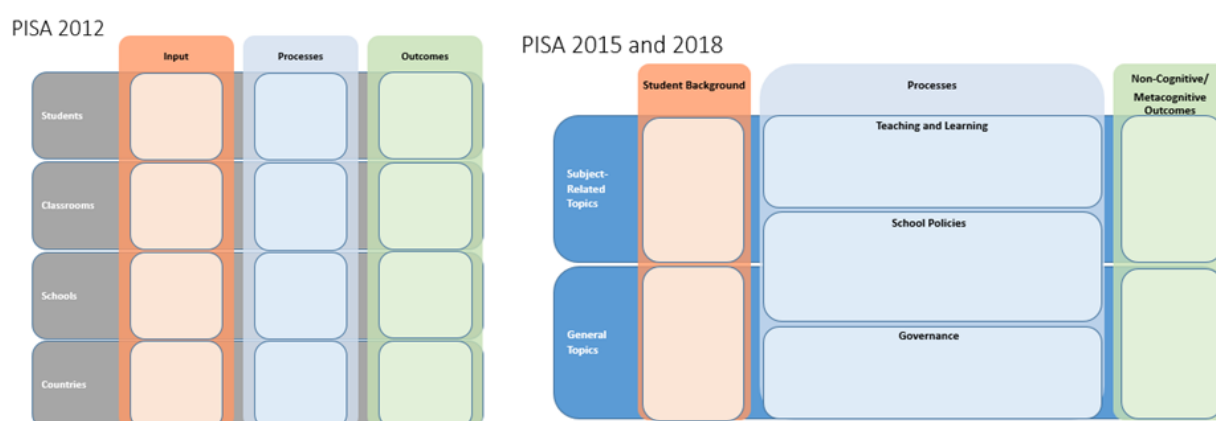
## PISA 2025 questionnaire framework taxonomy

PISA 2025 continued to use the two-dimensional questionnaire framework taxonomy introduced in PISA 2022, which refined the taxonomies used in previous PISA cycles.

Beginning with the questionnaire framework used for the PISA 2009 assessment, questionnaire content was explicitly linked to different levels of the education system: the student level, level of instruction in the classroom, school level, and system level (Jude, 2016<sup>[12]</sup>). The questionnaire framework used for PISA 2012, and subsequently refined for PISA 2015 and 2018, further underscored the importance of collecting information on learning contexts for comparative system monitoring. These frameworks outlined an overarching two-dimensional structure of high-level questionnaire content areas to be measured and kept comparable across assessment cycles (OECD, 2013<sup>[13]</sup>).

The theoretical foundation of the 2012 overarching framework is based on Purves' (1987<sup>[14]</sup>) Context-Input-Process-Outcome (CIPO) model. In the CIPO model, contextual variables for understanding education systems are conceptualised as a series of inputs (i.e. student background), processes (i.e. teaching and learning, school policies, governance), and outcomes (i.e. performance and non-cognitive outcomes) shaped at the student, classroom, school, and country levels. Starting with PISA 2015 and 2018, an additional dimension further classified questions more explicitly into domain-specific and domain-general modules. Domain-specific modules represent the set of constructs with strong expected relationships to student experiences, outcomes, and teaching and learning factors tied to a specific content area (e.g. reading, mathematics or science). Domain-general modules represent the set of constructs that are important for understanding differences in achievement that are not tied to a specific subject-area. Figure 4.1 illustrates the high-level structures of the questionnaire frameworks from PISA 2012, 2015 and 2018.

**Figure 4.1. Framework structures of PISA 2012, 2015, and 2018**

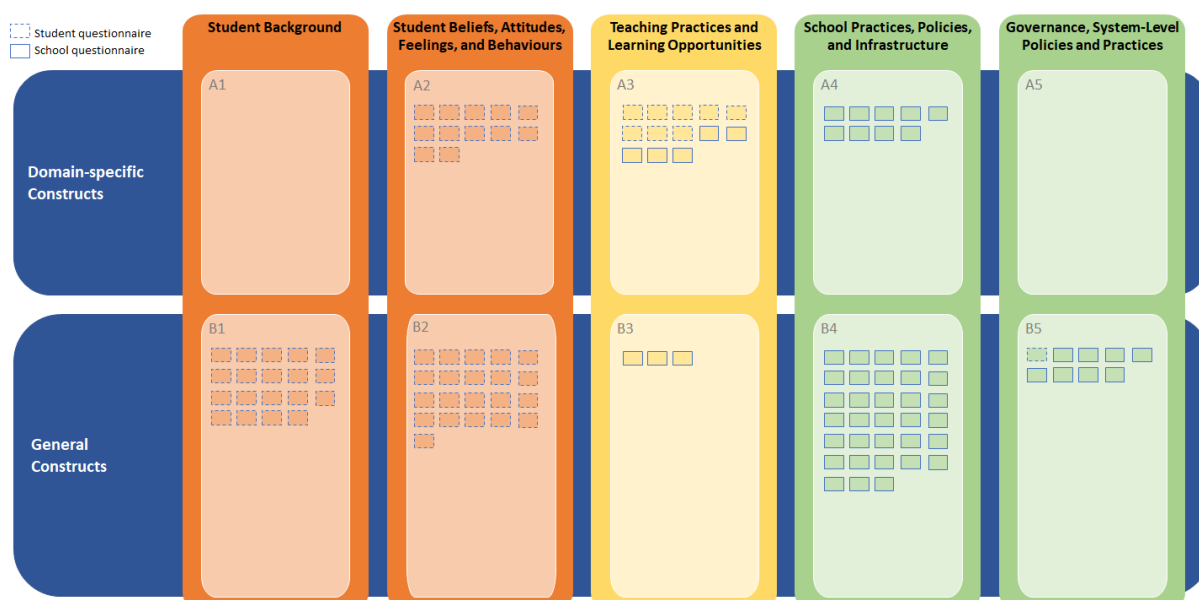


In keeping with the long-term goal of balancing continuity with innovation, the PISA 2022 questionnaires framework retained key framework elements from previous cycles as a foundation, and introduced refinements that facilitate the strategic development of new constructs and move toward improved measurement. This updated framework structure is illustrated in Figure 4.2 below. Please note, while performance and contextual variables have been classified as “outcomes” in previous PISA frameworks per the CIPO model, both variables also constitute possible inputs (OECD, 2013<sup>[13]</sup>). For instance, a student’s prior achievement and his/her curiosity, perseverance, achievement motivation or confidence will likely impact the student’s future achievement as well as his/her future development of social and emotional characteristics. Due to the cross-sectional nature of PISA, variables collected through the questionnaires cannot be clearly assigned a single “role”. While the CIPO model remains useful to describe an actionable

policy perspective and serve as a helpful theoretical perspective for researchers on the variables measured with the PISA questionnaires, it seems less useful as a guide to classify and prioritise variables for instrument development. Due to the ambiguity in classifying variables, constructs are not classified as inputs, processes or outcomes in the PISA 2022 framework taxonomy. Instead, we allude to the possible roles each variable might play in the detailed descriptions of each module. Further description of the framework dimensions and the modules is provided in subsequent sections of this framework.

Across the two overarching (vertical) framework dimensions and the five (horizontal) policy focus areas shown in Figure 4.2, a total of 19 modules are specified (see Table 4.4), which each consist of groupings of two or more related constructs that mark a key topic or theme relevant for education research, practice and policymaking. In addition to these 19 modules, a separate module focused on contextual questions specific to LDW was included in PISA 2025. Constructs in this module are described in the LDW framework. The small boxes in the taxonomy below indicate the relative distribution of constructs in the PISA 2025 MS across all 19 modules described in this framework.

**Figure 4.2. PISA 2025 Two-dimensional questionnaire framework taxonomy**



Note: Domain-specific constructs related to Learning in the Digital World are not reflected in this chart.

The 19 modules included in this PISA 2025 Questionnaire Framework are similar but not identical to those included in PISA 2022 Questionnaire Framework. The most important difference is that PISA 2022 included three mathematics-specific modules; in PISA 2025, these modules are specifically referred to in science education, reflecting the innovations in the PISA Science Framework. In addition, PISA 2022 included a module on “Health and well-being”; in PISA 2025, that module was deleted but the constructs within the module were relocated to other modules (measures of well-being are now included in Module 8, and health-related measures in Module 10). Finally, the specific constructs within modules have changed in some cases, as described in the detailed overview provided in this chapter.

When reading about these modules, bear in mind that PISA results do not establish causality. Rather, PISA identifies empirical correlations between student achievement and the characteristics of students, schools and education systems that are identified as relevant constructs in this framework.

Table 4.4. Modules included in the PISA 2025 questionnaire framework

|    |                                                              |
|----|--------------------------------------------------------------|
| 1  | Basic demographics                                           |
| 2  | Economic, social and cultural status (ESCS)                  |
| 3  | Educational career                                           |
| 4  | Migration and language exposure                              |
| 5  | PISA preparation and effort                                  |
| 6  | School culture and climate                                   |
| 7  | Science identity                                             |
| 8  | Curiosity, perseverance, mindset and well-being              |
| 9  | Post-secondary preparedness and aspirations                  |
| 10 | Experiences outside of regular school hours                  |
| 11 | School type and resources                                    |
| 12 | Selection and enrolment                                      |
| 13 | School autonomy                                              |
| 14 | Organisation of student learning at school                   |
| 15 | Exposure to science content                                  |
| 16 | Science teacher behaviours                                   |
| 17 | Teacher qualification, training and professional development |
| 18 | Assessment, evaluation and accountability                    |
| 19 | Parental involvement and support                             |

Table 4.5. Constructs in PISA 2025 included in previous PISA cycles (trend constructs)

| A. PISA 2025 Main Study Student Questionnaire |                                                 |                                                |
|-----------------------------------------------|-------------------------------------------------|------------------------------------------------|
| PISA 2025 questionnaire                       |                                                 | Administered in previous PISA cycles?          |
| Module number                                 | Student constructs                              |                                                |
| 1                                             | Grade                                           | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                                               | Date of birth                                   | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                                               | Gender                                          | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                                               | Home possessions                                | 2022                                           |
| 2                                             | Digital devices in the home                     | 2022                                           |
|                                               | Books in the home                               | 2022                                           |
|                                               | Food insecurity <sup>1</sup>                    | 2022                                           |
| 3                                             | Current study programme                         | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                                               | Age started ISCED 0                             | 2015, 2018, 2022                               |
|                                               | Grade repetition                                | 2003, 2009, 2012, 2015, 2018, 2022             |
|                                               | Time attended current school                    | 2022                                           |
| 4                                             | Country of birth of student and parents         | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                                               | Age at arrival in country of test               | 2015, 2018, 2022                               |
|                                               | Primary home language                           | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
| 5                                             | Effort in PISA test and questionnaires          | 2022                                           |
| 6                                             | Truancy                                         | 2012, 2015, 2018, 2022                         |
|                                               | Sense of belonging                              | 2012, 2015, 2018, 2022                         |
|                                               | Being bullied                                   | 2015, 2018, 2022                               |
|                                               | Personal safety                                 | 2022                                           |
| 7                                             | Science self-efficacy                           | 2006, 2015                                     |
|                                               | Enjoyment of science                            | 2006, 2015                                     |
|                                               | Epistemic beliefs                               | 2015                                           |
|                                               | Environmental awareness                         | 2006, 2015                                     |
| 8                                             | Curiosity                                       | 2022                                           |
|                                               | Perseverance (persistence)                      | 2022                                           |
|                                               | Students' growth mindset (general intelligence) | 2018, 2022                                     |

|    |                                                                 |                              |
|----|-----------------------------------------------------------------|------------------------------|
|    | Overall life satisfaction                                       | 2015, 2018, 2022             |
| 9  | Outlook on future education and career                          | 2022                         |
|    | Expected educational level                                      | 2022                         |
|    | Expected occupation                                             | 2000, 2006, 2015, 2018, 2022 |
| 10 | Digital device usage behaviours outside of regular school hours | 2022                         |
|    | Time spent on digital devices outside of regular school hours   | 2022                         |
|    | Activities before school                                        | 2022                         |
|    | Activities after school                                         | 2022                         |
|    | Time spent on homework and private study                        | 2022                         |
|    | Type of additional science instruction                          | 2022                         |
| 14 | Class periods per week in selected subjects and all subjects    | 2018                         |
|    | Digital device usage behaviours during regular school hours     | 2022                         |
|    | Time spent on digital devices during regular school hours       | 2022                         |
| 16 | Disciplinary climate in science                                 | 2015                         |
|    | Science Teacher Support                                         | 2015                         |
|    | Science Teacher Feedback                                        | 2015                         |

### B. PISA 2025 Main Study School Questionnaire

| PISA 2025 questionnaire |                                                                                                                                           | Administered in previous PISA cycles?          |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Module number           | School constructs                                                                                                                         |                                                |
| 4                       | Number of foreign instructional languages taught at school                                                                                | 2022                                           |
|                         | Instructional options for students whose heritage language is not the school's primary language of instruction                            | 2018                                           |
| 5                       | Principal's support of PISA administration / encouragement of teachers and students to take PISA test seriously and/or prep for PISA test | 2022                                           |
| 6                       | School diversity/multi-cultural views                                                                                                     | 2022                                           |
|                         | Disorder and delinquent behaviour at school                                                                                               | 2022                                           |
|                         | Factors hindering instruction                                                                                                             | 2012, 2015, 2018, 2022                         |
|                         | School demographic profile                                                                                                                | 2022                                           |
| 8                       | Social and emotional learning policies                                                                                                    | 2022                                           |
| 9                       | Career guidance is offered by the school                                                                                                  | 2022                                           |
|                         | Career guidance is voluntary or compulsory                                                                                                | 2022                                           |
|                         | Career guidance responsibility at school                                                                                                  | 2018, 2022                                     |
|                         | Information for students about future educational/work/career paths                                                                       | 2022                                           |
| 10                      | Extracurricular activities offered                                                                                                        | 2009, 2012, 2015, 2018, 2022                   |
|                         | School offering additional science instruction                                                                                            | 2012, 2022                                     |
|                         | School offering help with homework and study                                                                                              | 2015, 2022                                     |
| 11                      | Number of science teachers in school                                                                                                      | 2015                                           |
|                         | School location                                                                                                                           | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                         | School type                                                                                                                               | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                         | School funding sources                                                                                                                    | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                         | Number of students                                                                                                                        | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                         | Number of non-teaching staff                                                                                                              | 2022                                           |
|                         | Availability of digital devices                                                                                                           | 2015, 2018, 2022                               |
|                         | Shortage of teaching and non-teaching staff                                                                                               | 2015, 2018, 2022                               |
|                         | Shortage of educational material and physical infrastructure                                                                              | 2015, 2018, 2022                               |
|                         | Shortage of digital resources                                                                                                             | 2022                                           |
|                         | Number and work schedule of teachers in school                                                                                            | 2003, 2006, 2009, 2012, 2015, 2018, 2022       |
|                         | Science-related resources in school                                                                                                       | 2015                                           |
|                         | Preparedness for digital learning                                                                                                         | 2018, 2022                                     |
|                         | School competition for students                                                                                                           | 2006, 2009, 2012, 2018, 2022                   |
| 12                      | Academic school selectivity                                                                                                               | 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
|                         | Transfer policy                                                                                                                           | 2000, 2012, 2022                               |
| 13                      | Decentralisation of responsibility for decision making                                                                                    | 2022                                           |
|                         | School management team behaviours                                                                                                         | 2022                                           |
| 14                      | Ability grouping for all subjects                                                                                                         | 2006, 2015, 2018, 2022                         |
|                         | Average time in class period                                                                                                              | 2022                                           |

|    |                                                          |                                          |
|----|----------------------------------------------------------|------------------------------------------|
|    | Average number of students in test language classes      | 2006, 2012, 2015, 2018, 2022             |
|    | Digital device/ICT policies                              | 2022                                     |
|    | Global issues curriculum                                 | 2018                                     |
| 17 | Teacher professional development                         | 2015, 2018, 2022                         |
|    | Teacher qualifications                                   | 2003, 2006, 2009, 2012, 2015, 2018, 2022 |
| 18 | School evaluation                                        | 2012, 2015, 2018, 2022                   |
|    | Monitoring teacher practices                             | 2003, 2009, 2012, 2015, 2022             |
|    | Feedback to teachers                                     | 2022                                     |
|    | Use of assessments                                       | 2015, 2022                               |
|    | Use of achievement data in accountability systems        | 2006, 2009, 2012, 2015, 2018             |
| 19 | Parent participation in school activities                | 2012, 2015, 2018, 2022                   |
|    | Schools' encouragement of parent or guardian involvement | 2022                                     |

Note: <sup>1</sup> The item measuring food insecurity was excluded from the PISA 2025 international database due to a technical issue.

**Table 4.6. Science-specific constructs in PISA 2025 Main Study Student and School Questionnaires**

| Module number | Science-specific Constructs in PISA 2025                                                                                                                                                                                              |                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Student questionnaire                                                                                                                                                                                                                 | School questionnaire                                                                                                                                                        |
| 7             | Science self-efficacy<br>Enjoyment of science<br>Epistemic beliefs<br>Environmental awareness<br>Environmental agency<br>Science capital<br>Collective efficacy regarding environmental issues<br>Hope regarding environmental issues | School activities for environmental agency                                                                                                                                  |
| 8             | Students' growth mindset (general intelligence and in science)                                                                                                                                                                        | -                                                                                                                                                                           |
| 10            | Time spent on homework and private study<br>Type of additional science instruction<br>Time spent in additional instruction                                                                                                            | School offering additional science instruction<br>Purpose of additional science instruction offered at school<br>School offering help with homework and study               |
| 11            |                                                                                                                                                                                                                                       | Number of science teachers in school<br>Science-related resources in school                                                                                                 |
| 14            | Class periods per week in core subjects and all subjects                                                                                                                                                                              | Ability grouping in science classes<br>Average time in class period<br>Average number of students in science classes<br>Average number of students in test language classes |
| 15            | Engagement with science practices                                                                                                                                                                                                     | Science courses offered to students<br>Freedom in science curriculum choice                                                                                                 |
| 16            | Disciplinary climate in science<br>Science teacher support<br>Science teacher feedback<br>Explicit instruction<br>Scientific thinking for everyday life                                                                               |                                                                                                                                                                             |
| 17            |                                                                                                                                                                                                                                       | Science teacher qualifications<br>Areas of professional development for science teachers                                                                                    |

**Table 4.7. Constructs recommended for inclusion in the PISA 2025 Field Trial but not implemented in the main survey**

| Module | Construct                                     | Questionnaire |
|--------|-----------------------------------------------|---------------|
| 1      | Students' home                                | STQ           |
|        | Number of siblings                            |               |
| 2      | Mother's and fathers' education <sup>1</sup>  | STQ           |
|        | Mother's and fathers' occupation <sup>1</sup> |               |
|        | Subjective socio-economic status              |               |
| 3      | Age started ISCED 1                           | STQ           |
| 6      | School safety risks                           | STQ           |
|        | Quality of student-teacher relationships      | SCQ           |
|        | Gender composition of science teachers        |               |
| 7      | Science self-concept                          | STQ           |
|        | Instrumental motivation                       |               |
|        | Environmental concern                         |               |
| 8      | Perspective taking                            | STQ           |
|        | Assertiveness                                 |               |
|        | Self-control                                  |               |
|        | Empathy                                       |               |
|        | Emotional control                             |               |

| Module | Construct                                                                                | Questionnaire |
|--------|------------------------------------------------------------------------------------------|---------------|
|        | Trust                                                                                    |               |
|        | Co-operation                                                                             |               |
|        | Stress resistance                                                                        |               |
|        | Parents' failure mindsets                                                                |               |
|        | Parents' feedback to students                                                            |               |
|        | Growth mindset meaning system                                                            |               |
|        | Teachers' growth mindsets                                                                |               |
|        | Peer norms about academic effort and performance                                         |               |
| 9      | Educational requirement of jobs (perceived by student)                                   | STQ           |
|        | Information-seeking about future study or work                                           |               |
|        | Skills to find information on jobs and student financing                                 |               |
|        | Financial cost of education (perceived by student)                                       |               |
|        | Availability of student aid (perceived by student)                                       |               |
|        | Economic returns of education (perceived by student)                                     |               |
|        | Self-perceived academic capacity to complete higher education levels                     |               |
| 10     | Years attending additional instruction                                                   | STQ           |
|        | Purpose of participation in additional science instruction                               |               |
|        | Providers of additional instruction                                                      |               |
|        | Financial cost of additional instruction for families                                    |               |
|        | Location of additional instruction                                                       |               |
|        | Use of homework in science classes                                                       |               |
|        | Timing of additional instruction                                                         |               |
|        | Science activities                                                                       |               |
| 14     | Required or optional course-taking                                                       | SCQ           |
|        | Responsibility for selection of science courses                                          |               |
|        | Percentage of students who received marks below, at/above the pass mark in science class |               |
| 15     | Science courses taken                                                                    | STQ           |
|        | Time spent in each science course taken                                                  |               |
|        | Freedom in science curriculum choice                                                     |               |
|        | Exposure to major scientific topics                                                      |               |
| 16     | Perceived quality of science instruction                                                 | STQ           |
|        | Structure of science instruction                                                         |               |
|        | Adaptation of instruction and differentiation                                            |               |
|        | Procedural domain: Practical investigation                                               |               |
|        | Scientific epistemic reasoning                                                           |               |
|        | Scientific conceptual reasoning                                                          |               |
| 17     | In-house professional development                                                        | SCQ           |
|        | In-house professional development (for science teachers)                                 |               |
| 20     | Type/duration of school closure                                                          | STQ           |
|        | School actions/activities to sustain learning                                            |               |
|        | Type of digital device used for schoolwork while school was closed                       |               |
|        | Subjective impression of learning during school closure                                  |               |
|        | Types of learning resources used while school was closed                                 |               |
|        | Problems with self-learning                                                              |               |
|        | Family support for self-learning                                                         |               |
|        | Feelings about learning during school closure                                            |               |
|        | Self-learning self-efficacy                                                              |               |
|        | Feeling of preparedness for future school closures                                       |               |
|        | School closure                                                                           | SCQ           |
|        | School closure instruction                                                               |               |
|        | School closure resources                                                                 |               |

| Module | Construct                                                                                    | Questionnaire |
|--------|----------------------------------------------------------------------------------------------|---------------|
|        | Problems with schools' capacity to provide remote instruction                                |               |
|        | School closure communication; School closure encouragement of parent or guardian involvement |               |
|        | School closure attendance                                                                    |               |
|        | School closure preparation                                                                   |               |
|        | School closure support                                                                       |               |
|        | Feeling of preparedness for future school closures                                           |               |

Notes. <sup>1</sup> For PISA 2025, the definition of family and household composition was broadened to include not only parents but also guardians responsible for taking care of the student on a regular basis. As a result, constructs that previously referred specifically to mothers' and fathers' education and occupation now refer to the education and occupation of parents or guardians.

<sup>2</sup> Module 20 in the PISA 2025 field trial was an updated version of the Global Crises module, originally developed for PISA 2022 in response to COVID-19. None of the items from this module were included in the PISA 2025 Main Study.

## Detailed overview of PISA 2025 modules

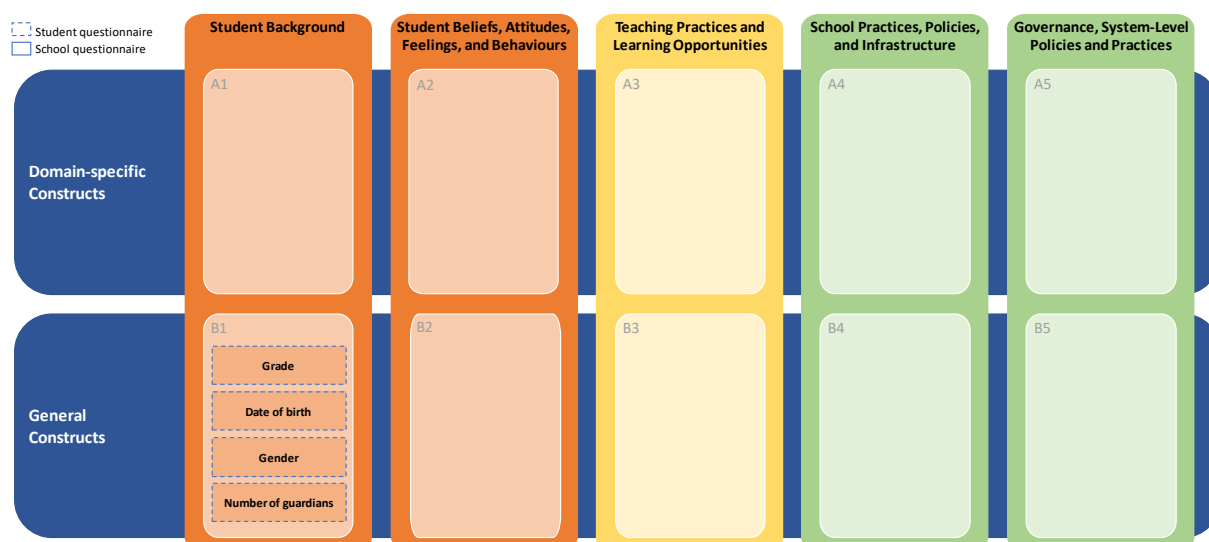
### Module 1: Basic demographics

The PISA 2025 STQ continued to include questions on students' gender, date of birth, and grade. In PISA 2025, the definition of family and household composition was broadened to include not only parents but also guardians such as step-parents, foster parents, grandparents, adult siblings and other caregivers.

Regarding gender, countries had the option to include a third gender category beyond the binary framework. This innovation was introduced in PISA 2025.

Figure 4.3 illustrates how all proposed constructs in this module map on the taxonomy.

Figure 4.3. Module 1: Basic demographics

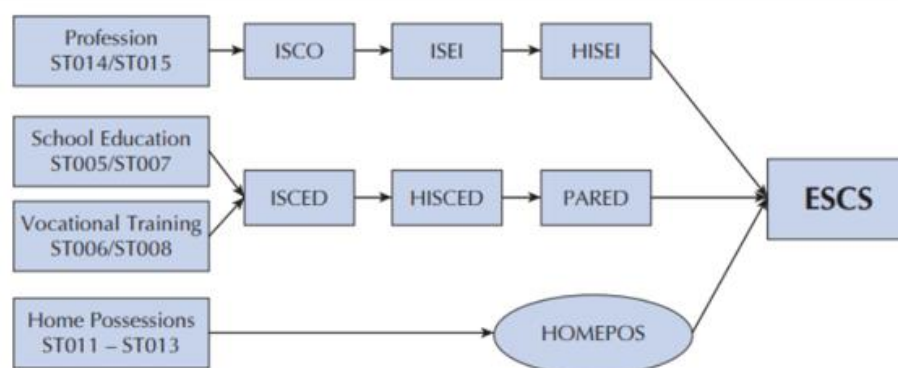


### Module 2: Economic, social and cultural status (ESCS)

Over the past seven PISA cycles, significant efforts went into the definition and operationalisation of individual student background indicators, leading to the establishment of an integrated indicator for students' economic, social and cultural status (ESCS index) (Willms, 2006<sup>[15]</sup>; Lee, Zhang and Stankov, 2019<sup>[16]</sup>). Figure 4.4 displays how ESCS was created in the three most recent PISA cycles. The PISA

ESCS index was derived from three variables related to family background: parents' highest level of education (PARED), parents' highest occupational status (HISEI), and home possessions (HOMEPOS), including books in the home.

**Figure 4.4. Computation of ESCS index in PISA 2015, 2018 and 2022**



Source: PISA 2015 Technical Report.

Notes: ISCO: International Standard Classification of Occupations; ISEI: occupational status of mother and father; HISEI: highest parental occupational status; ISCED: International Standard Classification of Education; HISCED: Highest education of parents (ISCED); PARED: Index for highest parental education in years of schooling; HOMEPOS: Index of home possessions (WLE); ESCS: Index of economic, social and cultural status.

The PISA ESCS index is considered internationally as a gold standard measure of socio-economic status (SES) in large-scale assessments (LSAs) (e.g. (Cowan et al., 2012<sup>[17]</sup>)). To examine trends over time and comparisons with previous PISA data on the ESCS index, it is crucial to establish minimal stability in assessing the three components (Avvisati, 2020<sup>[18]</sup>). While well established, the ESCS index has also been criticised in recent years (e.g. (Rutkowski and Rutkowski, 2013<sup>[19]</sup>; Willms and Tramonte, 2015<sup>[20]</sup>; Pokropek, Borgonovi and McCormick, 2017<sup>[21]</sup>)), calling for revisions and extensions of the index. In response, the PISA Secretariat conducted a comprehensive review of the ESCS index to assess how effectively it captures students' socio-economic backgrounds (see Box 4.3).

### Box 4.3. The measurement of socio-economic status in PISA

The PISA Secretariat conducted a comprehensive review of the ESCS index to evaluate how effectively it captures students' socio-economic backgrounds (Avvisati and Wuyts, 2024<sup>[22]</sup>). This review highlighted the need to balance multiple goals – such as accuracy, relevance and international comparability – while adapting to societal changes. Over time, the ESCS index has evolved through modifications in the questionnaires, scaling procedures and the inclusion of new variables. These changes were necessary to maintain the index's utility but also introduced complexities in ensuring consistency across survey cycles. Key conclusions and recommendations regarding each component of the index are the following:

- Occupational status, one of the three components of the ESCS index, suffers from high and increasing rates of missing data across countries and assessment cycles. Rather than changing the measurement instrument, improvements should focus on enhancing coding quality, through better tools, coder training, and documentation. With this purpose, PISA's Research, Development and Innovation (RDI) programme developed Risco, a browser-based interface for automatically coding open-text responses on parental occupation. During the 2025 Main Survey, about half of the national centres responsible for PISA data collection relied on Risco for occupational coding. Moreover, strategies to reduce non-response include repositioning occupation questions earlier in the questionnaire, using soft reminders, and avoiding multiple-choice formats. The latter are poor substitutes for open-ended responses, as they significantly reduce the response accuracy.
- Evidence on the quality of PISA's parental education measure (PAREDINT) is mixed. While student-parent agreement is low when using the detailed ISCED 2011 classification, it improves when responses are grouped into three broad categories: less than upper secondary, upper secondary, and tertiary education. At the country level, PISA data correlates well with external sources like labour-force surveys, both showing rising shares of tertiary-educated parents – especially mothers – though the magnitude of change varies across countries. Despite concerns about the validity of student-reported data, no clear alternative has emerged, and the current approach is recommended to continue. An experiment in PISA 2022 showed that adding a “don't know” option could disrupt trend measurement and is therefore not advised.
- In PISA 2022, the measurement of household possessions (HOMEPOS) underwent notable changes, including revised item wording, shorter batteries and new response formats. These adjustments improved clarity but complicated trend comparability. Despite being based on observable items, student-parent agreement was low, especially for books and digital devices, raising concerns about reliability. The average index values declined overall, mainly due to fewer reported books. Some items were excluded due to poor scaling fit, a challenge that has also been observed in previous PISA cycles. Similarly, the influence of household composition on responses – such as whether a student has a room of their own – has been a recurring issue, affecting some items more than others. To improve future assessments, a more theory-driven approach is needed, focusing on basic needs and incorporating household composition data.

As part of the second strand of the RDI project on measuring ESCS, the PISA 2025 Field Trial tested 21 new household possession items, mostly inspired by “country-specific” items in use in earlier cycles. Instead of adding items to the international list, countries were invited to select from this longer list; by co-ordinating their selections of household possession items from a fixed menu, comparability could improve. The list of items subject to selection, and the number of items that each country could include from this list, was somewhat reduced in the Main Survey based on Field Trial analyses and to avoid redundancy.

The Field Trial also tested new indicators of deprivation such as access to clothing, adequate meals, and leisure. While the new items generally supported a robust family resources scale, some food-related deprivation items were found to be problematic for cross-country comparisons, and only the trend (food insecurity) measure was retained for the Main Survey<sup>1</sup>.

Note: <sup>1</sup> The item measuring food insecurity was excluded from the PISA 2025 international database due to a technical issue.

Figure 4.5 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.5. Module 2: Economic, social and cultural status (ESCS)**



Note: On constructs related to parental/guardian education and occupation: In the PISA 2025 student questionnaire, students first report how many parents or guardians they have. Students who indicate one parent or guardian provide information on the education and occupation of that parent or guardian, while students who indicate two or more provide this information for two parents or guardians.

### Module 3: Educational career

PISA gathers retrospective and prospective information about students' early years, education pathways and careers across their lifespan. Research suggests that a variety of outcomes can be improved by high-quality pre-primary education, including children's cognitive development and later school achievement (Heckman, 2006<sup>[23]</sup>; OECD, 2018<sup>[24]</sup>); (Duncan et al., 2007<sup>[25]</sup>; Berlinski, Galiani and Gertler, 2009<sup>[26]</sup>). Additionally, attending pre-primary education can improve students' behaviour, attention, effort, class participation and discipline in primary school (Berlinski, Galiani and Gertler, 2009<sup>[26]</sup>). According to Bakken, Brown and Downing (2017<sup>[27]</sup>), early childhood education has long-term benefits on performance and socio-emotional skills, especially for economically disadvantaged children. In line with this, a study by Suziedelyte and Zhu (2015<sup>[28]</sup>) suggests that the benefits of attending pre-primary education tend to be greater for socio-economically disadvantaged children. Moreover, free pre-primary education is associated with increased primary school completion (Earle, Milovantseva and Heymann, 2018<sup>[29]</sup>). PISA 2018 also shows that students who had not attended or had attended pre-primary education for less than one year scored lower in reading at the age of 15 than their peers who attended between one and three years, before and after accounting for students' and schools' socio-economic profile (OECD, 2020<sup>[30]</sup>).

In secondary education, the most common form of horizontal stratification between schools, typically known as tracking, consists of sorting students into different instructional programmes (OECD, 2020<sup>[30]</sup>).

In education systems that employ tracking, some students choose or are selected into more academically demanding programmes while other students choose or are selected for vocational or technical programmes. In PISA 2018, students in general (academic) programmes scored almost 30 points higher in reading than those in vocational programmes (OECD, 2020<sup>[30]</sup>). In addition to performance gaps between students in academic and vocational programmes (Chmielewski, 2014<sup>[31]</sup>), research suggests that early tracking increases educational inequalities in terms of dispersion of student test scores and of opportunity by social background and race/ethnicity (Hanushek and Wößmann, 2006<sup>[32]</sup>; Van de Werfhorst and Mijs, 2010<sup>[33]</sup>). Tracking is also associated with student attitudes; for example, students in academic tracks tend to exert more effort in school (Carbonaro, 2005<sup>[34]</sup>) and, by contrast, students in vocational programmes tend to have lower academic self-concept (Marsh et al., 2018<sup>[35]</sup>).

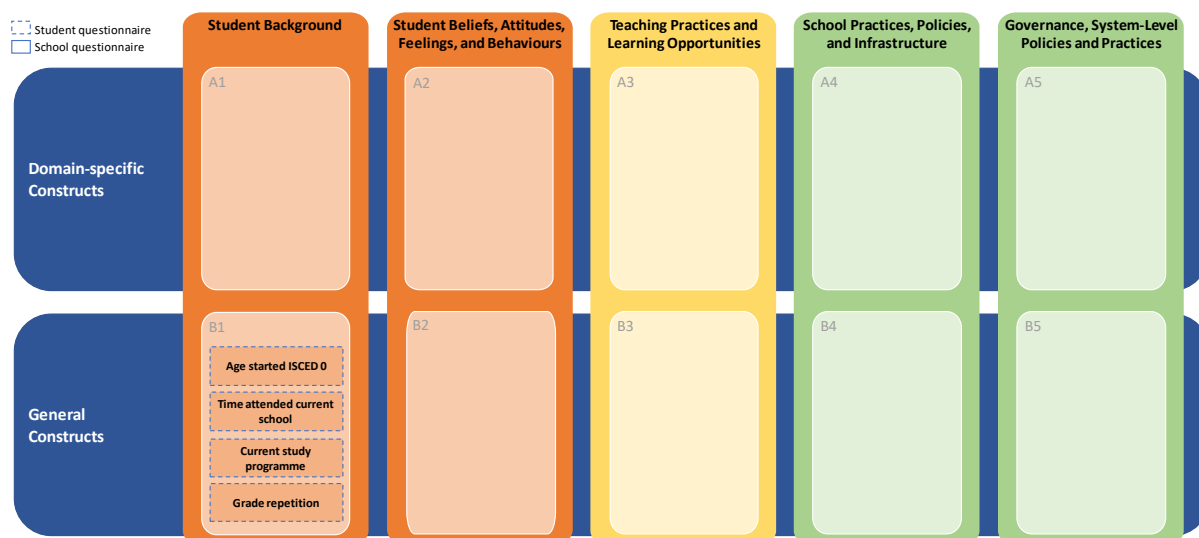
In some education systems, students repeat a grade when they do not perform well academically or for other reasons. The intended purpose of grade repetition is to give students a “second chance” to master the knowledge and skills appropriate for their grade level (OECD, 2020<sup>[30]</sup>). Nevertheless, the evidence on the benefits of grade repetition is mixed (Allen et al., 2009<sup>[36]</sup>; Martorell and Mariano, 2018<sup>[37]</sup>). Some longitudinal studies have found that grade retention harms individual careers and outcomes (Griffith et al., 2010<sup>[38]</sup>; Ou and Reynolds, 2010<sup>[39]</sup>), as well as student behaviour and well-being (Crothers et al., 2010<sup>[40]</sup>), while other research finds positive effects (Marsh et al., 2017<sup>[41]</sup>; Babcock and Bedard, 2011<sup>[42]</sup>; Greene and Winters, 2009<sup>[43]</sup>); for example, as an alternative to moving into a lower track in a highly tracked system (Kloosterman and de Graaf, 2010<sup>[44]</sup>). Some research finds that after repeating a grade, students’ academic performance does not typically improve at later grades (McCoy and Reynolds, 1999<sup>[45]</sup>) and that these students tend to retain more negative attitudes towards school than students who did not repeat a grade (Jimerson, Anderson and Whipple, 2002<sup>[46]</sup>; Ikeda and Gracia, 2014<sup>[47]</sup>). Moreover, students who repeated a grade are more likely to drop out of school (Stearns et al., 2007<sup>[48]</sup>; Jimerson, Anderson and Whipple, 2002<sup>[46]</sup>; Hughes et al., 2018<sup>[49]</sup>). Socio-economically disadvantaged students are more likely to repeat a grade than advantaged students even if both students score the same on the PISA reading test (OECD, 2020<sup>[30]</sup>). According to a study by Hughes et al. (2018<sup>[49]</sup>) the negative effects of grade repetition on school completion are the largest for minority students.

The module also collects data on the amount of time students have attended their current school. This information is important because PISA samples 15-year-old students, therefore it assesses cumulative learning from early childhood through primary and secondary schooling. It is unlikely that the school characteristics of students’ current school are the real explanatory factor of student achievement among students who have been in their current school for less than a year; inversely, the more time students have spent at their current school, the more sound it is to use school characteristics to account for student performance.

Constructs included in the PISA 2025 STQ under this module are considered primarily as general constructs: age started ISCED 0 and 1; time attended current school; current study programme; and grade repetition.

Figure 4.6 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.6. Module 3: Educational career



#### Module 4: Migration and language exposure

Linguistic and cultural diversity are basic facts of life in many regions of the world. Many nations are home to several sub-populations with different languages and cultures. International migration disseminates this diversity. In PISA, immigrant students are defined as students whose parents or guardians were born in a country/economy other than that where the student sat the PISA test. Amongst immigrant students, a distinction is made between first- and second-generation students, based on whether the student was born in or outside the country/economy of assessment. On average across OECD countries, first- and second-generation immigrant students currently comprise about 13% of the 15-year-old student population (OECD, 2019<sup>[50]</sup>). Immigration is currently a critical topic in many countries, particularly those with traditionally larger immigrant populations (e.g. the United States, Canada) as well as countries facing new challenges due to new populations of refugees (e.g. most Central European countries) (Bansak, Hainmueller and Hangartner, 2016<sup>[51]</sup>; Wike, Stokes and Simmons, 2016<sup>[52]</sup>).

Research shows that education outcomes of immigrant students and students from linguistic minority groups are shaped both by the resources, skills and dispositions of individual students; their families and immigrant communities; and by the social and educational policies and attitudes towards immigrants in the country of destination (Buchmann and Parrado, 2006<sup>[53]</sup>). Growing up bilingually may foster the development of language awareness and facilitate the learning of additional foreign languages (Hesse, Göbel and Hartig, 2008<sup>[54]</sup>). Students with an immigrant minority background often have stronger aspirations and higher motivation than their peers from ethnic majority families (Kao and Tienda, 1998<sup>[55]</sup>; Krahn and Taylor, 2005<sup>[56]</sup>; Stanat and Christensen, 2006<sup>[57]</sup>). In addition, PISA 2018 found that immigrant students reported a greater capacity than native-born students to understand the perspectives of others in ten countries (Australia, Austria, Canada, Hong Kong (China), Ireland, Macao (China), New Zealand, Scotland (United Kingdom) and the United Arab Emirates) whereas the reverse was observed only in Montenegro (OECD, 2020<sup>[58]</sup>). One possible explanation is that immigrants have to deal with at least two cultural perspectives in their daily lives, that of their country of immigration and that of their country of origin.

However, their possibly poorer competency in the language of instruction during the first years of schooling can also negatively affect learning in other subjects, and in the long run be disadvantageous for students' educational pathways (Schmid, 2001<sup>[59]</sup>; Stanat and Christensen, 2006<sup>[57]</sup>). In a number of education systems, students with an immigrant background perform at significantly lower levels than their native

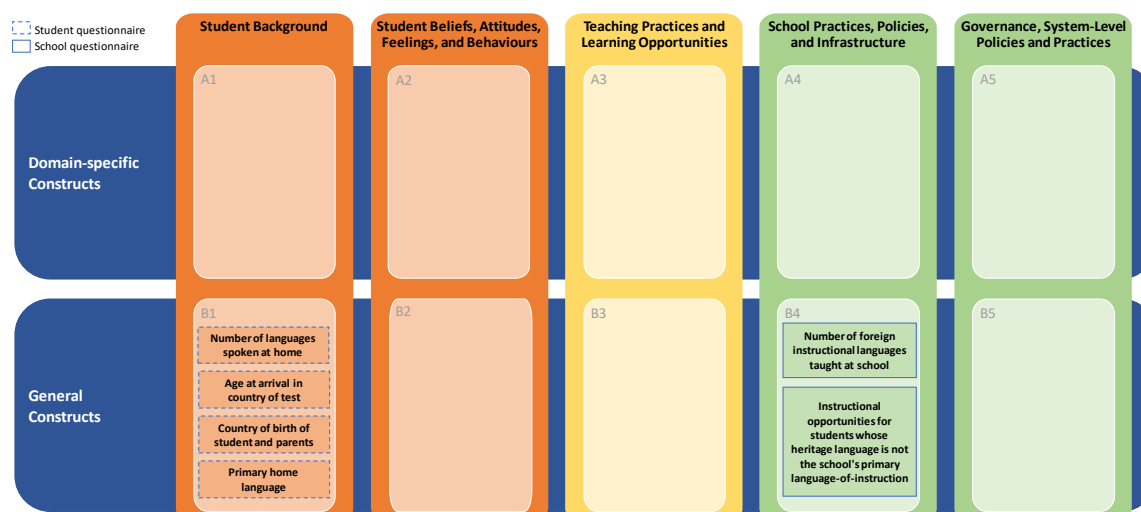
peers in key school subjects and are often faced with overt or covert discrimination with potentially detrimental consequences for their psychological development and well-being (Brown and Chu, 2012<sup>[60]</sup>; Amiot and Bourhis, 2005<sup>[61]</sup>).

Thus, providing students from different linguistic and cultural backgrounds with equal opportunities is often considered one of the central challenges for education systems in the 21<sup>st</sup> Century (OECD, 2021<sup>[62]</sup>). Students whose heritage language (e.g. language spoken at home) is not the school's primary language of instruction might benefit from preparatory programmes or additional periods of instruction aimed at developing language skills in the school's primary language of instruction, instruction in school subjects in their heritage language aimed at developing proficiency in both languages, or other language programmes (Trifonas and Aravossitas, 2018<sup>[63]</sup>). Issues regarding the student's experience of a school climate that is accepting of diversity and multi-culturalism are relevant to this module and overlap with content examined in Module 6 on school culture and climate.

The PISA 2025 STQ continued to administer the questions on students' immigration background and language exposure that have been included in previous cycles. Constructs on students' migration backgrounds include the country of birth of students and parents, and the age at which students born in a foreign country arrived in the host country (i.e. the country of the test). Constructs related to language exposure include the primary language students speak at home and the number of languages spoken at home (multilingualism). General constructs in the SCQ include the number of foreign instructional languages being taught at the school and instructional opportunities for students whose heritage language is not the school's primary language of instruction.

Figure 4.7 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.7. Module 4: Migration and language exposure**



### Module 5: PISA preparation and effort

Several researchers have investigated whether test-taker effort on low-stakes large-scale assessments may impact achievement results, or have asked whether differential effort may play a role in explaining score differences between student groups or educational systems (Debeer et al., 2014<sup>[64]</sup>; Eklöf, Pavešič and Grønmo, 2014<sup>[65]</sup>; Hopfenbeck and Kjærnsli, 2016<sup>[66]</sup>; Jerrim, 2015<sup>[67]</sup>; Pools and Monseur, 2021<sup>[68]</sup>; Rios, 2021<sup>[69]</sup>). Research findings show that low-stakes assessments elicit lower levels of engagement (Wolf and Smith, 1995<sup>[70]</sup>; Finn, 2015<sup>[71]</sup>). Lower levels of engagement, in turn, are associated with lower

performance on tests (Pintrich and de Groot, 1990<sup>[72]</sup>; Wise and DeMars, 2005<sup>[73]</sup>; Kuhfeld and Soland, 2019<sup>[74]</sup>).

In an attempt to inform educational policy with regard to test-taker effort in PISA, this module covers students' subjective perceptions of how much effort they applied when answering PISA test questions in mathematics, reading or science, as well as in filling out the STQ. Questions draw on the idea of the "effort thermometer" introduced in PISA 2003 (Butler and Adams, 2007<sup>[75]</sup>).

To complement questions examining students' perceptions of effort, a school question examined administrators' support of the PISA test administration; their communication with staff, students, parents or guardians, and teachers about PISA; and their encouragement of students to take the PISA test seriously. There is evidence that some countries have made efforts to systematically improve their test results in PISA. For example, Canadian school teachers were given a handbook urging them to "encourage (students) to take the assessment seriously and strive for excellence" (Akyol, Krishna and Wang, 2021<sup>[76]</sup>).

Figure 4.8 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.8. Module 5: PISA preparation and effort**



## Module 6: School culture and climate

School climate encompasses the atmosphere, norms and values, and social networks of a school (Loukas and Murphy, 2007<sup>[77]</sup>). A positive school climate is present when students feel supported by their teachers, have positive relations between each other and are satisfied with their classes and the general school environment (Jia et al., 2009<sup>[78]</sup>; Loukas and Murphy, 2007<sup>[77]</sup>). Research has shown that a positive school climate contributes to student achievement (Jia et al., 2009<sup>[78]</sup>; Hoy, Hannum and Tschannen-Moran, 1998<sup>[79]</sup>; MacNeil, Prater and Busch, 2009<sup>[80]</sup>; Kutsyruba, Klinger and Hussain, 2015<sup>[81]</sup>) and students' attitudes such as self-esteem (Way, Reddy and Rhodes, 2007<sup>[82]</sup>). It also protects students from deviant or risky behaviours such as truancy, smoking, drinking and using drugs (LaRusso, Romer and Selman, 2008<sup>[83]</sup>; Gase et al., 2017<sup>[84]</sup>). Furthermore, a positive school climate has the potential to mitigate the negative association between socio-economic disadvantage and academic achievement (Berkowitz et al., 2017<sup>[85]</sup>).

School safety is an important facet of school climate. An orderly, safe and supportive learning environment is crucial for high attendance and the effective use of learning time. In contrast to this, when students do

not feel safe in their school, it can hinder their learning and distract from educational goals (Perumean-Chaney and Sutton, 2013<sup>[86]</sup>). Schools without supportive relationships between students and teachers and with more organisational dysfunction (i.e. shortage of teachers) are more likely to experience violence and victimisation, which is often associated with reduced academic achievement (Astor, Guerra and Van Acker, 2010<sup>[87]</sup>).

Sense of belonging at school reflects how accepted, valued and supported students feel within the school community (Slaten et al., 2016<sup>[88]</sup>; Goodenow and Grady, 1993<sup>[89]</sup>). Students' sense of belonging at school is positively associated with a positive disciplinary school climate (Ma, 2003<sup>[90]</sup>); as well as with parental, teacher and peer support; school safety (Slaten et al., 2016<sup>[88]</sup>); and participation in extracurricular activities (Dotterer, McHale and Crouter, 2007<sup>[91]</sup>). Moreover, students reporting a greater sense of belonging at school tend to display higher academic motivation and achievement (Wang and Holcombe, 2010<sup>[92]</sup>; Arslan, 2021<sup>[93]</sup>). Students who feel they belong at school also have fewer mental health problems in the face of school bullying experiences (Arslan, 2021<sup>[93]</sup>).

Bullying is characterised by a systematic abuse of power and an unequal power relationship between the bully and the victim. It can be physical (hitting, punching and kicking), verbal (name-calling and mocking) and relational (spreading gossip and engaging in other forms of public humiliation, shaming and social exclusion) (Woods and Wolke, 2004<sup>[94]</sup>). Research indicates that both aggressors and victims tend to skip classes, drop out of school more often and perform worse academically than peers not involved in bullying (Juvonen, Yueyan Wang and Espinoza, 2011<sup>[95]</sup>; Konishi et al., 2010<sup>[96]</sup>). Additionally, bullying at school can affect any student and is a major risk factor for adolescents' mental and physical health (Wolke and Lereya, 2015<sup>[97]</sup>) that can have long-term consequences for students (Phillips, 2007<sup>[98]</sup>). Research suggests that a positive school climate, more specifically, one that is characterised by transparent rules and clear disapproval of harassments is linked to a lower prevalence of bullying (Låftman, Östberg and Modin, 2017<sup>[99]</sup>).

The quality of student-teacher relationships plays a crucial role in students' learning. Students need to feel that their teachers support them and care about them and their achievement to fully engage in learning activities and perform at their best (Federici and Skaalvik, 2013<sup>[100]</sup>). Previous research suggests that supportive teacher-student relations positively affect student achievement (Lee, 2012<sup>[101]</sup>), motivation (Pitzer and Skinner, 2017<sup>[102]</sup>), engagement in learning (Ruzek et al., 2016<sup>[103]</sup>) and well-being (Suldo et al., 2009<sup>[104]</sup>). A quality student-teacher relationship can also potentially protect students from behavioural disengagement at school (Archambault, Vandenbossche-Makombo and Fraser, 2017<sup>[105]</sup>) while perceived conflict with teachers has a significant positive effect on students' engagement in active bullying of other students (Longobardi et al., 2018<sup>[106]</sup>).

Truancy, which includes skipping or arriving late to school, implies that learning opportunities are being missed. Research has identified several risk factors for school absenteeism such as externalising problems (e.g. conduct problems), mental health problems, parental unemployment and illness or chronic disease (Ingul et al., 2012<sup>[107]</sup>). On the school level, a poor school climate (Brookmeyer, Fanti and Henrich, 2006<sup>[108]</sup>) and diminished sense of belonging at school (Shochet et al., 2006<sup>[109]</sup>) can increase truancy. Previous research has highlighted that students with lower socio-economic backgrounds are most likely to suffer repeated absences. This is particularly important since they benefit most from school attendance because of the equalizing effects of schooling on cognitive abilities (Ready, 2010<sup>[110]</sup>). In line with this, a study based on the United States by Gershenson, Jacknowitz and Brannegan (2017<sup>[111]</sup>) found that the harm associated with student absences is greater among low-income and English-as-a-second-language students, particularly for reading performance. In general, absenteeism reduces math and reading outcomes, and educational and social engagement at school (Gottfried, 2014<sup>[112]</sup>; Smerillo et al., 2018<sup>[113]</sup>). Moreover, unexcused absences were found to be twice as harmful for achievement as excused absences (Gershenson, Jacknowitz and Brannegan, 2017<sup>[111]</sup>).

Disorder and delinquent behaviour at school, including cheating, profanity, vandalism, theft, injury or verbal abuse, can be a significant barrier to learning (Kutsyuruba, Klinger and Hussain, 2015<sup>[81]</sup>). Delinquent behaviours at school threaten the health, well-being, social relations and academic achievement of adolescents (Masten et al., 2008<sup>[114]</sup>). Socio-economically disadvantaged schools experience more disorder and student delinquency in terms of theft, physical injury or vandalism than advantaged schools (Gottfredson et al., 2005<sup>[115]</sup>). Previous research has shown that rule clarity or teacher support can play a role in decreasing the prevalence of bullying and delinquent behaviours (Aldridge, McChesney and Afari, 2018<sup>[116]</sup>). Additionally, schools characterised by high teacher morale, strong leadership and high teacher involvement tend to be more protected from school crime (Gottfredson et al., 2005<sup>[115]</sup>).

Factors such as student truancy; absence of respect for teachers; drug use; teacher absenteeism; staff resisting change; and teachers not being prepared for class can hinder instruction and impact the effective use of learning time. When teachers miss work, the learning process is disrupted and student achievement can be reduced, particularly when the absences are unexpected (Miller, Murnane and Willett, 2008<sup>[117]</sup>). Teachers' resistance to change is another form of behaviour that can hinder student learning, as it is a major factor in reform failure (Zimmerman, 2006<sup>[118]</sup>). When teachers feel they are not respected by students, this can increase their experience of stress, which is a risk factor for teacher burnout (Hakanen, Bakker and Schaufeli, 2006<sup>[119]</sup>).

Demographic diversity in the student population has increased in many countries. According to Gurin et al., (2002<sup>[120]</sup>), fostering diversity in education not only increases access for greater numbers of students but boosts students' academic and social development. In a study by Juvonen, Kogachi and Graham (2018<sup>[121]</sup>), school-level ethnic diversity was associated with students feeling safer, less victimised and less lonely. It also decreases out-group distance; that is, it reduces students' preference for the ethnic in-group over their out-group peers. Additionally, students' attitudes towards diversity at school is connected to perceptions of teachers' fair and equal treatment of different ethnic groups. In a sample of Mexican-American adolescents, perceived poor school quality was associated with perceived discrimination, whereas positive student-teacher relationships protected against discrimination (Stone and Han, 2005<sup>[122]</sup>). To measure the extent to which schools recognise and encourage the diversity of students' backgrounds, PISA 2025 continued to collect data on the construct "school diversity/multi-cultural views".

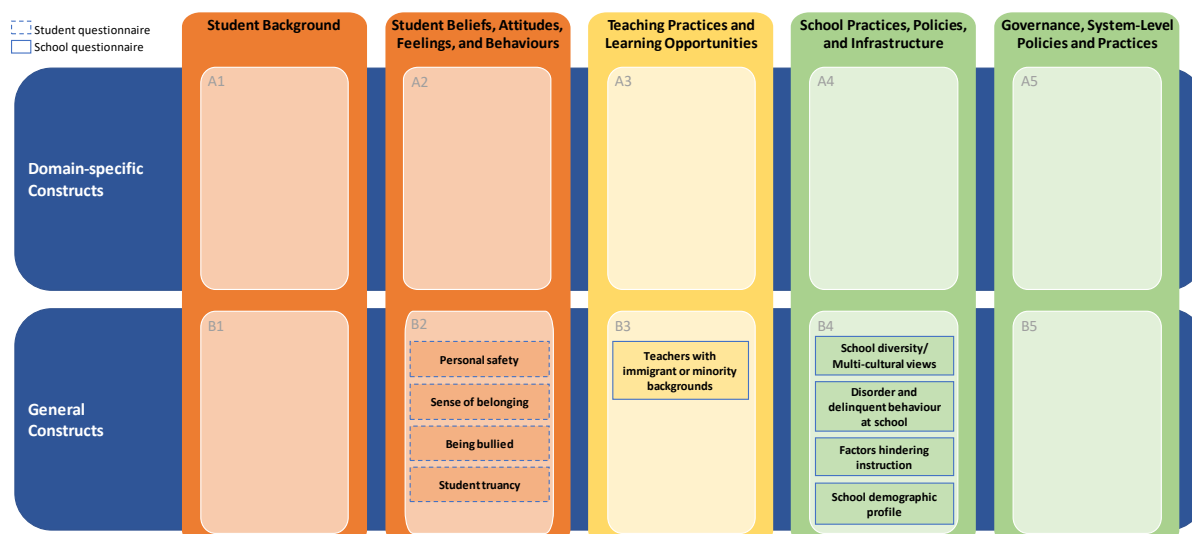
The diversity and heterogeneity found in student populations should be mirrored in the teaching workforce by attracting and retaining more diverse teachers (Schleicher, 2020<sup>[123]</sup>). The teaching workforce remains ethnically homogeneous in many OECD countries (Forghani-Arani, Cerna and Bannon, 2019<sup>[124]</sup>). Teacher diversity can have an impact on student learning and well-being (Brussino, 2021<sup>[125]</sup>). For example, students who have similar identities to teachers, including ethnicity and gender, have better academic outcomes (Dee, 2005<sup>[126]</sup>; Goldhaber, Theobald and Tien, 2019<sup>[127]</sup>; Woodson and Bristol, 2020<sup>[128]</sup>) and fewer absences and suspensions (Holt and Gershenson, 2019<sup>[129]</sup>). They are also more likely to challenge gender stereotypes (Hutchings et al., 2008<sup>[130]</sup>). For these reasons, a new development in PISA 2025 was the construct "teachers with immigrant/minority background"; this new information will allow PISA to carry out analysis on the relationship between teacher diversity at school, and student performance and psychological and social well-being at school.

General constructs in the PISA 2025 STQ include student truancy and personal safety, bullying experiences, and students' sense of belonging. General constructs in the PISA 2025 SCQ include a school's efforts to promote school diversity/multi-cultural views; disorder and delinquent behaviour at school; and factors hindering instruction and school demographic profile. Teachers with immigrant or minority background is a new general construct for PISA 2025.

Questions in this module show some conceptual overlap with domain-specific questions in other modules (e.g. disciplinary climate in Module 16; well-being in Module 8).

Figure 4.9 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.9. Module 6: School culture and climate



## Module 7: Science identity

“Science identity” refers to a person’s sense of being a “science person”; that is, a science learner “who knows about, uses, and sometimes contributes to science” (Bell et al., 2018<sup>[131]</sup>) and who is recognised as such by others (Carlone and Johnson, 2007<sup>[132]</sup>). A science identity comprises science-related agency, attitudes, beliefs, behaviours and experiences, and reflects a meaningful and personal connection between the self and science.

Supporting young people’s science identities has been identified as an important goal for science education (OECD, 2020<sup>[133]</sup>). It is an important way to enable young people and their communities and societies to tackle the many challenges that they and the world will face in the coming decades. Young people need to be able to engage critically with and use scientific knowledge and competencies in their lives beyond the classroom; not just for their own personal benefit but for the health, fairness and prosperity of societies globally. From an international policy perspective, the shortage of skilled workers in technical and science professions, and the relatively narrow demographic profile of those who continue into science, technology, engineering and mathematics (STEM) degrees and careers is an enduring concern (e.g. (European Union, 2015<sup>[134]</sup>; OECD, 2019<sup>[135]</sup>)). Therefore, PISA aims to understand the extent to which students identify and feel connected with science.

As demonstrated by extensive research, identity is associated with science learning, engagement and participation. It shapes the extent to which scientific knowledge and competencies are developed and the likelihood that a young person will go on to study science at degree level and/or enter science occupations (e.g. (Archer et al., 2010<sup>[136]</sup>; Calabrese Barton and Tan, 2019<sup>[137]</sup>; Carlone, Scott and Lowder, 2014<sup>[138]</sup>; Danielsson, 2009<sup>[139]</sup>; Pea., 1993<sup>[140]</sup>; Barron, 2006<sup>[141]</sup>; Nardi and O’Day, 1999<sup>[142]</sup>; Dweck and Leggett, 1988<sup>[143]</sup>; Nasir and Hand, 2008<sup>[144]</sup>; Shanahan, 2009<sup>[145]</sup>)).

Science self-perception is a key dimension of science identity. In PISA 2025, science self-perception was measured by science self-efficacy. Science self-efficacy, as defined in PISA, refers to students’ beliefs in their ability to perform science-related tasks in their everyday lives (Bandura, 1997<sup>[146]</sup>; Schunk, 1991<sup>[147]</sup>). Positive self-efficacy belief is highly related to motivation, learning behaviour, general expectations for the future and students’ performance (OECD, 2007<sup>[148]</sup>).

In addition to self-perception, science capital is an important dimension for science identity. Science capital refers to science-related forms of cultural and social capital (Archer et al., 2015<sup>[149]</sup>), namely science-

related knowledge, attitudes, dispositions, resources, behaviours and social contacts. Science capital is a composite construct that has been shown to significantly relate to young people's science identity and participation in post-compulsory science (Archer et al., 2015<sup>[149]</sup>; Moote et al., 2019<sup>[150]</sup>; Moote et al., 2020<sup>[11]</sup>; Gonsalves et al., 2021<sup>[151]</sup>; Turnbull et al., 2020<sup>[152]</sup>). This construct includes dimensions that have been shown by wider research to be important factors relating to science identity and participation, including science attitudes, behaviours and motivation from significant others (e.g. (Mujtaba and Reiss, 2013<sup>[153]</sup>)).

In relation to students' interests, attitudes and motivation to study science, this framework considers enjoyment of science, a construct measured in PISA 2006 and PISA 2015. Research suggests that motivation to learn science and affirmative beliefs about oneself as a science learner are positively associated with students' performance in science (OECD, 2007<sup>[148]</sup>). This relationship is reciprocal: science-related beliefs and attitudes can be both a consequence and a driver of science performance. Motivation to learn that is based on interest and enjoyment is experienced as self-determinate and intrinsic (Krapp and Prenzel, 2011<sup>[154]</sup>). It affects student engagement, learning activities, performance and career choices, and can be shaped by classroom instructions and parental motivation practices (Gottfried et al., 2009<sup>[155]</sup>; Kunter, 2005<sup>[156]</sup>; Rakoczy et al., 2008<sup>[157]</sup>; Ryan and Deci, 2000<sup>[158]</sup>).

The environmental crisis is a pressing global concern. Human impact has increased exponentially since 1950 and led to significant disruptions to the systems within the biosphere, hydrosphere, geosphere and atmosphere (Lewis and Maslin, 2015<sup>[159]</sup>; IPCC, 2021<sup>[160]</sup>). Threats to the environment are prominently discussed in the media, and students are challenged to understand complex environmental issues such as climate change, its causes and consequences. To prepare young people to address contemporary environmental challenges, young people need a series of attitudes and dispositions that affect their willingness to work individually, with others, and across generations, which, in turn, affect the world's sustainability. Therefore, the following constructs were included in PISA 2025:

- Student environmental awareness (construct based on PISA 2006/2015 with updated items): PISA 2006 showed that students from more socio-economically advantaged backgrounds reported higher levels of awareness of environmental issues and that this construct is linked with students' science performance (OECD, 2007<sup>[148]</sup>).
- Student environmental agency: Agency is defined as a person's ability to "positively influence their own lives and the world around them as well as the capacity to set a goal, reflect and act responsibly to effect change" (OECD, 2019<sup>[161]</sup>). A new construct on environmental agency in PISA 2025 focused on personal decision making in relation to environmental issues. Research underlines the importance of supporting young people's environmental agency to enable them to act in ways that support environmental sustainability (e.g. (Bigger and Webb, 2010<sup>[162]</sup>; Wals and Jickling, 2000<sup>[163]</sup>)).
- Collective efficacy regarding environmental issues: To develop agency, young people must demonstrate a sense of self (individual) and collective efficacy, working across generations, guided by a sense of actionable hope (Ajzen, 1985<sup>[164]</sup>; Ryan and Deci, 2000<sup>[165]</sup>; National Research Council, 2002<sup>[166]</sup>; Schultz, 2002<sup>[167]</sup>). Collective efficacy is believing that one's group can meet their goals. Those who believe their group can work effectively tend to have a greater sense of their own self-efficacy (Jugert et al., 2016<sup>[168]</sup>).
- Hope regarding environmental issues: Hope is the sense that there is a way towards a possible future that is worth achieving. Without hope, the belief that the current environmental predicament will not change or improve may result in anxiety, depression and helplessness (Peterson, Maier and Seligman, 1996<sup>[169]</sup>; Li and Monroe, 2017<sup>[170]</sup>).

School activities might contribute to students' environmental agency. Research suggests that school activities such as student participation in school's decisions or in initiatives for improving the life conditions of others can enable student agency on topics like climate change and environmental sustainability

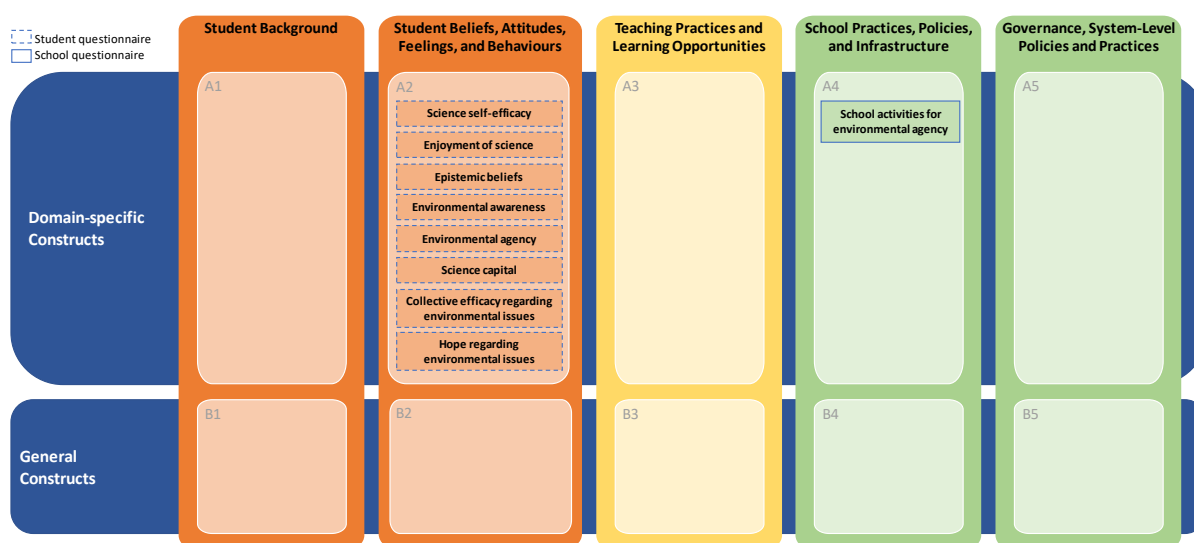
development (Environmental Science Expert Group, 2022<sup>[171]</sup>). Policymakers wish to know the effects of school activities on student agency; this was a new development in the PISA 2025 STQ.

Finally, epistemic beliefs are closely related to students' general values of science and scientific inquiry (Fleener, 1996<sup>[172]</sup>; Hofer and Pintrich, 2002<sup>[173]</sup>). They include beliefs about science as an evolving and changing subject, and how individuals justify knowledge (Conley and et al., 2004<sup>[174]</sup>). PISA measured epistemic beliefs in 2015. Since then, students' beliefs about science and the judgements they make about expertise have become ever more important (McIntyre, 2018<sup>[175]</sup>; Sinatra and Hofer, 2021<sup>[176]</sup>). Of particular concern is whether they are absolutists (believing knowledge is fixed and certain), multiplists (believing knowledge is subjective and there are no criteria for making comparative judgements), or evaluativists (believing there are criteria for judging and comparing knowledge claims) when it comes to making decisions about knowledge claims (Kuhn, 1999<sup>[177]</sup>). Of particular interest here is the extent to which they place trust in claims made by scientists and domain-specific experts compared to other sources of information (Kozyreva, Lewandowsky and Hertwig, 2020<sup>[178]</sup>).

Domain-specific constructs included in the PISA 2025 STQ are science self-efficacy, science capital, enjoyment of science, environmental awareness, environmental agency, collective efficacy regarding environmental issues, hope regarding environmental issues, and epistemic beliefs. The domain-specific construct suggested for inclusion in the PISA 2025 SCQ is school activities for environmental agency.

Figure 4.10 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.10. Module 7: Science identity**



### **Module 8: Curiosity, perseverance, mindset and well-being**

Social and emotional skills refer to the abilities to regulate one's thoughts, emotions and behaviour (OECD, 2017<sup>[179]</sup>). Following the framework of the OECD's Survey of Social and Emotional Skills (SSES), social and emotional characteristics can be defined as individual capacities that: a) are manifested in consistent patterns of thoughts, feelings and behaviours; b) can be developed through formal and informal learning experiences; and c) influence important socio-economic outcomes throughout an individual's life (OECD, 2017<sup>[179]</sup>). Besides cognitive skills, school achievement is also dependent on a number of social and emotional skills such as perseverance, self-control, responsibility, curiosity and emotional stability (Van Eijck and De Graaf, 2004<sup>[180]</sup>; Lechner, Anger and Rammstedt, 2019<sup>[181]</sup>). For example, social and emotional skills have been found to be almost as influential as cognitive skills for occupational outcomes such as income, unemployment and occupational stability (Roberts et al., 2007<sup>[182]</sup>). Teaching and learning

social and emotional skills by schools (i.e. social and emotional learning policies) can foster students' social and emotional skills, well-being (Taylor et al., 2017<sup>[183]</sup>) and engagement at school (Yang, Bear and May, 2018<sup>[184]</sup>).

PISA 2022 examined a broad range of social and emotional characteristics using measures grounded in the Big Five Model of Personality (Costa and McCrae, 1992<sup>[185]</sup>), a framework that identifies five broad personality dimensions: openness to experience, conscientiousness, extraversion, agreeableness and neuroticism. The PISA 2025 cycle focused on two key constructs – perseverance and curiosity – which closely align with “task performance” (conscientiousness) and “open-mindedness” (openness to experience), respectively. They were selected for having the strongest association with student achievement in PISA 2022 (OECD, 2023<sup>[186]</sup>).

“Task performance” describes different aspects of students' conscientiousness, including perseverance and self-control; for example, setting high standards for themselves and working hard to meet them; fulfilling commitments and being reliable; and being able to avoid distractions and focus attention on tasks (OECD, 2017<sup>[179]</sup>). According to Nofle and Robins (2007<sup>[187]</sup>), conscientiousness is the strongest predictor of high school and college grades among the Big Five. Additionally, self-control is associated with a lower likelihood of substance use and violence (Roberts and Bogg, 2004<sup>[188]</sup>) and greater academic attainment (Duckworth and Gross, 2014<sup>[189]</sup>; Ivcevic and Brackett, 2014<sup>[190]</sup>).

“Open-mindedness” encompasses students' openness to new experiences, including curiosity and perspective-taking (i.e. the ability to put oneself in another person's place), as well as enjoyment of generating novel ideas or visions. According to a study by Nofle and Robins (2007<sup>[187]</sup>), openness is the strongest predictor of SAT verbal scores among the Big Five. Furthermore, fostering curiosity in students can enhance school achievement (Shah et al., 2018<sup>[191]</sup>; Lechner, Miyamoto and Knopf, 2019<sup>[192]</sup>).

Social and emotional skills, and overall life satisfaction are positively related (OECD, 2017<sup>[179]</sup>). PISA defines “life satisfaction” as an overall evaluation that an individual makes about his or her perceived quality of life according to his or her chosen criteria (Shin and Johnson, 1978<sup>[193]</sup>; OECD, 2019<sup>[194]</sup>). Life satisfaction is closely related to happiness and can foster success in academic achievement and work performance (Lyubomirsky, King and Diener, 2005<sup>[195]</sup>; Park, 2004<sup>[196]</sup>). Asking students to report on their satisfaction with life is one way to measure the positive development of young people (Park, 2004<sup>[196]</sup>). Adolescents' subjective well-being is related to health and behaviour that may persist into adulthood (Currie et al., 2009<sup>[197]</sup>; Elgar et al., 2017<sup>[198]</sup>).

This module also covers growth mindset, a topic measured in PISA 2018 and 2022. Implicit theories of intelligence or mindsets (Dweck and Leggett, 1988<sup>[199]</sup>) play an important role in students' motivation and performance in school. A growth mindset, as opposed to a fixed mindset, is the belief in the malleability of ability and intelligence and is one possible explanation as to why some people fulfil their potential while others do not (Dweck, 2006<sup>[200]</sup>). Growth mindset interventions aim to foster a growth mindset and in this way, to reduce students' negative effort beliefs (the belief that having to try hard or ask for help means you lack ability), fixed-trait attributions (the attribution that failure stems from low ability) and performance avoidance goals (the goal of never looking stupid) (Yeager et al., 2019<sup>[201]</sup>). The standard approach for these interventions is teaching scientific information about the brain and its malleability; for example, that the brain consists of networks of neurons that become “stronger” and more efficient when people learn new and challenging material. A commonly used metaphor is that the brain is like a “muscle” that gets stronger with exercise. The information is often followed by a “saying-is-believing” task that asks participant to use what they learned to write mentoring letters to future struggling students. Typical growth mindset interventions can be delivered online and take less than one hour (Dweck and Yeager, 2019<sup>[202]</sup>; Paunesku et al., 2015<sup>[203]</sup>; Yeager et al., 2019<sup>[201]</sup>; Yeager et al., 2016<sup>[204]</sup>). Holding a growth mindset is associated with a number of positive attributes in the learning context such as setting more challenging learning goals; appreciating effort as an inherent part of the process; less fear of failure; higher levels of well-being; and more perseverance in the face of setbacks (Dweck, 2006<sup>[200]</sup>).

In the last decade, the importance of mindsets in learning contexts has received increasing attention and students' growth mindsets were included in PISA 2018 for the first time. In line with past research, PISA 2018 data also showed a positive relationship between students' growth mindsets and performance as well as with motivational measures (OECD, 2021<sup>[205]</sup>). These findings are of potential importance for policymakers. Growth mindset interventions can increase performance and reduce achievement gaps between advantaged and disadvantaged students. Despite a great heterogeneity of effects in growth mindset interventions, the effect of interventions is larger for disadvantaged students (Paunesku et al., 2015<sup>[203]</sup>; Broda et al., 2018<sup>[206]</sup>; Yeager et al., 2019<sup>[201]</sup>). Interventions can potentially foster greater equity in education if implemented on a larger scale (Broda et al., 2018<sup>[206]</sup>; Rattan et al., 2015<sup>[207]</sup>).

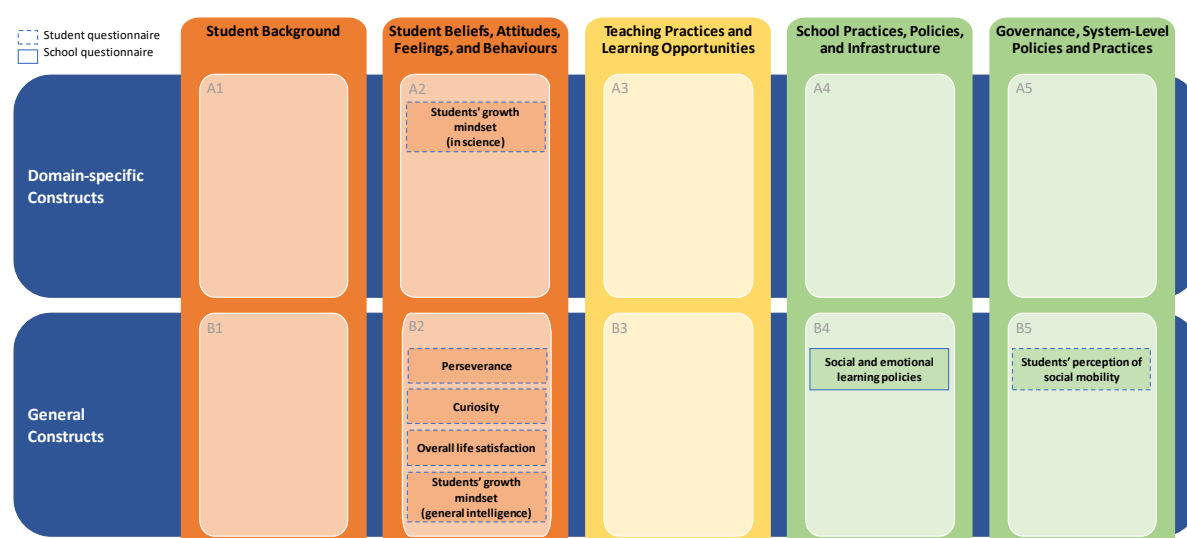
Growth mindsets are most effective in a context that supports the message that individuals can grow their abilities (Walton and Yeager, 2020<sup>[208]</sup>). The effectiveness of growth mindsets depends on features of contexts that make an adaptive perspective possible. Walton and Yeager (2020<sup>[208]</sup>) argue that mindset interventions offer people a perspective that they test out (e.g. a growth mindset); but if this perspective is not supported in daily life, if it feels inauthentic or misfitting, its impact may fade. To capture a supportive context for growth mindsets at the country level, student's perception of social mobility was included as a new development in PISA 2025.

Social mobility might play a role in the development and effectiveness of students' growth mindsets. In both an analysis of PISA 2018 data and an experiment, Jia et al. (2021<sup>[209]</sup>) observed that in low-mobility environments, growth mindsets are less potent and beneficial than in environments with high educational mobility. Even though students from lower-income families are less likely to hold a growth mindset, a growth mindset can buffer against the negative effects of poverty on school achievement, especially in contexts where students perceive there are opportunities for upward mobility (Claro, Paunesku and Dweck, 2016<sup>[210]</sup>).

General constructs included in the PISA 2025 STQ are perseverance, curiosity, overall life satisfaction, students' growth mindset (general intelligence), and students' perception of social mobility, while the domain-specific construct included is students' growth mindsets in science. The PISA 2025 SCQ includes a single general construct on social and emotional learning policies.

Figure 4.11 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.11. Module 8: Curiosity, perseverance, mindset and well-being**



## **Module 9: Post-secondary preparedness and aspirations**

In addition to collecting data on students' early educational careers, previous PISA cycles have gathered prospective information about students' educational and occupational expectations for the future. In this module, expectations are defined as the level of education or occupational prestige that students expect to complete given their life circumstances at age 15 as opposed to what they wished they could complete or what they feel able to complete.

Students' expectations can play a large role in shaping their actual future achievements (Nurmi, 2013<sup>[211]</sup>). Guyon and Huillery (2020<sup>[212]</sup>) find that among students in the same classroom with similar academic abilities and family SES, students who aspire to academic high school and graduate studies at the beginning of Grade 9 (at age 14) have significantly better educational outcomes later on than students with lower expectations. Furthermore, longitudinal data from PISA and national follow-up surveys show that students in Australia, Canada and Denmark who held higher occupational expectations at age 15 were more likely to continue with their studies and attain high-skilled jobs at age 25 than students who had lower expectations, even after accounting for academic performance at age 15 (OECD, 2018<sup>[213]</sup>)<sup>1</sup>.

Socio-economically disadvantaged students are less likely to aspire to higher education and top educational programmes and jobs than their socio-economically advantaged peers, even when they have equal test scores (Sewell, Haller and Ohlendorf, 1970<sup>[214]</sup>; Jencks, Cruose and Mueser, 1983<sup>[215]</sup>; Hoxby and Avery, 2013<sup>[216]</sup>; La Ferrara, 2019<sup>[217]</sup>). In theory, if disadvantaged students are not as ambitious compared to what they could optimally aspire to, they might exert a sub-optimal level of effort, resulting in sub-optimal outcomes. This is concerning because it could create an expectations-based "poverty-trap" that reinforces social inequity (Appadurai, 2004<sup>[218]</sup>; Ray, 2006<sup>[219]</sup>; Dalton, Ghosal and Mani, 2016<sup>[220]</sup>; Hoff and Stiglitz, 2016<sup>[221]</sup>). Education policies should not only provide more resources to disadvantaged students but help students be more confident, hopeful and ambitious, and provide students with sufficient information to help them get a fuller picture of possible future careers, and the education and skills needed to pursue and succeed in them.

PISA 2018 showed that many students, especially disadvantaged students, hold lower expectations than would be expected given their academic achievement. However, the reasons for low expectations are not well understood so far. PISA 2025 collected data on a number of constructs that aim at understanding how expectations are formed and why low-SES students aspire to less ambitious educational programmes and jobs than their equally achieving high-SES counterparts. Because these constructs can reveal the factors that form educational and professional aspirations, they can be useful for designing nation-wide policies aimed at helping students to reach their true potential in education and work. For example, policymakers could implement advising programmes that are particularly sensitive to disadvantaged students who do not have accurate information about academic requirements, financial costs and student aid for higher education. Advising students in this way could help them to have a clearer idea of the paths available for them and empower them to make better choices for their future.

Expectations are the product of different considerations that students consciously or unconsciously have about their future. A first determinant of adolescents' post-secondary expectations is their knowledge of existing education programmes, jobs and links between education and jobs. Existing evidence suggests that young people often have little understanding of labour market demands (OECD, 2017<sup>[222]</sup>). Research also suggests that low-SES students, even the highest-achieving ones, are less aware of top educational pathways such as master's programmes than are their high-SES peers (Guyon and Huillery, 2021<sup>[223]</sup>). In addition, students' educational expectations are often inconsistent with their professional expectations: there is a considerable proportion of students who aspire to educational levels below the levels necessary to reach the occupation they aspire to. Thus, it is important for PISA 2025 to collect data on how students seek information about future study and work, and whether students feel they have the skills to find this information. This includes measuring how schools support students by providing them with study and career information, and guidance. A new question aimed at capturing students' knowledge of the

educational requirement of jobs was included in the PISA 2025 Field Trial but was ultimately not implemented in the main survey.

Socio-economic differences in expectations may also reflect the fact that low-SES students face external constraints. The cost of education may be a barrier for low-SES students whose families cannot afford the tuition fees and housing costs associated with higher education programmes. Research shows that lack of access to credit can make it difficult for the poor to invest in education (Dearden et al., 2009<sup>[224]</sup>; Kaufmann, 2014<sup>[225]</sup>). Lower economic returns on education might be another external constraint if discrimination in the job market can reduce the returns on education of minorities or disadvantaged students (Adida, Laitin and Valfort, 2014<sup>[226]</sup>). Other studies suggest that information about the net benefits from higher education is imperfect and that this contributes to discouraging low-SES students from applying to higher education (Bettinger et al., 2012<sup>[227]</sup>; Oreopoulos and Dunn, 2013<sup>[228]</sup>; Hoxby and Turner, 2015<sup>[229]</sup>).

Social background has a direct impact on students' post-secondary expectations because people use comparisons and similarities with peers and relatives to form their expectations (Ray, 2006<sup>[230]</sup>; Genicot and Ray, 2017<sup>[231]</sup>). By observing their peers, people obtain information about the existing pathways and form views about their own probability of succeeding in each pathway. Talented people from low-SES backgrounds generally have peers who are not following the pathways that would be optimal for them, i.e. that would allow them to reach their best possible outcomes. This leads talented low-SES individuals to form a restricted set of possible selves. Furthermore, choosing an identity-reinforcing attitude may be a way of limiting disruption and maintaining a sense of unity, avoiding cognitive dissonance or protecting against the threat of losing one's culture (Akerlof and Kranton, 2002<sup>[232]</sup>; Bénabou and Tirole, 2011<sup>[233]</sup>). Staying within one's SES may also allow people to secure social interactions and co-operation in the future (Austen-Smith and Fryer Jr, 2005<sup>[234]</sup>; Fang and Loury, 2005<sup>[235]</sup>; Fryer Jr, 2007<sup>[236]</sup>). Thus, socio-economically disadvantaged students might have reason to hold lower educational and occupational expectations to maximise the socio-emotional benefits associated with the protection of their social identity. Assessing student perceptions of their parents' and peers' expectations can shed light on observed socio-economic differences in post-secondary expectations. To this end, PISA 2025 introduced new constructs: expected educational level of peers in school (perceived by the student) and educational level expected by parents (perceived by the student).

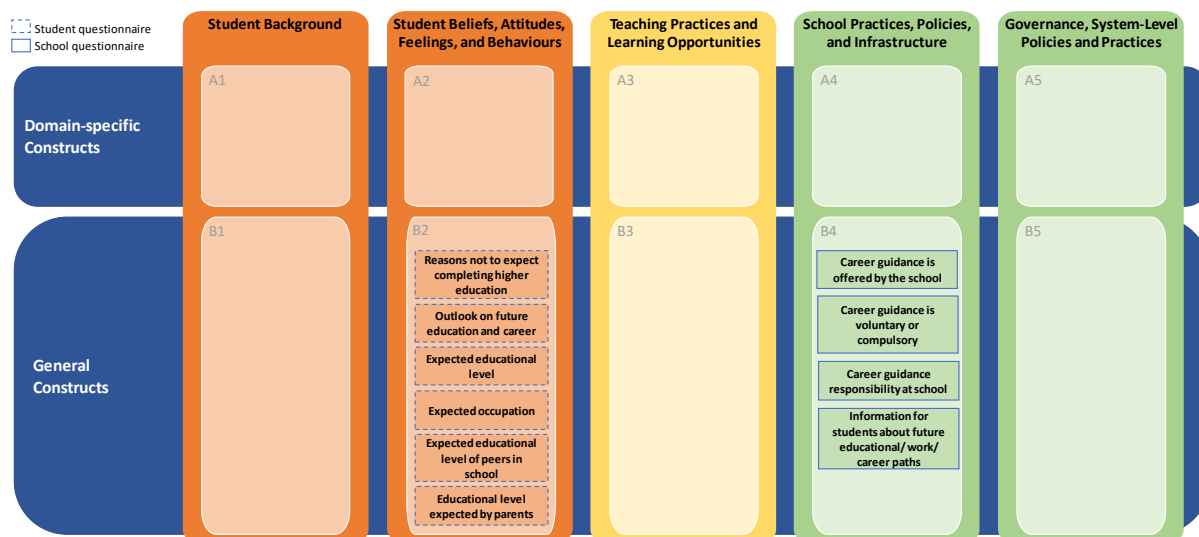
Finally, how adolescents perceive their ability and competence at academic schoolwork, their potential for improvement, and their future chances of success may be a crucial explanation of their educational and professional expectations. Guyon and Huillery (2021<sup>[223]</sup>) find that low-SES students underestimate their academic ability relative to equally achieving high-SES students; in their study, the gap in self-assessed academic ability explains the major part of the social gap in expectations once actual academic ability and classroom fixed effects are accounted for. Other research conjectures that beliefs related to social inferiority affect the perceived probability of success, i.e. self-confidence (Hoff and Stiglitz, 2016<sup>[221]</sup>). To capture this dimension, a new construct called "self-perceived academic capacity to complete higher education levels" was included in the PISA 2025 Field Trial but was ultimately not implemented in the main survey.

Finally, a special question was addressed to students who do not expect to complete higher education; this question asked students why they don't expect to complete higher education and included a list of possible reasons.

Constructs measured under this module are considered primarily as general constructs. Constructs in the STQ include expected educational level; expected occupational prestige; reasons not to expect completing higher education; outlook (i.e. self-reported preparedness and confidence) on future education and career; expected educational level of peers in school (perceived by student); and educational level expected by parents (perceived by student). The SCQ includes constructs on schools providing career guidance and information to students about future education and work.

Figure 4.12 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.12. Module 9: Post-secondary preparedness and aspirations**



### **Module 10: Experiences outside of regular school hours**

Experiences that occur outside of regular school hours shape students' learning, social and emotional skills, and their health and well-being (Park et al., 2016<sup>[237]</sup>; Suter, 2016<sup>[238]</sup>; Gromada and Shewbridge, 2016<sup>[239]</sup>). Furthermore, the study of socio-economic disparities in participation in activities out of regular school hours is important for understanding equity in education outcomes (Buchmann, Condon and Roscigno, 2010<sup>[240]</sup>; OECD, 2014<sup>[241]</sup>).

In the PISA 2022 Questionnaire Framework, this module was titled "Experiences out-of-school". That label was replaced by the current title to make it explicit that the experiences and activities considered in this module can happen either outside of school or at school (i.e. in the school building or premises that students attend) and before or after regular school hours (i.e. the period of the day when formal classroom lessons at school are carried out). This new title is consistent with the language used in PISA reports where these data have been recently reported under subtitles such as "learning time during regular school hours" or "additional lessons at school after regular school hours" (OECD, 2020<sup>[30]</sup>).

As in PISA 2022, this module considers students' academic and non-academic experiences outside of regular school hours. Academically-oriented experiences include additional instruction, homework and private study. Additional instruction is instruction on school topics received by the students in addition to and after (or before) regular classroom lessons. Thus, a basic feature of additional instruction is *supplementation*, i.e. instruction that covers subjects already covered in school (Bray, 1999<sup>[242]</sup>)<sup>2</sup>. Furthermore, additional instruction involves a teacher or instructor and is distinct from homework and private study, which can be conducted independently by students. Private study (a.k.a. self-study) is study that students carry on their own; it does not refer to private tutoring, which in this framework is considered as a form of additional instruction.

Each of these academically-oriented experiences outside of school hours matters to PISA because they are considered part of the overall amount of time students spend learning (i.e. student learning time), which also includes the time spent in regular classroom instruction (the latter is covered in Module 14) (Fuligni and Stevenson, 1995<sup>[243]</sup>; Fan et al., 2017<sup>[244]</sup>; Carroll, 1989<sup>[245]</sup>; OECD, 2016<sup>[246]</sup>). Learning time on additional instruction on the one hand, and learning time on homework and private study on the other,

should be measured separately and in a way that allows for adding them up and deriving the variable overall student learning time.

In addition to time spent in additional lessons, the module collects data on features that define different kinds of additional instruction received by students, including the type of additional lessons received by students (e.g. one-on-one tutor, in a group with a small or large number of students, online, etc.).

Homework and study time can have a positive influence on student achievement (Cooper, Robinson and Patall, 2006<sup>[247]</sup>). Previous PISA cycles have measured the amount of time spent on homework and private study and the extent to which schools and parents provide support for homework and study.

Non-academic experiences considered in this module are extracurricular activities (i.e. activities offered at school such as sports, art clubs, music bands, school plays, or volunteering) and other activities such as physical exercise, working for pay, and eating habits. Participation in extracurricular activities is associated with social and emotional skills such as task persistence, independence and group work (Covay and Carbonaro, 2010<sup>[248]</sup>). It is also related to lower dropout rates for at-risk students, improved grade point averages, and higher educational expectations and academic achievement (Broh, 2002<sup>[249]</sup>; Mahoney and Cairns, 1997<sup>[250]</sup>). In recent years, the role of extracurricular activities in students' academic performance has gained importance in the educational debate. In the Chinese context, for example, participation in extracurricular activities has been seen by parents as a way to boost academic performance or increase chances for elite school and university admission (see e.g. (Andersen and Jæger, 2015<sup>[251]</sup>; Liu and Bray, 2018<sup>[252]</sup>; Tan, Cai and Bodovski, 2021<sup>[253]</sup>)). Therefore, it is worthwhile to collect data and take a comparative look at both academically-oriented experiences of students and their participation in extracurricular activities.

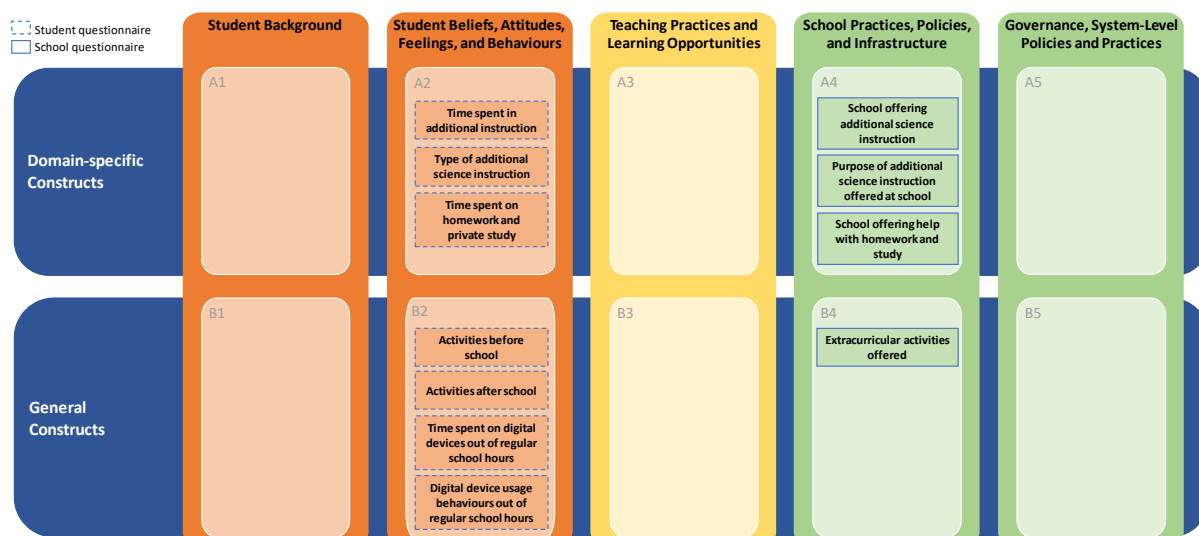
With regards to other activities students take part in before and after school, there is evidence that participating in physical activity reduces depression and anxiety disorders, and boosts self-esteem; it also appears to improve memory, perseverance and self-regulation (Biddle and Asare, 2011<sup>[254]</sup>). Similarly, eating habits, and eating breakfast in particular, are related to physical and psychological well-being, and also to better academic performance (Adolphus, Lawton and Dye, 2013<sup>[255]</sup>). Students who spend more of their leisure time working for pay can gain new experience and knowledge, explore career options, and earn money, but they may also spend less time on studying and on leisure activities (Robert Warren and Lee, 2003<sup>[256]</sup>; Kalenkoski and Pabilonia, 2012<sup>[257]</sup>).

In addition, students' use of digital devices outside of regular school hours for learning or leisure is considered in this module. Digital activities might be separated into those that might improve and those that hinder students' academic outcomes and well-being. Behaviours such as extreme video gaming, compulsive texting and overuse of smartphones can have negative consequences for students, including less physical activity, sleeping disorders and depression (Currie et al., 2012<sup>[258]</sup>; Kim et al., 2006<sup>[259]</sup>; Punamäki et al., 2007<sup>[260]</sup>; Smahel, Wright and Cernikova, 2015<sup>[261]</sup>; Fraillon et al., 2014<sup>[262]</sup>).

Domain-specific constructs included in the PISA 2025 STQ are time spent in additional instruction outside of regular school hours; time spent on homework and private study; providers of additional science instruction; and type of additional science instruction. The domain-specific construct included in the SCQ is the school offering additional lessons in science. General constructs included in the STQ cover activities before and after school, digital device usage behaviours outside of regular school hours and time spent on digital devices outside of regular school hours. General constructs included in the SCQ relate to school offering help with homework and private study, and extracurricular activities offered by the school.

Figure 4.13 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.13. Module 10: Experiences out of regular school hours



### Module 11: School type and resources

This module examines school-level characteristics that shape students' learning environment, including the school's geographic location (i.e. rural or urban), type (i.e. public or private), its sources of funding, and the availability of human resources (i.e. teachers and non-teaching staff) and material resources (i.e. physical infrastructure, educational materials and digital resources) in school.

In PISA, the geographic location of a school is defined in terms of the number of people living in a specific locality. Schools located in communities with fewer than 3 000 inhabitants are typically considered rural schools in PISA analysis whereas urban schools are those located in areas with more than 100 000 inhabitants. The evidence about whether rural or urban schools provide more opportunities or risk for students' academic performance is mixed (Reeves and Bylund, 2005<sup>[263]</sup>; Schafft and Jackson, 2010<sup>[264]</sup>). While some studies find that students in rural schools display less motivation for learning than those in urban schools (Iwaniec, 2020<sup>[265]</sup>), other research shows that rural schools have fewer disciplinary problems, greater participation in extracurricular activities and more time for individual student-teacher interactions than schools in urban settings (Freeman and Anderman, 2005<sup>[266]</sup>; Knoblauch and Woolfolk Hoy, 2008<sup>[267]</sup>). In PISA 2018, principals in urban schools were more likely than those in rural schools to report that teacher behaviours hinder learning, while shortages of educational staff were more prevalent in rural schools than in urban schools (OECD, 2020<sup>[268]</sup>).

School type is defined in PISA based on whether a school is public (i.e. managed by a public education authority, government agency or governing board appointed by the government or elected by public franchise) or private (i.e. managed by a non-government organisation such as a church, trade union, business or other private institution). In addition, PISA distinguishes between two types of private schools, based on their level of public funding. Government-independent private schools are those funded mainly through student fees or other private contributions (e.g. benefactors, donations); government-dependent private schools are privately managed schools that receive more than half of their funding from government sources (OECD, 2020<sup>[268]</sup>).

School choice policies have been the focus of debate in many countries (Adamson, Astrand and Darling-Hammond, 2016<sup>[269]</sup>; Koinzer, Nikolai and Waldow, 2017<sup>[270]</sup>). Advocates of school choice argue that competition between schools provides incentives for schools, both public and private, to improve their quality. However, school choice can increase pre-existing inequities in education when socio-economically disadvantaged families are more constrained in their choice of school than advantaged families (Rowe and

Lubienski, 2017<sup>[271]</sup>; OECD, 2020<sup>[268]</sup>). PISA 2018 found that students in public schools score higher in reading than students in private schools on average across OECD countries after socio-economic status is taken into account. Across all countries and economies, school systems with larger shares of students in private-independent schools tended to show lower mean performance in reading, mathematics and science after accounting for per capita GDP (OECD, 2020<sup>[268]</sup>).

Of all the school resources that are needed to foster students' learning and well-being, teachers are perhaps the most important (OECD, 2020<sup>[268]</sup>). In addition to teachers, a range of other professionals, including social workers or psychologists, play an important role in school life (OECD, 2019<sup>[272]</sup>). If schools do not have a sufficient number of teachers and educational staff, improving the quality of education is unlikely. Teacher shortage is associated with a decrease in student performance (Marks, 2010<sup>[273]</sup>; Sterling, 2004<sup>[274]</sup>). By contrast, effective teacher policies can be the foundation on which to build successful education systems (OECD, 2005<sup>[275]</sup>).

Material resources available to teachers and students are important components of a high-quality education. In order to enhance student learning, educational materials and school infrastructure need to meet at least three conditions: they have to be available, of appropriate quality and be used effectively (OECD, 2020<sup>[268]</sup>). Teachers need educational materials such as textbooks or computers to provide instruction that is up-to-date, challenging and responsive to students' needs (Murillo and Román, 2011<sup>[276]</sup>). In addition, a school environment that is conducive to teaching and learning requires adequate buildings, grounds, heating and cooling systems, lighting and acoustic systems, and ventilation (Conlin and Thompson, 2017<sup>[277]</sup>; Gunter and Shao, 2016<sup>[278]</sup>; Neilson and Zimmerman, 2014<sup>[279]</sup>).

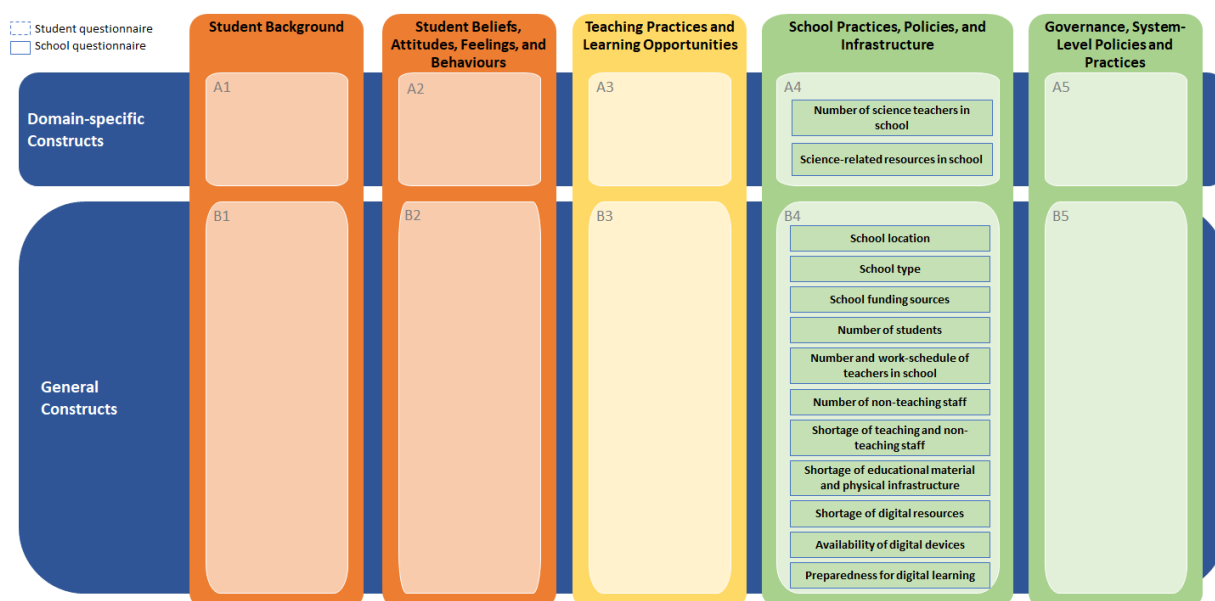
Computers and other digital devices are important pedagogical tools and help prepare students for today's digital world (Barrow, Markman and Rouse, 2009<sup>[280]</sup>). Nevertheless, the "digital divide" in education remains a concern for researchers and policymakers (Dolan, 2016<sup>[281]</sup>). Socio-economic disparities in access and use of digital devices were highlighted throughout the COVID-19 pandemic when schools had to shift to online learning. The movement online of essential activities during COVID-19 took inadequate Internet service from an inconvenience to a crisis for many households, especially socio-economically disadvantaged families (Lai and Widmar, 2021<sup>[282]</sup>; Nguyen, Hargittai and Marler, 2021<sup>[283]</sup>).

This module also includes the trend construct "preparedness for digital learning", which reflects schools' overall capacity to effectively support teaching and learning using digital devices. The construct encompasses the availability, quality and suitability of digital infrastructure (e.g. number of devices, Internet connectivity, computing power and software), as well as the organisational conditions that enable meaningful digital integration in instruction. These conditions include teachers' technical and pedagogical skills, time for preparation, access to professional learning resources, incentives for using digital tools, alignment of the curriculum with digital practices, and the availability of technical support staff. Taken together, these elements indicate how well a school is equipped – both materially and organisationally – to enhance teaching and learning through digital technologies.

General constructs in the PISA 2025 SCQ include school location, school type, school funding sources, number of students, number and work schedule of teachers, number of non-teaching staff, shortage of teaching and non-teaching staff, shortage of educational material and physical infrastructure, availability and shortage of digital devices and resources, and preparedness for digital learning. Domain-specific constructs in the SCQ include the number of science teachers and the availability of science-related resources at school.

Figure 4.14 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.14. Module 11: School type and resources



### Module 12: Selection and enrolment

The way in which students are channelled into different schools is an important aspect of school organisation and policy. The pros and cons of school selectivity based upon students' prior academic achievement or other factors has been debated in educational research (Jerrim and Sims, 2019<sup>[284]</sup>). In PISA, school selectivity encompasses the factors that are considered when students are admitted to the school (e.g. academic performance, residence, disciplinary record or cultural background) (OECD, 2020<sup>[268]</sup>; OECD, 2016<sup>[285]</sup>). A study by Jerrim and Sims (2019<sup>[284]</sup>) found that selective systems increase the socio-economic inequalities in educational expectations due to the divergence between those who are and are not admitted to the selective track. Furthermore, the mental health and well-being of young people may also suffer due to the stressful nature of the selection process; for example, going through an academic selection process may influence the way that students view their own abilities and the way they approach school (Ahmavaara and Houston, 2007<sup>[286]</sup>).

School competition, defined in PISA as the availability of different school options for students, has been controversial in many countries. While some argue that school competition provides incentive for school improvement, others find that it leads to school socio-economic segregation (Adamson, Astrand and Darling-Hammond, 2016<sup>[269]</sup>; Koinzer, Nikolai and Waldow, 2017<sup>[270]</sup>; Rowe and Lubienski, 2017<sup>[271]</sup>; OECD, 2020<sup>[268]</sup>). The PISA 2018 results show that schools whose principal reported that one or more schools in the same area compete for students tended to be socio-economically advantaged schools, located in urban areas and privately managed. In most countries, and on average across OECD countries, school competition was associated with higher reading scores, but this difference is not observed after accounting for students' and schools' socio-economic profile (OECD, 2020<sup>[268]</sup>).

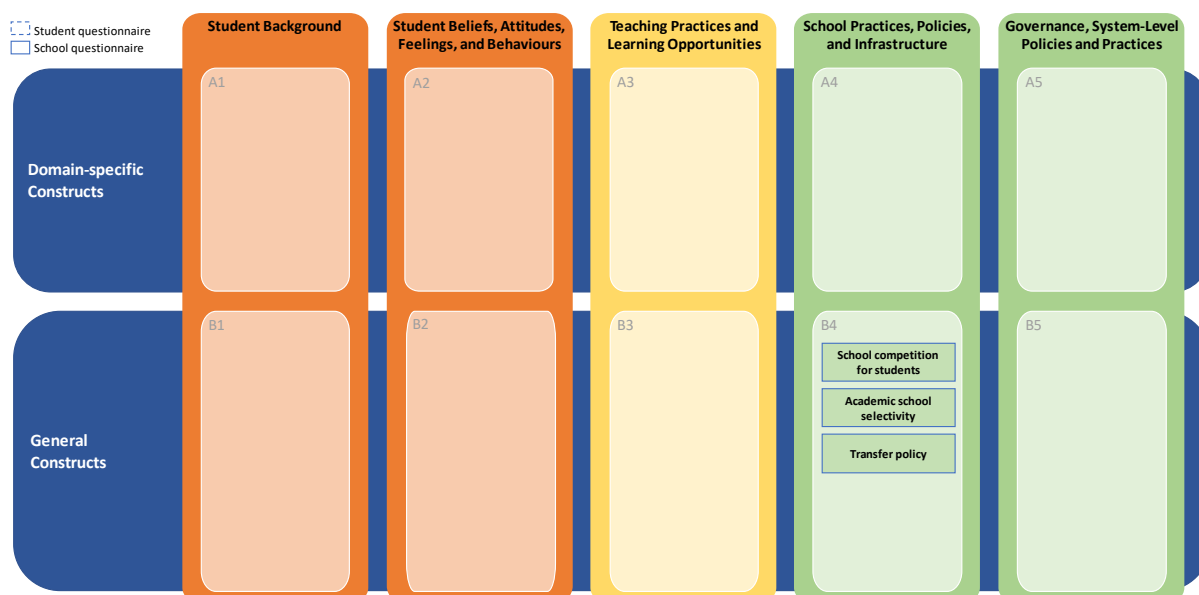
Transferring students out of school because of low academic achievement, behavioural problems or special learning needs is one way for schools to reduce heterogeneity in the learning environment and to facilitate instruction for the remaining students (OECD, 2013<sup>[287]</sup>). However, research suggests that student transfer policies and school segregation are closely interrelated (Allen and West, 2009<sup>[288]</sup>; Fernández Llera and Muniz Perez, 2012<sup>[289]</sup>). In PISA 2012, across OECD countries, systems with more education programmes, more students enrolled in vocational programmes, more academically selective schools and which select students into different programmes at an earlier age tended to have a greater incidence of

school transfers. In highly stratified systems (i.e. those with greater and earlier differentiation in academic programmes), there may be more incentives for schools to select the best students and fewer incentives to support difficult students if there is an option of transferring them to other schools (OECD, 2013<sup>[287]</sup>). In contrast, in comprehensive systems (i.e. those with smaller and later differentiation in academic programmes), schools must find ways of working with students from across the performance spectrum. These different incentive systems may help explain the greater level of equity achieved in systems that use stratification less (OECD, 2013<sup>[287]</sup>; OECD, 2011<sup>[290]</sup>).

General constructs in the SCQ include school competition for students, academic school selectivity, and transfer policy. This module overlaps conceptually with Module 3 (educational career) and Module 11 (school type and resources).

Figure 4.15 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.15. Module 12: Selection and enrolment**



### Module 13: School autonomy

Education systems differ in the degree of autonomy that is granted to schools. Since the early 1980s, many school systems such as those in Australia, Canada, Finland, Hong Kong (China), Israel, Singapore, Spain, Sweden and the United Kingdom have given schools greater authority to make decisions about curricula and resource allocation (Cheng, Ko and Lee, 2016<sup>[291]</sup>; Fuchs and Wößmann, 2007<sup>[292]</sup>; Wößmann, 2007<sup>[293]</sup>; Wang, 2013<sup>[294]</sup>; Wermke, Olason Rick and Salokangas, 2019<sup>[295]</sup>; Liu et al., 2020<sup>[296]</sup>). The underlying premise is that teachers, principals and other school staff are good at assessing their students' learning needs and can design and implement curricula, internal evaluations and accountability mechanisms for their schools (Caldwell and Spinks, 2013<sup>[297]</sup>; Hanushek, Link and Woessmann, 2013<sup>[298]</sup>). Such school-based autonomy often involves increasing principals' decision-making responsibility and accountability and, in some cases, the management responsibilities of teachers (OECD, 2016<sup>[285]</sup>).

In PISA, responsibility for school decisions is measured by asking school principals to report whether the teachers, the principal, the school's governing board, the regional or local education authorities, the national education authority or a combination of them have considerable responsibility for allocating

resources to schools (e.g. appointing and dismissing teachers; determining teachers' salaries) and for the school curriculum (e.g. choosing learning material; deciding which courses are offered).

In PISA 2015, students in education systems where principals or teachers hold greater autonomy in managing their schools score higher in science (particularly when they have greater responsibility for the curriculum) (OECD, 2016<sup>[285]</sup>). However, more school autonomy may not always be effective (Hanushek, Link and Woessmann, 2013<sup>[298]</sup>). For instance, PISA 2015 showed that more autonomy for schools and teachers was not positively associated with equity in science performance; in fact, results in science were more equitable when education authorities have greater responsibility for disciplinary policies (OECD, 2016<sup>[285]</sup>; Christ and Dobbins, 2016<sup>[299]</sup>).

Some studies find that in order to reap the full benefits of school autonomy, education systems need to have effective accountability systems to discourage opportunistic behaviour by school staff. It also requires highly qualified teachers and strong school leaders to design and implement internal evaluations and curricula (Hanushek, Link and Woessmann, 2013<sup>[298]</sup>; Torres, 2021<sup>[300]</sup>; OECD, 2011<sup>[301]</sup>). At the country level, more school autonomy in combination with high school accountability is associated with better student performance in PISA but when accountability levels are very low, more school autonomy is associated with worse student performance (Torres, 2021<sup>[300]</sup>). Accountability systems based on student assessments are used in many OECD countries (Rosenkvist, 2010<sup>[302]</sup>; OECD, 2013<sup>[303]</sup>).

General constructs in the PISA 2025 SCQ under this module include decentralisation of responsibility for decision making (including resource allocation and curriculum and assessment) and school management team behaviours. This module shows a conceptual overlap with Module 18 (assessment, evaluation and accountability).

Figure 4.16 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.16. Module 13: School autonomy**



## Module 14: Organisation of student learning at school

During time spent in the classroom, students are exposed to subject content, curriculum materials, instructional strategies, and a diversity of learning experiences. Learning time and the intended curriculum content in school have been found to be closely related to student outcomes (Abedi et al., 2006<sup>[304]</sup>; Scherff

and Piazza, 2008<sup>[305]</sup>; Schmidt and Maier, 2009<sup>[306]</sup>). Student learning time and achievement are correlated, yet there are heterogeneous effects across countries and within countries among different groups of students and schools (Cattaneo, Oggenfuss and Wolter, 2017<sup>[307]</sup>; Ghuman and Lloyd, 2010<sup>[308]</sup>; OECD, 2011<sup>[309]</sup>; Martin, Mullis and Foy, 2008<sup>[310]</sup>; Schmidt et al., 2001<sup>[311]</sup>; Schmidt and Burroughs, 2016<sup>[312]</sup>; Schmidt et al., 2015<sup>[313]</sup>). Furthermore, students of high ability might require less time to achieve the same knowledge as do those with lower ability (Carroll, 1989<sup>[245]</sup>). Research shows that additional learning time can have diminishing positive effects on student performance (Long, 2014<sup>[314]</sup>; Gromada and Shewbridge, 2016<sup>[239]</sup>). For example, PISA 2018 showed that the positive association between learning time in regular language-of-instruction lessons and reading performance weakened among students who spend more than three hours per week in these lessons (OECD, 2020<sup>[30]</sup>).

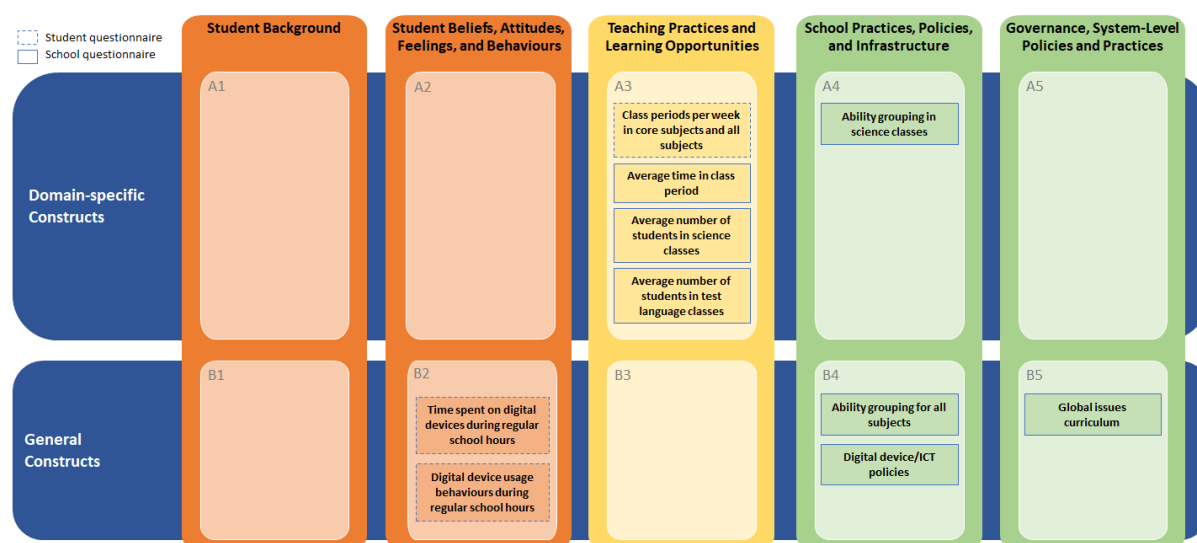
Intended learning content (i.e. the curriculum) may vary largely across grades, courses, schools and countries (Schmidt et al., 2001<sup>[311]</sup>; Martin, Mullis and Foy, 2008<sup>[310]</sup>). There may be misalignments between the curriculum designed at the system level, the curriculum communicated by the teacher or in the textbook, and the curriculum as understood by students and their parents. The implementation of the school's curriculum can be understood empirically by examining whether students are grouped between or within classes by ability; and how often schools hold back students in the same grade. In addition, PISA 2025 continues to examine the extent to which global issues are included in the school curriculum, a topic first introduced in PISA 2018. The school questionnaire includes items on climate change and global warming, migration, international conflicts, hunger and malnutrition, and the causes of poverty.

Domain-specific constructs in the STQ include students' class periods per week in selected subjects (i.e. science, mathematics, reading and foreign language) and all subjects. Domain-specific constructs in the SCQ include administrators' reports of the average time in a class period (in science and across all subjects); the average number of students in science classes and in test-language classes; the school policy on ability grouping for all subjects; and student-ability grouping in science classes. General constructs included in this module are, for the STQ, digital device usage behaviours during regular school hours, and, for the SCQ, ability grouping for all classes, digital policies and global issues curriculum.

This module complements Module 10 (experiences out of regular school hours), Module 15 (exposure to science content) and Module 16 (science teacher behaviours) in mapping out a broad view of students' opportunity to learn (OTL) at school and outside of school.

Figure 4.17 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.17. Module 14: Organisation of student learning at school**



### **Module 15: Exposure to science content**

This module focuses on students' exposure to relevant science content and scientific practices. As such, it focuses on the first two types of opportunity-to-learn-related variables described by Stevens (1993<sup>[315]</sup>): content coverage variables (i.e. which sciences students are taught and whether they are compulsory or optional); and content emphasis variables (i.e. which topics and practices within the curriculum are selected for emphasis and whether students receive instruction which emphasises lower-order skills (i.e. rote memorisation) or higher-order skills (i.e. critical problem-solving)).

With regards to science content coverage, a basic requirement is to ascertain what kind of science course students have taken, and whether students had a choice in the range of science courses. Thus, it is important for PISA 2025 to measure which sciences students are taught and whether they are compulsory or optional. "Science" is a generic term and may be offered by one teacher teaching all sciences in an integrated or co-ordinated manner or, alternatively, separate teachers teaching a range of courses e.g. physics, chemistry, biology, earth sciences, environmental science. In addition, some countries offer students the opportunity to choose one or more sciences at different ages while in other countries students study all sciences either separately or in an integrated or co-ordinated form. PISA 2025 included questions on these constructs in the school questionnaire. From these data, it will be possible to construct a picture of whether students are being offered an education across all the sciences or a curtailed offering limited to certain sciences.

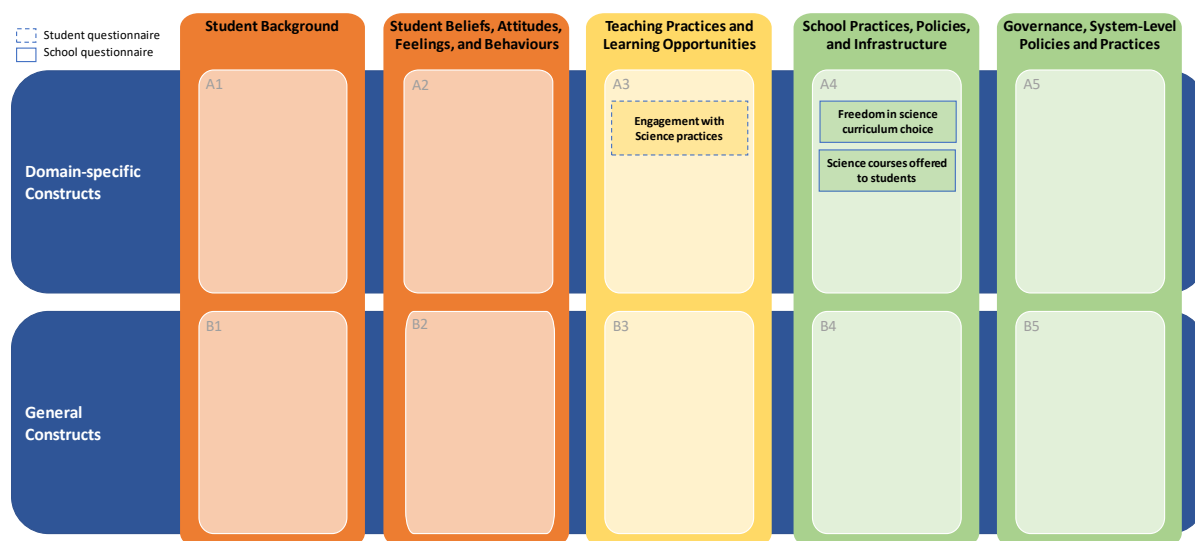
A specific area for new development in PISA 2025 is around students' opportunities to engage in scientific reasoning. In science courses, students can be exposed to activities such as analysing and interpreting data; constructing explanations; examining a scientific argument to identify a claim or the evidence used to support it. Only by engaging in such practices can students begin to become literate in science in the fundamental sense (Lemke, 1990<sup>[316]</sup>; Norris and Phillips, 2003<sup>[317]</sup>; van Lier, 1996<sup>[318]</sup>). Scientific practices that are demanding and require higher scientific reasoning are ones that require students to synthesise disparate facts to provide a coherent explanation of a phenomenon or identify flaws in an argument for an experimental design or the interpretation of a set of data. Science practices with lower reasoning are those that require the ability to recall relevant information. The goal of PISA here is to identify the extent to which students' learning of science has provided them with opportunities to engage in scientific practices and to use the knowledge that they have acquired.

Moreover, there is a fundamental aspect to scientific literacy that requires the acquisition of a fluency with a disciplinary language. Such language is only acquired through practice. PISA will seek to ascertain what kind of reading, writing and verbal opportunities are provided for students in their experience of learning science. This question will focus on whether students engage in language-based activities including, for example, turn and talk, organising presentations, writing for audiences other than the teacher, and obtaining and evaluating information from websites.

The domain-specific construct included in the STQ is engagement with scientific practices (including hands-on experiments, argumentation and disciplinary language). Domain-specific constructs in the SCQ include science courses offered to students and freedom in science curriculum choice.

Figure 4.18 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.18. Module 15: Exposure to science content



### Module 16: Science teacher behaviours

How student learning is organised (Module 14) and what content is being taught (Module 15) are conceptually distinct from constructs that capture teaching practices and behaviours (instructional quality). Teaching practices and behaviours can serve as vehicles through which different levels of content coverage and exposure may occur. What teachers do has the strongest direct school-based influence on student learning outcomes (Hattie, 2009<sup>[319]</sup>; Slavin and Lake, 2008<sup>[320]</sup>; Wang, Haertel and Walberg, 1993<sup>[321]</sup>).

Though understood differently across the field, there is general agreement that instructional quality is a multidimensional concept (Fauth et al., 2014<sup>[322]</sup>; Kane and Cantrell, 2010<sup>[323]</sup>). The 2018 OECD Teaching and Learning International Survey (TALIS) Framework (Ainley and Carstens, 2018<sup>[324]</sup>) identifies the following dimensions of teaching practices as having an influence on student achievement: classroom management (van Tartwijk and Hammerness, 2011<sup>[325]</sup>); teacher support (Klieme, Pauli and Reusser, 2009<sup>[326]</sup>); clarity of instruction (Hospel and Galand, 2016<sup>[327]</sup>; Kane and Cantrell, 2010<sup>[323]</sup>; Seidel, Rimmele and Prenzel, 2005<sup>[328]</sup>); cognitive activation (Lipowsky et al., 2009<sup>[329]</sup>); and instructional assessment and feedback (Hattie and Timperley, 2007<sup>[330]</sup>; Kyriakides and Creemers, 2008<sup>[331]</sup>; Scheerens, 2016<sup>[332]</sup>).

While TALIS has focused on measurement of general teaching practices, PISA complements these efforts by measuring closely aligned constructs that are domain-specific, as has been done in previous cycles. In PISA 2025, questions about instructional practices are framed within the context of school science, conceptualised as referring to one specific course. The PISA 2025 Science Framework emphasises that the purpose of science teaching is to foster students' capacity to explain phenomena scientifically; understand scientific enquiry and interpret scientific data and evidence critically; and research, evaluate and use scientific information for decision making. This questionnaire module is intended to probe the extent to which schools' and teachers' practice are consistent with these purposes. The aim is to be able to describe science teaching in the classroom to generate country-specific profiles of how science is taught, and to allow an investigation of the relationship between instructional practices in science classrooms and students' learning and affective outcomes.

Analyses of PISA results in 2006 and 2015, when science was the main domain of assessment, have shown that items and scales for science teaching practices can be used to provide a profile of the patterns of science teaching in the classroom. Moreover, student performance and affective outcomes such as students' interest in science topics can be related to particular profiles of teaching practices (DeNoel et al.,

2017<sup>[333]</sup>; Kobarg et al., 2011<sup>[334]</sup>; Prenzel, Seidel and Kobarg, 2011<sup>[335]</sup>; Jiang and McComas, 2015<sup>[336]</sup>; Kang and Keinonen, 2017<sup>[337]</sup>; Oliver, McConney and Woods-McConney, 2019<sup>[338]</sup>; Liou, 2020<sup>[339]</sup>).

Constructs developed for this module represent the broad areas described in the TALIS Framework. The construct “disciplinary climate in science” examines disciplinary issues that hinder science learning in the classroom, complementing the TALIS dimension of classroom management. The construct “science teacher support” measures how often science teachers help students with their learning and whether they show an interest in every student’s learning (OECD, 2004<sup>[340]</sup>). The construct “science teacher feedback” is conceptually similar to the TALIS dimension of instructional assessment and feedback.

For the measure of cognitive activation, it is assumed that the level of cognitive challenge is determined by the type of problem and the way it is presented in the lesson. A series of constructs linked to features of inquiry emphasised by teachers were recommended for PISA 2025. These constructs draw on Furtak et al.’s (2012<sup>[341]</sup>) procedural, epistemic, conceptual and social inquiry domains<sup>3</sup>:

- Collaborative reasoning: The extent to which the experience of teaching makes use of collaborative discussion and decision making based on small group work.
- Procedural domain: Work that focuses on investigative procedures and reporting and presenting the findings. (This construct was not implemented in the PISA 2025 Main Study.)
- Scientific epistemic reasoning: Work that focuses on a critical examination of evidence and its support or not for scientific claims. (This construct was not implemented in the PISA 2025 Main Study.)
- Scientific conceptual reasoning: Work that focuses on support for problem solving, constructing explanations, developing models and reasoning using scientific concepts. (This construct was not implemented in the PISA 2025 Main Study.)

In addition, to what extent does the scientific thinking relate to everyday life contexts and contemporary issues such energy production and use, resource extraction and depletion, lifestyle choices or any other science that students perceive to be important to their lives (Scientific thinking for everyday life).

Engaging students in undertaking scientific practices plays a significant role in the quality of science education. Research shows that guided inquiry-based instruction can not only improve student achievement (Blanchard et al., 2010<sup>[342]</sup>) but attitudes such as engagement with the subject (Furtak et al., 2012<sup>[341]</sup>; Potvin and Hasni, 2014<sup>[343]</sup>; Savelsbergh et al., 2016<sup>[344]</sup>; Cairns and Areepattamannil, 2017<sup>[345]</sup>) and transferable critical thinking skills (Hattie, 2009<sup>[319]</sup>) as well. In PISA 2015, the OECD analysis suggested that greater exposure to inquiry-based instruction was negatively associated with science performance in most participating countries and economies, even after accounting for student and school socio-economic profile (yet caution was advised in interpreting this finding because selection bias or other factors could be playing a role). Other systematic research suggests that the optimum performance was achieved with a mix of “direct” instruction and “guided inquiry” (Denoël et al., 2017<sup>[346]</sup>; Teig, Scherer and Nilsen, 2018<sup>[347]</sup>; Forbes, Neumann and Schiepe-Tiska, 2020<sup>[348]</sup>).

Moreover, research suggests that inquiry-based instruction and teacher-directed instruction should not be measured as categorically different constructs because they have in common a number of instructional practices including guided classroom discussion, discussion of student questions, and guided practical investigative work (Lau and Lam, 2017<sup>[349]</sup>; Liou, 2020<sup>[339]</sup>; Oliver, McConney and Woods-McConney, 2019<sup>[338]</sup>; Lazonder and Harmsen, 2016<sup>[350]</sup>). In PISA 2025, teacher guidance of student inquiry and reasoning is represented by the four distinct constructs on procedural, epistemic, conceptual and social inquiry domains mentioned above.<sup>3</sup> This allows a closer analysis of the effects of particular instructional approaches without the implication that these can be combined into a singular “inquiry” construct distinct from “direct instruction”.

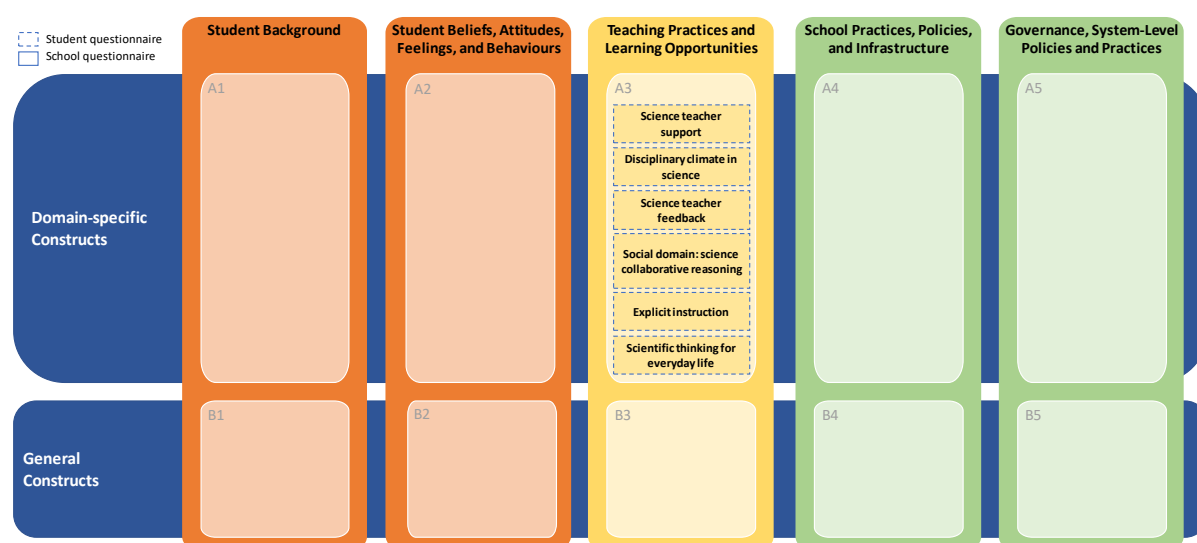
There is a continuing interest in embedding science teaching and learning in contexts that are real and meaningful for learners (Fensham, 2009<sup>[351]</sup>). Scientific argumentation as a central goal of science

education (Osborne, 2011<sup>[352]</sup>) requires classroom situations with sufficient opportunities for social interaction. Instruction that emphasises students' active thinking and drawing conclusions from data seems to be particularly beneficial for students' development (Minner, Levy and Century, 2010<sup>[353]</sup>).

Domain-specific constructs in the STQ include the following: disciplinary climate in science; scientific thinking for everyday life; science teacher support; science teacher feedback; social domain: collaborative reasoning; and explicit instruction.

Figure 4.19 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.19. Module 16: Science teacher behaviours**



### **Module 17: Teacher qualification, training and professional development**

The OECD's annual International Summit on the Teaching Profession (ISTP) (Schleicher, 2021<sup>[354]</sup>) has exemplified the continuously growing focus on teacher-related policies for improving the quality of teachers, teaching and learning. In addition to teacher's professional behaviour (e.g. interactions with students in the classroom and with their parents or guardians), the composition of the teaching force in terms of age and educational level, their initial education and qualifications, their individual beliefs and competencies, as well as professional practices at the school level (e.g. professional development, interactions with parents) have been topics of education policy discussions.

A number of studies have demonstrated a clear influence of teacher-related factors on student learning and outcomes (e.g. (Schmidt et al., 2016<sup>[355]</sup>; Hattie, 2009<sup>[319]</sup>; Ing et al., 2015<sup>[356]</sup>)). Shulman's model has been most influential in research on teachers (e.g. (Hill, Rowan and Ball, 2005<sup>[357]</sup>; Kunter et al., 2013<sup>[358]</sup>; Blömeke, Olsen and Suhl, 2016<sup>[359]</sup>)). Several studies and reviews show positive relationships between teachers' initial education and their teaching effectiveness (e.g. (Boyd et al., 2009<sup>[360]</sup>; Darling-Hammond et al., 2005<sup>[361]</sup>)). Research has shown that when teachers have opportunities to expand and develop their teaching practices and their knowledge of instructional approaches, they are more likely to provide a broader range of learning opportunities for students and be more effective in improving students' learning outcomes (Heller et al., 2012<sup>[362]</sup>; Roth et al., 2018<sup>[363]</sup>)).

General constructs in the SCQ include administrators' reports of teacher qualifications and professional development opportunities. Domain-specific constructs include science teacher qualifications and areas of professional development specifically for science teachers.

Figure 4.20 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.20. Module 17: Teacher qualification, training, and professional development



### Module 18: Assessment, evaluation and accountability

Evaluation and assessment practices are crucial for both improvement and accountability in school systems. They are used to understand how well students are learning, provide information about educational performance to parents and society, and to improve schools and teaching practices (OECD, 2013<sup>[303]</sup>). Reporting and sharing data from assessments and evaluations with different stakeholders provides multiple opportunities for monitoring, feedback, quality management and improvement.

Student assessments can have different purposes (OECD, 2013<sup>[303]</sup>; OECD, 2020<sup>[268]</sup>; Looney, 2011<sup>[364]</sup>). Formative assessments, also known as assessments for student learning, do not have consequences for students' grades but instead aim to provide feedback to improve student learning (Hattie and Timperley, 2007<sup>[330]</sup>; Shepard, 2000<sup>[365]</sup>). The use of formative assessment has been shown to enhance students' achievement (Baird et al., 2014<sup>[366]</sup>; Black, 2015<sup>[367]</sup>; Hattie, 2009<sup>[319]</sup>); (Black and Wiliam, 1998<sup>[368]</sup>; Andersson and Palm, 2017<sup>[369]</sup>; Cauley and McMillan, 2010<sup>[370]</sup>). However, there is a large variation in the implementation of formative assessment practices as reported in recent studies in the United States, Canada, Sweden, Scotland and Norway, among others (DeLuca et al., 2015<sup>[371]</sup>; Hopfenbeck, Flórez Petour and Tolo, 2015<sup>[372]</sup>; Jonsson, Lundahl and Holmgren, 2015<sup>[373]</sup>; Hayward, 2015<sup>[374]</sup>; Wylie and Lyon, 2015<sup>[375]</sup>). Formative assessment can also take the form of low-stakes national assessments; however, there are very few low-income countries that have a national assessment system that can evaluate learning in a standardised way to provide feedback for education policies and programmes (Birdsall, Bruns and Madan, 2016<sup>[376]</sup>).

Traditionally, educational evaluation primarily focused on the assessment of students. Many countries are developing broader evaluation and assessment of schools (OECD, 2013<sup>[303]</sup>). School evaluation is used as a means of ensuring transparency, facilitating policy decisions and guiding overall school development (Faubert, 2009<sup>[377]</sup>). Evaluation can either be external (i.e. the process is controlled and headed by an external body and the school does not define the areas that are judged) or internal (i.e. the process is controlled by the school itself and the school defines the areas that are judged) (Berkemeyer and Müller, 2010<sup>[378]</sup>).

In many countries, evaluation of individual teachers and principals, separate from school-wide evaluation, is also common (Faubert, 2009<sup>[377]</sup>). Feedback to teachers can help them identify how to better shape and improve their teaching practice (OECD, 2013<sup>[303]</sup>). Research shows that providing teachers with constructive feedback based on their teaching might have the largest impact on student performance of

any school intervention (Hattie, 2009<sup>[319]</sup>). In line with this, TALIS 2018 finds that almost three-quarters of teachers who received feedback on their work found it useful for improving their practice. These results suggest that a culture of regular feedback can help teacher and school improvement. Appraisal systems (such as financial bonus, learning opportunities or other rewards) can ensure that teachers feel recognised for their efforts (Schleicher, 2020<sup>[379]</sup>; OECD, 2020<sup>[380]</sup>).

Accountability systems based on assessment instruments are increasingly common in OECD countries (Rosenkvist, 2010<sup>[302]</sup>; Scheerens, 2004<sup>[381]</sup>). PISA asks school principals whether their school uses achievement data for accountability purposes, and if so, how. Three forms of school accountability are considered: providing achievement data to parents directly; posting achievement data publicly (e.g. in the media); and tracking achievement data over time by an administrative authority. Accountability systems are an important means of ensuring school quality, especially when granting more autonomy to schools (Torres, 2021<sup>[382]</sup>; OECD, 2011<sup>[301]</sup>).

Domain-specific constructs in the PISA 2025 SCQ include principals' reports of the use of science achievement data in accountability systems, whereas general constructs include use of assessments; school evaluation; monitoring teacher practices; feedback to teachers; use of social and emotional learning assessments; and the use of social and emotional learning data in accountability systems. Constructs in this module are closely related to Module 13 (school autonomy).

Figure 4.21 below illustrates how all constructs in this module map on the taxonomy.

**Figure 4.21. Module 18: Assessment, evaluation and accountability**



### **Module 19: Parental involvement and support**

Parental involvement in education has been conceptualised as parents' interactions with schools and their children to encourage academic success (Hill and Tyson, 2009<sup>[383]</sup>). This includes school-based involvement (e.g. attending parent-teacher meetings or volunteering at school), home-based involvement (e.g. assisting with homework), and academic socialisation (e.g. parents' or guardians' educational goals and expectations for their children) (LaRocque, Kleiman and Darling, 2011<sup>[384]</sup>; Epstein, 2018<sup>[385]</sup>; Kim and Hill, 2015<sup>[386]</sup>; Hill and Tyson, 2009<sup>[383]</sup>). Getting involved at school allows parents to obtain information on their child's learning environment and process as well as to learn how to navigate the education system. It also demonstrates to their child that education is important (Grolnick and Slowiack, 1994<sup>[387]</sup>; OECD, 2019<sup>[194]</sup>).

Previous studies have found that parental involvement in their child's education has a positive effect on student achievement (Castro et al., 2015<sup>[388]</sup>; Hill and Taylor, 2004<sup>[389]</sup>; Jeynes, 2012<sup>[390]</sup>; Lara and Saracostti, 2019<sup>[391]</sup>), social skills (El Nokali, Bachman and Votruba-Drzal, 2010<sup>[392]</sup>), attendance (Avvisati et al., 2014<sup>[393]</sup>), reduced behavioural problems (Domina, 2005<sup>[394]</sup>; El Nokali, Bachman and Votruba-Drzal, 2010<sup>[392]</sup>), relationships with peers (Garbacz et al., 2018<sup>[395]</sup>) and mental health; more specifically, reduced depressive symptoms (Wang and Sheikh-Khalil, 2014<sup>[396]</sup>), anxiety, attention difficulties and eating disorders (Wang et al., 2019<sup>[397]</sup>).

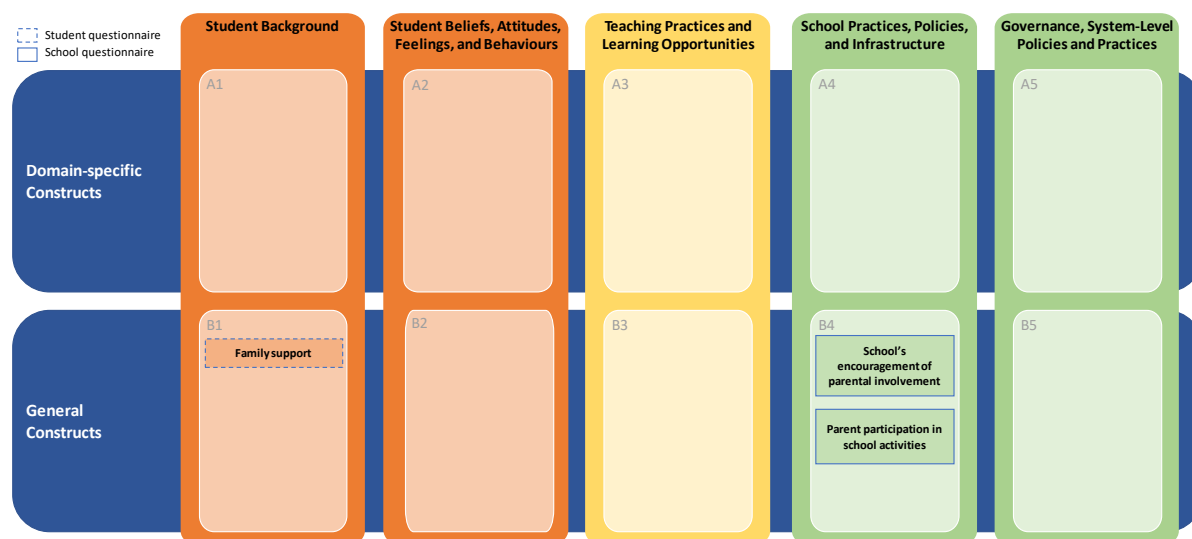
Socio-economically advantaged schools tend to have more parental involvement at school. However, the incidence of parental involvement varies by whether the participation is initiated by parents or school teachers. PISA 2018 found that, on average across OECD countries, parents discussing their child's progress was more common in socio-economically advantaged schools when the initiative was taken by parents and in disadvantaged schools when the initiative was taken by teachers (OECD, 2019<sup>[194]</sup>). Research shows that efforts to foster parental involvement at school can bolster student achievement in socio-economically disadvantaged schools (Avvisati et al., 2014<sup>[393]</sup>; Domina, 2005<sup>[394]</sup>; Gubbins and Otero, 2020<sup>[398]</sup>).

In addition to parents' involvement in school activities, family support plays an important role in fostering student learning. In PISA, family support encompasses discussing how well the child is doing at school; eating the main meal together; talking about the importance of finishing school and about problems at school; showing interest in the child's grades and learning; talking about future education; and asking about what they did at school that day. Parents' engagement with their 15-year-olds is strongly associated with better performance and more academic resilience in PISA (OECD, 2019<sup>[399]</sup>). Several meta-analyses show a positive relationship between parental involvement in education and student achievement, and parents' academic socialisation of their children was found to have a strong positive relationship with achievement (Fan and Chen, 2001<sup>[400]</sup>; Hill and Tyson, 2009<sup>[383]</sup>; Kim and Hill, 2015<sup>[386]</sup>). This correlation generally holds across race and ethnicity and when accounting for socio-economic differences (Jeynes, 2012<sup>[390]</sup>; Kim and Hill, 2015<sup>[386]</sup>). Meta-analytic evidence further showed that the strongest associations between parental involvement and student achievement are found when the families have high academic expectations for their children<sup>4</sup>, develop and maintain communication with them about school activities, and help them to develop reading habits (Castro et al., 2015<sup>[388]</sup>).

General constructs in the PISA 2025 SCQ include school's encouragement of parental involvement and parent participation in school activities. The general construct in the STQ under this module is (students' perceptions of) family support.

Figure 4.22 below illustrates how all constructs in this module map on the taxonomy.

Figure 4.22. Module 19: Parental involvement and support



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## Notes

<sup>1</sup> In Switzerland, the difference is not statistically significant after accounting for academic performance at age 15.

<sup>2</sup> A qualitative study of the role of private tutoring among higher secondary students in the science instructional stream in urban Maharashtra, India shows that some forms of tutoring have moved beyond supplementation to become substitutes that supplant regular schooling (Bhorkar and Bray, 2018<sup>[401]</sup>). According to the authors, “decisions regarding private supplementary tutoring (PST) began to take precedence, and students spent more time in coaching classes than in their mainstream institutions. Also, curricular content and subjects available in PST but not mainstream institutions, particularly concerning the entrance examinations to tertiary institutions, gained more focus”. According to some studies, a similar phenomenon of substitution or supplantation is occurring in countries such as Egypt, Myanmar and Türkiye (Hartmann, 2008<sup>[402]</sup>; Liu and Bray, 2020<sup>[405]</sup>; Altinyelken, 2013<sup>[403]</sup>). While this was not an aspect recommended for data collection in PISA 2025, these studies show that findings on additional instruction and private tutoring should be interpreted carefully, considering the purposes of additional instruction for students in different settings.

<sup>3</sup> Of these four domains, only “social domain: science collaborative reasoning” was implemented in the PISA 2025 Main Study.

<sup>4</sup> Murayama et al. (2016<sup>[404]</sup>) found that parental expectation and children’s achievement are linked positively, but unrealistically high parental expectations were detrimental for children’s achievement.

## Annex A. PISA Mathematics Framework

The PISA mathematics framework received its most recent major update when mathematics was the major domain of assessment in PISA 2022. This updated framework refined and expanded the constructs developed for PISA 2012, which also informed the assessments in 2015 and 2018.

PISA's mathematics assessment measures individuals' capacity to reason mathematically and to formulate, employ and interpret mathematics to solve problems in a variety of real-world contexts. The assessment includes concepts, procedures, facts and tools to describe, explain and predict phenomena. It helps individuals recognise the role that mathematics plays in the world and make the well-founded judgements and decisions needed by constructive, engaged and reflective 21<sup>st</sup>-century citizens.

The updated framework for PISA 2022 aims to consider mathematics in a rapidly changing world driven by new technologies and trends in which citizens are creative and engaged, making non-routine judgements for themselves and the society in which they live. This brings into focus the ability to reason mathematically, which has always been a part of the PISA framework. This technology change is also creating the need for students to understand the computational thinking concepts that are part of mathematical literacy. Finally, the framework recognises that improved computer-based assessment is available to most students within PISA.

The full version of the mathematics framework can be found in the [\*PISA 2022 Assessment and Analytical Framework\*](#).

## Annex B. PISA Reading Framework

The PISA reading framework received its most recent major updates when it was the major domain of assessment in PISA 2018. It retained aspects of the PISA 2009 and 2015 frameworks that were still relevant to PISA 2018 and 2022. However, for PISA 2018 and onwards, the framework was enhanced and revised in the following ways:

- The framework fully integrated reading in a traditional sense with the new forms of reading that have emerged over the past decades and that continue to emerge due to the spread of digital devices and digital texts.
- The framework incorporated constructs involved in basic reading processes. These constructs, such as fluent reading, literal interpretation, inter-sentence integration, extraction of the central themes and drawing inferences, are critical skills for processing complex or multiple texts for specific purposes. If students fail at performing higher-level text processing functions, it is critical to know whether the failure was due to difficulties in these basic underlying skills in order to provide appropriate support to these students.
- The framework revisited the way in which the domain is organised to incorporate reading processes such as evaluating the veracity of texts, seeking information, reading from multiple sources and integrating/synthesising information across sources. The revision rebalances the prominence of different reading processes to reflect the global importance of the different constructs, while ensuring there is a link to the prior frameworks in order to be able to measure trends in achievement.
- The revision considered how new technology options and the use of scenarios involving print and digital text can be harnessed to achieve a more authentic assessment of reading, consistent with the current use of texts around the world.

The full version of the reading framework can be found in the [\*PISA 2018 Assessment and Analytical Framework\*](#).

## Annex C. PISA Foreign Language Assessment Framework

For the first time, the PISA 2025 cycle included an optional assessment of foreign language, which will provide policymakers and educators with comparable results of their students' foreign language competence and allow them to gain insights into the best practices and policies for teaching and learning a foreign language. This assessment will be offered every two PISA cycles, to allow for the analyses of trends. The first cycle started with the assessment of English as a foreign language and focused on three skills: reading, speaking and listening. The coverage of languages and skills may progressively widen in future cycles.

The PISA foreign language assessment framework was developed as the conceptual foundation for this new optional domain. It discusses the relevance of assessing foreign languages and provides the definition of foreign language proficiency that is used to guide the assessment, which underlines the ability to use a foreign language to communicate effectively, requiring a combination of communicative language competences and general competences that allow students to perform communicative language activities (reception, production, interaction and mediation), which involve one or a combination of the skills of reading, listening, speaking or writing. It also requires the activation of appropriate language strategies.

The framework also presents the theory underlying the tests to assess foreign language proficiency by describing the two existing frameworks that the cognitive framework builds on: the Common European Framework of Reference for Languages (CEFR) and the socio-cognitive model of language use based on Weir's 2005 model of test development and validation. It also describes the cognitive processes that learners employ when using a foreign language according to their proficiency level, and discusses how the skills of reading, listening and speaking are assessed.

Further, it outlines the framework for the various questionnaires distributed to students, school principals, parents and teachers, as well as for the foreign language assessment system-level questionnaire. It concludes with the description of how the assessment results will be reported.

The full version of the foreign language assessment framework, that was published in 2021 and was used to guide instrument development, is available [here](#). An updated version will be made available by May 2027 on the PISA website.

## Annex D. PISA 2025 Expert Groups

### Science Expert Group

- Jonathan Osborne, Stanford University, United States (Chair)
- Doris Jorde, University of Oslo, Norway
- Joe Krajcik, Michigan State University, United States
- Yew Jin Lee, National Institute of Education, Singapore
- Knut Neuman, Leibniz-Institute for Maths & Science Education, Germany
- Nascira Ramia, Universidad San Francisco de Quito, Ecuador
- Russell Tytler, Deakin University, Australia
- Young Shin Park, Chosun University, South Korea

### Environmental Sciences Expert Group

- Peta White, Deakin University, Australia (Chair)
- Nicole Ardoin, Stanford University, United States
- Chris Eames, University of Waikato, New Zealand
- Martha Monroe, University of Florida, United States

### Extended Science Expert Group

- Marilar Jiménez Aleixandre, University de Santiago de Compostela, Spain
- Jan Alexis Nielsen, University of Copenhagen, Denmark
- Louise Archer, University College London, United Kingdom
- Maurice Cheng, University of Waikato, New Zealand/Hong Kong University, Hong Kong (China)
- Cary Sneider, Portland State University, United States
- Charles Tracy, Institute of Physics, United Kingdom
- Erin Furtak, University of Colorado, Boulder, United States

### Peripheral Science Expert Group

- Marianne Cutler, The Association for Science Education, United Kingdom
- Peter Finegold, The Royal Society, United Kingdom
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- Florence Le Hebel, Researcher, University of Lyon, France
- Gia Khatisashvili, National Assessment and Examinations Centre of Georgia, Georgia

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- Magnus Oskarsson, Mid Sweden University, Sweden
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## Oxford University Press

- Dave Leach (Project Director)
- Alexandra Tomescu (Science Content Lead) & Amie Hewish (Science Content Lead)
- Uma S (Assessment & Framework Design Consultant)
- Akansha Yadav (Item Test Developer)
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## Learning in the Digital World Expert Group

- Gautam Biswas, Vanderbilt University, United States
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- Sanna Järvelä, University of Oulu, Finland
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## Questionnaire Expert Group

- Therese Hopfenbeck (Chair), Professor in Educational Assessment and Director of the Assessment and Evaluation Research Centre, at the Melbourne Graduate School of Education, Australia
- Jonas Bertling, Education Testing Service
- David Kaplan, Professor, Educational Psychology, School of Education, University of Wisconsin-Madison, United States
- Nordin Abd Razak, Associate Professor at the School of Educational Studies at Universiti Sains, Malaysia
- Nina Jude, Professor of Education Science at Ruprecht-Karls-Universität Heidelberg, Germany
- Jonathan Osbourne (Chair of Science Expert Group)

This report presents the conceptual foundations of the OECD Programme for International Student Assessment (PISA), now in its ninth cycle of comprehensive and rigorous international surveys of student knowledge and skills that are essential for full participation in modern societies. As in previous cycles, the PISA 2025 assessment covered reading, mathematics and science, this time with a major focus on science. For the first time, PISA 2025 also assessed Learning in the Digital World as an innovative domain and included an optional assessment of English as a foreign language focused on reading, speaking and listening skills. This report includes the frameworks for assessing science and the innovative domain of Learning in the Digital World. These chapters describe the content knowledge and skills that students need to acquire in each domain, how each domain is assessed, and the context in which this knowledge and these skills are applied. The report also outlines the framework for the questionnaires distributed to students and schools. It concludes with references to the most recent frameworks for mathematics, reading and foreign language assessment.

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